Publication Report for scheduling Survey

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1 Introduction

This report is a companion document to the main report generated for the extracted information used in the survey of CP and Scheduling. This document is concerned with some of the summary statistics, and with data quality issues that nre highlighted for correction by the authors.

2 Data Quality

This section gives an overall overview of the works covered by the survey. We first look at all works, and consider which entries cannot be full analyzed. We consider the following status outcomes: no DOI, the bib entry does not give a DOI, this typically means that we cannot find the citation and reference counts for the work. A special case is the Thesis type, which typically do not have a DOI assigned by the university. Even entries with a DOI may not be covered, we distinguish entries that are covered by neither Crossref nor Scopus, or entries which are covered by one, but not the other. The OK status indicates that we can find the entry in all our sources.

Note that OpenCitations does not distinguish between a DOI that is not covered, and a DOI for which there are no references or citations. In both cases, an empty list is returned by the query.

We may be able to repair some of the entries by finding a DOI for entries which miss them, or by correcting a mistake in a DOI, where neither Crossref nor Scopus recognizes the entry. Note that the system responses are cached, and missing entries are not repeatedly queried by the system. This means that additions or corrections in the databases that occur after we first queried them for a specific entry are not automatically taken into account. It may be good practice to re-run all queries from time to time to reflect updates in the databases.

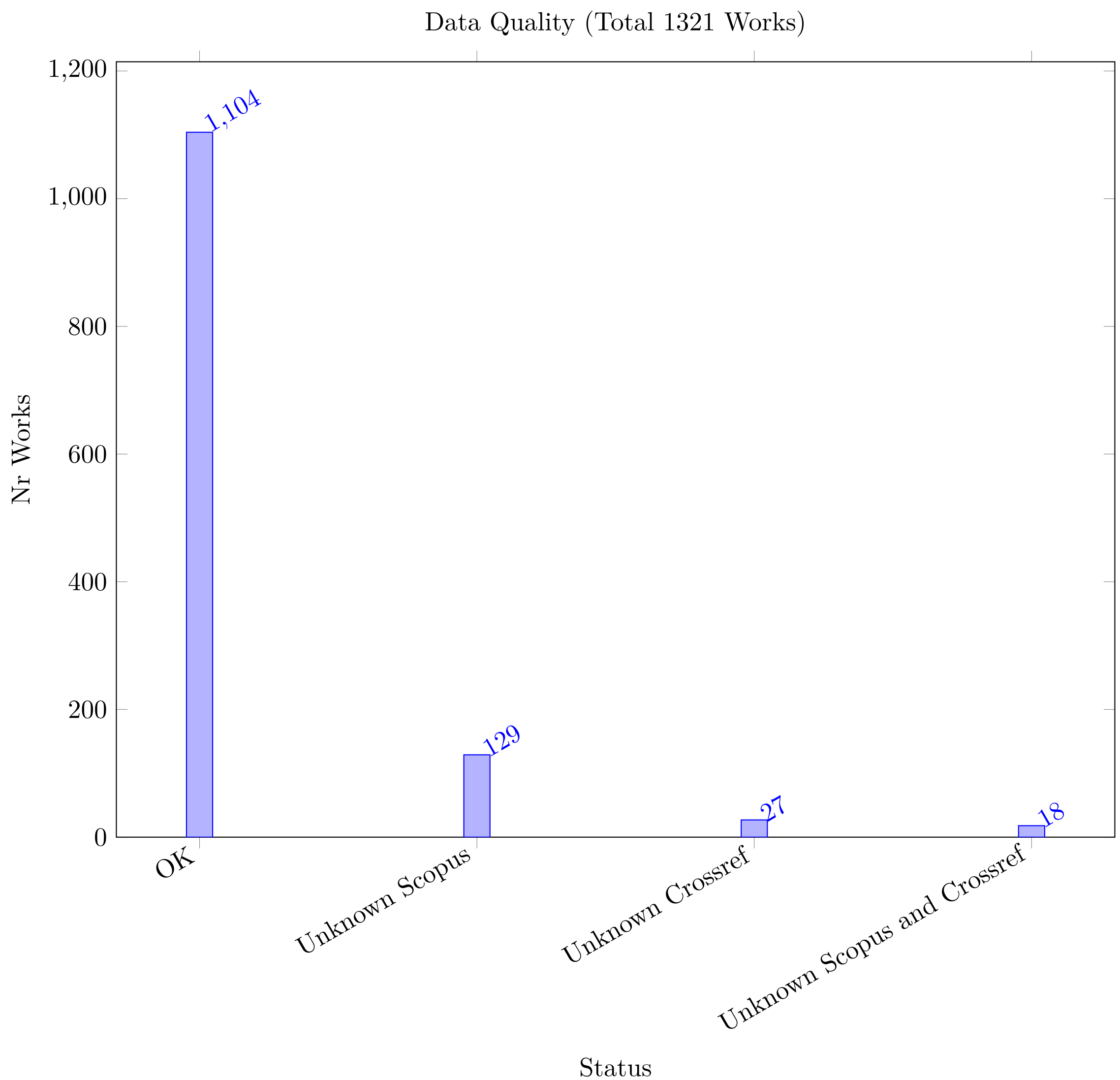

Data Quality (Total 1321 Works)
Nr Works
1,200
1,000
800
600
400
200
0
1,104
129
27
18
OK
Unknown Scopus
Unknown Crossref
Unknown Scopus and Crossref
Status

Table 1: Works Unknown to Crossref and Scopus

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
abs-2402-00459	10.48550/arxiv.2402.00459	Preprint	2024	0	0	0	null	0	NaN
Cloutier24	10.4230/lipics.cp.2024.7	CP	2024	0	0	0	null	0	NaN
Green24	10.4230/lipics.cp.2024.12	CP	2024	0	0	0	null	0	NaN
Le24	10.4230/lipics.cp.2024.18	CP	2024	0	0	0	null	0	NaN
Pucel24	10.4230/lipics.cp.2024.23	CP	2024	0	0	0	null	0	NaN
Verhaeghe24	10.4230/lipics.cp.2024.30	CP	2024	0	0	0	null	0	NaN
Cherif24	10.4230/lipics.cp.2024.34	CP	2024	0	0	0	null	0	NaN
abs-2305-19888	10.48550/arxiv.2305.19888	Preprint	2023	0	0	0	null	0	NaN
abs-2306-05747	10.48550/arxiv.2306.05747	Preprint	2023	0	0	0	null	0	NaN
abs-2312-13682	10.48550/arxiv.2312.13682	Preprint	2023	0	0	0	null	0	NaN
GokPTGO23	10.1007/s10479-022-04547-	ORJournal	2023	0	0	0	null	0	NaN
abs-2211-14492	10.48550/arxiv.2211.14492	Preprint	2022	0	0	0	null	0	NaN
OrnekOS20	10.1007/s12351-020-00563-	ORJournal	2022	0	0	0	null	0	NaN
OrnekO16	10.23055/ijietap.2016.23.1.1930	OtherJournal	2016	0	0	0	null	0	NaN
OddiRCS11	10.5591/978-1-57735-516-8/ijcai11-332	IJCAI	2011	0	0	0	null	0	NaN
AronssonBK09	10.4230/oasics.atmos.2009.2141	OtherConf	2009	0	0	0	null	0	NaN
KanetAG04	10.1201/9780203489802.ch47	Incoll	2004	0	0	0	null	0	NaN
BeckF98	10.1609/aimag.v19i4.1426	AIJournal	1998	0	0	0	null	0	NaN

Table 2: Works Unknown to Crossref

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
Antuori2025	10.4230/lipics.cp.2025.3	CP	2025	0	0	0	null	0	NaN
Bleukx2025	10.4230/lipics.cp.2025.6	CP	2025	0	0	0	null	0	NaN
Hebrard2025	10.4230/lipics.cp.2025.13	CP	2025	0	0	0	null	0	NaN
Le2025	10.4230/lipics.cp.2025.25	CP	2025	0	0	0	null	0	NaN
Luchterhand2025	10.4230/lipics.cp.2025.27	CP	2025	0	0	0	null	0	NaN
Sidorov2025	10.4230/lipics.cp.2025.35	CP	2025	0	0	0	null	0	NaN
Weidmann2025	10.4230/lipics.cp.2025.37	CP	2025	0	0	0	null	0	NaN
Prumm2025	10.4230/lipics.cp.2025.43	CP	2025	0	0	0	null	0	NaN
JuvinHHL23	10.4230/lipics.cp.2023.19	CP	2023	0	0	0	null	0	NaN
PovedaAA23	10.4230/lipics.cp.2023.31	CP	2023	0	0	0	null	0	NaN
AalianPG23	10.4230/lipics.cp.2023.6	CP	2023	0	0	0	null	0	NaN
KameugneFND23	10.4230/lipics.cp.2023.20	CP	2023	0	0	0	null	0	NaN
BoudreaultSLQ22	10.4230/lipics.cp.2022.10	CP	2022	1	0	0	null	1	100.00
PopovicCGNC22	10.4230/lipics.cp.2022.34	CP	2022	1	0	0	null	1	100.00
WinterMMW22	10.4230/lipics.cp.2022.41	CP	2022	0	0	0	null	0	NaN
ArmstrongGOS21	10.4230/lipics.cp.2021.16	CP	2021	1	0	1	null	1	100.00
AntuoriHHEN21	10.4230/lipics.cp.2021.14	CP	2021	0	0	1	null	1	100.00
KovacsTKSG21	10.4230/lipics.cp.2021.36	CP	2021	0	0	4	null	4	100.00
LacknerMMWW21	10.4230/lipics.cp.2021.37	CP	2021	0	0	3	null	3	100.00
WangB20	10.3233/faia200114	ECAI	2020	0	0	0	null	0	NaN
BarzegaranZP20	10.4230/oasics.fog-iot.2020.3	OtherConf	2020	0	0	0	null	0	NaN
BridiLBBM16	10.3233/978-1-61499-672-9-1598	ECAI	2016	0	0	0	null	0	NaN
BartakV15	10.5220/0005215701190130	OtherConf	2015	0	0	1	null	1	100.00
TranB12	10.3233/978-1-61499-098-7-774	ECAI	2012	0	0	30	null	30	100.00

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
PacinoH11	10.5591/978-1-57735-516-8/ijcai11-333	IJCAI	2011	0	0	0	null	0	NaN
OddiRC10	10.3233/978-1-60750-606-5-967	ECAI	2010	0	0	2	null	2	100.00
Hunsberger08	10.3233/978-1-58603-891-5-553	ECAI	2008	0	0	1	null	1	100.00

Table 3: Works Unknown to Scopus

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
Euler2024	10.1007/978-3-031-60597-0_17	CPAIOR	2024	0	0	0	null	0	NaN
Barral2024	10.1007/978-3-031-60597-0_3	CPAIOR	2024	0	0	0	null	0	NaN
Thomas2024	10.1007/978-3-031-60599-4_13	CPAIOR	2024	0	0	0	null	0	NaN
Houten2024	10.1007/978-3-031-60599-4_15	CPAIOR	2024	0	0	0	null	0	NaN
Infantes2024	10.1007/978-3-031-60597-0_21	CPAIOR	2024	0	0	0	null	0	NaN
Zhang2024	10.1049/tje2.12368	OtherJournal	2024	0	1	0	null	1	100.00
Caballero23	10.1007/s10601-023-09357-0	Constraints	2023	0	0	0	null	0	NaN
NaderiBZ23	10.2139/ssrn.4494381	Preprint	2023	0	0	0	null	0	NaN
GunerGSKD23	10.1080/00207543.2023.2226772	OtherJournal	2023	9	3	0	null	9	100.00
IklassovMR023	10.24963/ijcai.2023/594	IJCAI	2023	0	0	0	null	0	NaN
Lyons2023	10.3390/analytics2030036	OtherJournal	2023	1	0	0	null	1	100.00
Bley2023	10.1007/978-3-031-24907-5_68	OtherConf	2023	0	0	0	null	0	NaN
Akan2023	10.33714/masteb.1324266	OtherJournal	2023	0	0	0	null	0	NaN
Abreu2023	10.1007/978-3-031-36121-0_9	OtherConf	2023	0	0	0	null	0	NaN
HebrardALLCMR22	10.24963/ijcai.2022/643	IJCAI	2022	0	0	0	null	0	NaN
NaderiBZ22	10.2139/ssrn.4140716	Preprint	2022	1	0	0	null	1	100.00
JuvinHL22	10.2139/ssrn.4068164	Preprint	2022	0	0	0	null	0	NaN
NaderiR22	10.1287/ijoo.2021.0056	ORJournal	2022	11	7	0	null	11	100.00
KotaryFH22	10.1609/aaai.v36i7.20685	AAAI	2022	5	2	0	null	5	100.00
Ouellet2022	10.1609/aaai.v36i4.20296	AAAI	2022	1	0	0	null	1	100.00
QinWSLS21	10.1109/tase.2019.2947398	OtherJournal	2021	35	19	0	null	35	100.00
GeibingerMM21	10.1609/aaai.v35i7.16789	AAAI	2021	1	1	0	null	1	100.00
KletzanderMH21	10.1609/aaai.v35i13.17408	AAAI	2021	3	2	0	null	3	100.00
Pinarbasi21	10.1080/0305215x.2021.1921171	OtherJournal	2021	9	6	0	null	9	100.00
Strak2021	10.5937/tehnika2102239s	OtherJournal	2021	0	0	0	null	0	NaN
Eiter2021	10.24963/kr.2021/27	OtherConf	2021	7	7	0	null	7	100.00
Rodler2021	10.24963/kr.2021/72	OtherConf	2021	1	1	0	null	1	100.00
GodetLHS20	10.1609/aaai.v34i02.5510	AAAI	2020	1	1	0	null	1	100.00
FallahiAC20	10.1504/ijams.2020.10026882	OtherJournal	2020	0	0	0	null	0	NaN
KletzanderM20	10.1609/icaps.v30i1.6688	ICAPS	2020	2	2	0	null	2	100.00
AbidinK20	10.1016/j.cor.2020.105069	ORJournal	2020	18	14	0	null	18	100.00
Danzinger2020	10.1609/icaps.v30i1.6681	ICAPS	2020	2	2	0	null	2	100.00
NishikawaSTT19	10.15803/ijnc.9.2_131	OtherJournal	2019	3	3	0	null	3	100.00
BlazewiczEP19	10.1007/978-3-319-99849-7	Incoll	2019	51	38	0	null	51	100.00
SenderovichBB19	10.1609/icaps.v29i1.3504	ICAPS	2019	5	2	0	null	5	100.00
PinarbasiAY19	10.1108/aa-12-2018-0262	OtherJournal	2019	21	18	0	null	21	100.00
AlakaPY19	10.1007/s00500-019-04294-8	OtherJournal	2019	19	0	0	null	19	100.00
PachecoPR19	10.24963/ijcai.2019/161	IJCAI	2019	1	1	0	null	1	100.00
BhatnagarKL19	10.24963/ijcai.2019/803	IJCAI	2019	2	1	0	null	2	100.00
RiahiNS018	10.1609/icaps.v28i1.13895	ICAPS	2018	4	4	0	null	4	100.00
AgussurjaKL18	10.1609/aaai.v32i1.12086	AAAI	2018	7	4	0	null	7	100.00

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
TranVNB17a	10.24963/ijcai.2017/726	IJCAI	2017	1	1	0	null	1	100.00
Laborie2017	10.1609/icaps.v27i1.13844	ICAPS	2017	2	2	0	null	2	100.00
Gonzlez2017	10.1609/icaps.v27i1.13809	ICAPS	2017	14	12	0	null	14	100.00
Bonfietti16	10.3233/ia-160095	AIJournal	2016	0	0	0	null	0	NaN
TranDRFWOVB16	10.1609/socs.v7i1.18390	OtherConf	2016	15	9	0	null	15	100.00
FrankDT16	10.1609/icaps.v26i1.13780	ICAPS	2016	5	5	0	null	5	100.00
KinsellaS0OS16	10.1609/aaai.v30i2.19079	AAAI	2016	2	2	0	null	2	100.00
Abdul-Niby2016	10.48084/etasr.627	OtherJournal	2016	8	4	0	null	8	100.00
Siala15	10.1007/s10601-015-9213-y	Constraints	2015	4	3	0	null	4	100.00
Kameugne15	10.1007/s10601-015-9227-5	Constraints	2015	0	0	0	null	0	NaN
LimBTBB15a	10.1609/aaai.v29i1.9236	AAAI	2015	4	3	0	null	4	100.00
Oliveira2015	10.14807/ijmp.v6i1.262	OtherJournal	2015	2	1	0	null	2	100.00
Bzdyra2015	10.4028/www.scientific.net/amm.791.70	OtherJournal	2015	5	5	0	null	5	100.00
FriedrichFMRSST14	10.1007/978-3-319-28697-6_23	OtherConf	2014	3	3	0	null	3	100.00
LipovetzkyBPS14	10.1609/icaps.v24i1.13666	ICAPS	2014	6	5	0	null	6	100.00
LudwigKRBMS14	10.1609/aaai.v28i2.19030	AAAI	2014	1	1	0	null	1	100.00
ChunS14	10.1609/aaai.v28i2.19013	AAAI	2014	4	3	0	null	4	100.00
Silva2014	10.1590/2238-1031.jtl.v8n4a9	OtherJournal	2014	2	2	0	null	2	100.00
Levine2014	10.1609/icaps.v24i1.13672	ICAPS	2014	20	20	0	null	20	100.00
Lozano2014	10.1145/2666357.2597815	OtherJournal	2014	3	2	0	null	3	100.00
Banaszak2014	10.1515/fman-2015-0014	OtherJournal	2014	9	8	0	null	9	100.00
BonfiettiLM13	10.1609/icaps.v23i1.13608	ICAPS	2013	1	1	0	null	1	100.00
LombardiM13	10.1609/icaps.v23i1.13580	ICAPS	2013	3	0	0	null	3	100.00
TranTDB13	10.1609/icaps.v23i1.13552	ICAPS	2013	2	2	0	null	2	100.00
MalapertCGJLR13	10.1609/icaps.v23i1.13575	ICAPS	2013	0	0	0	null	0	NaN
Zoulfaghari2013	10.4018/jaec.2013040103	OtherJournal	2013	5	5	0	null	5	100.00
Guimarans2013	10.4018/978-1-4666-2461-0.ch007	Inbook	2013	1	1	0	null	1	100.00
Janosikova2013	10.26552/com.c.2013.1.39-43	OtherJournal	2013	0	0	0	null	0	NaN
Kelareva2012	10.1609/icaps.v22i1.13494	ICAPS	2012	16	14	0	null	16	100.00
BajestaniB11	10.1609/icaps.v21i1.13450	ICAPS	2011	2	2	0	null	2	100.00
Milano11	10.1002/9780470400531.eorms0473	Inbook	2011	0	0	0	null	0	NaN
Lizarralde2011	10.3917/proj.007.0089	OtherJournal	2011	1	1	0	null	1	100.00
Laborie2011	10.1007/978-3-642-23592-4_6	Inbook	2011	3	2	0	null	3	100.00
Baptiste09	10.1007/978-3-642-04244-7_1	CP	2009	0	0	0	null	0	NaN
MonetteDH09	10.1609/icaps.v19i1.13356	ICAPS	2009	10	10	0	null	10	100.00
Lorterapong2009	10.4203/ccp.74.8	OtherConf	2009	2	2	0	null	2	100.00
MercierH08	10.1287/ijoc.1070.0226	InformsJC	2008	35	33	0	null	35	100.00
AggounMV08	10.1007/978-0-387-74759-0_396	Inbook	2008	0	0	0	null	0	NaN
Terashima-Marn2008a	10.1007/978-3-540-88636-5_39	OtherConf	2008	5	5	0	null	5	100.00
Banaszak2008	10.7494/dmms.2008.2.2.5	OtherJournal	2008	4	4	0	null	4	100.00
Limtanyakul07	10.1007/978-3-540-77903-2_65	OtherConf	2007	2	2	0	null	2	100.00
2007	10.1007/978-3-540-32220-7_13	Inbook	2007	0	0	0	null	0	NaN
NeronABCDD06	10.1007/978-0-387-33768-5_7	Inbook	2006	3	3	0	null	3	100.00
RussellU06	10.1016/j.cor.2004.09.029	ORJournal	2006	22	22	0	null	22	100.00
Trilling2006	10.3182/20060517-3-fr-2903.00340	OtherJournal	2006	31	25	0	null	31	100.00
OddiPCC05	10.1007/0-387-27744-7_7	OtherConf	2005	3	3	0	null	3	100.00
Bartak2005	10.4018/978-1-59140-450-7.ch010	Inbook	2005	4	3	0	null	4	100.00
Vazacopoulos2005	10.1007/0-387-26281-4_12	Inbook	2005	3	3	0	null	3	100.00
Zhang2005	10.1109/icmlc.2004.1380769	OtherConf	2005	1	0	0	null	1	100.00
DannaP04	10.1007/978-1-4419-8917-8_2	Inbook	2004	2	2	0	null	2	100.00
AjiliW04	10.1007/978-1-4419-8917-8_6	Inbook	2004	4	4	0	null	4	100.00
AggounV04	10.1007/978-3-540-24734-0_15	Inbook	2004	7	7	0	null	7	100.00
HenzMT04	10.1016/s0377-2217(03)00101-2	EJOR	2004	49	47	0	null	49	100.00

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
Tsang03	10.1023/a:1024016929283	OtherJournal	2003	1	0	0	null	1	100.00
DomdorfPH03	10.1007/978-3-642-18965-4_31	Inbook	2003	0	0	0	null	0	NaN
Apt03	10.1017/cbo9780511615320	Background	2003	406	374	0	null	406	100.00
Sadykov2003	10.2139/ssrn.988640	Preprint	2003	3	3	0	null	3	100.00
Timpe2003	10.1007/978-3-662-05607-3_5	Inbook	2003	2	2	0	null	2	100.00
ElkhyariGJ02	10.1007/3-540-46135-3_49	CP	2002	1	1	0	null	1	100.00
ZhuS02	10.1007/3-540-47961-9_69	OtherConf	2002	0	0	0	null	0	NaN
MilanoORT02	10.1287/ijoc.14.4.387.2830	InformsJC	2002	14	14	0	null	14	100.00
Hooker02	10.1287/ijoc.14.4.295.2828	InformsJC	2002	101	93	0	null	101	100.00
Hentenryck02	10.1287/ijoc.14.4.345.2826	Background	2002	53	50	0	null	53	100.00
EastonNT02	10.1007/978-3-540-45157-0_6	OtherConf	2002	51	50	0	null	51	100.00
Varnier2002	10.1109/icsmc.1996.561432	OtherConf	2002	0	0	0	null	0	NaN
Richard2002	10.1109/etfa.1995.496763	OtherConf	2002	4	4	0	null	4	100.00
Petith2002	10.1109/etfa.1995.496657	OtherConf	2002	0	0	0	null	0	NaN
BaptistePN01	10.1007/978-1-4615-1479-4	Book	2001	313	302	0	null	313	100.00
BosiM2001	10.1002/1097-024x(200101)31:1<17::aid-spe355>3.0.co;2-l	OtherJournal	2001	3	3	0	null	3	100.00
Henz01	10.1287/opre.49.1.163.11193	ORJournal	2001	68	68	0	null	68	100.00
Rgin2001	10.1090/dimacs/057/07	Inbook	2001	29	29	0	null	29	100.00
Baptiste2001	10.1007/978-1-4615-1479-4_2	Inbook	2001	1	1	0	null	1	100.00
LopezAKYG00	10.1016/s0947-3580(00)71114-9	OtherJournal	2000	0	0	0	null	0	NaN
Hooker00	10.1002/9781118033036	Book	2000	193	186	0	null	193	100.00
Simonis99	10.1007/3-540-45406-3_6	OtherConf	1999	6	5	0	null	6	100.00
DorndorfPH99	10.1007/978-3-642-58409-1_35	OtherConf	1999	0	0	0	null	0	NaN
DorndorfHP99	10.1007/978-1-4615-5533-9_10	Inbook	1999	17	18	0	null	18	100.00
CarlssonKA99	10.1007/3-540-49201-1_23	OtherConf	1999	1	1	0	null	1	100.00
PembertonG98	10.1090/dimacs/057/06	OtherConf	1998	30	0	0	null	30	100.00
MarriottS98	10.7551/mitpress/5625.001.0001	Background	1998	423	423	0	null	423	100.00
BeckDDF98	10.1002/(sici)1099-1425(199808)1:2<89::aid-jos9>3.0.co;2-h	OtherJournal	1998	9	8	0	null	9	100.00
Jaffar1998	10.1093/oso/9780198537922.003.0012	Inbook	1998	3	3	0	null	3	100.00
Mesghouni1997	10.1007/978-0-387-35086-8_12	Inbook	1997	2	2	0	null	2	100.00
Simonis95a	10.1007/3-540-60794-3_11	OtherConf	1995	1	1	0	null	1	100.00
Schiex1994	10.1142/s0218213094000108	OtherJournal	1994	66	66	0	null	66	100.00
Freuder1994	10.7551/mitpress/2122.001.0001	Book	1994	24	22	0	null	24	100.00
Icmeli1993	10.1108/01443579310046454	OtherJournal	1993	100	99	0	null	100	100.00
Barber1993	10.1145/152947.152955	OtherJournal	1993	13	13	0	null	13	100.00
BaptisteLV92	10.1109/robot.1992.220195	OtherConf	1992	16	11	0	null	16	100.00
Demeulemeester1992	10.1287/mnsc.38.12.1803	ORJournal	1992	394	387	0	null	394	100.00
Elmaghraby1992	10.1287/mnsc.38.9.1245	ORJournal	1992	122	121	0	null	122	100.00
CarlierP90	10.1007/bf03543071	Background	1990	112	114	0	null	114	100.00
CarlierP89	10.1287/mnsc.35.2.164	Background	1989	537	524	0	null	537	100.00
PritskerWW69	10.1287/mnsc.16.1.93	Background	1969	547	518	0	null	547	100.00

2.1 Range of Citation Counts

We get citation counts for the works included in the survey from different sources. OpenCitations provides the set of papers citing a reference, but only if both have DOIs. Crossref gives a count of how many papers cite a reference, they include some papers without DOI. Scopus gives a citation count, but does not give access to the actual citations. In this table we show the works with the largest range of citation count, excluding all background works. A typical issue is that one source does not cover the work, and has a zero count. An alternative is where papers with many citations give a slightly different count depending on which links are included in their database.

The results seem to indicate the using multiple sources is required, to avoid leaving out works that are not covered by one specific source. Note that the WoS numbers are only present for a few works, we show them, but do not include them in computing range.

Table 4: Works with largest Range of Citation Counts

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
Demeulemeester1992	10.1287/mnsc.38.12.1803	ORJournal	1992	394	387	0	null	394	100.00
BaptistePN01	10.1007/978-1-4615-1479-4	Book	2001	313	302	0	null	313	100.00
Hooker00	10.1002/9781118033036	Book	2000	193	186	0	null	193	100.00
BensanaLV99	10.1023/a:1026488509554	Constraints	1999	107	0	150	null	150	100.00
JainM99	10.1016/s0377-2217(98)00113-1	EJOR	1999	532	503	630	null	127	20.16
Elmaghraby1992	10.1287/mnsc.38.9.1245	ORJournal	1992	122	121	0	null	122	100.00
SakkoutW00	10.1023/a:1009856210543	Constraints	2000	77	0	105	null	105	100.00
Hooker02	10.1287/ijoc.14.4.295.2828	InformsJC	2002	101	93	0	null	101	100.00
Icmeli1993	10.1108/01443579310046454	OtherJournal	1993	100	99	0	null	100	100.00
Smith-Miles2009	10.1145/1456650.1456656	OtherJournal	2009	325	307	395	null	88	22.28
BaptistePN99	10.1023/a:1018995000688	ORJournal	1999	77	0	85	null	85	100.00
MintonJPL92	10.1016/0004-3702(92)90007-k	AIJournal	1992	440	440	525	null	85	16.19
ArtiguesDN08	10.1002/9780470611227	Book	2008	73	60	0	null	73	100.00
Younes2003	10.1613/jair.1136	OtherJournal	2003	56	55	128	null	73	57.03
OhrimenkoSC09	10.1007/s10601-008-9064-x	Constraints	2009	137	128	198	null	70	35.35
MengZRZL20	10.1016/j.cie.2020.106347	OtherJournal	2020	201	133	152	null	68	33.83
Henz01	10.1287/opre.49.1.163.11193	ORJournal	2001	68	68	0	null	68	100.00
RodosekWH99	10.1023/a:1018904229454	ORJournal	1999	54	0	67	null	67	100.00
Schiex1994	10.1142/s0218213094000108	OtherJournal	1994	66	66	0	null	66	100.00
BaptisteP00	10.1023/a:1009822502231	Constraints	2000	49	0	62	null	62	100.00
BlazewiczDP96	10.1016/0377-2217(95)00362-2	EJOR	1996	362	357	412	null	55	13.35
BeldiceanuC94	10.1016/0895-7177(94)90127-9	OtherJournal	1994	178	169	223	null	54	24.22
BlazewiczEP19	10.1007/978-3-319-99849-7	Incoll	2019	51	38	0	null	51	100.00
EastonNT02	10.1007/978-3-540-45157-0_6	OtherConf	2002	51	50	0	null	51	100.00
NuijtenP98	10.1023/a:1009687210594	OtherJournal	1998	41	0	50	null	50	100.00
Wallace96	10.1007/bf00143881	Constraints	1996	93	89	138	null	49	35.51
HenzMT04	10.1016/s0377-2217(03)00101-2	EJOR	2004	49	47	0	null	49	100.00
Yang2000	10.1109/72.839016	OtherJournal	2000	37	0	48	null	48	100.00
Ham18	10.1016/j.trc.2018.03.025	OtherJournal	2018	238	192	197	null	46	19.33
Laborie03	10.1016/s0004-3702(02)00362-4	AIJournal	2003	133	129	175	null	46	26.29
BeckR03	10.1023/a:1021849405707	ORJournal	2003	32	0	45	null	45	100.00
AchterbergBKW08	10.1007/978-3-540-68155-7_4	CPAIOR	2008	85	80	125	null	45	36.00
Fisher1985	10.1287/inte.15.2.10	OtherJournal	1985	484	473	517	null	44	8.51
HookerO03	10.1007/s10107-003-0375-9	OtherJournal	2003	375	333	371	null	42	11.20
He2019	10.1007/s10845-019-01518-4	OtherJournal	2019	42	0	36	null	42	100.00
Laborie09	10.1007/978-3-642-01929-6_12	CPAIOR	2009	57	52	91	null	39	42.86
Michel2012	10.1007/978-3-642-29828-8_15	CPAIOR	2012	49	48	87	null	39	44.83
JainG01	10.1287/ijoc.13.4.258.9733	InformsJC	2001	294	284	321	null	37	11.53
Gent1996	10.1007/3-540-61551-2_74	CP	1996	57	56	93	null	37	39.78
QinWSLS21	10.1109/tase.2019.2947398	OtherJournal	2021	35	19	0	null	35	100.00
MercierH08	10.1287/ijoc.1070.0226	InformsJC	2008	35	33	0	null	35	100.00
SadehF96	10.1016/0004-3702(95)00098-4	AIJournal	1996	97	97	131	null	34	25.95
KendallKRU10	10.1016/j.cor.2009.05.013	ORJournal	2010	196	186	220	161	34	15.45
PerronSF04	10.1007/978-3-540-30201-8_35	CP	2004	35	34	67	null	33	49.25
Ham20	10.1080/00207543.2019.1709671	OtherJournal	2020	52	19	37	null	33	63.46
Zhu2006	10.1287/ijoc.1040.0121	InformsJC	2006	88	85	118	null	33	27.97
SchildW00	10.1023/a:1009804226473	Constraints	2000	24	0	32	null	32	100.00

Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
Trilling2006	10.3182/20060517-3-fr-2903.00340	OtherJournal	2006	31	25	0	null	31	100.00
PembertonG98	10.1090/dimacs/057/06	OtherConf	1998	30	0	0	null	30	100.00
TranB12	10.3233/978-1-61499-098-7-774	ECAI	2012	0	0	30	null	30	100.00

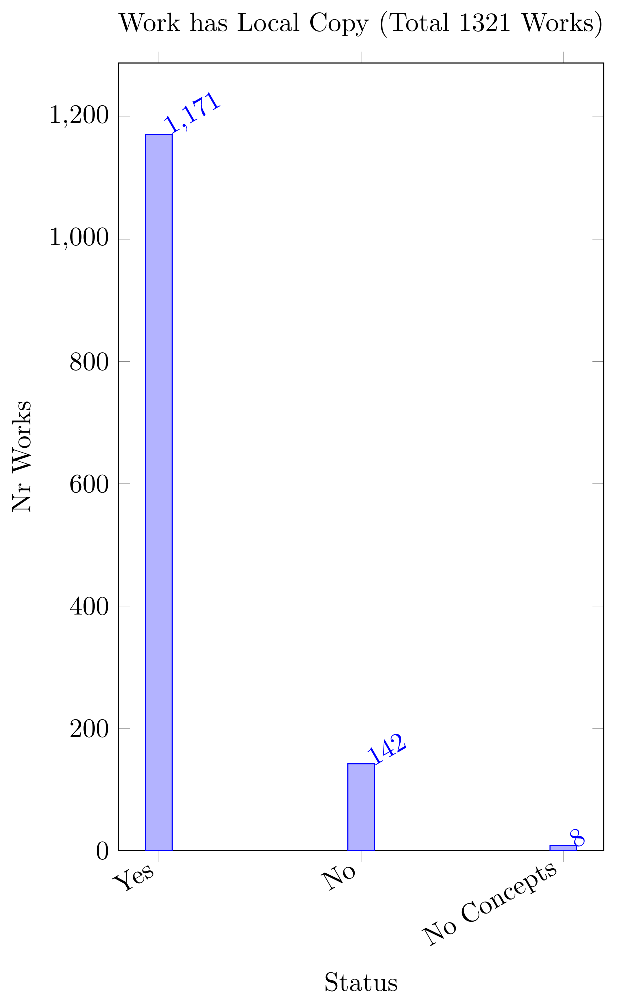
We only have Web of Science data in a few bibtex entries, we here try to evaluate their citation numbers on those bib entries which are from WoS.

Table 5: Works with WoS Citation Counts

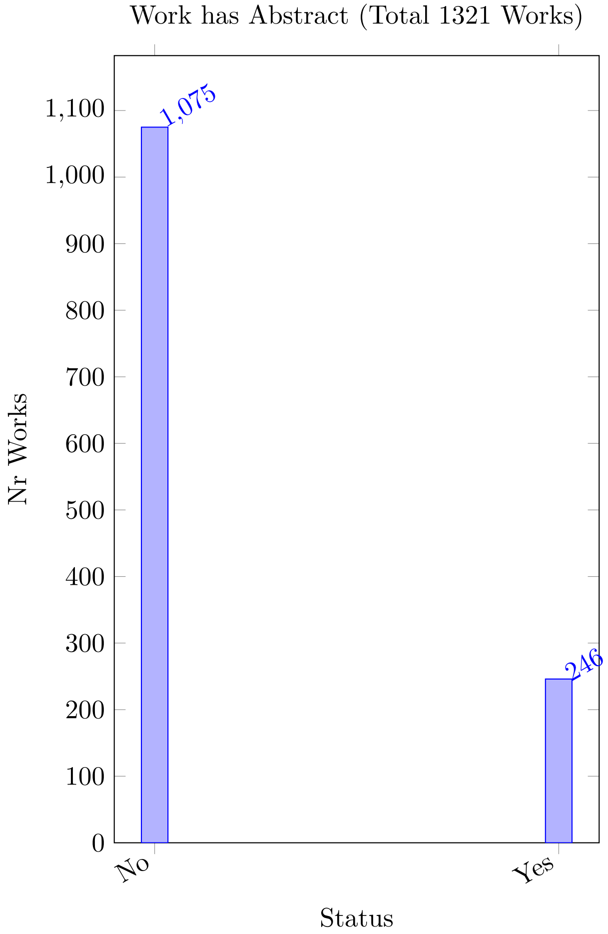
Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage
KendallKRU10	10.1016/j.cor.2009.05.013	ORJournal	2010	196	186	220	161	34	15.45
MeskensDL13	10.1016/j.dss.2012.10.019	OtherJournal	2013	109	102	116	103	14	12.07
RasmussenT07	10.1016/j.ejor.2005.10.063	EJOR	2007	62	62	71	53	9	12.68
Ribeiro12	10.1111/j.1475-3995.2011.00819.x	OtherJournal	2012	56	52	54	41	4	7.14
ElfJR03	10.1016/s0167-6377(03)00025-7	OtherJournal	2003	45	41	45	34	4	8.89
Trick03	10.1007/978-3-540-45157-0_4	OtherConf	2003	22	24	39	34	17	43.59
RasmussenT06	10.1007/11757375_15	CPAIOR	2006	11	12	19	11	8	42.11
FelizariAL09	10.1016/s1570-7946(05)80013-6	OtherConf	2009	7	7	12	1	5	41.67
MagataoAN05	10.1016/s1570-7946(05)80013-6	OtherConf	2005	7	7	12	12	5	41.67
RasmussenT09	10.1007/s10479-008-0384-4	ORJournal	2009	10	9	9	8	1	10.00
LiuLH19a	10.5220/0007252300290039	OtherConf	2019	6	3	4	4	3	50.00
GhandehariK22	10.1016/j.apm.2022.01.001	OtherJournal	2022	6	4	4	3	2	33.33
Trick11	10.1007/978-1-4419-1644-0_15	Incoll	2011	3	2	5	5	3	60.00
BulckG22	10.1007/s10951-021-00717-3	OtherJournal	2022	5	3	3	3	2	40.00
SuCC13	10.1016/j.cie.2013.02.021	OtherJournal	2013	2	2	4	1	2	50.00
ZengM12	10.1016/j.cor.2011.10.004	ORJournal	2012	4	3	4	3	1	25.00
Perron05	10.1007/11564751_67	CP	2005	1	1	2	1	1	50.00
LiuLH18	10.1007/978-3-030-05918-7_7	OtherConf	2018	2	2	1	1	1	50.00
MeskensDHG11	n/a	OtherConf	2011	0	0	0	null	0	NaN
NaqviAIAAA22	10.32604/cmc.2022.019653	OtherJournal	2022	0	0	0	0	0	NaN
KonowalenkoMM19	10.1109/tla.2019.8932340	OtherJournal	2019	0	0	0	0	0	NaN

2.2 Local Copies

The tool relies on local pdf copies of works to perform a detailed analysis of the content of the work. We have collected our own private copies of works for that purpose. The following plot shows how many entries do not have a local copy, or which do not extract any concepts from the local copy. A detailed list of all missing entries is given in the main report. Note that in some cases we use an open access version of the work, which might differ slightly from the published version.


Work has Local Copy (Total 1321 Works)
Nr Works
1,200
1,000
800
600
400
200
0
1,171
142
8
Yes
No
No Concepts
Status

2.3 Presence of Abstracts



2.4 Orphan Files

The following list shows entries for which we have a pdf file in the works directory, but the name of hte file does not match any key in the bibliography. These orphans should be resolved, either by correcting the name, or adding a bib entry for the work, or by removing the file, if it is not required.

If there are no files listed, then all pdf files in the works directory correspond to a bib entry, and no clean-up is required.

Table 6: Orphan Files

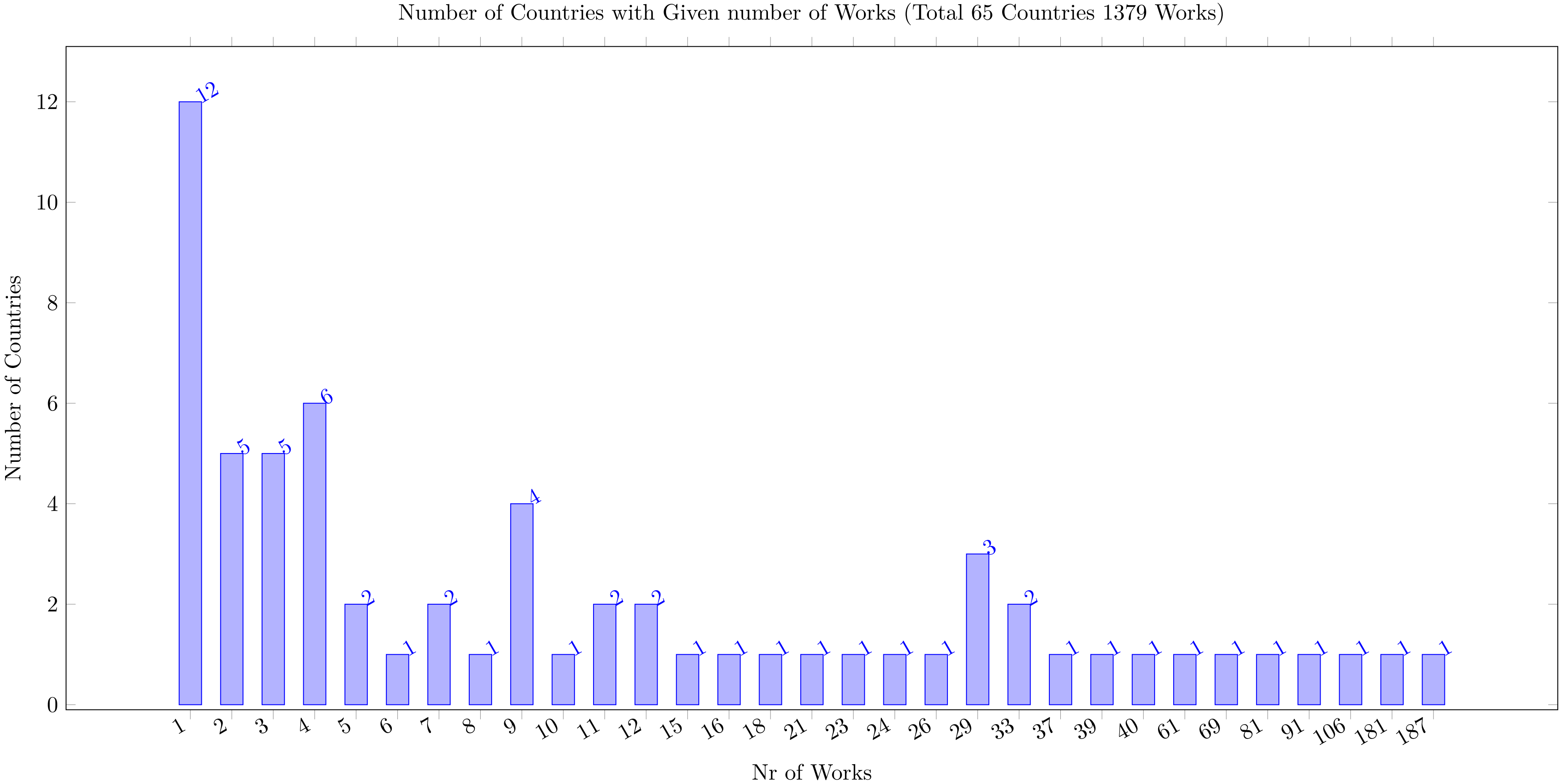
Key	File

2.5 Missing Publisher

Table 7: Missing Publisher

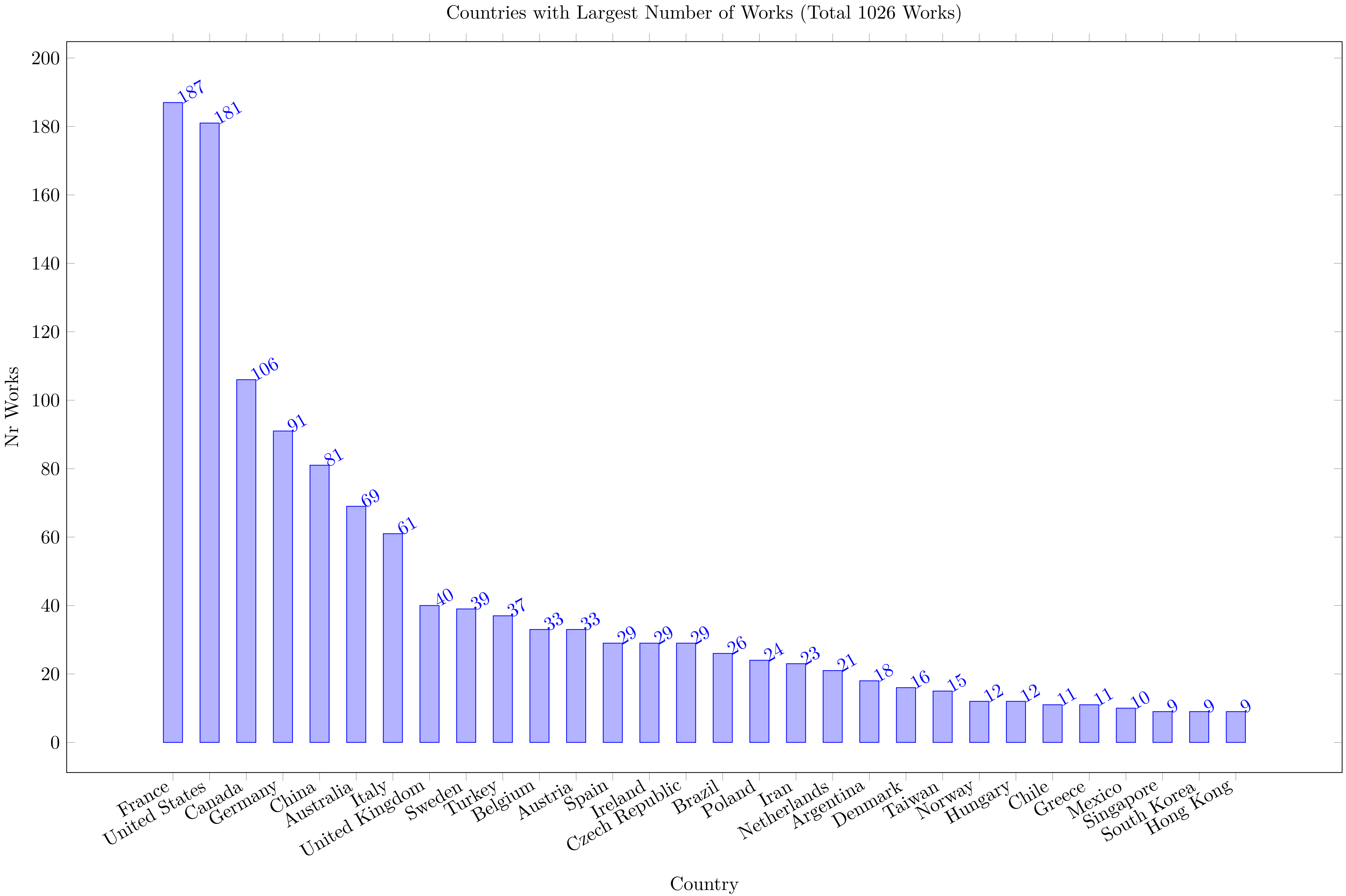
Key	DOI	Source Group	Year	Nr Citations	Crossref Citations	Scopus Citations	WoS Citations	Range Citations	Range Percentage

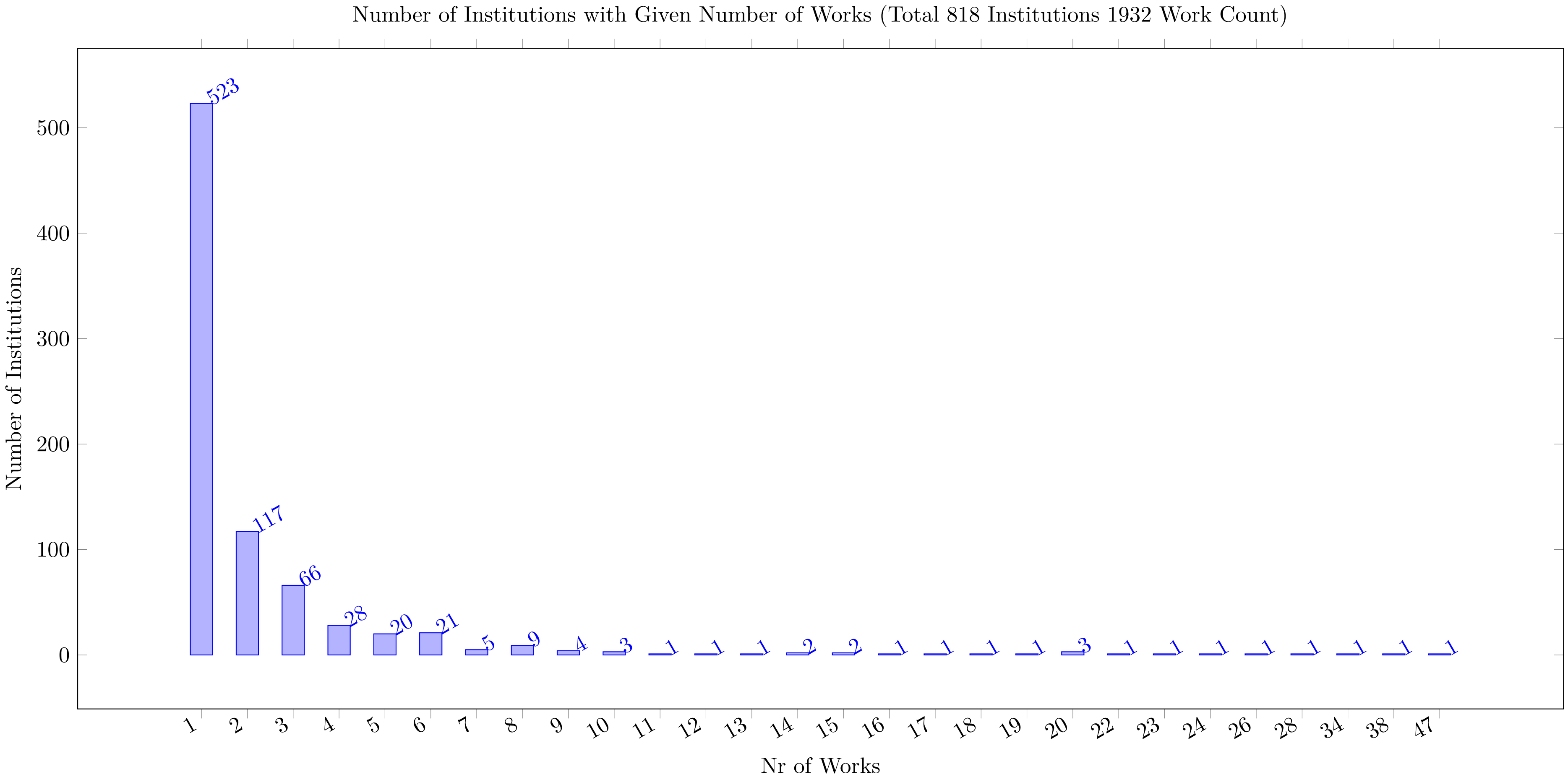
3 Works by Location



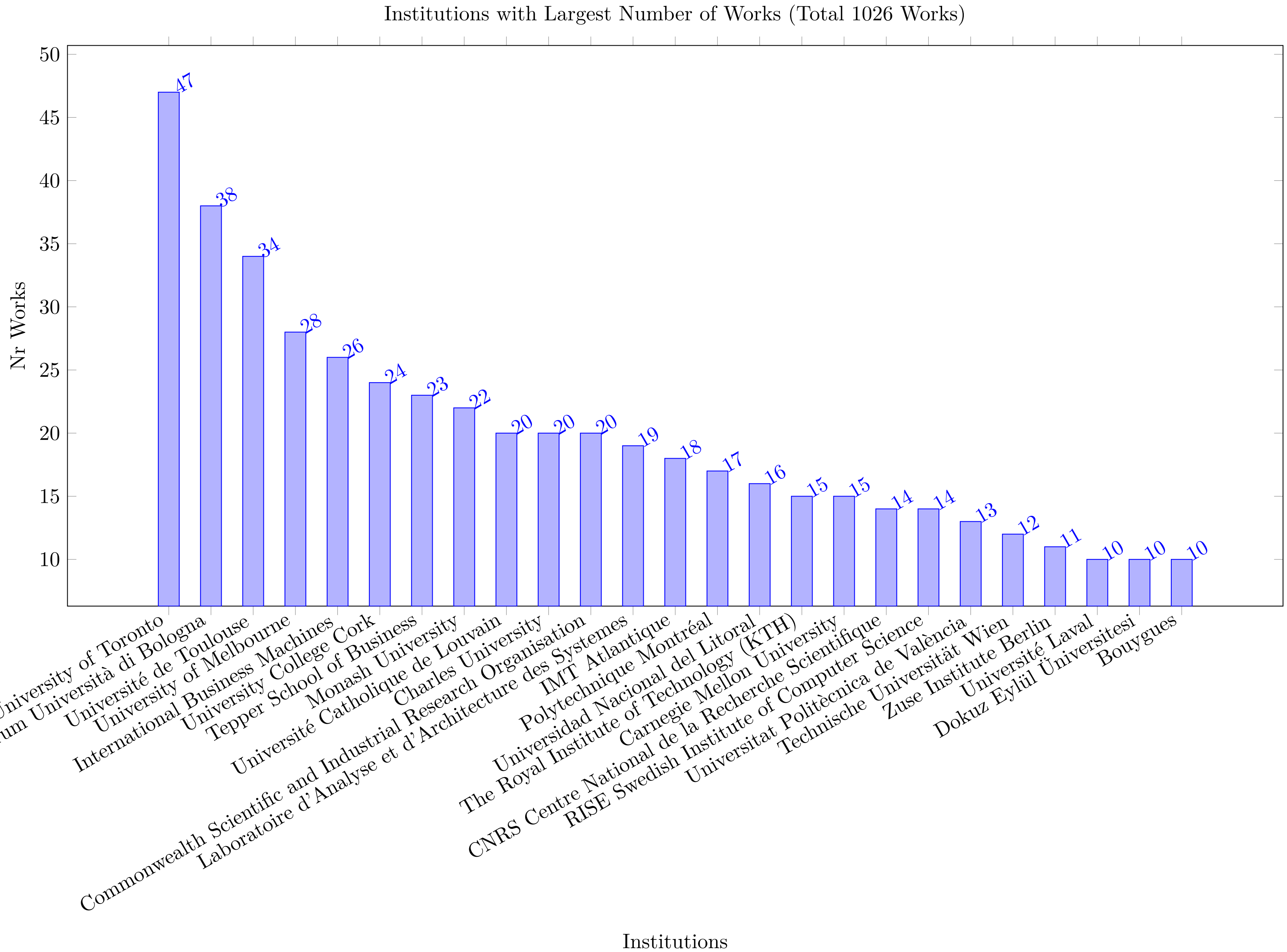
This section analyzes papers by affiliation, which is given by the Scopus data only. Only works which are covered by Scopus are included. We first present the number of papers by country. A paper is counted in this analysis (once), if at least one of the affiliations is from the country. Multiple affiliations from the same country only count once. The 30 countries with the largest counts are shown.

Note that one work will be counted for multiple countries, if the affiliations are from different countries. So the sum of the bar heights typically exceeds the total number of works considered.


Countries with Largest Number of Works (Total 1026 Works)
Nr Works
Country
0
20
40
60
80
100
120
140
160
180
200
187
181
106
91
81
69
61
40
39
37
33
33
29
29
29
26
24
23
21
18
16
15
12
12
11
11
10
9
9
9
France
United States
Canada
Germany
China
Australia
Italy
United Kingdom
Sweden
Turkey
Belgium
Austria
Spain
Ireland
Czech Republic
Brazil
Poland
Iran
Netherlands
Argentina
Denmark
Taiwan
Norway
Hungary
Chile
Greece
Mexico
Singapore
South Korea
Hong Kong

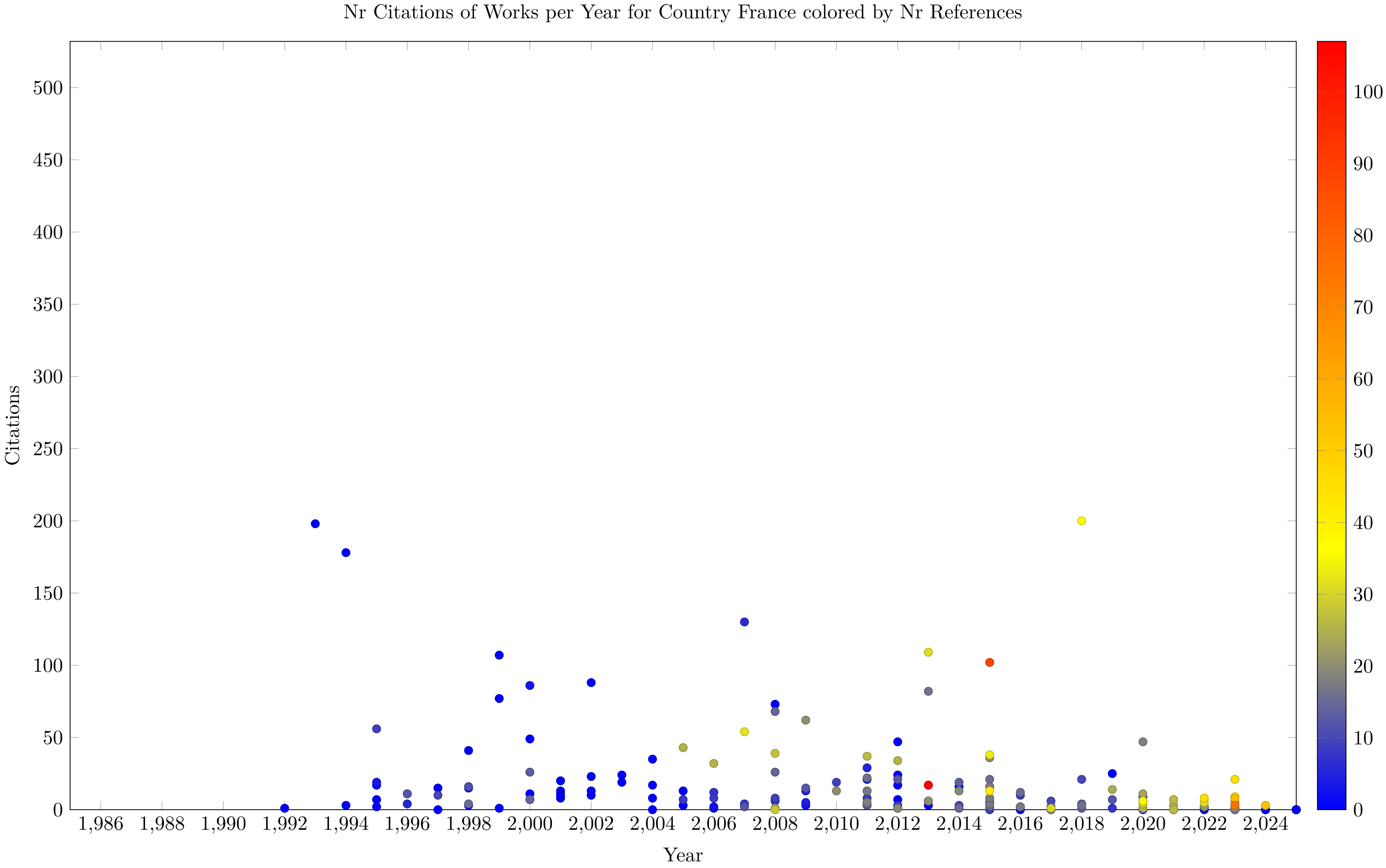


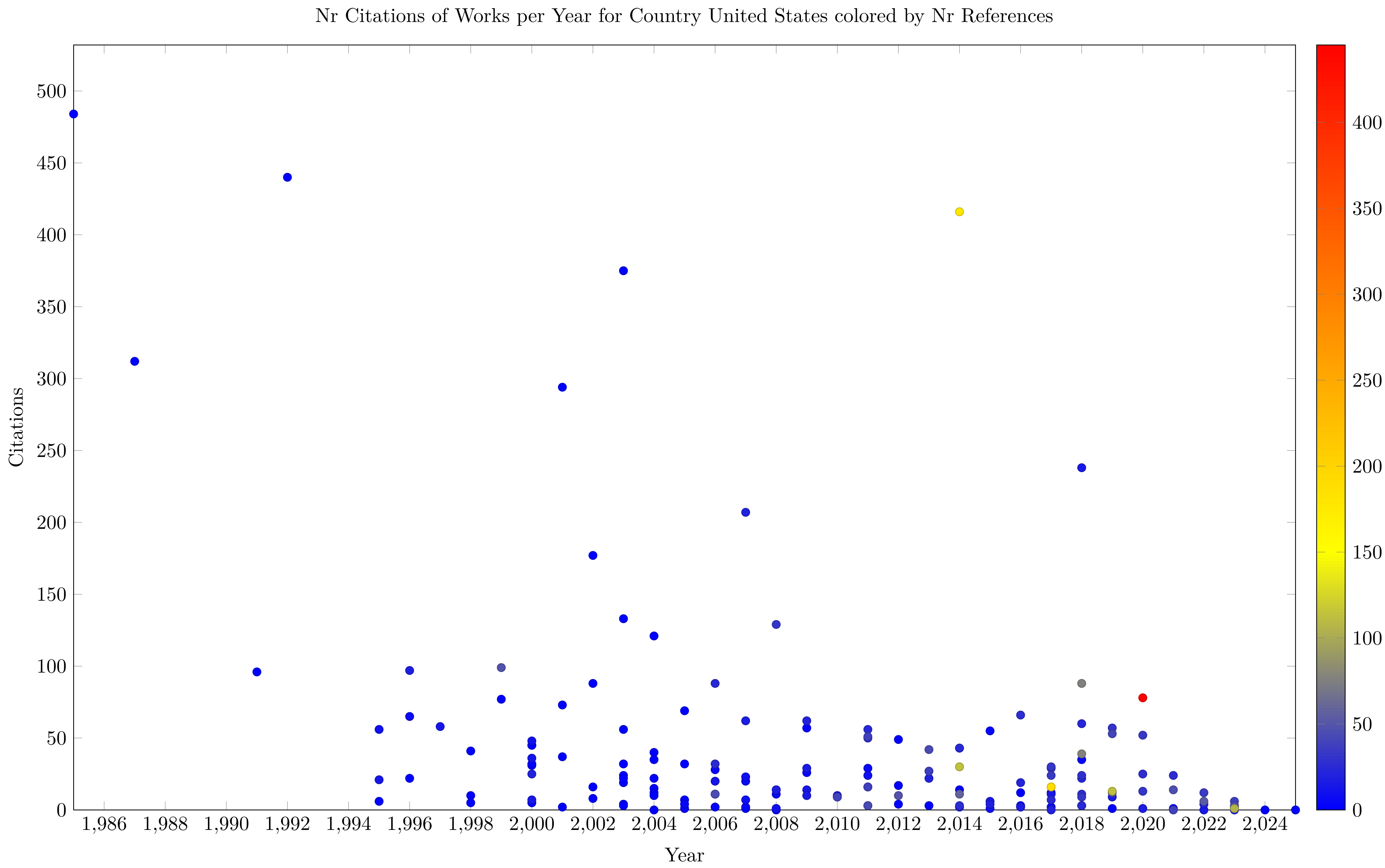
The next plot shows the number of papers associated to institutions, as stated in the Scopus affiliation. A work is counted, if at least one of the affiliations is from a given institution. Due to the format of the Scopus data, we cannot fractionally assign a paper based on the author affiliations, each paper is counted one for every institution for which an affiliation is given. If some author has multiple affiliations listed, we (mis)count the work for each of them.

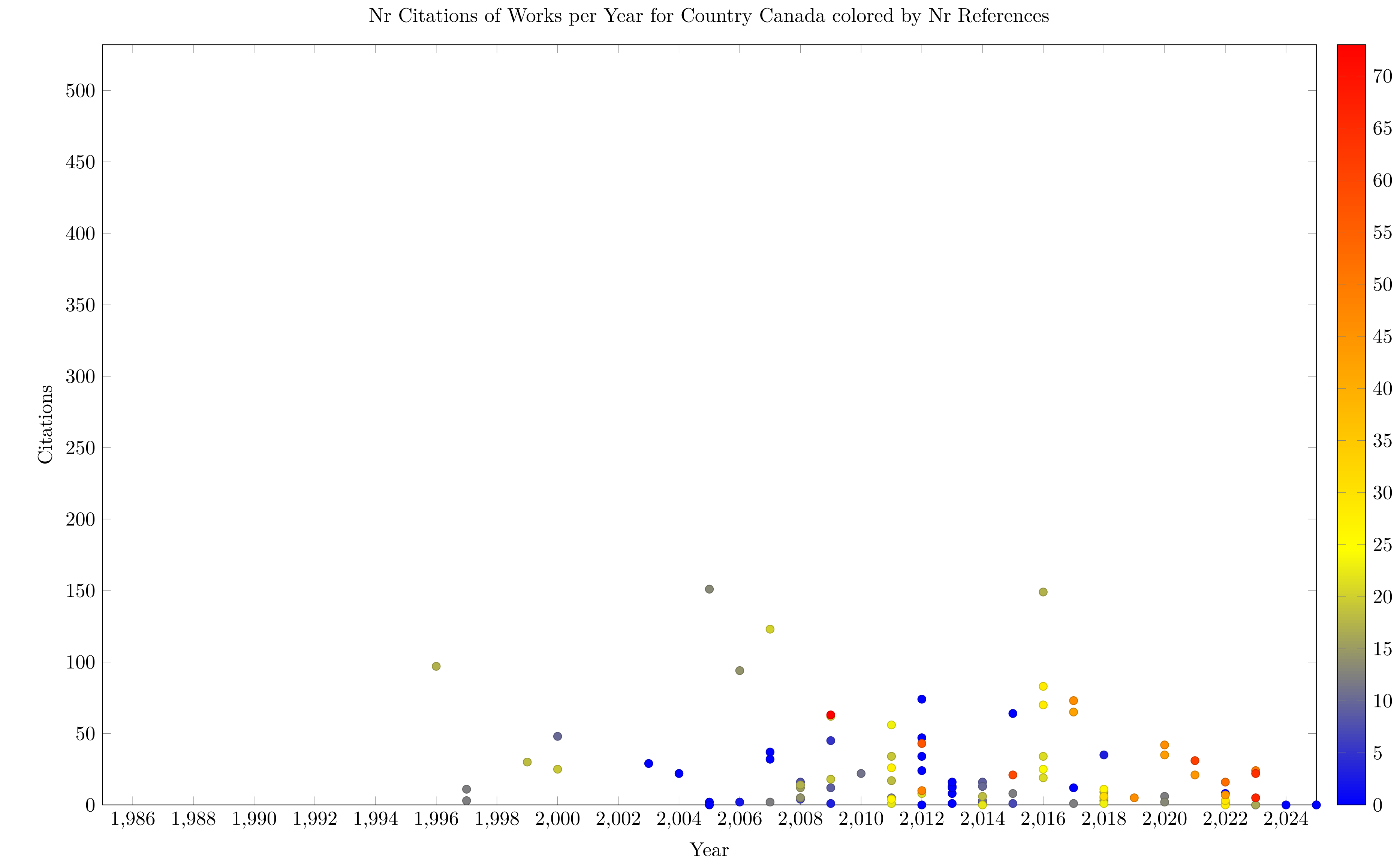

Institutions with Largest Number of Works (Total 1026 Works)
Nr Works
Institutions
University of Toronto 47
Alma Mater Studiorum Università di Bologna 38
Université de Toulouse 34
University of Melbourne 28
International Business Machines 26
University College Cork 24
Tepper School of Business 23
Monash University 22
Université Catholique de Louvain 20
Charles University 20
Commonwealth Scientific and Industrial Research Organisation 20
Laboratoire d'Analyse et d'Architecture des Systemes 19
IMT Atlantique 18
Polytechnique Montréal 17
Universidad Nacional del Litoral 16
The Royal Institute of Technology (KTH) 15
Carnegie Mellon University 15
CNRS Centre National de la Recherche Scientifique 14
RISE Swedish Institute of Computer Science 14
Universitat Politècnica de València 13
Technische Universität Wien 12
Zuse Institute Berlin 11
Université Laval 10
Dokuz Eylül Üniversitesi 10
Bouygues 10

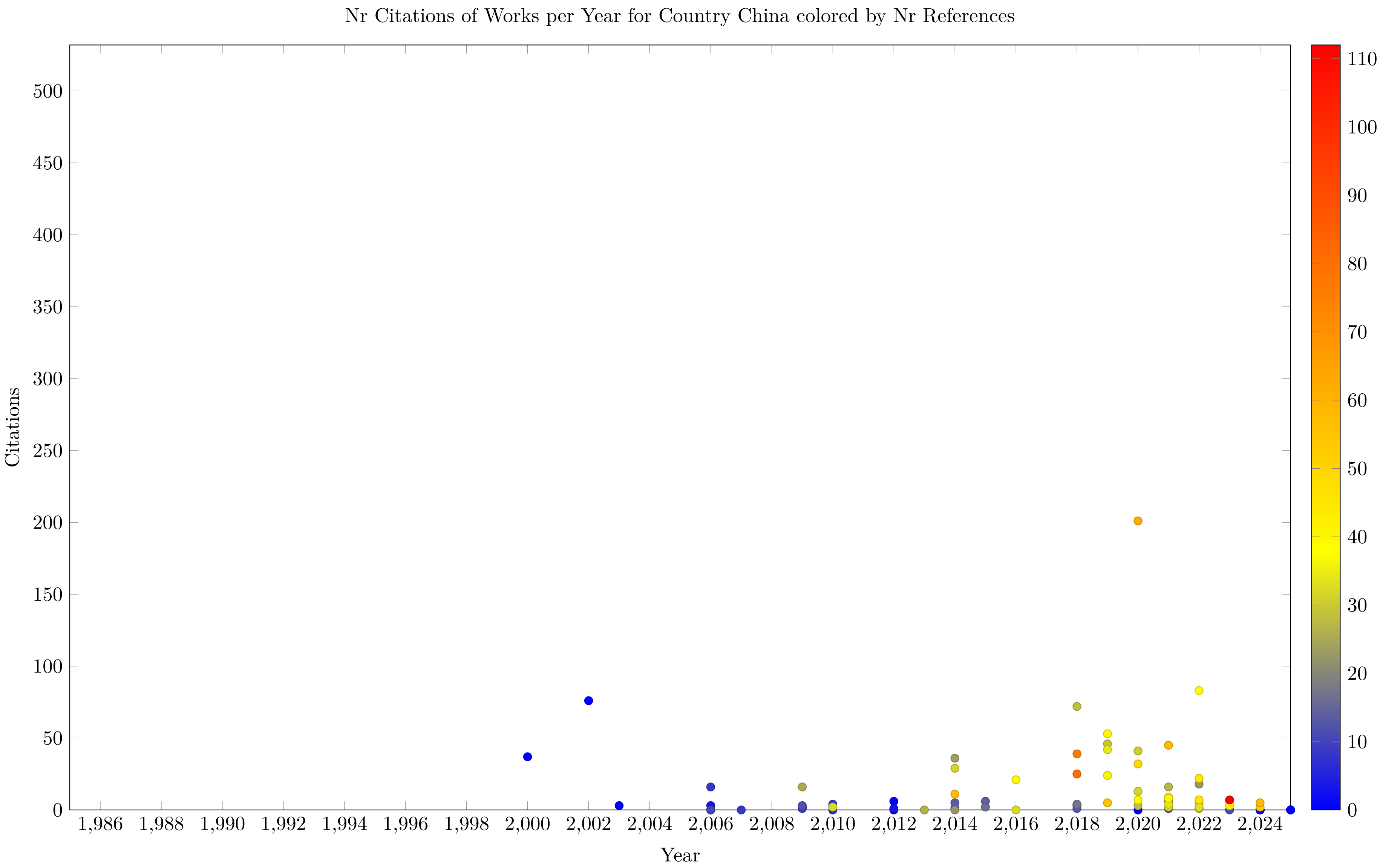
The following plots show for the top 30 countries when the works included were published, and how many citations (OpenCitation count) each paper had. The scatter plots are colored by the number of references (OpenCitation count), this help to identify surveys more easily. The plot gives an indication in which period the work from the country falls, and how influential the published works are. The x and y ranges of all plots are uniform to allow comparison between plots.

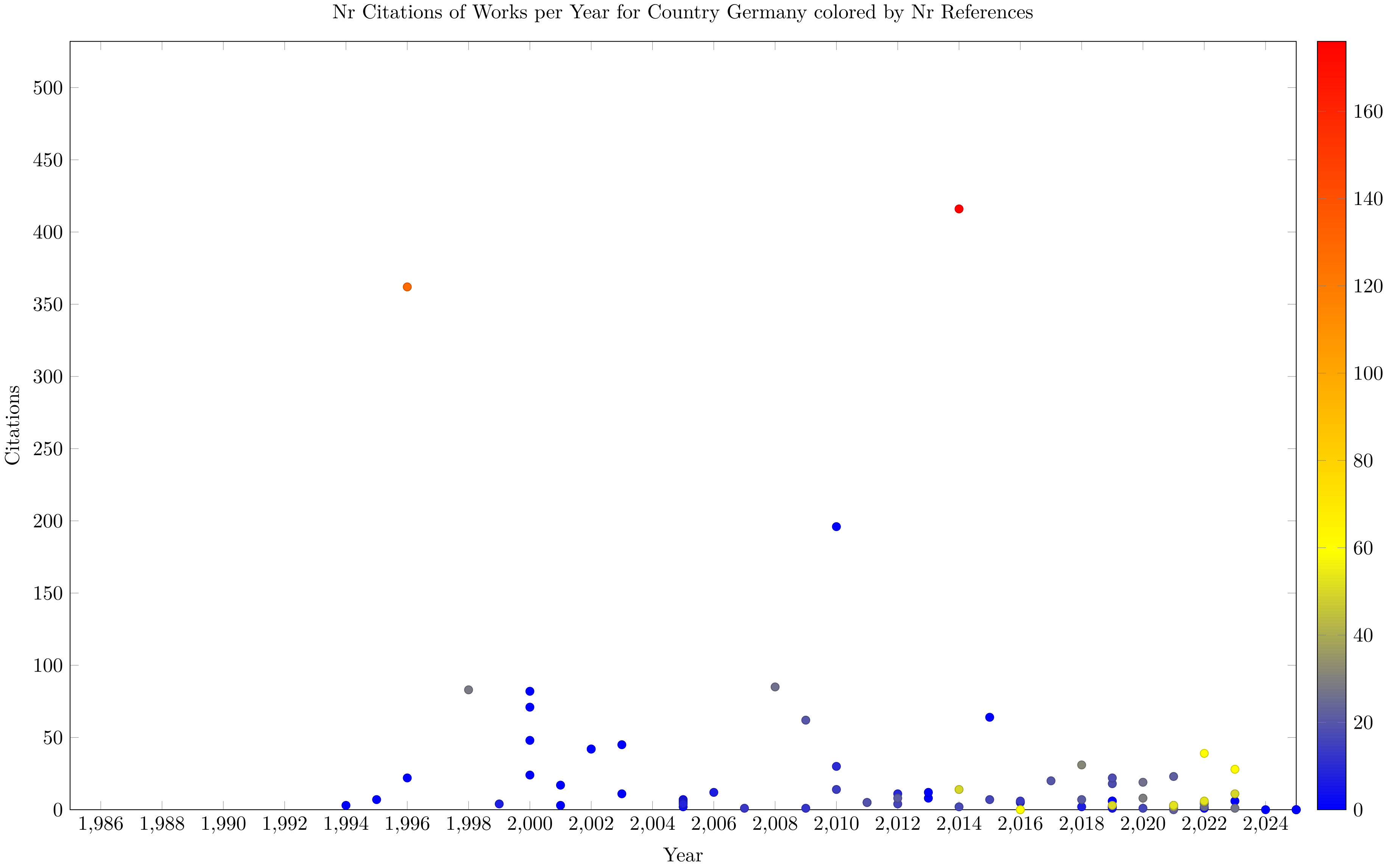
It would be nice to have tooltips on the plots, so identify specific works in the plots. This is currently not supported by the framework library used.

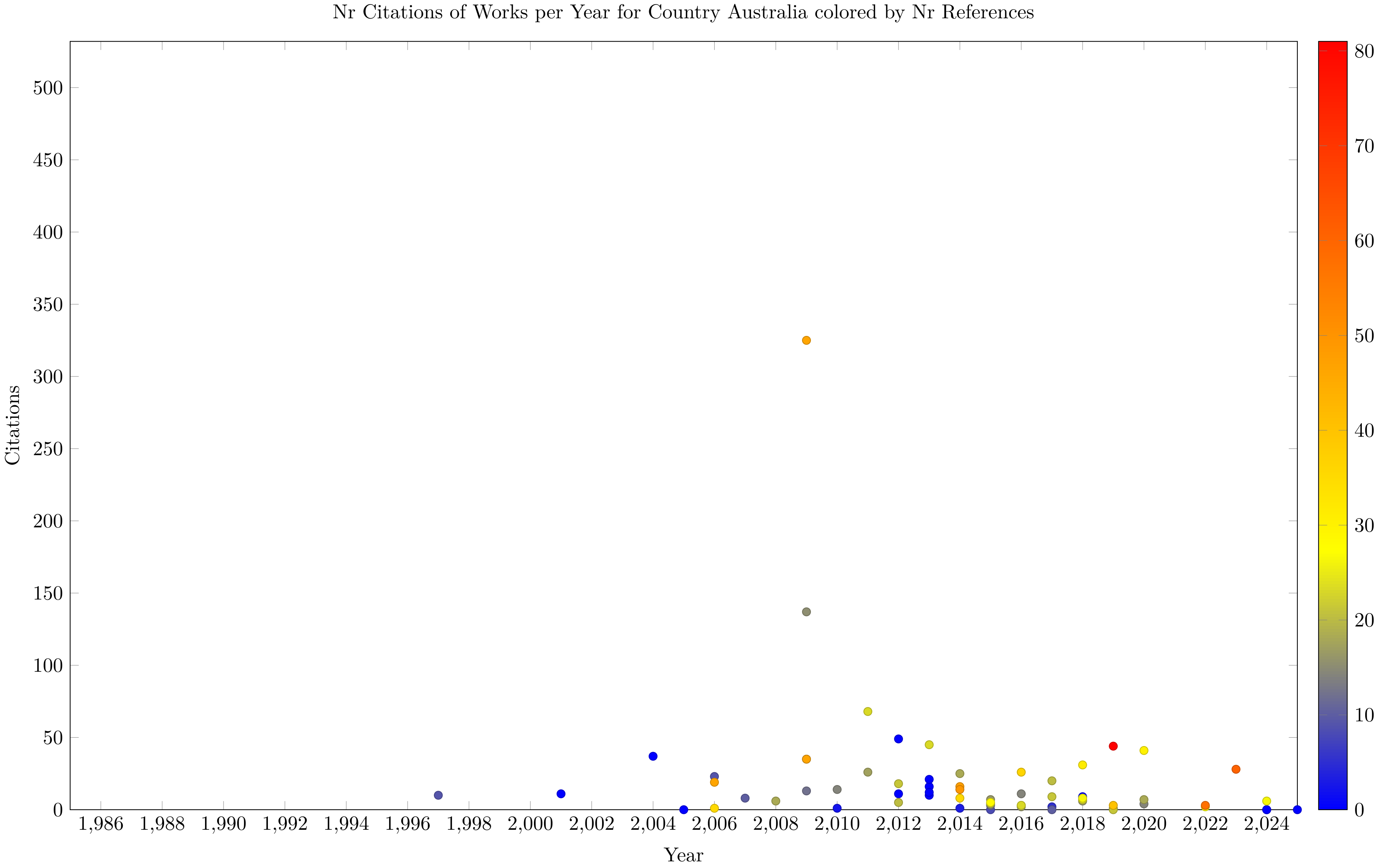

Nr Citations of Works per Year for Country France colored by Nr References
Citations
Year
0
50
100
150
200
250
300
350
400
450
500
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
0
10
20
30
40
50
60
70
80
90
100

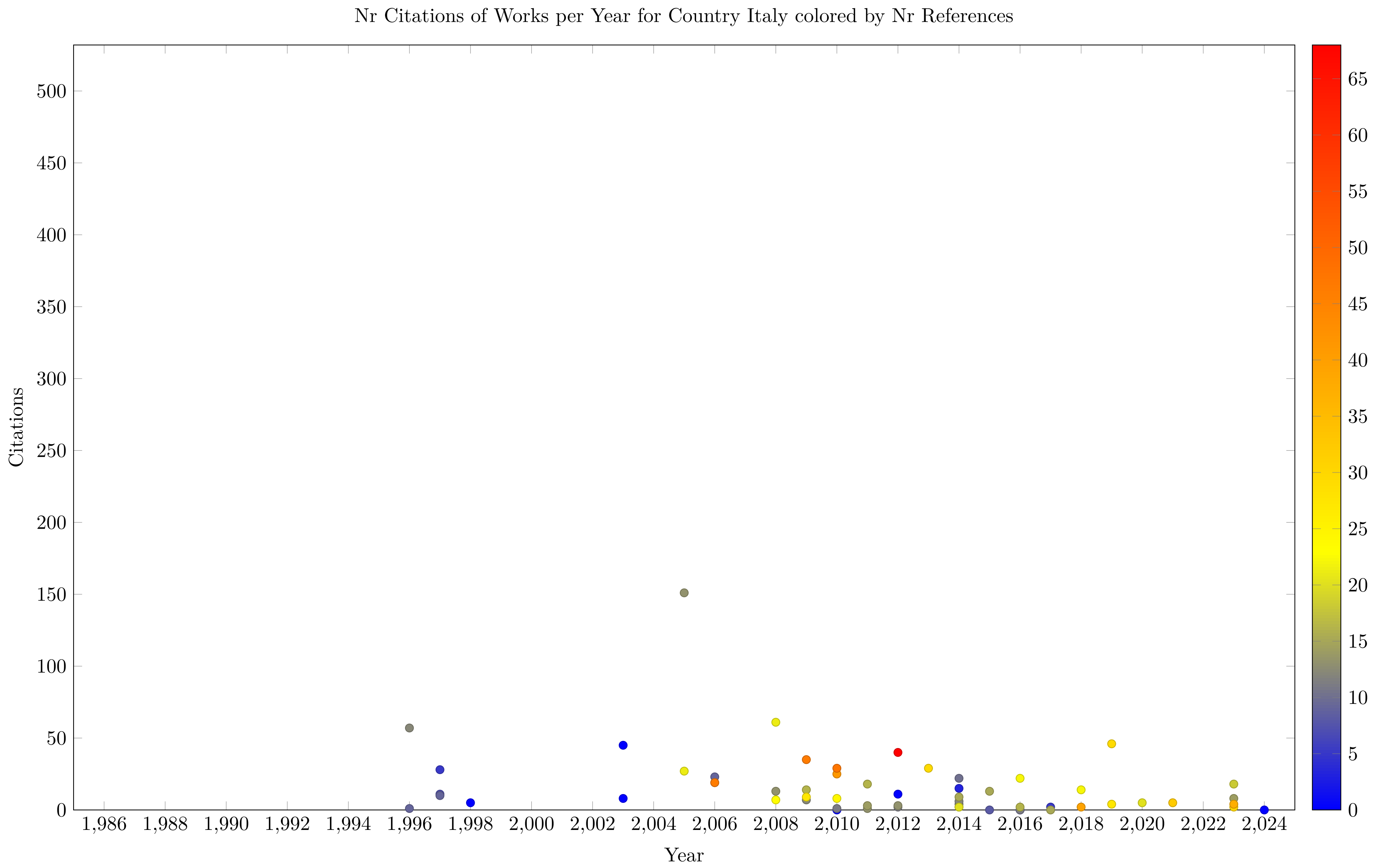

Nr Citations of Works per Year for Country United States colored by Nr References
Citations
Year
0
50
100
150
200
250
300
350
400
450
500
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
0
50
100
150
200
250
300
350
400

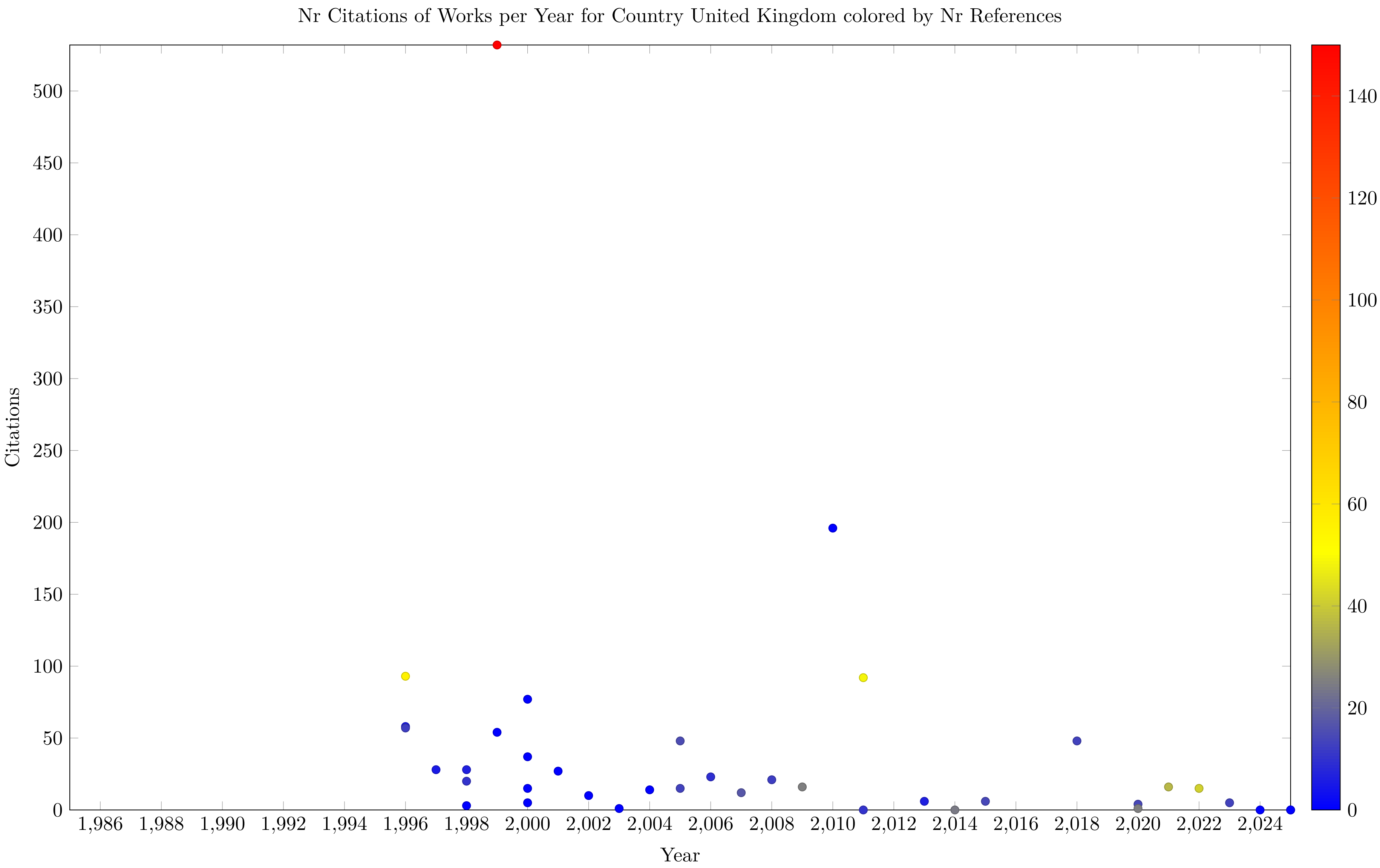

Nr Citations of Works per Year for Country Canada colored by Nr References
Citations
Year
0
50
100
150
200
250
300
350
400
450
500
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70

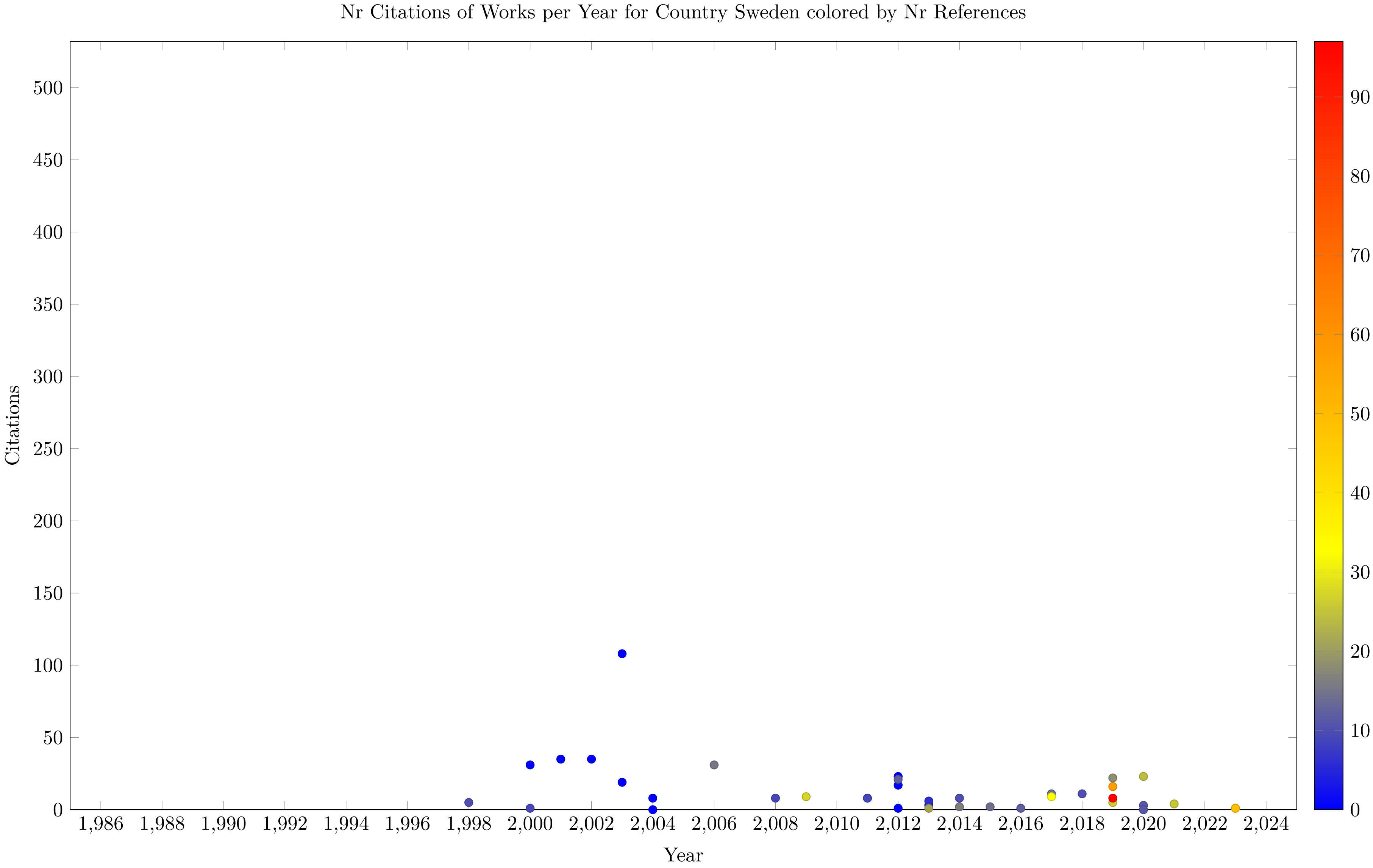

Nr Citations of Works per Year for Country China colored by Nr References
Citations
0
50
100
150
200
250
300
350
400
450
500
Year
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
0
10
20
30
40
50
60
70
80
90
100
110

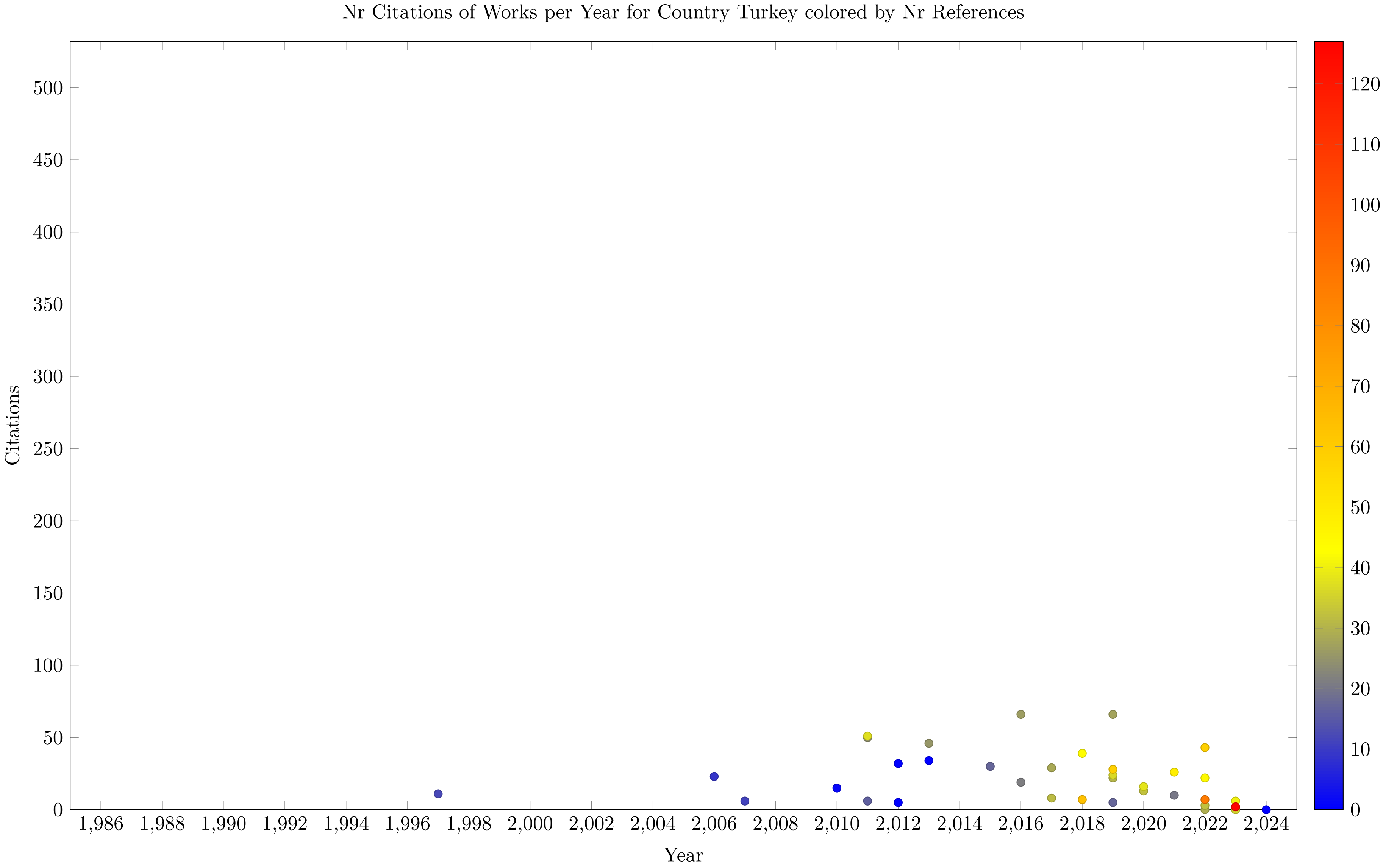

Nr Citations of Works per Year for Country Germany colored by Nr References
Citations
Year
0
50
100
150
200
250
300
350
400
450
500
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
0
20
40
60
80
100
120
140
160

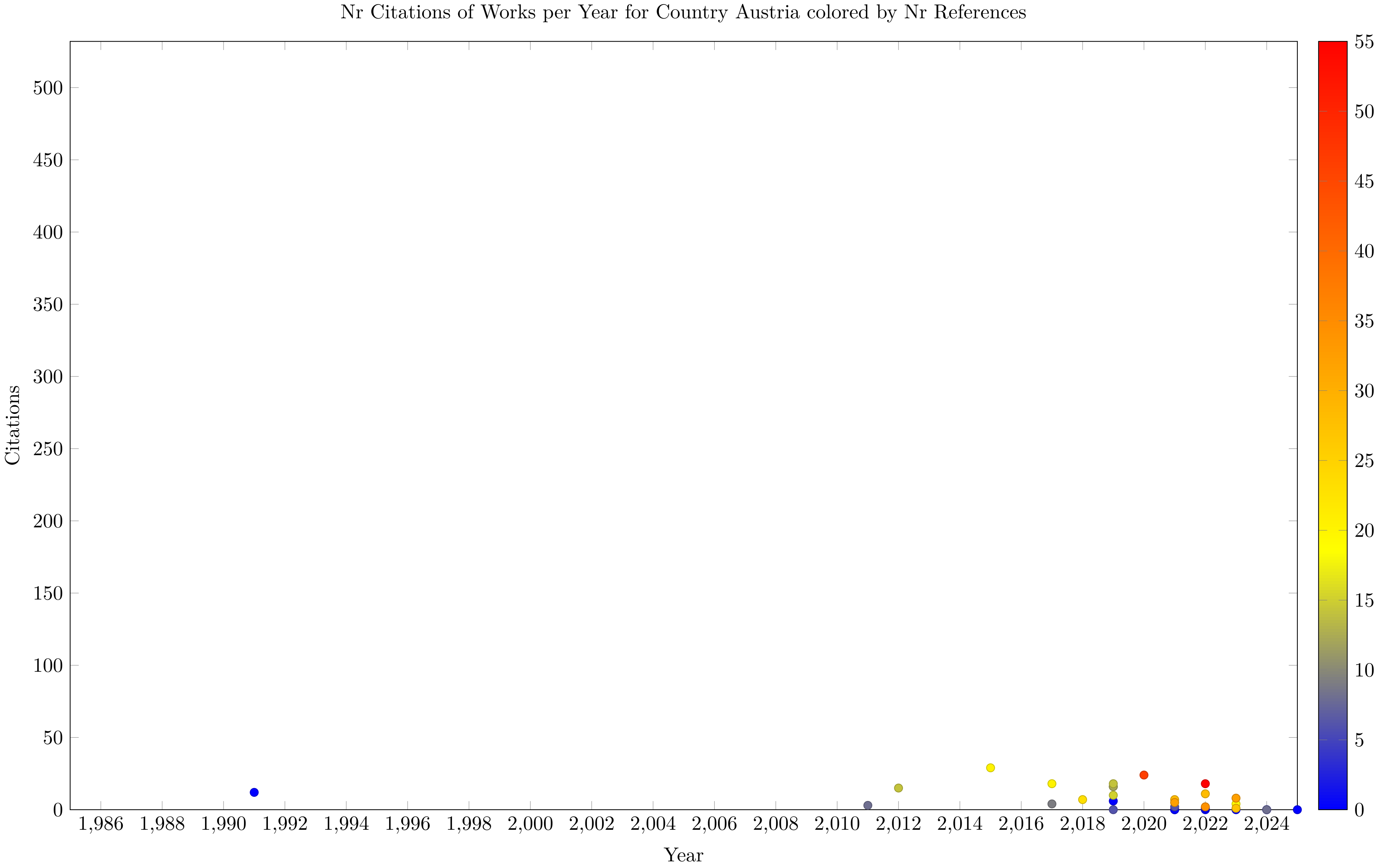

Nr Citations of Works per Year for Country Australia colored by Nr References
Citations
0
50
100
150
200
250
300
350
400
450
500
Year
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
0
10
20
30
40
50
60
70
80


Nr Citations of Works per Year for Country Italy colored by Nr References
Citations
0
50
100
150
200
250
300
350
400
450
500
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
Year
0
5
10
15
20
25
30
35
40
45
50
55
60
65


Nr Citations of Works per Year for Country United Kingdom colored by Nr References
Citations
0
50
100
150
200
250
300
350
400
450
500
Year
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
0
20
40
60
80
100
120
140


Nr Citations of Works per Year for Country Sweden colored by Nr References
Citations
Year
0
50
100
150
200
250
300
350
400
450
500
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
0
10
20
30
40
50
60
70
80
90


Nr Citations of Works per Year for Country Turkey colored by Nr References
Citations
Year
0
50
100
150
200
250
300
350
400
450
500
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
0
10
20
30
40
50
60
70
80
90
100
110
120


Nr Citations of Works per Year for Country Austria colored by Nr References
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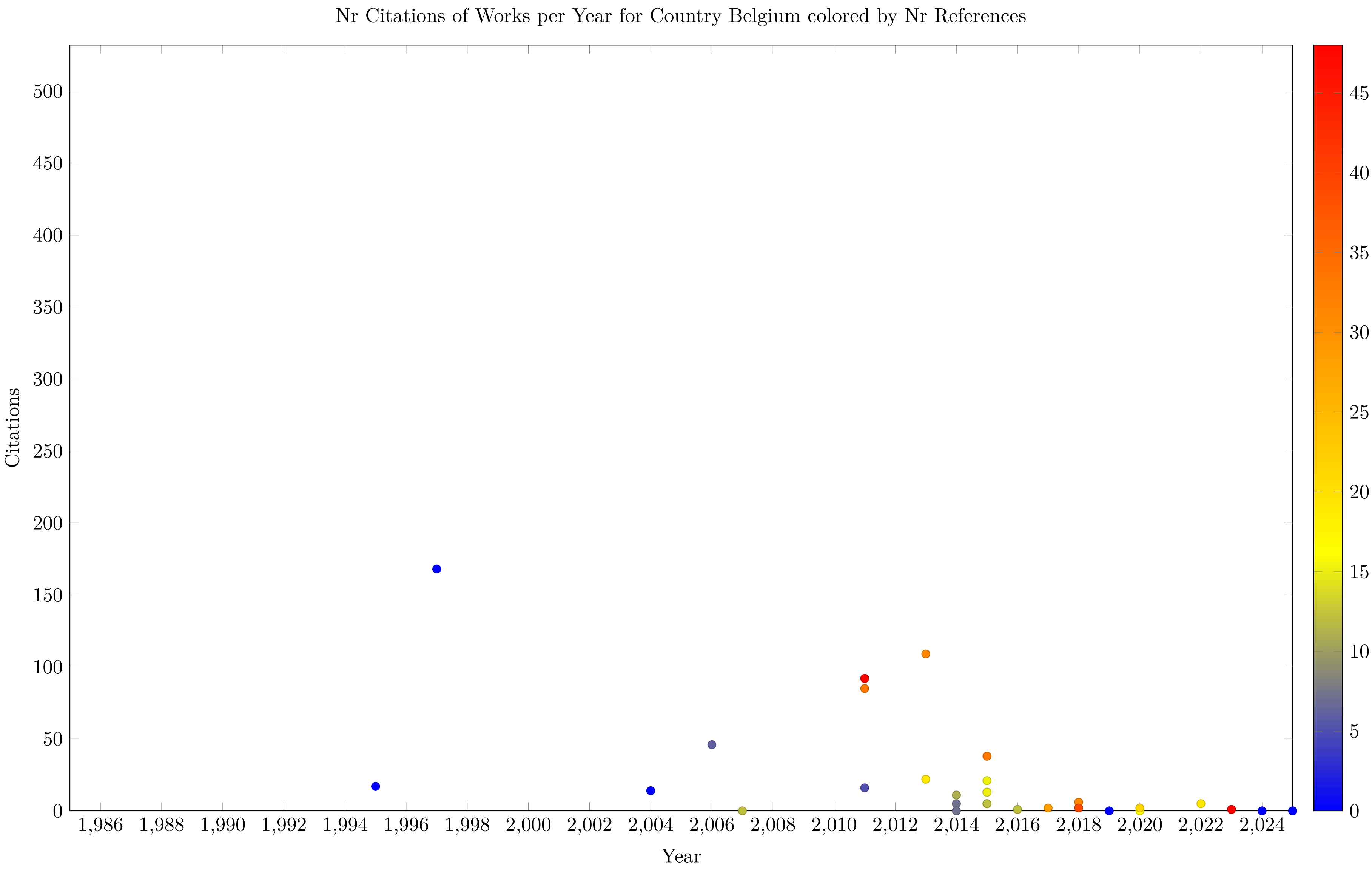
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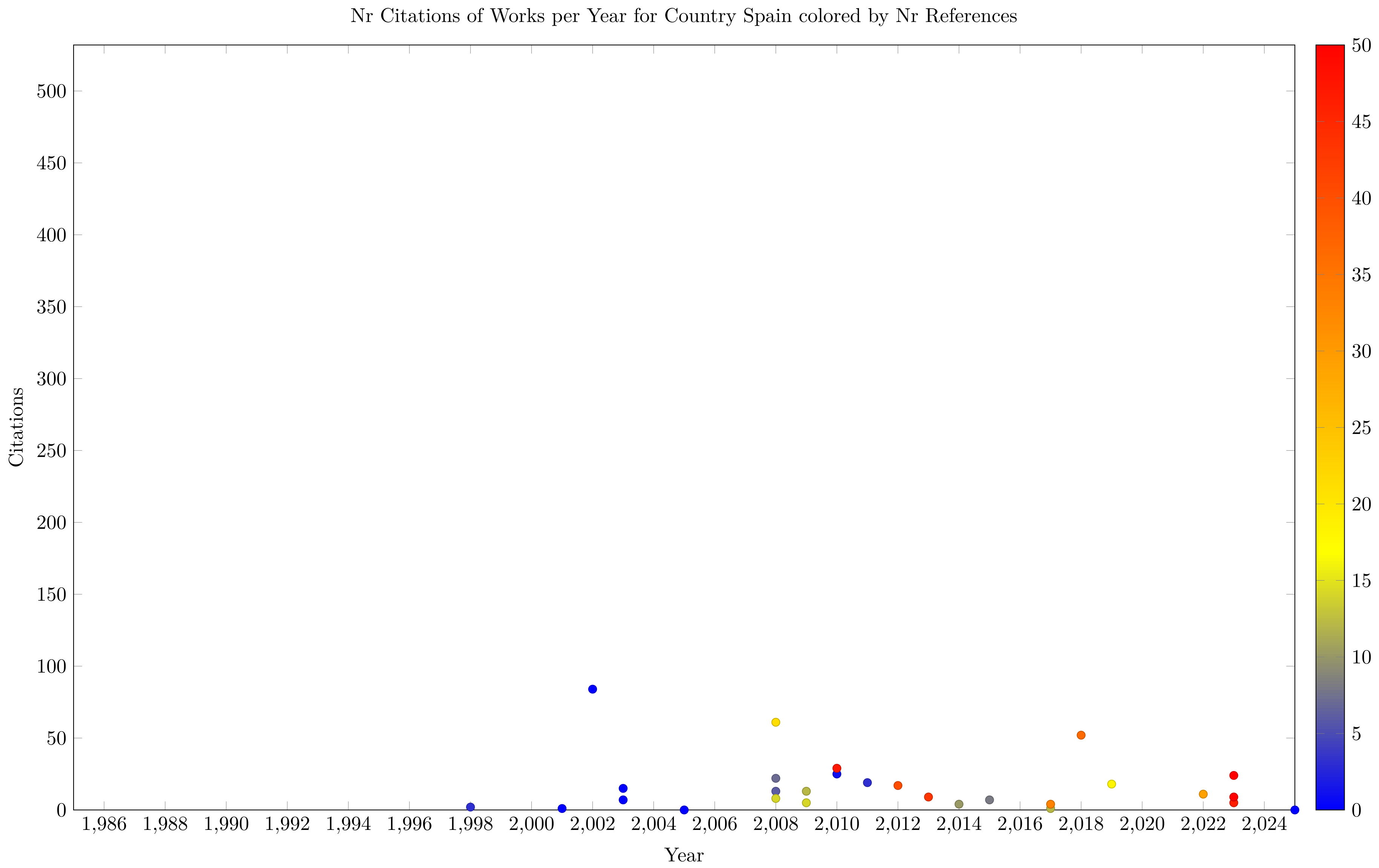
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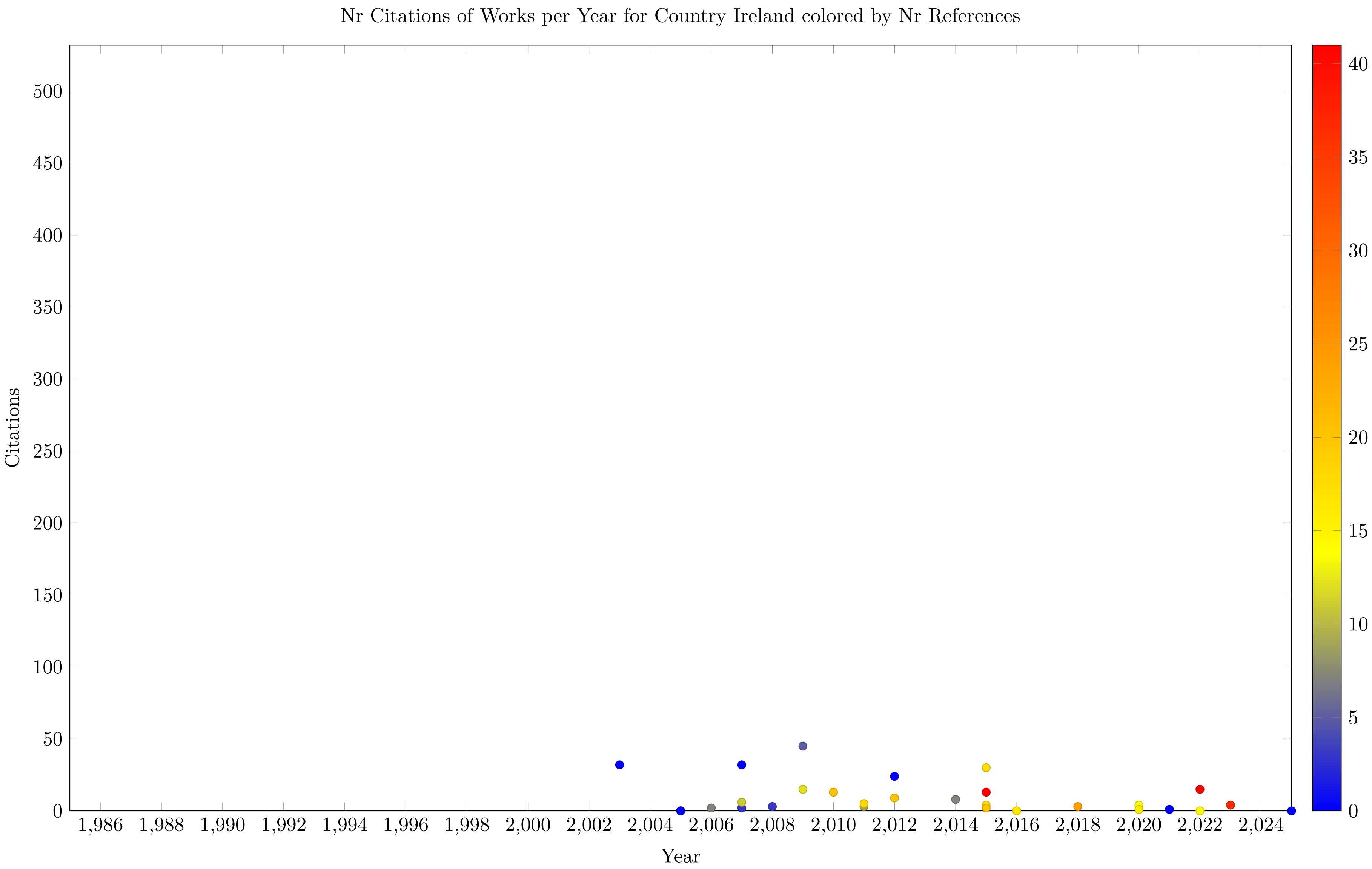
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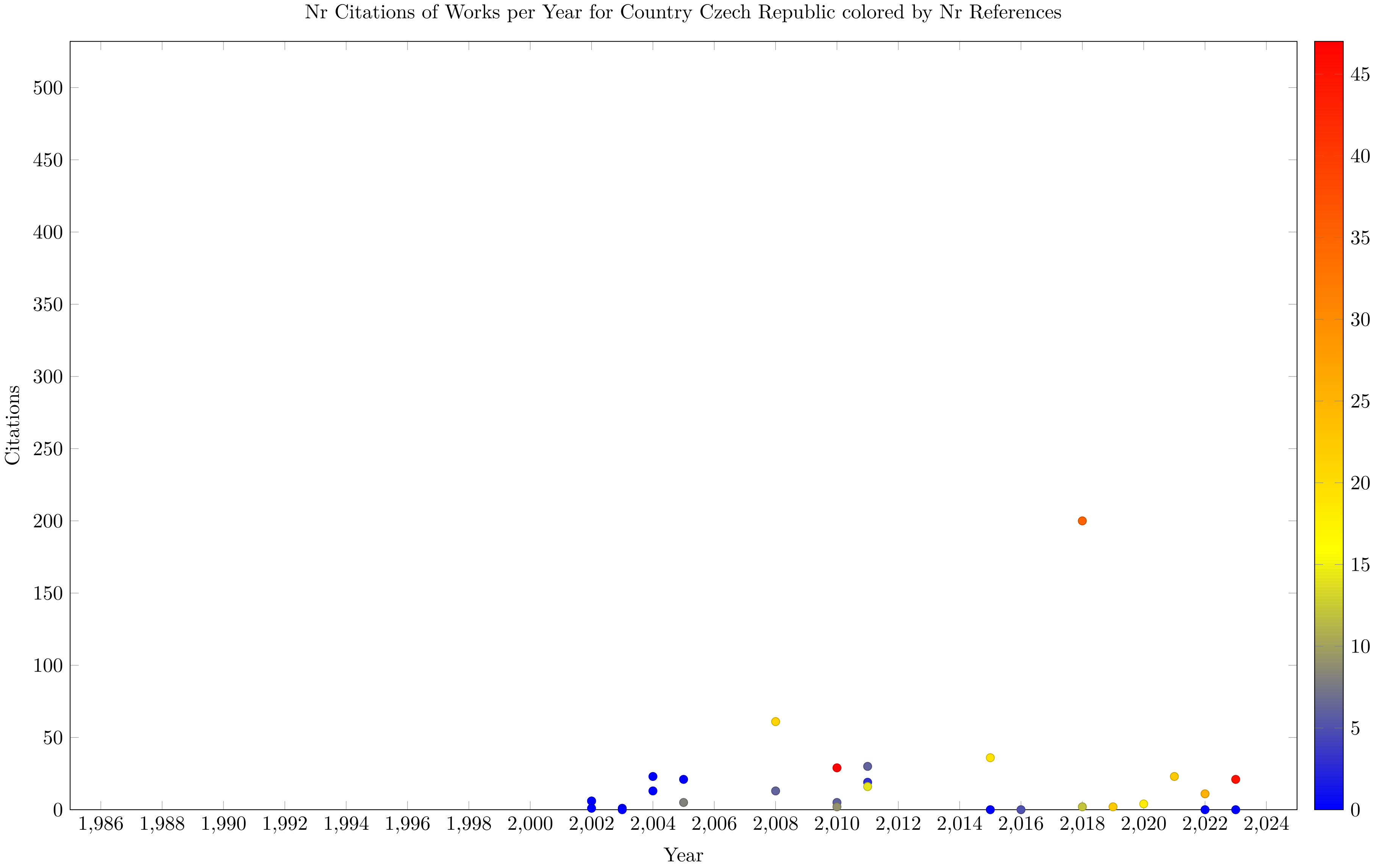
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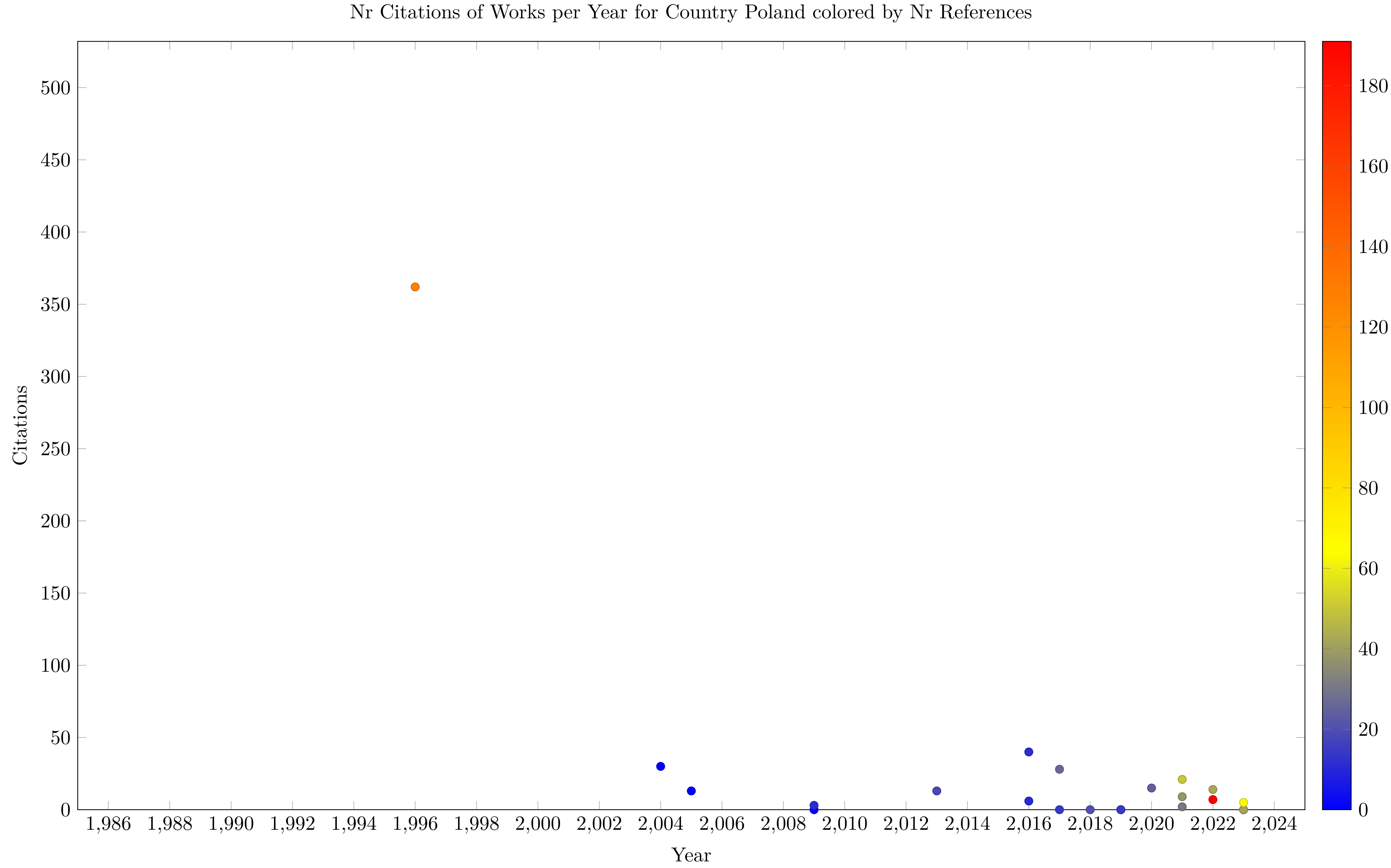
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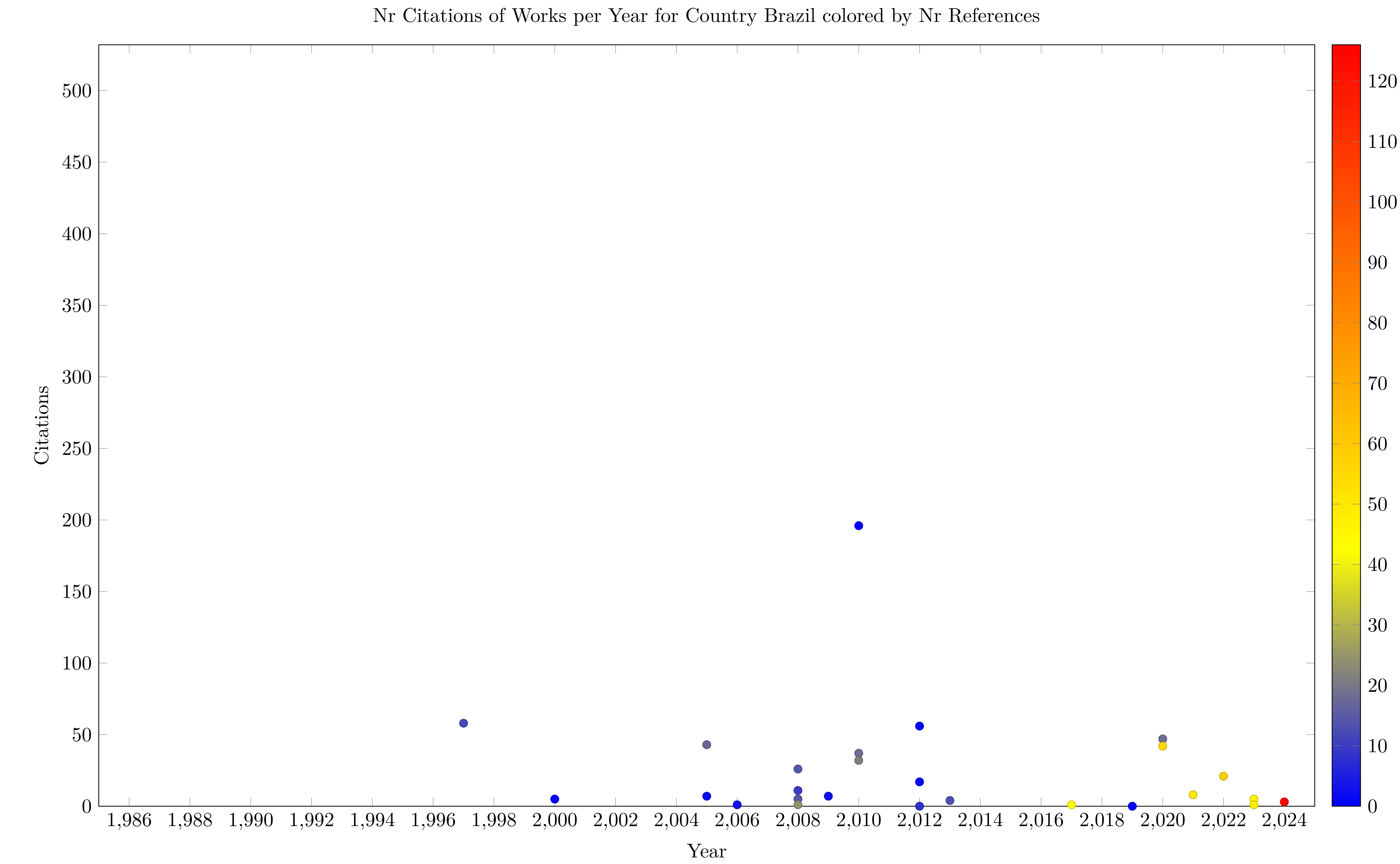
Nr Citations of Works per Year for Country Brazil colored by Nr References
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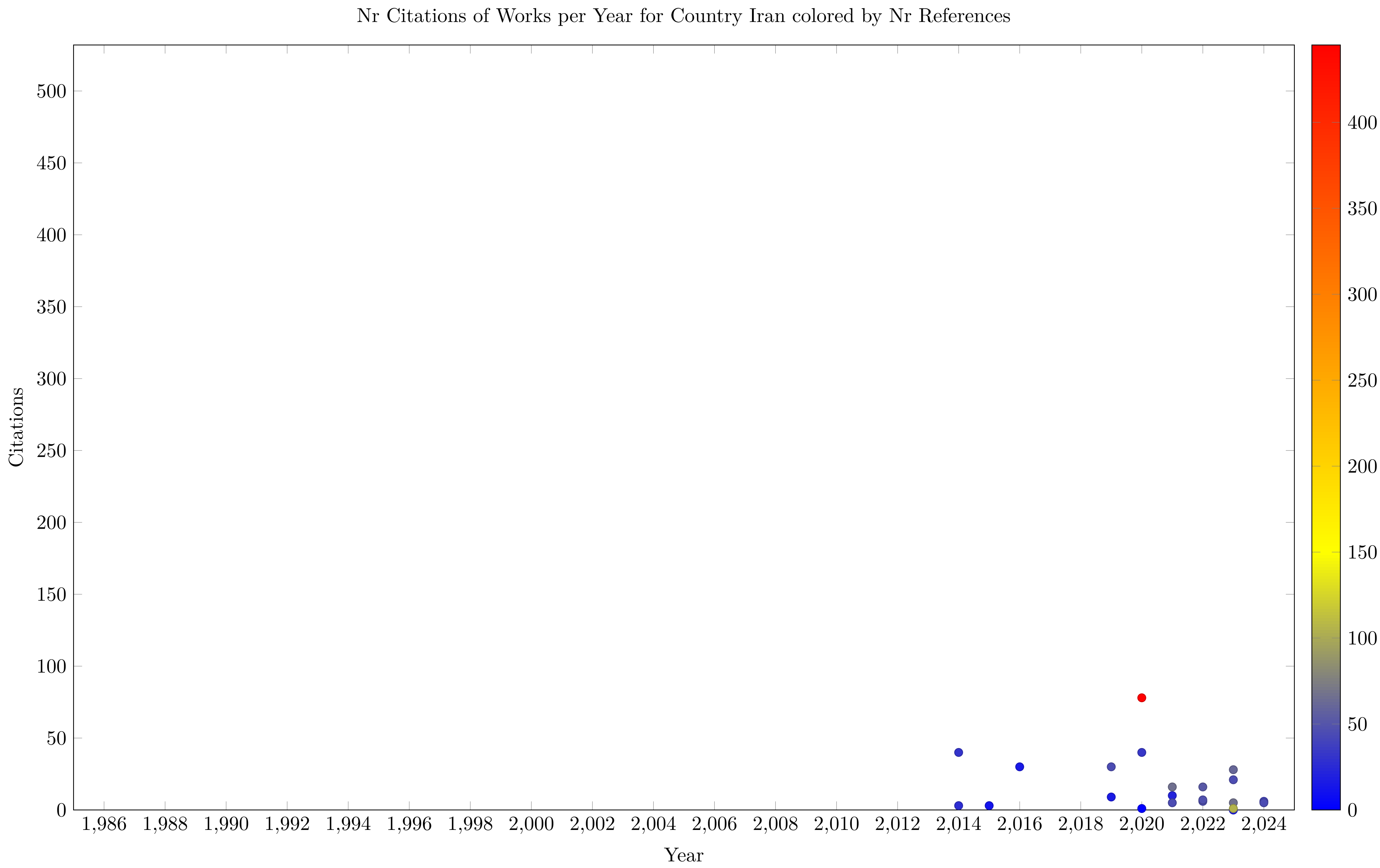
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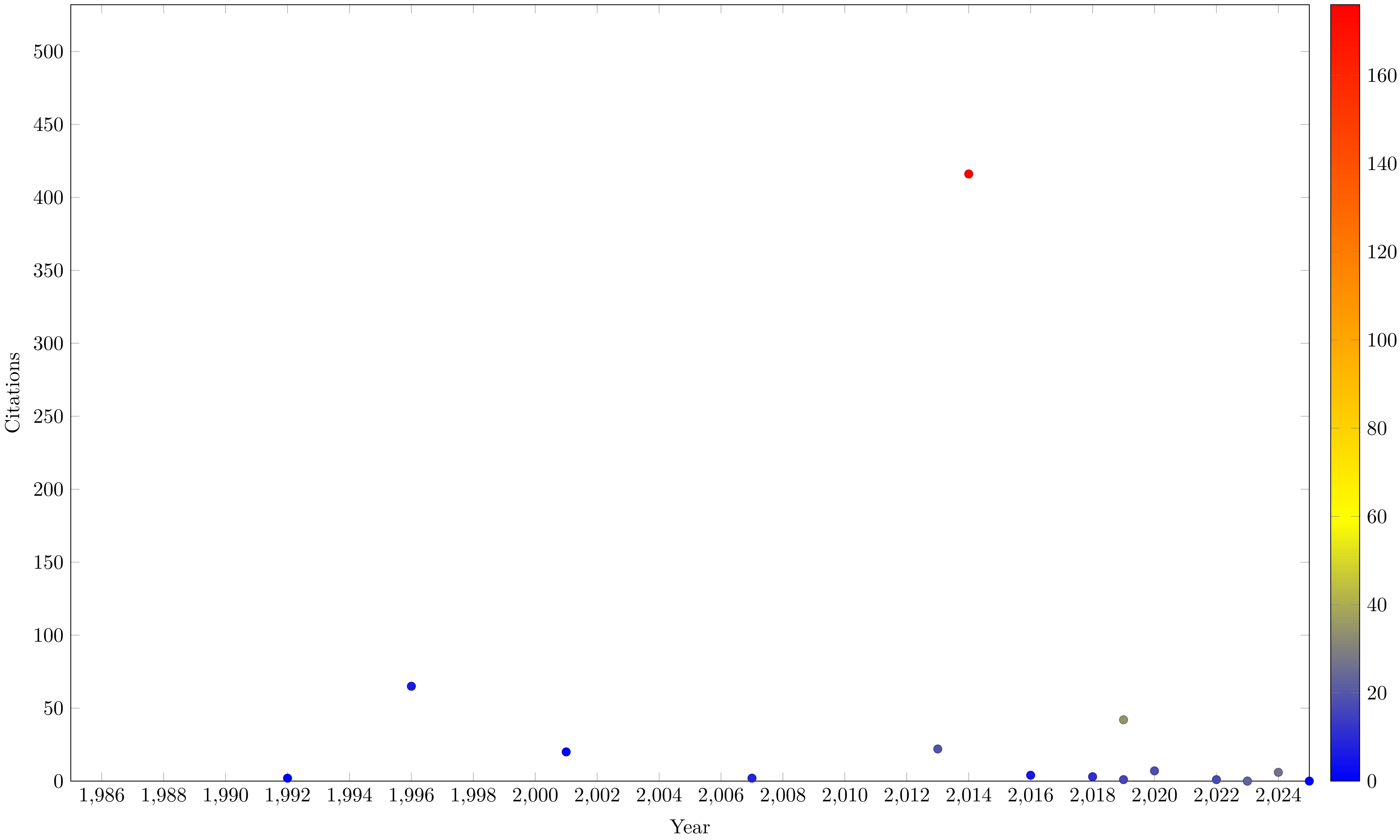
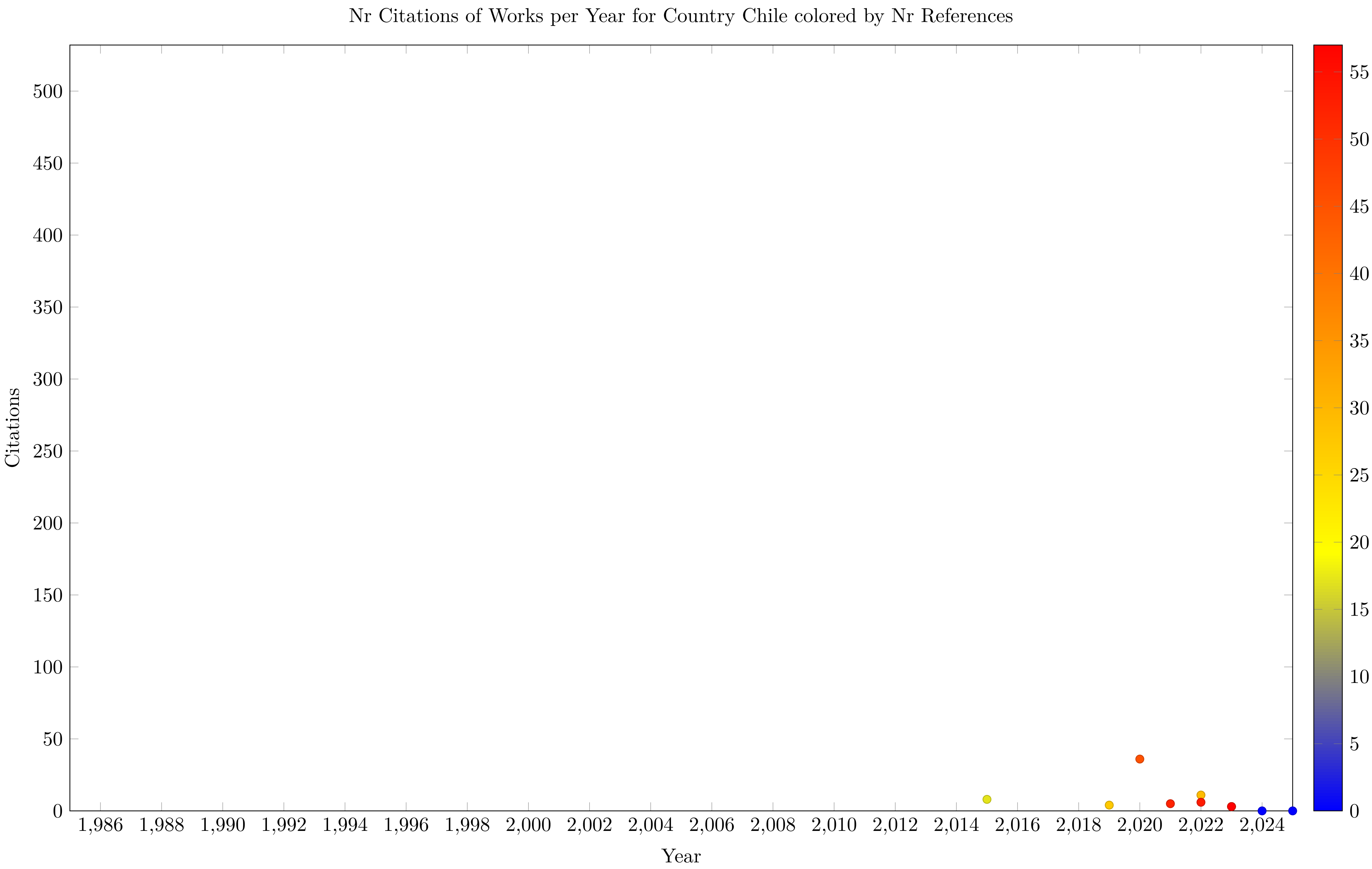
Nr Citations of Works per Year for Country Netherlands colored by Nr References

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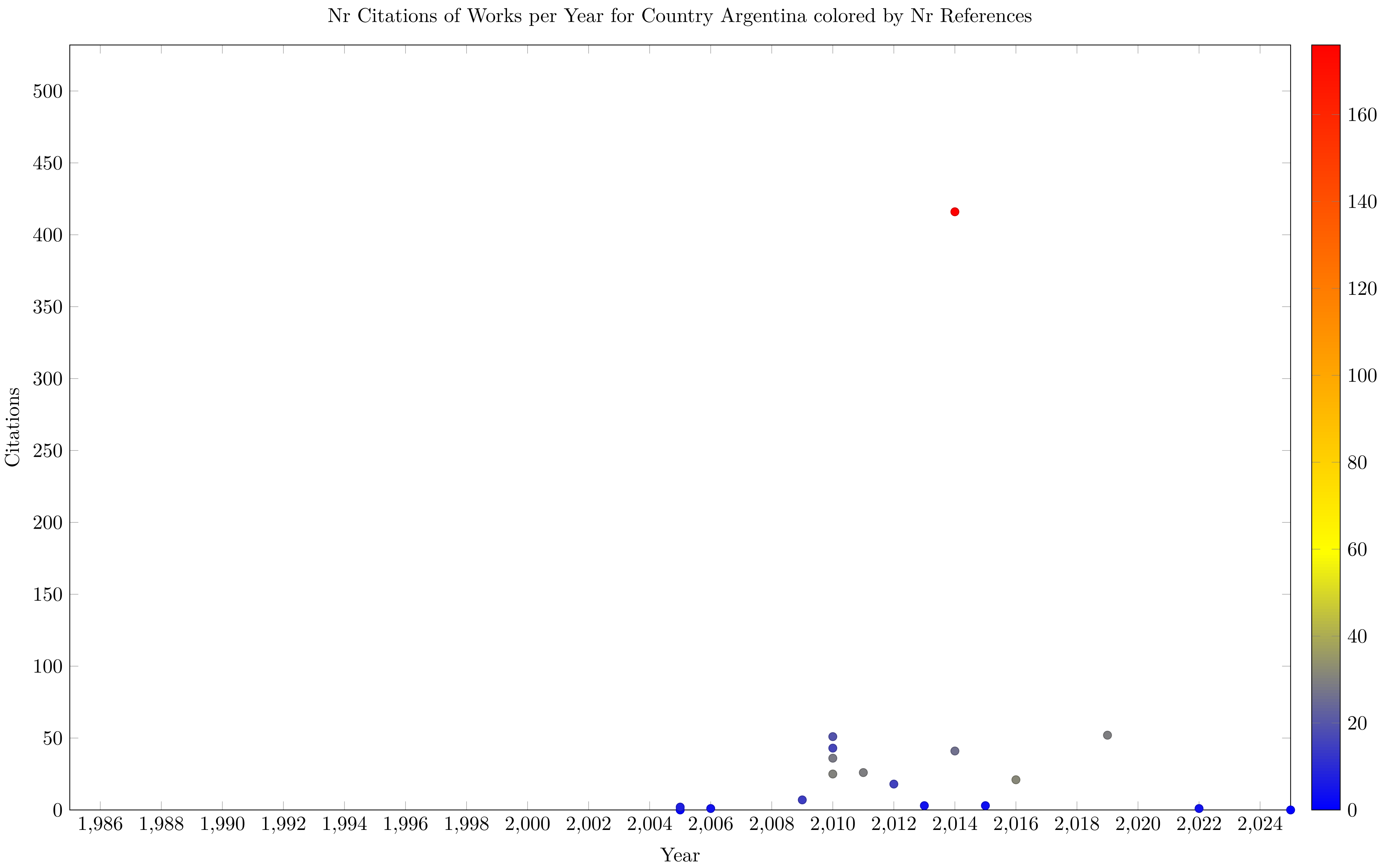
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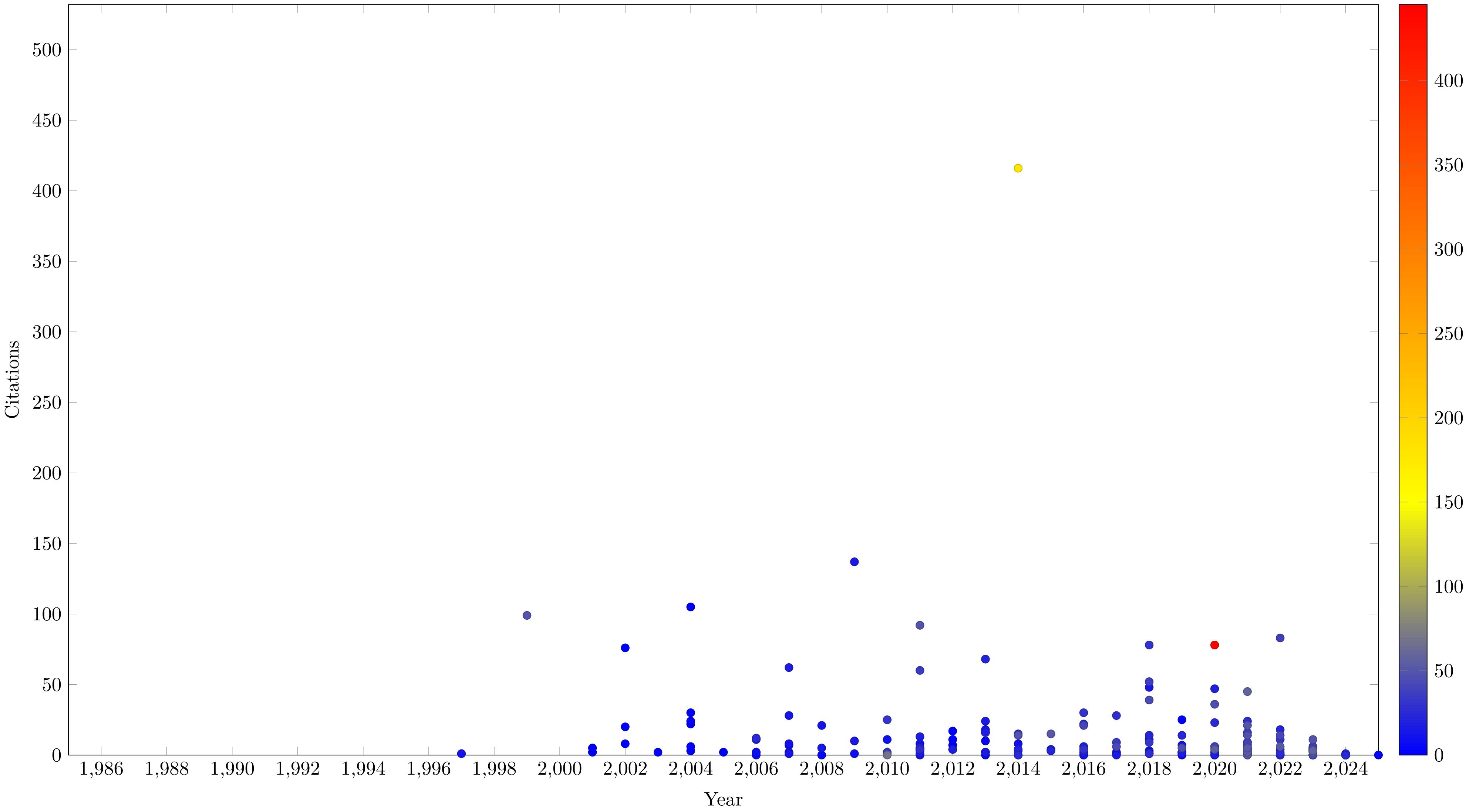
Nr Citations of Works per Year for Other Countries colored by Nr References



4 Collaborations

This section shows data about collaborations between multiple affiliations for the same work. This is based on Scopus data, which associates the affiliation with the work, not with each author of the work. The analysis excludes background work.

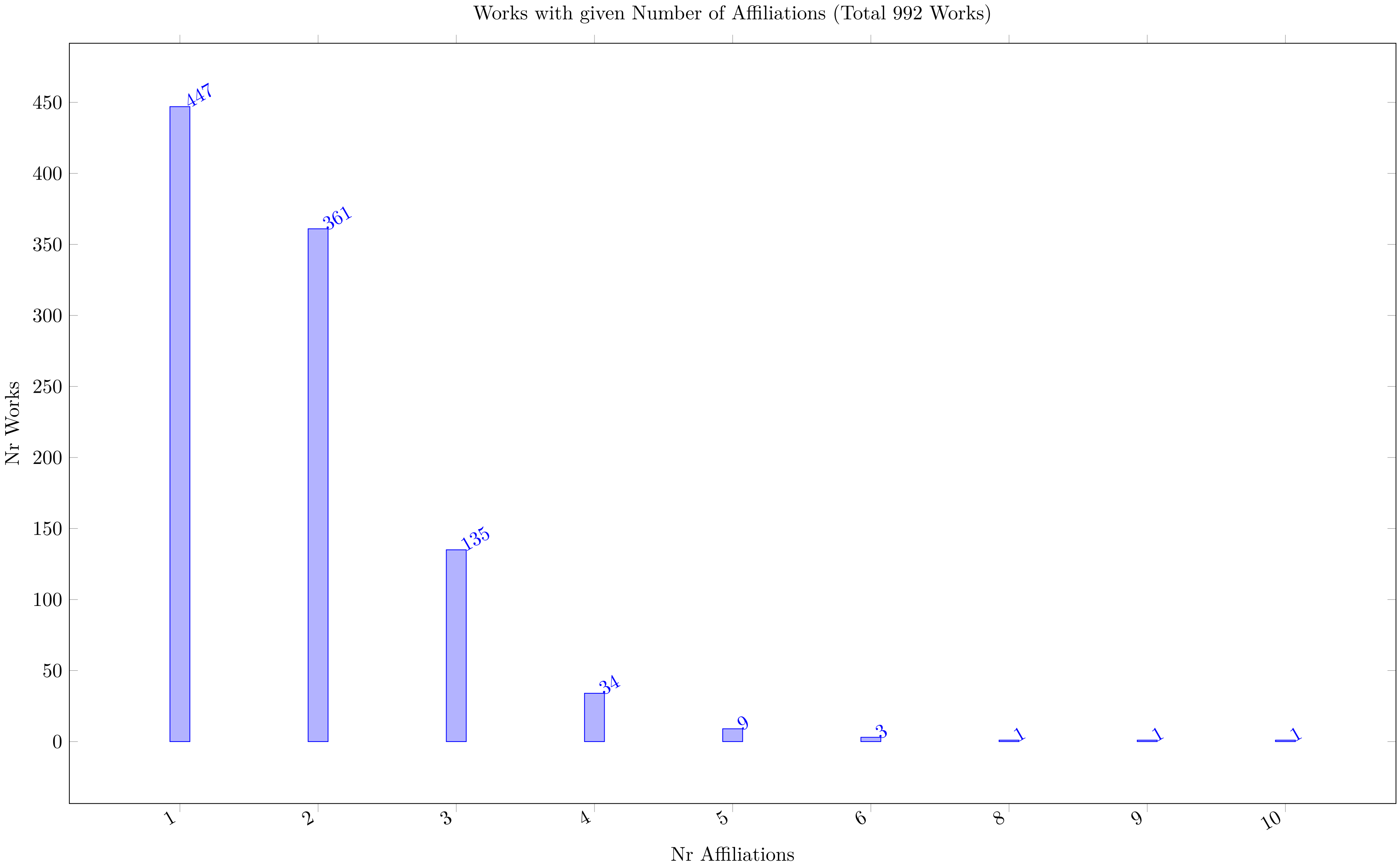

Works with given Number of Affiliations (Total 992 Works)
Nr Works
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Table 8: Collaboration Data (Top 45 Inst by Decreasing Collab Fraction)

Inst	Nr Works	Collab Count	Domestic Collab	International Collab	Collab Fraction	Domestic Fraction	International Fraction	Collab Percentage
University of Toronto, Toronto, Canada	47	46	18	28	29.00	8.50	20.50	61.70
Université de Toulouse, Toulouse, France	34	44	23	21	26.00	18.33	7.67	76.47
University of Melbourne, Melbourne, Australia	28	34	23	11	22.00	15.00	7.00	78.57
Commonwealth Scientific and Industrial Research Organisation, Canberra, Australia	20	37	25	12	20.00	15.33	4.67	100.00
Monash University, Clayton, Australia	22	28	19	9	19.00	13.50	5.50	86.36
University College Cork, Cork, Ireland	24	33	7	26	18.00	2.26	15.74	75.00
Laboratoire d'Analyse et d'Architecture des Systemes, Toulouse, France	19	25	19	6	17.00	13.50	3.50	89.47
Alma Mater Studiorum Università di Bologna, Bologna, Italy	38	26	6	20	16.00	3.83	12.17	42.11
International Business Machines, Armonk, United States	26	20	2	18	15.00	2.00	13.00	57.69
The Royal Institute of Technology (KTH), Stockholm, Sweden	15	25	18	7	14.00	10.67	3.33	93.33
IMT Atlantique, Nantes, France	18	17	7	10	14.00	5.00	9.00	77.78
RISE, Swedish Institute of Computer Science, Kista, Sweden	14	16	5	11	12.00	4.00	8.00	85.71
Polytechnique Montréal, Montreal, Canada	17	15	10	5	11.00	8.00	3.00	64.71
Université Catholique de Louvain, Louvain-la-Neuve, Belgium	20	13	2	11	10.00	1.33	8.67	50.00
CNRS Centre National de la Recherche Scientifique, Paris, France	14	14	9	5	10.00	7.00	3.00	71.43
Tepper School of Business, Pittsburgh, United States	23	19	7	12	10.00	3.33	6.67	43.48
Technische Universität Wien, Vienna, Austria	12	10	6	4	9.00	6.00	3.00	75.00
Charles University, Prague, Czech Republic	20	12	5	7	9.00	4.50	4.50	45.00
University of Connecticut, Storrs, United States	8	12	8	4	8.00	5.83	2.17	100.00
Rotman School of Management, Toronto, Canada	8	19	15	4	8.00	6.17	1.83	100.00
Universidade de São Paulo, Sao Paulo, Brazil	8	9	5	4	7.00	4.50	2.50	87.50
Dokuz Eylül Üniversitesi, Izmir, Turkey	10	8	6	2	7.00	5.00	2.00	70.00
Universitat Politècnica de València, Valencia, Spain	13	10	1	9	7.00	1.00	6.00	53.85
Zuse Institute Berlin, Berlin, Germany	11	9	6	3	7.00	4.50	2.50	63.64
Bouygues, Paris, France	10	8	6	2	7.00	5.00	2.00	70.00
Politechnika Koszalinska, Koszalin, Poland	8	11	8	3	7.00	5.00	2.00	87.50
Université d'Avignon et des Pays du Vaucluse, Avignon, France	8	10	8	2	7.00	5.00	2.00	87.50
Brown University, Providence, United States	8	13	7	6	6.00	4.53	1.47	75.00
Universidad Andrés Bello, Santiago, Chile	6	14	6	8	6.00	2.40	3.60	100.00
Université de Maroua, Maroua, Cameroon	6	10	6	4	6.00	3.67	2.33	100.00
University of Windsor, Windsor, Canada	6	13	11	2	6.00	5.17	0.83	100.00
ABB Corporate Research, Vasteras, Vasteras, Sweden	6	12	10	2	6.00	5.00	1.00	100.00
Izmir Ekonomi Universitesi, Izmir, Turkey	8	12	5	7	6.00	3.50	2.50	75.00
Universidad Nacional del Litoral, Santa Fe, Argentina	16	14	4	10	6.00	4.00	2.00	37.50
Magyar Tudomanyos Akademia, Budapest, Hungary	9	7	1	6	6.00	1.00	5.00	66.67
Technische Universität Berlin, Berlin, Germany	6	13	5	8	5.00	3.50	1.50	83.33
Universidad de La Sabana, Chía, Colombia	5	12	0	12	5.00	0.00	5.00	100.00
Huazhong University of Science and Technology, Wuhan, China	7	13	9	4	5.00	3.50	1.50	71.43
The Australian National University, Canberra, Australia	6	10	4	6	5.00	1.83	3.17	83.33
Centre Interuniversitaire de Recherche sur les Réseaux dEntreprise, la Logistique et le Transport, Montreal, Canada	9	7	6	1	5.00	4.50	0.50	55.56
National University of Singapore, Singapore City, Singapore	5	7	1	6	5.00	0.50	4.50	100.00
Czech Institute of Informatics, Robotics and Cybernetics, Prague, Czech Republic	5	7	3	4	5.00	2.50	2.50	100.00
University of Tehran, Tehran, Iran	7	7	1	6	5.00	1.00	4.00	71.43
École des Mines de Saint-Étienne, Saint-Etienne, France	5	12	8	4	5.00	4.00	1.00	100.00
Aalborg University, Aalborg, Denmark	5	8	0	8	5.00	0.00	5.00	100.00

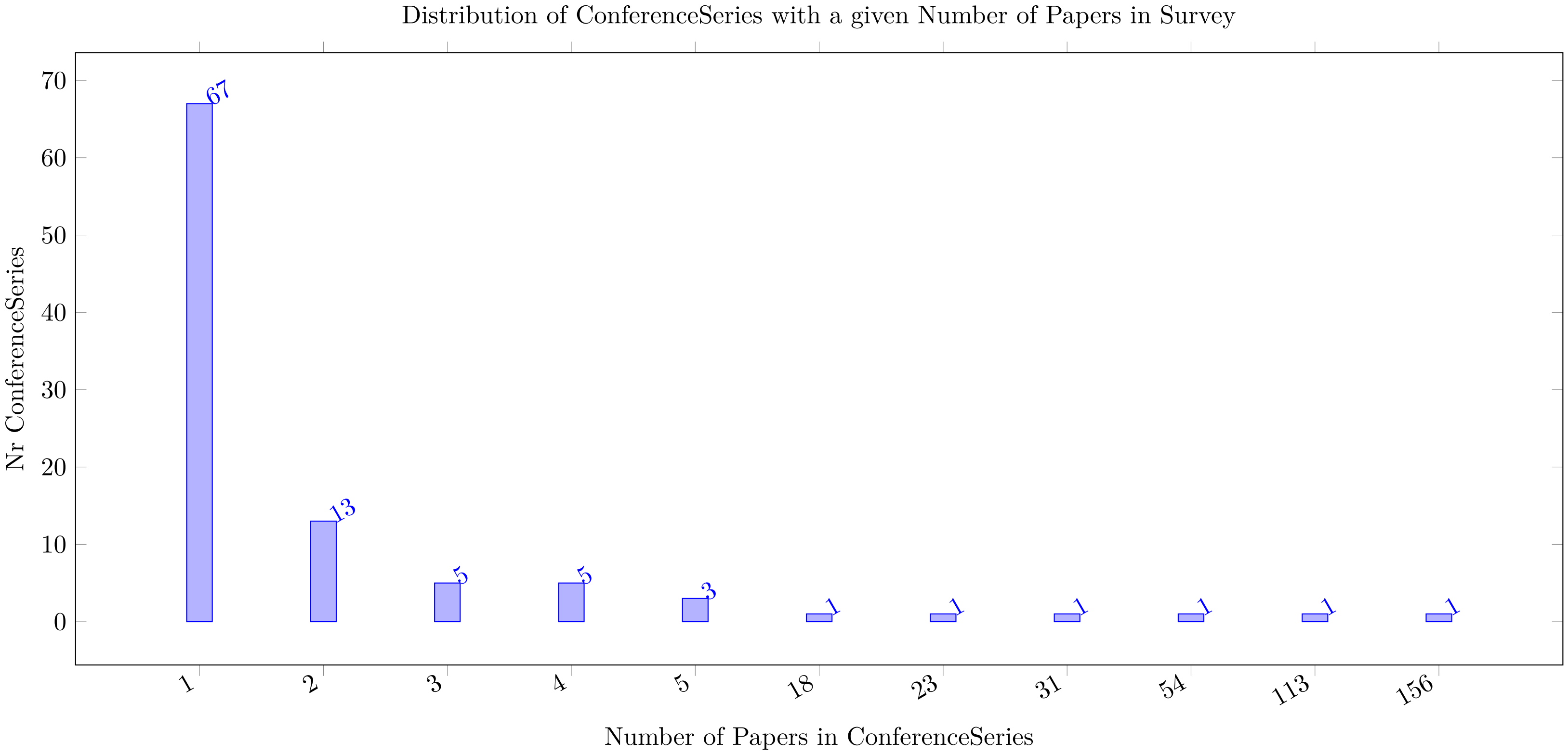
Table 9: Heat Map based on Collaboration between Countries (Fractional Count)

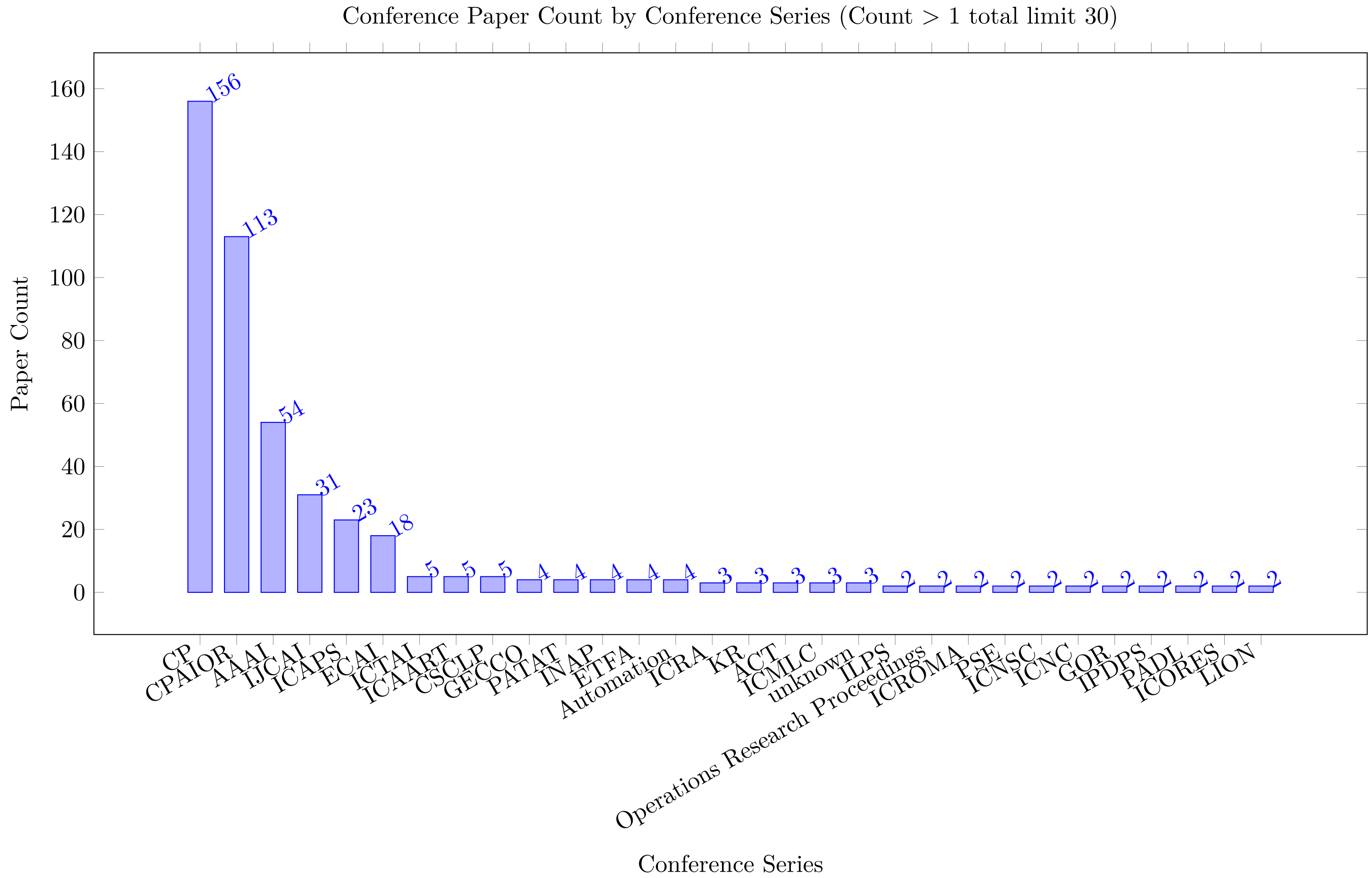
From/To	Total	France	United States	Canada	China	Australia	Germany	Sweden	Italy	Turkey	Austria	United Kingdom	Ireland	Poland	Chile	Belgium	Czech Republic	Spain	Brazil	Iran	Denmark	Taiwan	Norway	Singapore	Greece	Argentina	Other
Total		218.00	161.00	102.00	97.00	91.00	70.00	50.00	41.00	38.00	33.00	33.00	31.00	25.00	24.00	24.00	22.00	22.00	19.00	17.00	15.00	15.00	13.00	13.00	13.00	13.00	
France	218.00	150.33	11.12	4.50	2.00	2.00	3.98	9.50	0.83	0.00	0.00	3.00	6.07	1.00	1.00	5.83	2.50	0.00	1.50	0.50	1.00	0.50	2.58	0.00	0.25	0.00	8.00
United States	161.00	11.12	79.69	10.83	6.67	2.67	3.01	4.00	3.17	4.50	0.00	0.83	6.98	0.00	1.00	2.53	0.00	0.00	1.50	3.50	4.00	1.00	0.00	0.00	0.00	0.44	13.57
Canada	102.00	4.50	10.83	65.00	1.00	0.33	4.33	0.00	1.00	1.50	0.00	0.00	3.50	0.00	0.00	1.00	0.00	1.00	0.00	3.00	0.00	0.00	0.00	0.00	0.00	0.00	5.00
China	97.00	2.00	6.67	1.00	69.00	1.00	0.00	0.00	1.00	2.33	0.00	3.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5.83	0.00	0.00	4.50
Australia	91.00	2.00	2.67	0.33	1.00	63.67	5.17	0.00	4.92	0.25	1.00	1.00	0.50	0.00	0.00	0.00	0.00	1.00	0.00	0.50	1.67	0.00	0.33	0.00	0.00	0.00	5.00
Germany	70.00	3.98	3.01	4.33	0.00	5.17	33.22	1.00	1.50	1.00	2.00	1.33	0.57	1.00	0.00	0.50	1.00	1.00	0.67	0.50	1.33	0.00	0.00	0.00	0.00	0.22	6.67
Sweden	50.00	9.50	4.00	0.00	0.00	0.00	1.00	26.67	1.33	0.00	0.00	0.50	0.00	0.00	0.00	2.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	4.00
Italy	41.00	0.83	3.17	1.00	1.00	4.92	1.50	1.33	12.67	0.25	1.00	2.50	1.00	0.00	1.00	2.00	1.00	1.00	0.00	0.00	0.00	0.00	0.83	0.00	0.00	0.00	4.00
Turkey	38.00	0.00	4.50	1.50	2.33	0.25	1.00	0.00	0.25	24.67	0.00	0.50	2.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00
Austria	33.00	0.00	0.00	0.00	0.00	1.00	2.00	0.00	1.00	0.00	29.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
United Kingdom	33.00	3.00	0.83	0.00	3.67	1.00	1.33	0.50	2.50	0.50	0.00	9.00	1.50	0.00	0.00	0.50	0.00	0.00	1.17	0.00	1.00	0.00	0.50	0.00	1.50	0.00	4.50
Ireland	31.00	6.07	6.98	3.50	0.00	0.50	0.57	0.00	1.00	2.00	0.00	1.50	7.38	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.50
Poland	25.00	1.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	17.00	0.00	0.00	0.00	0.00	0.00	0.00	4.00	0.00	0.00	0.00	0.00	0.00	2.00
Chile	24.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	13.20	0.00	0.00	1.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.50	6.10
Belgium	24.00	5.83	2.53	1.00	0.00	0.00	0.50	2.00	2.00	0.00	0.00	0.50	0.00	0.00	0.00	5.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	3.60
Czech Republic	22.00	2.50	0.00	0.00	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	13.00	3.00	0.00	1.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Spain	22.00	0.00	0.00	1.00	0.00	1.00	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	1.20	0.00	3.00	11.40	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.40
Brazil	19.00	1.50	1.50	0.00	0.00	0.00	0.67	0.00	0.00	0.00	0.00	1.17	0.00	0.00	0.00	0.00	0.00	1.00	12.67	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.50
Iran	17.00	0.50	3.50	3.00	0.00	0.50	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.50	0.00	0.00	7.00	0.00	0.00	0.00	0.00	0.00	0.00	0.50
Denmark	15.00	1.00	4.00	0.00	0.00	1.67	1.33	0.00	0.00	0.00	0.00	1.00	0.00	4.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	1.00
Taiwan	15.00	0.50	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.00	0.50	0.00	0.00	0.00	5.00
Norway	13.00	2.58	0.00	0.00	0.00	0.33	0.00	1.00	0.83	0.00	0.00	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.50	4.00	0.00	0.25	0.00	3.00
Singapore	13.00	0.00	0.00	0.00	5.83	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.67	0.00	0.00	2.50
Greece	13.00	0.25	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.50	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.25	0.00	9.00	0.00	1.00
Argentina	13.00	0.00	0.44	0.00	0.00	0.00	0.22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10.00	1.83
Cameroon	12.00	0.00	0.00	1.00	0.00	0.00	0.00	2.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.00
South Korea	12.00	0.00	3.67	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	6.33
Tunisia	11.00	3.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	7.00
Netherlands	11.00	1.00	3.29	0.00	1.00	2.00	0.44	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.60	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.22	2.44
Hungary	10.00	1.50	0.00	2.50	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.00
Switzerland	7.00	0.00	0.00	0.00	0.00	0.00	1.00	2.00	4.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Pakistan	7.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6.00
Japan	7.00	0.00	0.50	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6.00
Mexico	6.00	0.00	1.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.00	0.00	0.00	2.00	0.00	0.00	0.00	0.00	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Hong Kong	6.00	0.00	1.00	0.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	2.00	0.00	0.00	0.00
Colombia	6.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.10	0.00	0.00	1.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.50	0.00
Luxembourg	5.00	0.50	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.50	0.00	0.00	0.00	3.00	0.00	0.00	0.00	0.00
Portugal	4.00	0.00	0.44	0.00	0.00	0.00	0.22	0.00	0.00	0.00	0.00	0.50	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.11	0.72
Indonesia	4.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.00	0.00	0.00	0.00	0.00	0.00
Serbia	4.00	0.00	0.00	0.00	0.00	0.00	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.50
Other		2.00	2.17	1.00	1.50	2.00	2.50	0.00	0.00	0.00	0.00	2.00	0.50	2.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.50	1.00	0.00	

Table 10: Heat Map based on Collaboration between Countries (Integer Count)

From/To	Total	United States	France	Germany	Canada	Australia	Italy	United Kingdom	Sweden	Belgium	Turkey	Norway	Denmark	Netherlands	Spain	Poland	China	Switzerland	Hong Kong	Ireland	Austria	Finland	Macao	Peru	Cyprus	Iran	Other
Total		123.00	109.00	72.00	66.00	59.00	58.00	54.00	47.00	41.00	37.00	36.00	36.00	35.00	34.00	30.00	29.00	28.00	27.00	27.00	25.00	23.00	22.00	21.00	21.00	13.00	
United States	123.00	0.00	16.00	7.00	14.00	5.00	5.00	3.00	6.00	3.00	6.00	1.00	5.00	6.00	1.00	1.00	7.00	1.00	2.00	7.00	1.00	2.00	1.00	1.00	1.00	4.00	17.00
France	109.00	16.00	0.00	7.00	8.00	4.00	3.00	5.00	12.00	7.00	1.00	6.00	2.00	2.00	1.00	2.00	2.00	1.00	1.00	7.00	1.00	1.00	1.00	1.00	1.00	1.00	16.00
Germany	72.00	7.00	7.00	0.00	6.00	8.00	3.00	3.00	2.00	2.00	2.00	1.00	3.00	2.00	2.00	2.00	0.00	2.00	1.00	1.00	3.00	1.00	1.00	1.00	1.00	1.00	10.00
Canada	66.00	14.00	8.00	6.00	0.00	2.00	2.00	1.00	1.00	2.00	3.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00	1.00	4.00	1.00	1.00	1.00	1.00	1.00	3.00	6.00
Australia	59.00	5.00	4.00	8.00	2.00	0.00	8.00	3.00	1.00	1.00	2.00	2.00	3.00	3.00	2.00	1.00	1.00	1.00	2.00	1.00	2.00	1.00	1.00	1.00	1.00	1.00	2.00
Italy	58.00	5.00	3.00	3.00	2.00	8.00	0.00	4.00	2.00	3.00	2.00	3.00	1.00	1.00	3.00	1.00	1.00	5.00	1.00	1.00	2.00	1.00	1.00	1.00	1.00	0.00	3.00
United Kingdom	54.00	3.00	5.00	3.00	1.00	3.00	4.00	0.00	2.00	2.00	2.00	2.00	2.00	1.00	1.00	1.00	4.00	1.00	1.00	2.00	1.00	1.00	1.00	1.00	1.00	0.00	9.00
Sweden	47.00	6.00	12.00	2.00	1.00	1.00	2.00	2.00	0.00	3.00	1.00	2.00	1.00	1.00	1.00	1.00	0.00	3.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	0.00	2.00
Belgium	41.00	3.00	7.00	2.00	2.00	1.00	3.00	2.00	3.00	0.00	1.00	1.00	1.00	2.00	1.00	1.00	0.00	1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	0.00	4.00
Turkey	37.00	6.00	1.00	2.00	3.00	2.00	2.00	2.00	1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	2.00	1.00	1.00	1.00	1.00	1.00	0.00	1.00
Norway	36.00	1.00	6.00	1.00	1.00	2.00	3.00	2.00	2.00	1.00	1.00	0.00	1.00	1.00	1.00	1.00	0.00	1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	0.00	5.00
Denmark	36.00	5.00	2.00	3.00	1.00	3.00	1.00	2.00	1.00	1.00	1.00	1.00	0.00	1.00	1.00	5.00	0.00	1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	0.00	1.00
Netherlands	35.00	6.00	2.00	2.00	1.00	3.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00	0.00	1.00	2.00	1.00	1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	0.00	2.00
Spain	34.00	1.00	1.00	2.00	2.00	2.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.00	1.00	0.00	1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	0.00	8.00
Poland	30.00	1.00	2.00	2.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	5.00	2.00	1.00	0.00	0.00	1.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	0.00	2.00
China	29.00	7.00	2.00	0.00	1.00	1.00	1.00	4.00	0.00	0.00	2.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	8.00
Switzerland	28.00	1.00	1.00	2.00	1.00	1.00	5.00	1.00	3.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.00	0.00	1.00	0.00	1.00	1.00	1.00	1.00	1.00	0.00	0.00
Hong Kong	27.00	2.00	1.00	1.00	1.00	2.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.00	0.00	1.00	1.00	1.00	1.00	1.00	0.00	3.00
Ireland	27.00	7.00	7.00	1.00	4.00	1.00	1.00	2.00	0.00	0.00	2.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	1.00
Austria	25.00	1.00	1.00	3.00	1.00	2.00	2.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.00	1.00	1.00	0.00	0.00	1.00	1.00	1.00	1.00	0.00	0.00
Finland	23.00	2.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.00	1.00	1.00	1.00	1.00	0.00	1.00	1.00	1.00	0.00	0.00
Macao	22.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.00	1.00	1.00	0.00	1.00	1.00	0.00	0.00
Peru	21.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.00	1.00	1.00	0.00	1.00	1.00	1.00	0.00	1.00	0.00	0.00
Cyprus	21.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.00	1.00	1.00	0.00	1.00	1.00	1.00	1.00	0.00	0.00	0.00
Iran	13.00	4.00	1.00	1.00	3.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.00
Chile	12.00	1.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8.00
Czech Republic	12.00	0.00	3.00	1.00	0.00	0.00	2.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.00	0.00
South Korea	10.00	4.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.00
Singapore	9.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6.00	0.00	2.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00
Brazil	9.00	2.00	2.00	1.00	0.00	0.00	0.00	2.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00
Taiwan	8.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.00
Colombia	8.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6.00
Portugal	7.00	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.00
Mexico	7.00	2.00	0.00	0.00	0.00	0.00	0.00	2.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	2.00
Hungary	7.00	0.00	2.00	1.00	3.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Argentina	7.00	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.00
Luxembourg	6.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00
Tunisia	6.00	0.00	3.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.00
Greece	6.00	0.00	1.00	0.00	0.00	0.00	0.00	2.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00
United Arab Emirates	5.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.00
Other		4.00	2.00	4.00	3.00	2.00	0.00	2.00	2.00	2.00	0.00	0.00	0.00	0.00	0.00	2.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	

5 Conference Papers by Most Common Conference Series

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CP
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ICAART
CSCLP
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PATAT
INAP
ETFA
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ICRA
KR
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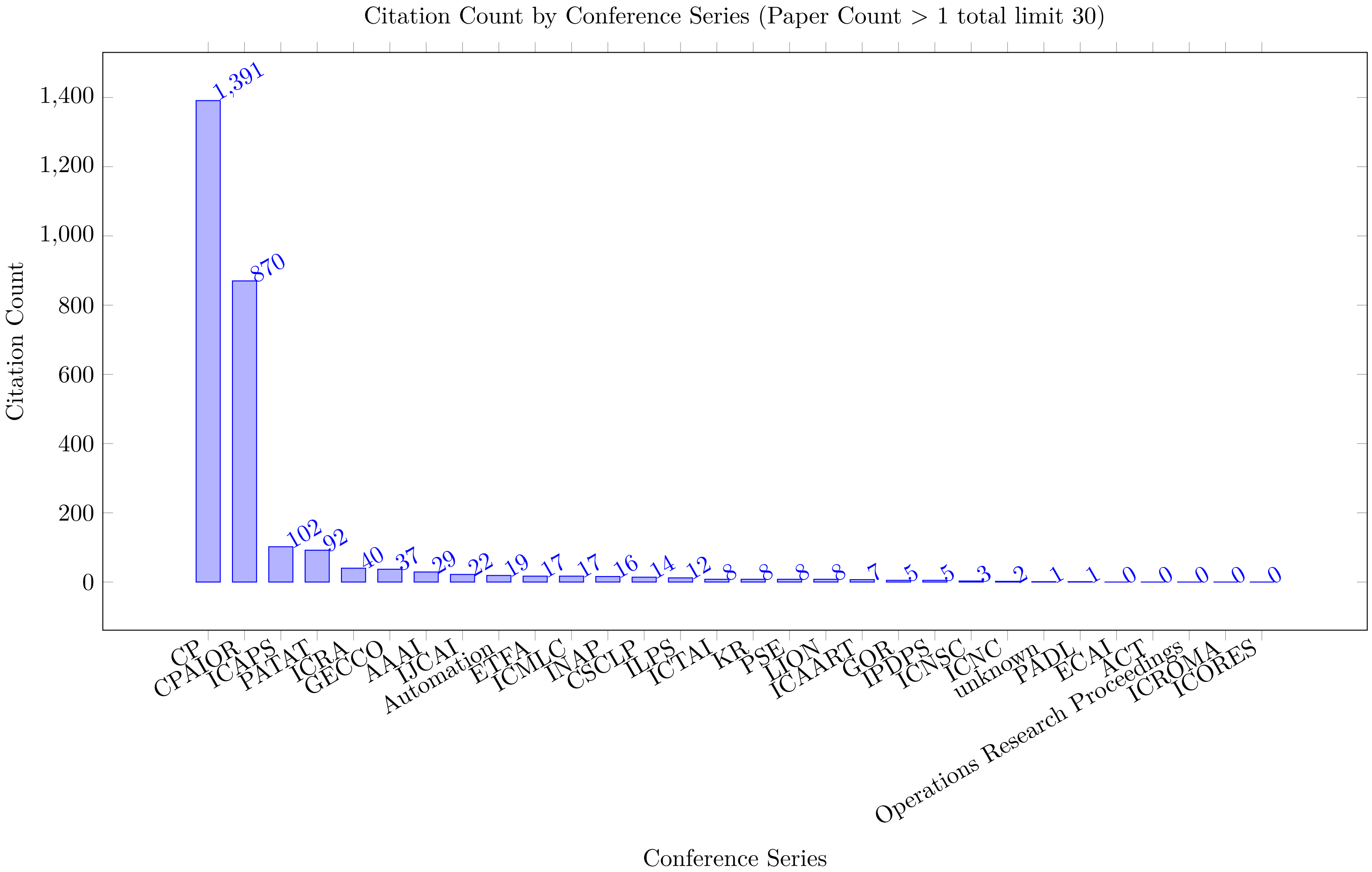
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CP
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ICAPS
PATAT
ICRA
GECCO
AAAI
IJCAI
Automation
ETFA
ICMLC
INAP
CSCLP
ILPS
ICTAI
KR
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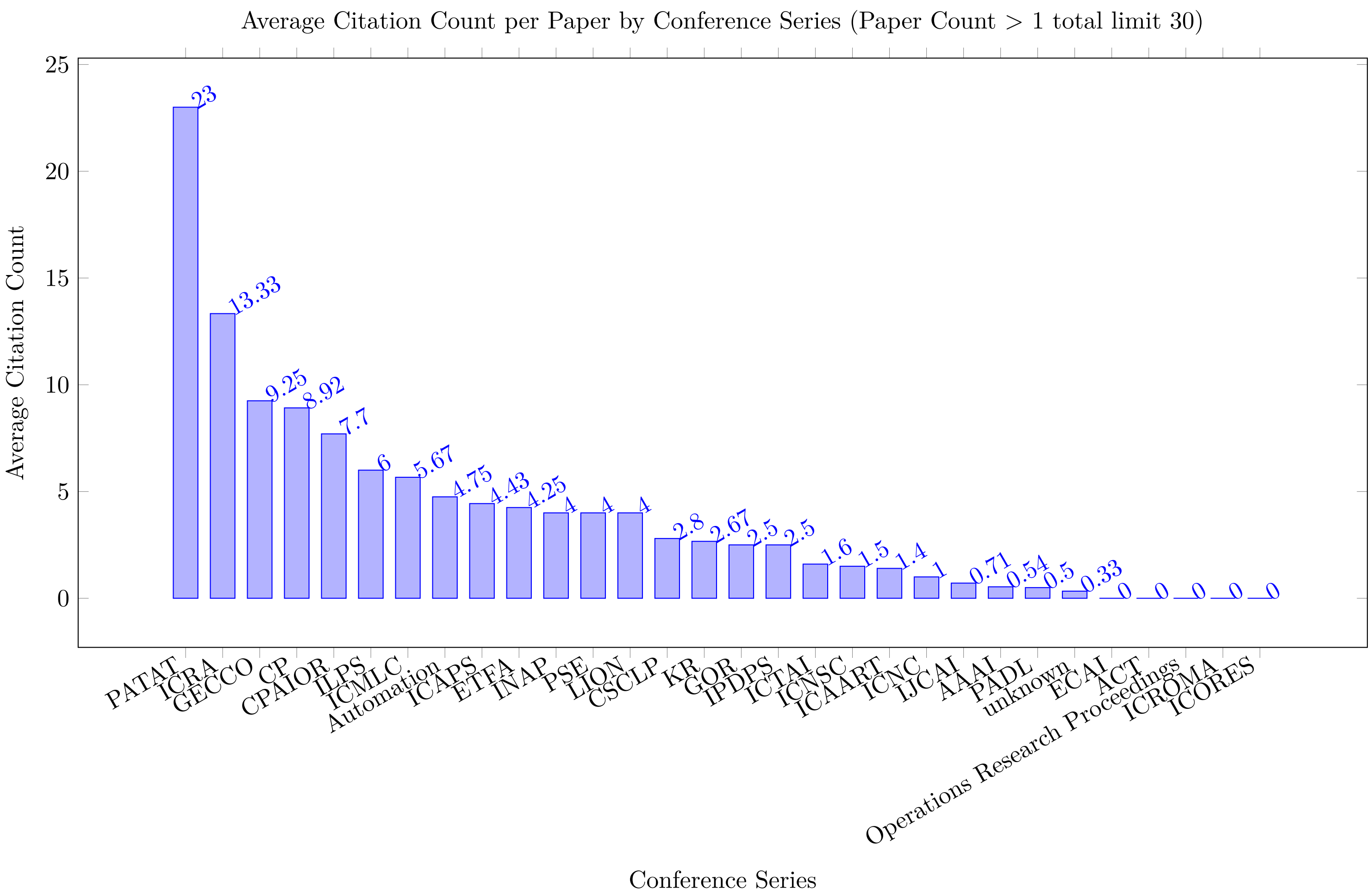
Average Citation Count per Paper by Conference Series (Paper Count > 1 total limit 30)
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PATAT
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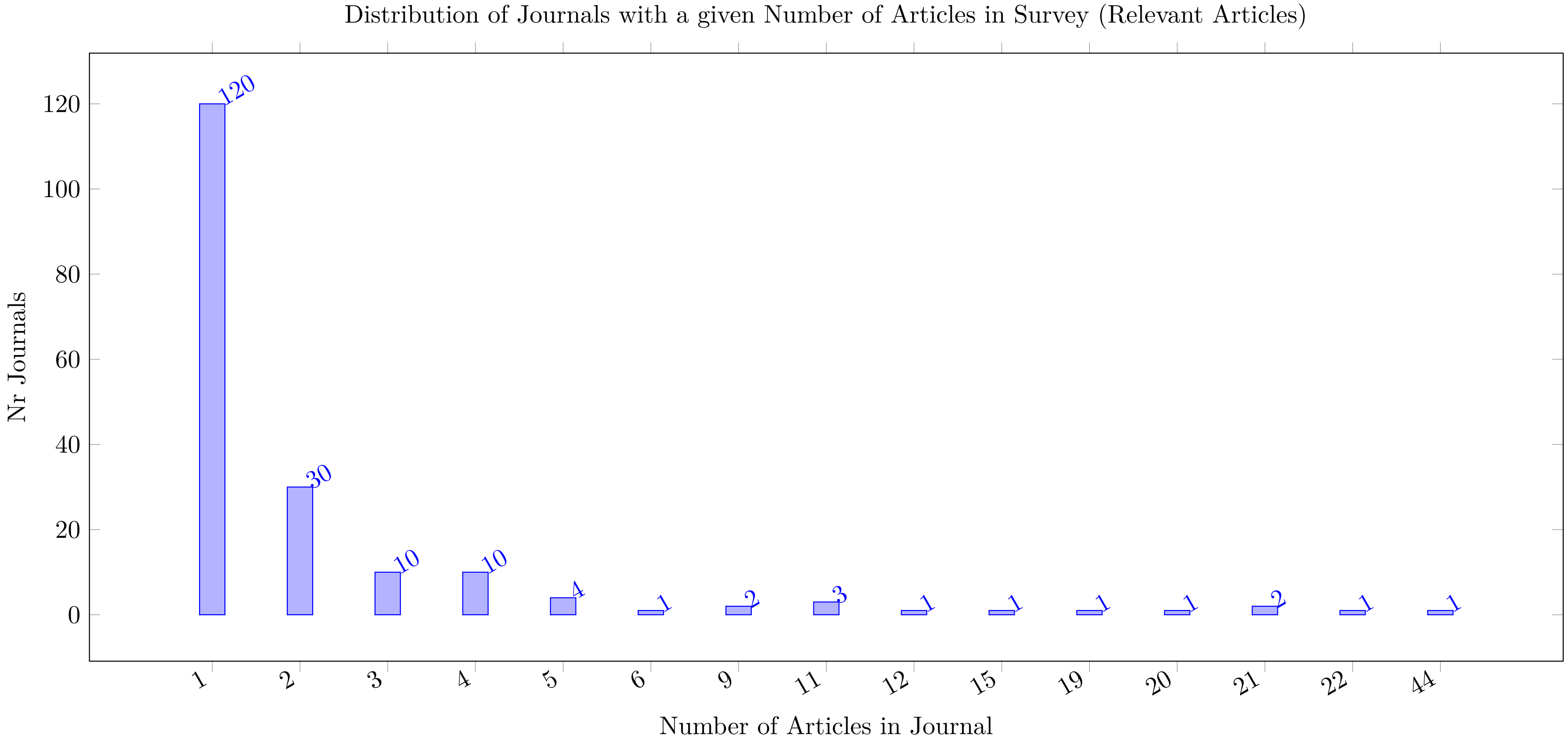
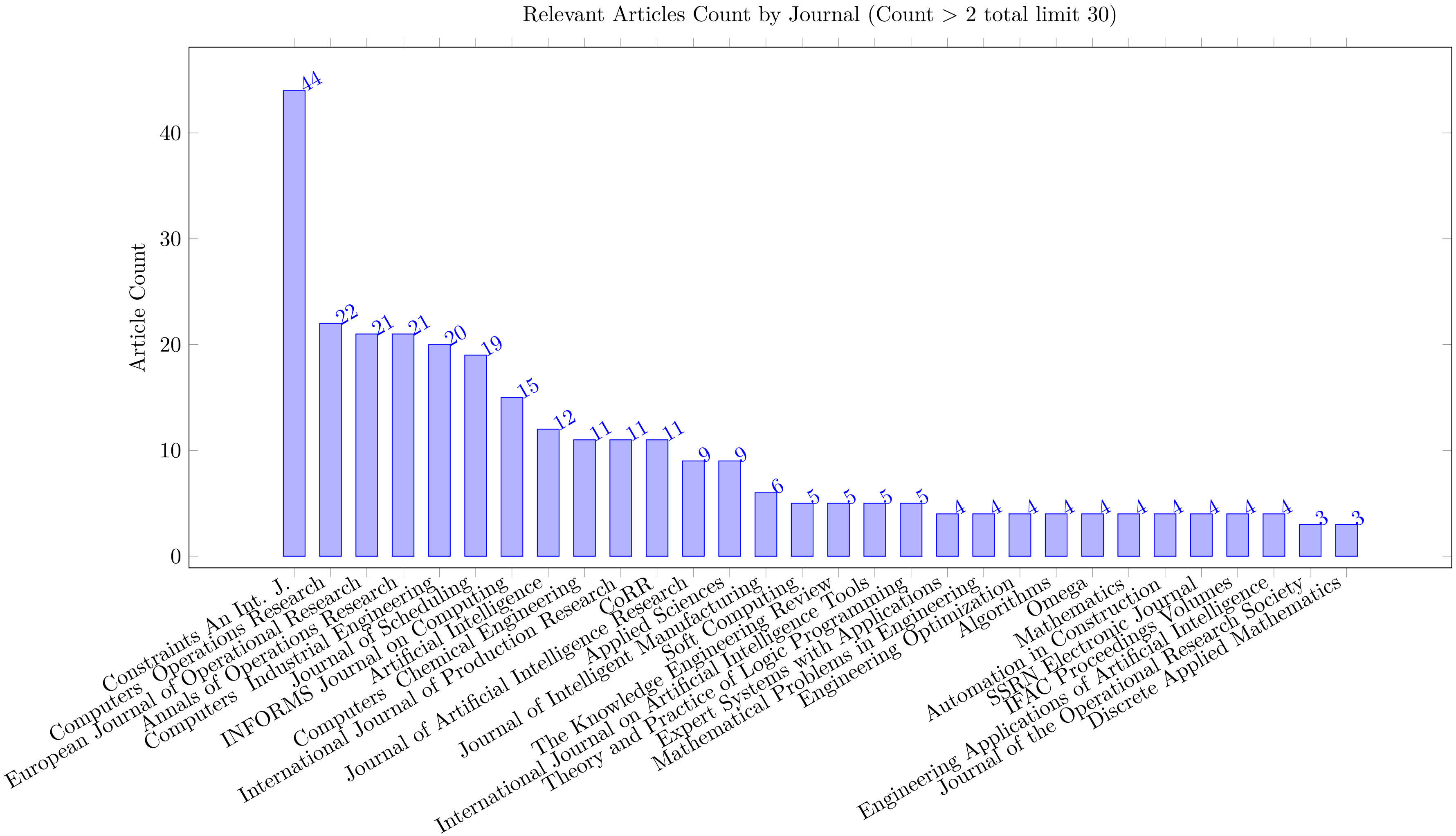
GECCO
CP
CPAIOR
ILPS
ICMLC
Automation
ICAPS
ETFA
INAP
PSE
LION
CSCLP
KR
GOR
IPDPS
ICTAI
ICNSC
ICAART
ICNC
IJCAI
AAAI
PADL
unknown
ECAI
ACT
Operations Research Proceedings
ICROMA
ICORES

6 Journal Articles by Most Common Journals

6.1 Relevant Articles




Relevant Articles Count by Journal (Count > 2 total limit 30)
Article Count
Journal
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30
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44
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20
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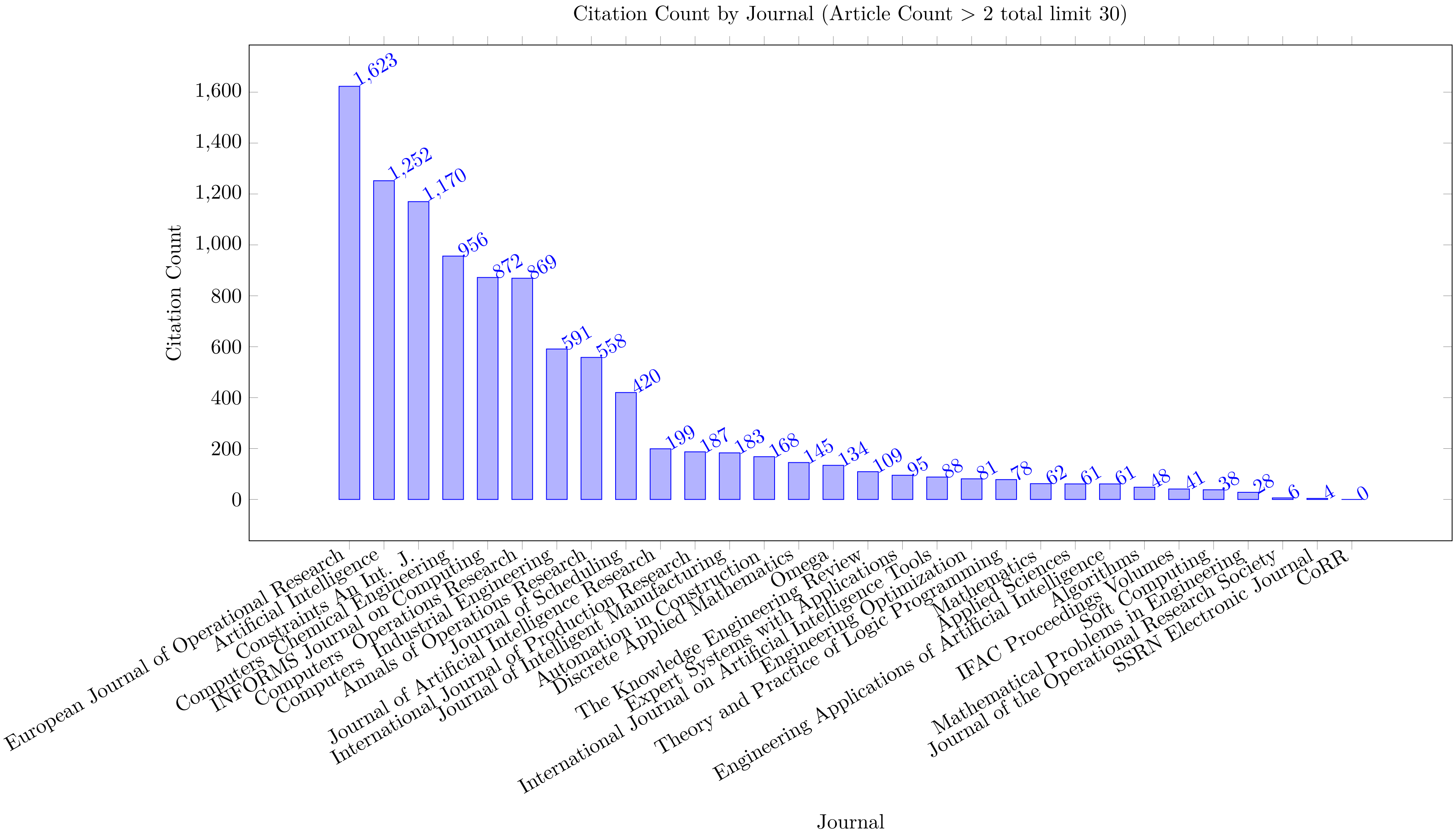
11
11
9
9
6
5
5
5
5
4
4
4
4
4
4
4
4
4
4
3
3
Constraints An Int. J.
Computers Operations Research
European Journal of Operational Research
Annals of Operations Research
Computers Industrial Engineering
Journal of Scheduling
INFORMS Journal on Computing
Artificial Intelligence
Computers Chemical Engineering
International Journal of Production Research
CoRR
Journal of Artificial Intelligence Research
Applied Sciences
Journal of Intelligent Manufacturing
Soft Computing
The Knowledge Engineering Review
International Journal on Artificial Intelligence Tools
Theory and Practice of Logic Programming
Expert Systems with Applications
Mathematical Problems in Engineering
Engineering Optimization
Algorithms
Omega
Mathematics
Automation in Construction
SSRN Electronic Journal
IFAC Proceedings Volumes
Engineering Applications of Artificial Intelligence
Journal of the Operational Research Society
Discrete Applied Mathematics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183
168
145
134
109
95
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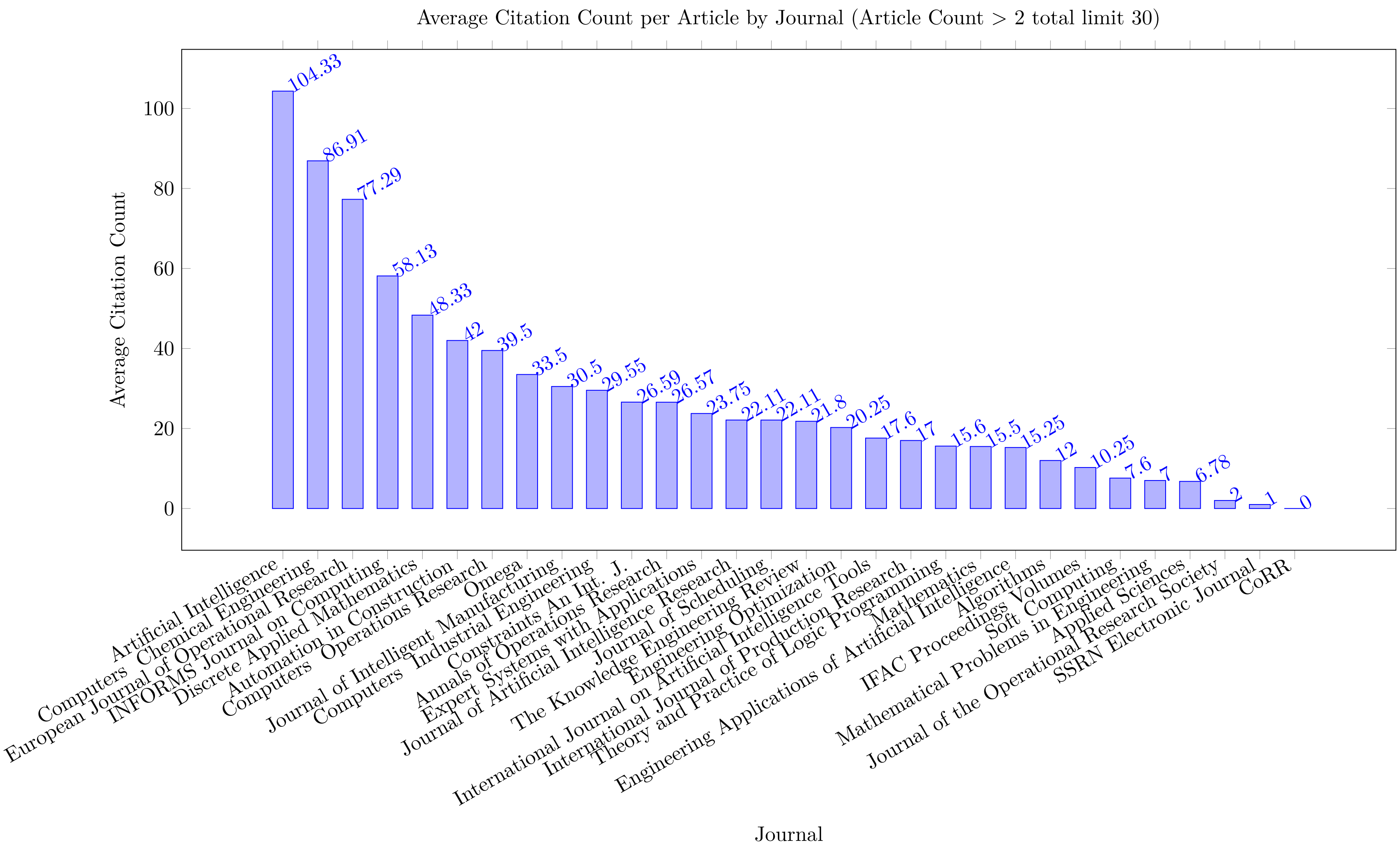
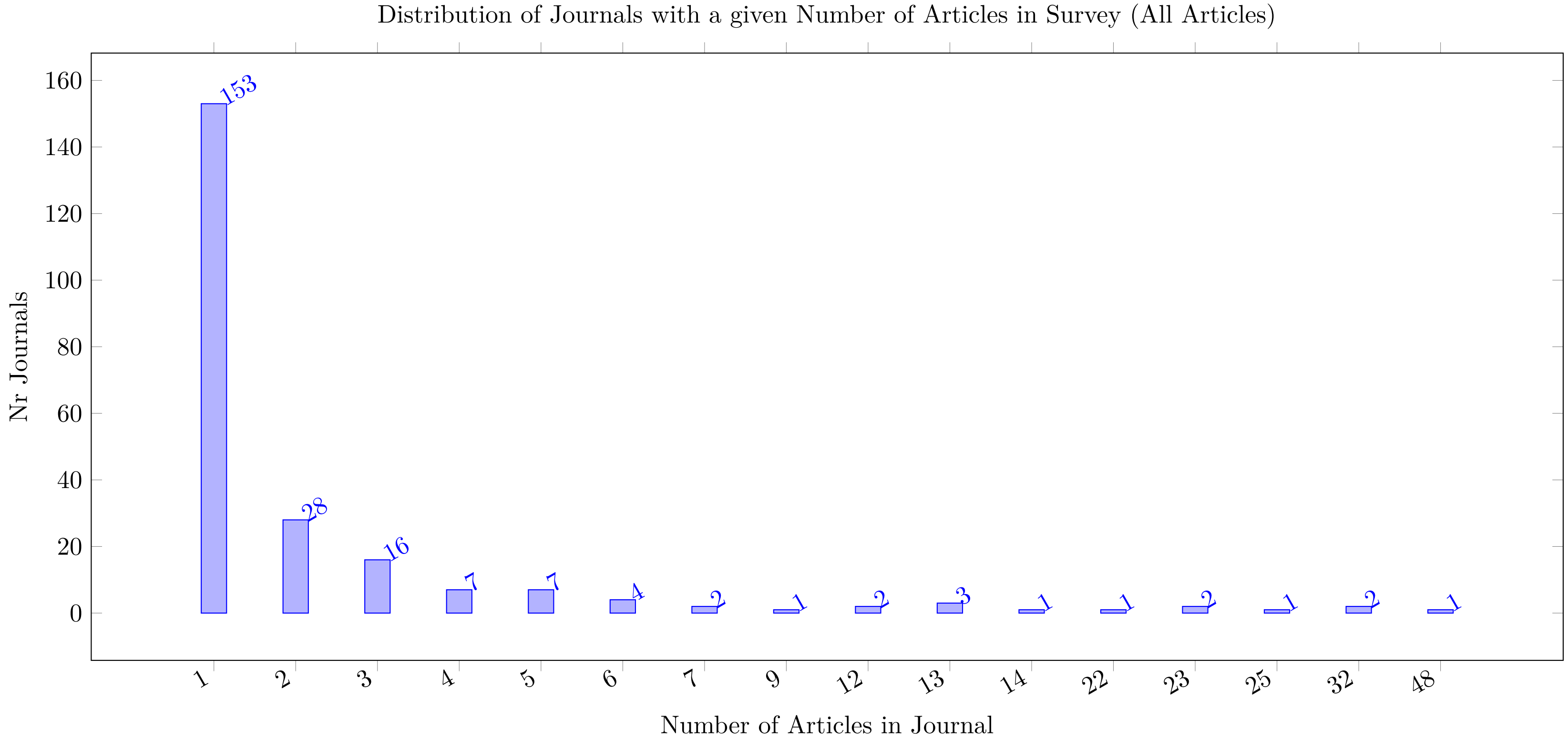
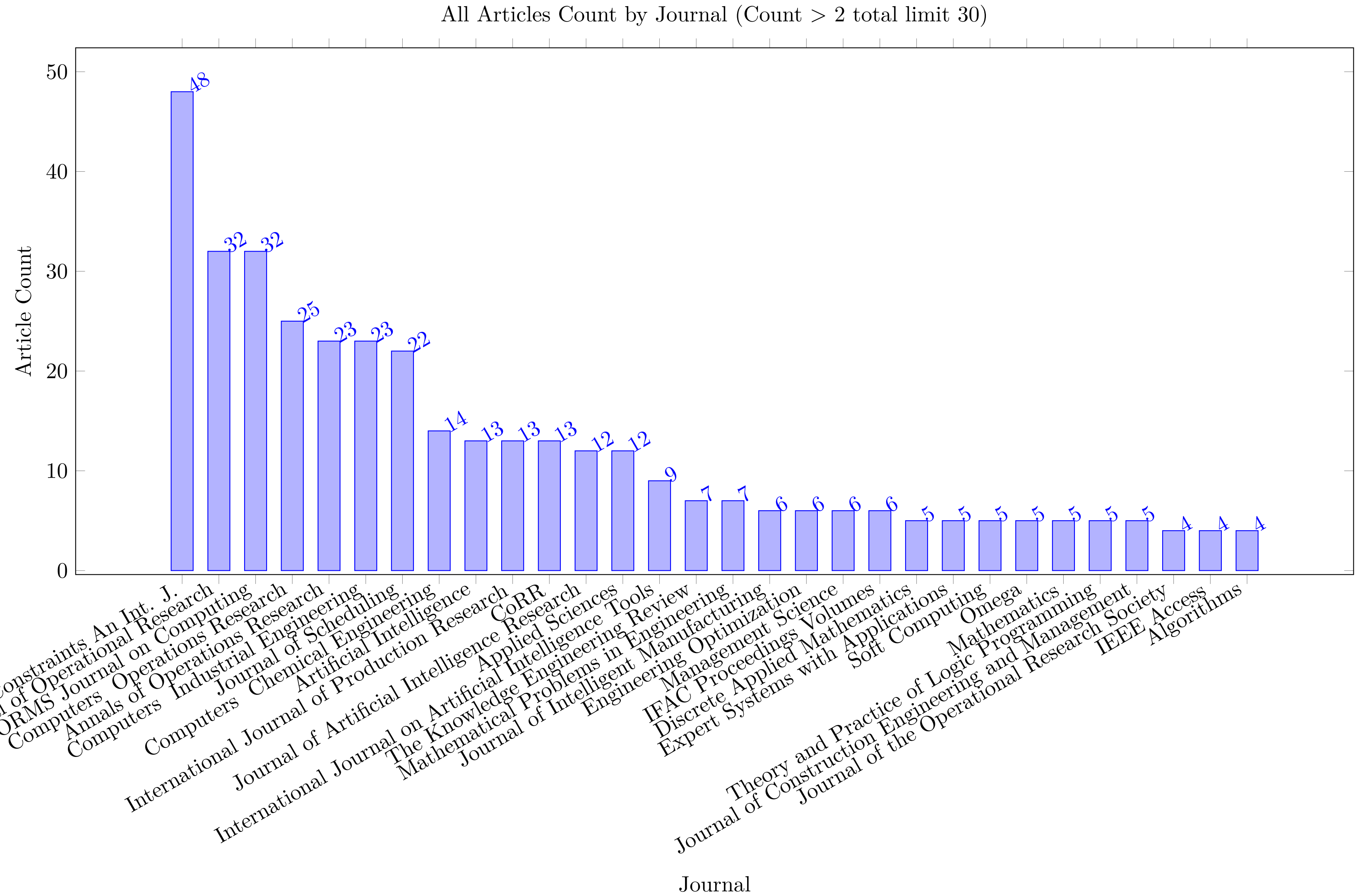
81
78
62
61
61
48
41
38
28
6
4
0
European Journal of Operational Research
Artificial Intelligence
Constraints An Int. J.
Computers Chemical Engineering
INFORMS Journal on Computing
Computers Operations Research
Computers Industrial Engineering
Annals of Operations Research
Journal of Scheduling
Journal of Artificial Intelligence Research
International Journal of Production Research
Journal of Intelligent Manufacturing
Automation in Construction
Discrete Applied Mathematics
Omega
The Knowledge Engineering Review
Expert Systems with Applications
International Journal on Artificial Intelligence Tools
Engineering Optimization
Theory and Practice of Logic Programming
Mathematics
Applied Sciences
Engineering Applications of Artificial Intelligence
Algorithms
IFAC Proceedings Volumes
Soft Computing
Mathematical Problems in Engineering
Journal of the Operational Research Society
SSRN Electronic Journal
CoRR

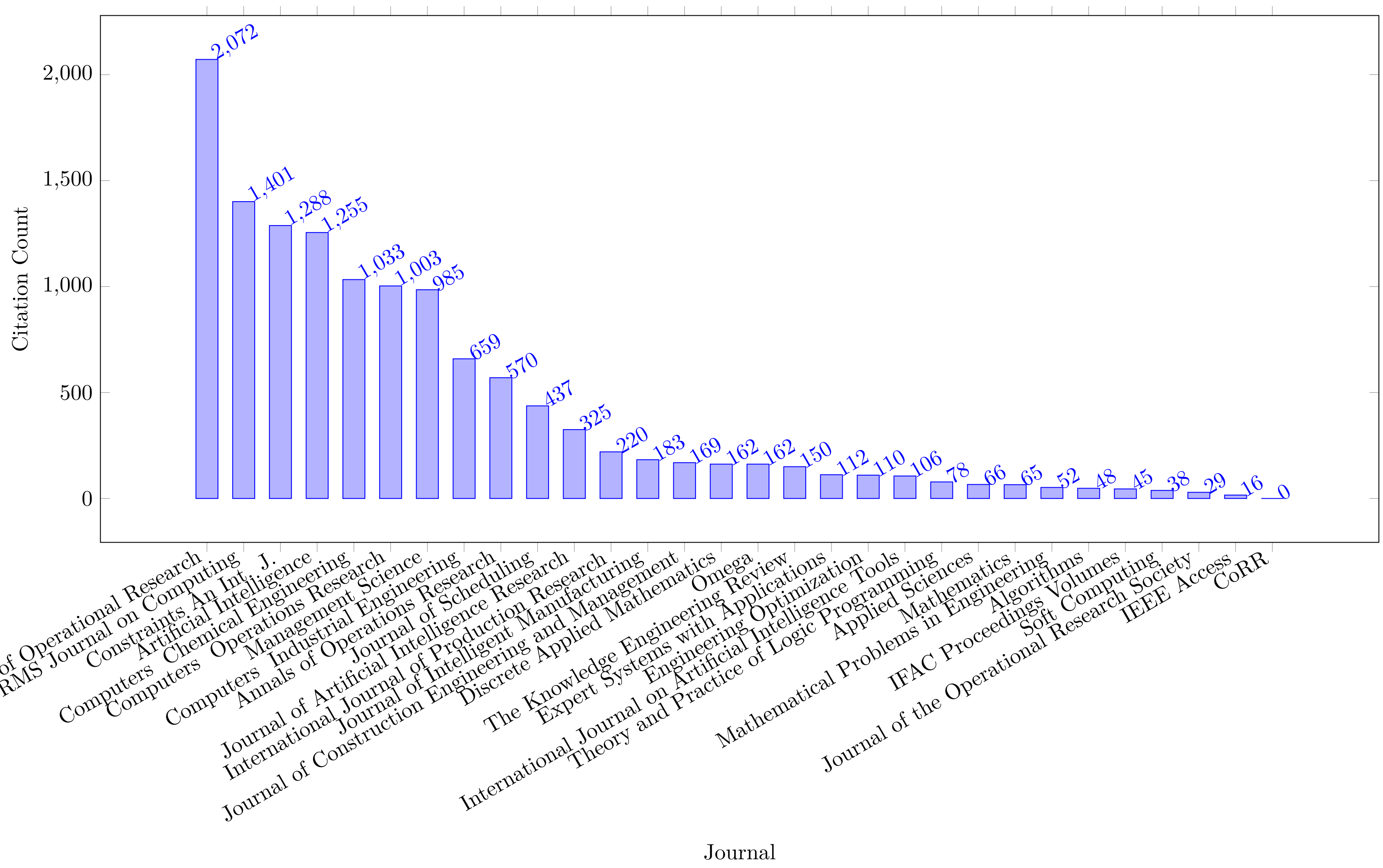

Average Citation Count per Article by Journal (Article Count > 2 total limit 30)
Average Citation Count
Journal
Artificial Intelligence 104.33
Computers Chemical Engineering 86.91
European Journal of Operational Research 77.29
INFORMS Journal on Computing 58.13
Discrete Applied Mathematics 48.33
Automation in Construction 42
Computers Operations Research 39.5
Omega 33.5
Journal of Intelligent Manufacturing 30.5
Computers Industrial Engineering 29.55
Constraints An Int. J. 26.59
Annals of Operations Research 26.57
Expert Systems with Applications 23.75
Journal of Artificial Intelligence Research 22.11
Journal of Scheduling 22.11
The Knowledge Engineering Review 21.8
Engineering Optimization 20.25
International Journal on Artificial Intelligence Tools 17.6
International Journal of Production Research 17
Theory and Practice of Logic Programming 15.6
Mathematics 15.5
Engineering Applications of Artificial Intelligence 15.25
Algorithms 12
IFAC Proceedings Volumes 10.25
Soft Computing 7.6
Mathematical Problems in Engineering 7
Applied Sciences 6.78
Journal of the Operational Research Society 2
SSRN Electronic Journal 1
CoRR 0

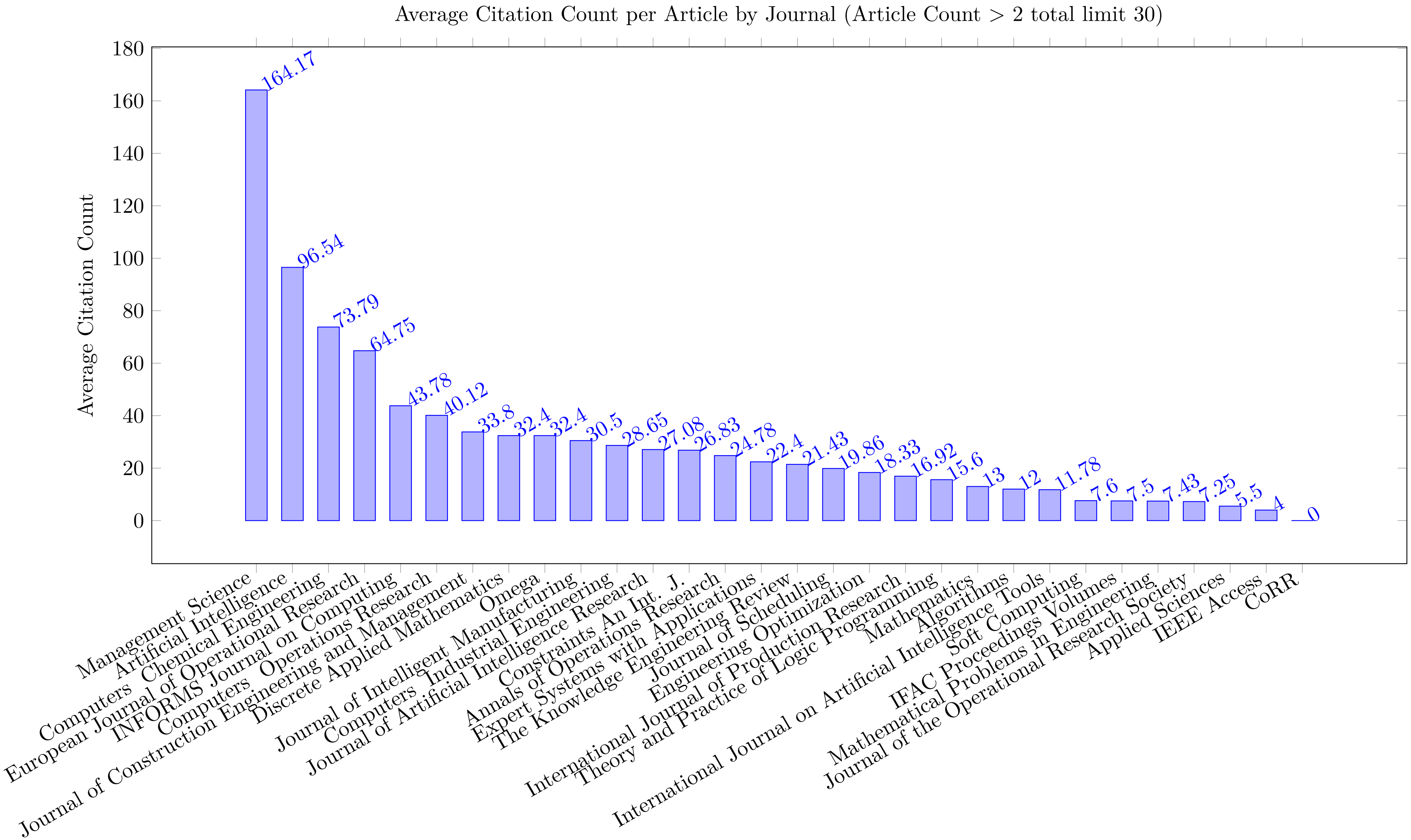
6.2 All Articles




All Articles Count by Journal (Count > 2 total limit 30)
Article Count
Journal
Constraints An Int. J. 48
European Journal of Operational Research 32
INFORMS Journal on Computing 32
Computers Operations Research 25
Annals of Operations Research 23
Computers Industrial Engineering 23
Journal of Scheduling 22
Computers Chemical Engineering 14
Artificial Intelligence 13
International Journal of Production Research 13
CoRR 13
Journal of Artificial Intelligence Research 12
Applied Sciences 12
International Journal on Artificial Intelligence Tools 9
The Knowledge Engineering Review 7
Mathematical Problems in Engineering 7
Journal of Intelligent Manufacturing 6
Engineering Optimization 6
Management Science 6
IFAC Proceedings Volumes 6
Discrete Applied Mathematics 5
Expert Systems with Applications 5
Soft Computing 5
Omega 5
Mathematics 5
Theory and Practice of Logic Programming 5
Journal of Construction Engineering and Management 5
Journal of the Operational Research Society 4
IEEE Access 4
Algorithms 4

Citation Count by Journal (Article Count > 2 total limit 30)

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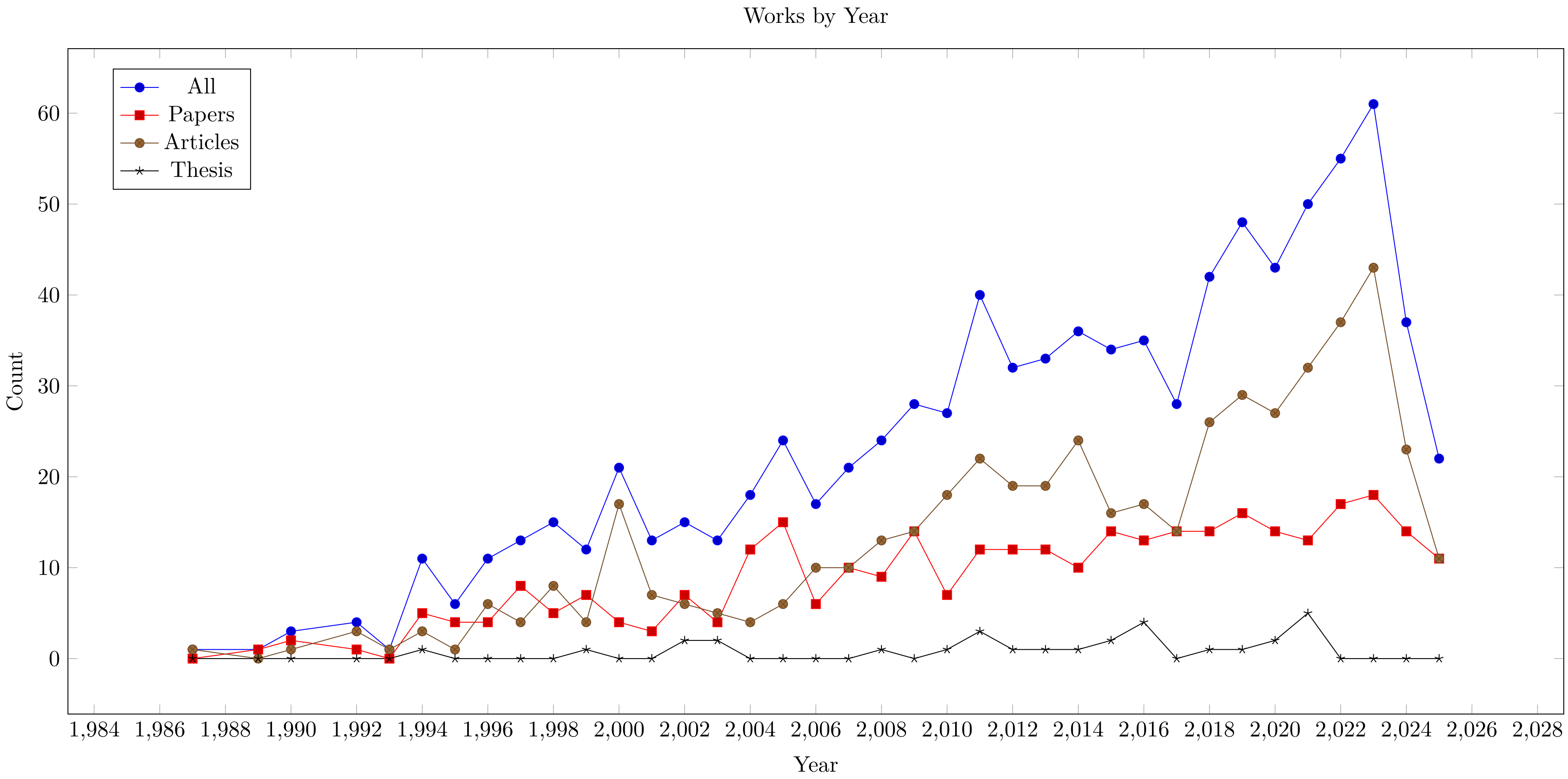
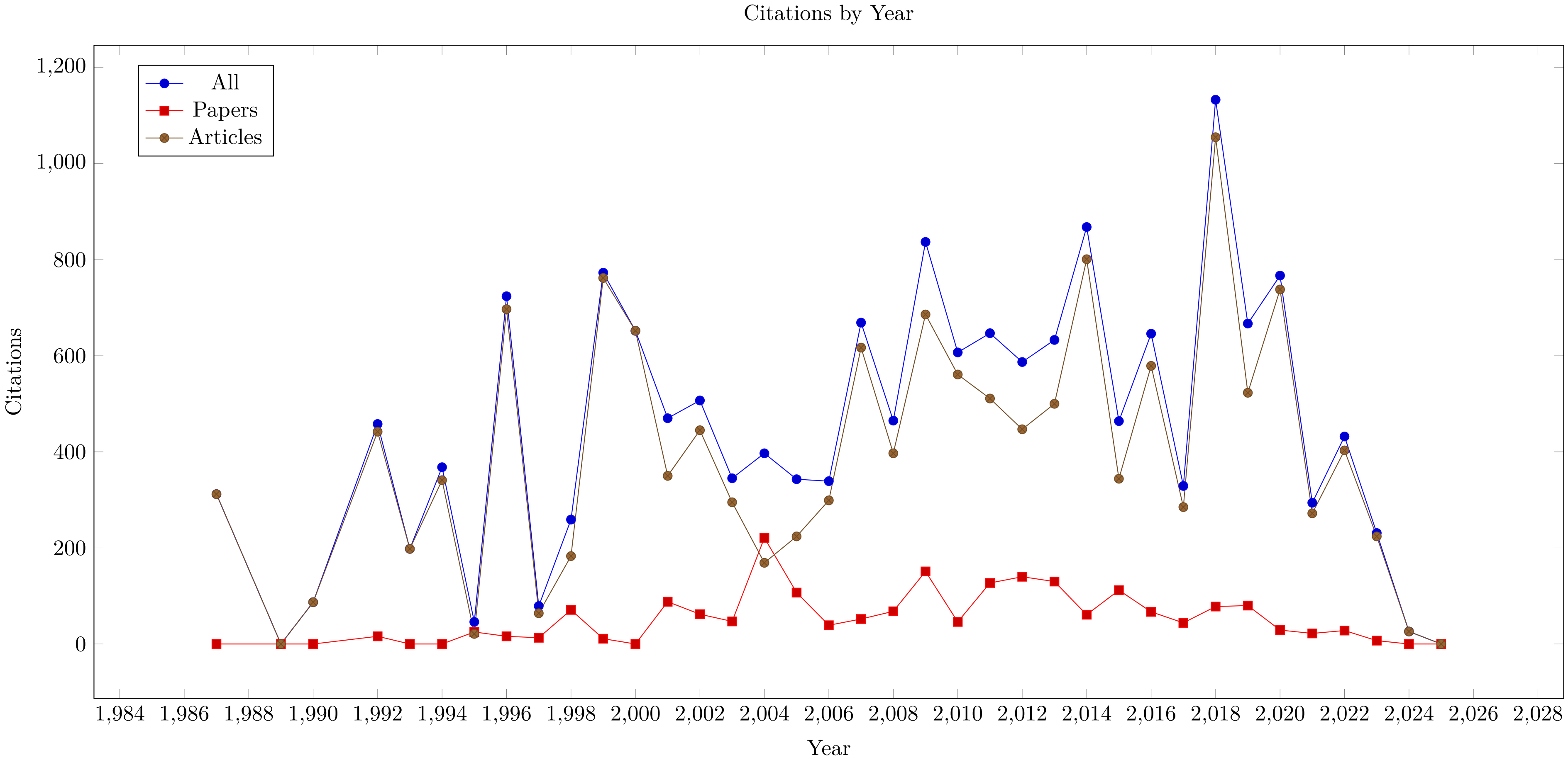
Management Science 164.17
Artificial Intelligence 96.54
Computers Chemical Engineering 73.79
European Journal of Operational Research 64.75
INFORMS Journal on Computing 43.78
Computers Operations Research 40.12
Journal of Construction Engineering and Management 33.8
Discrete Applied Mathematics 32.4
Omega 32.4
Journal of Intelligent Manufacturing 30.5
Computers Industrial Engineering 28.65
Journal of Artificial Intelligence Research 27.08
Constraints An Int. J. 26.83
Annals of Operations Research 24.78
Expert Systems with Applications 22.4
The Knowledge Engineering Review 21.43
Journal of Scheduling 19.86
Engineering Optimization 18.33
International Journal of Production Research 16.92
Theory and Practice of Logic Programming 15.6
Mathematics 13
Algorithms 12
International Journal on Artificial Intelligence Tools 11.78
Soft Computing 7.6
IFAC Proceedings Volumes 7.5
Mathematical Problems in Engineering 7.43
Journal of the Operational Research Society 7.25
Applied Sciences 5.5
IEEE Access 4
CoRR 0

7 Works by Year

7.1 Relevant Works




Citations by Year
All
Papers
Articles
Citations
Year
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600
800
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1,200
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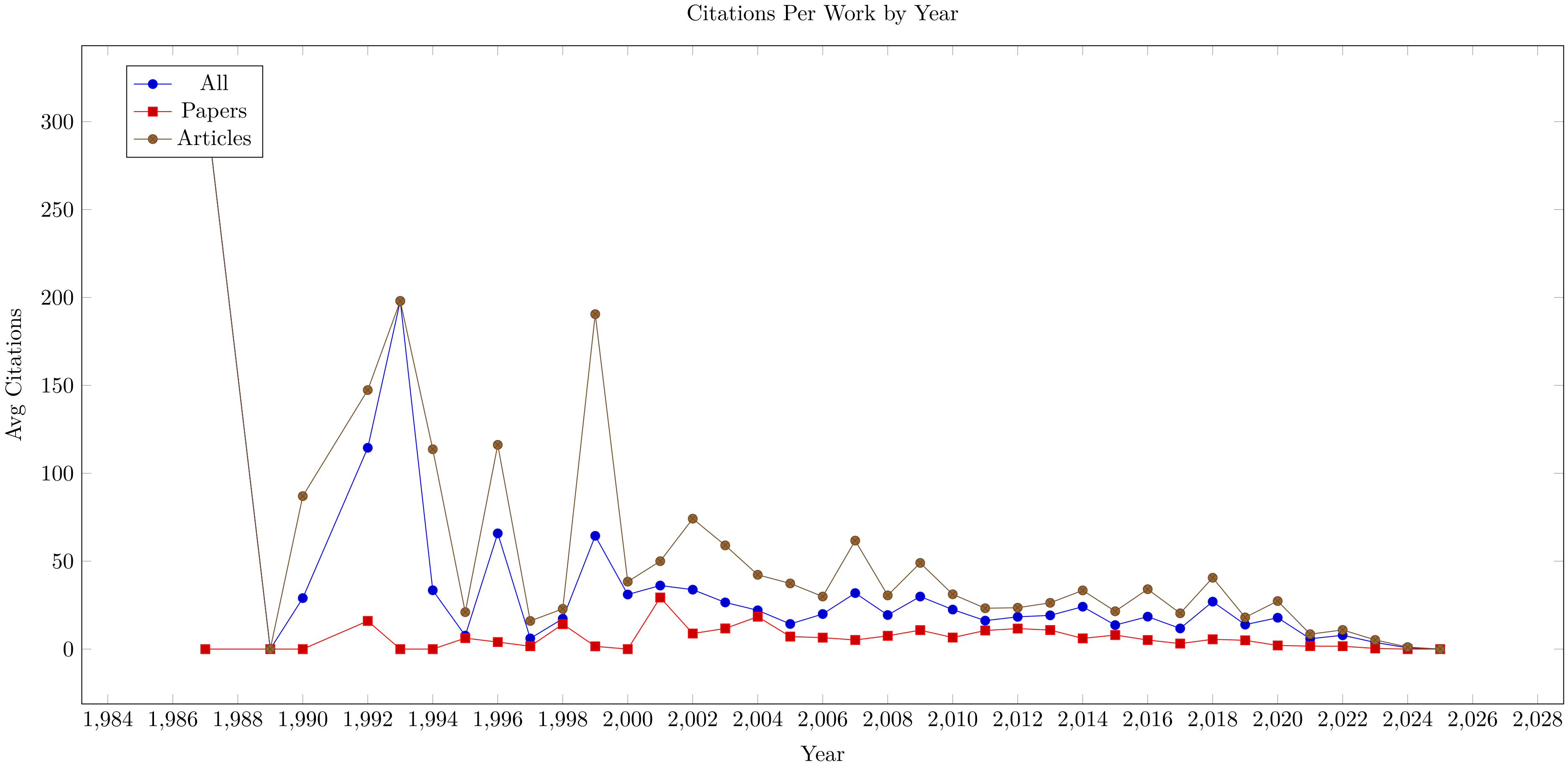
Citations Per Work by Year
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Papers
Articles
Avg Citations
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150
200
250
300
Year
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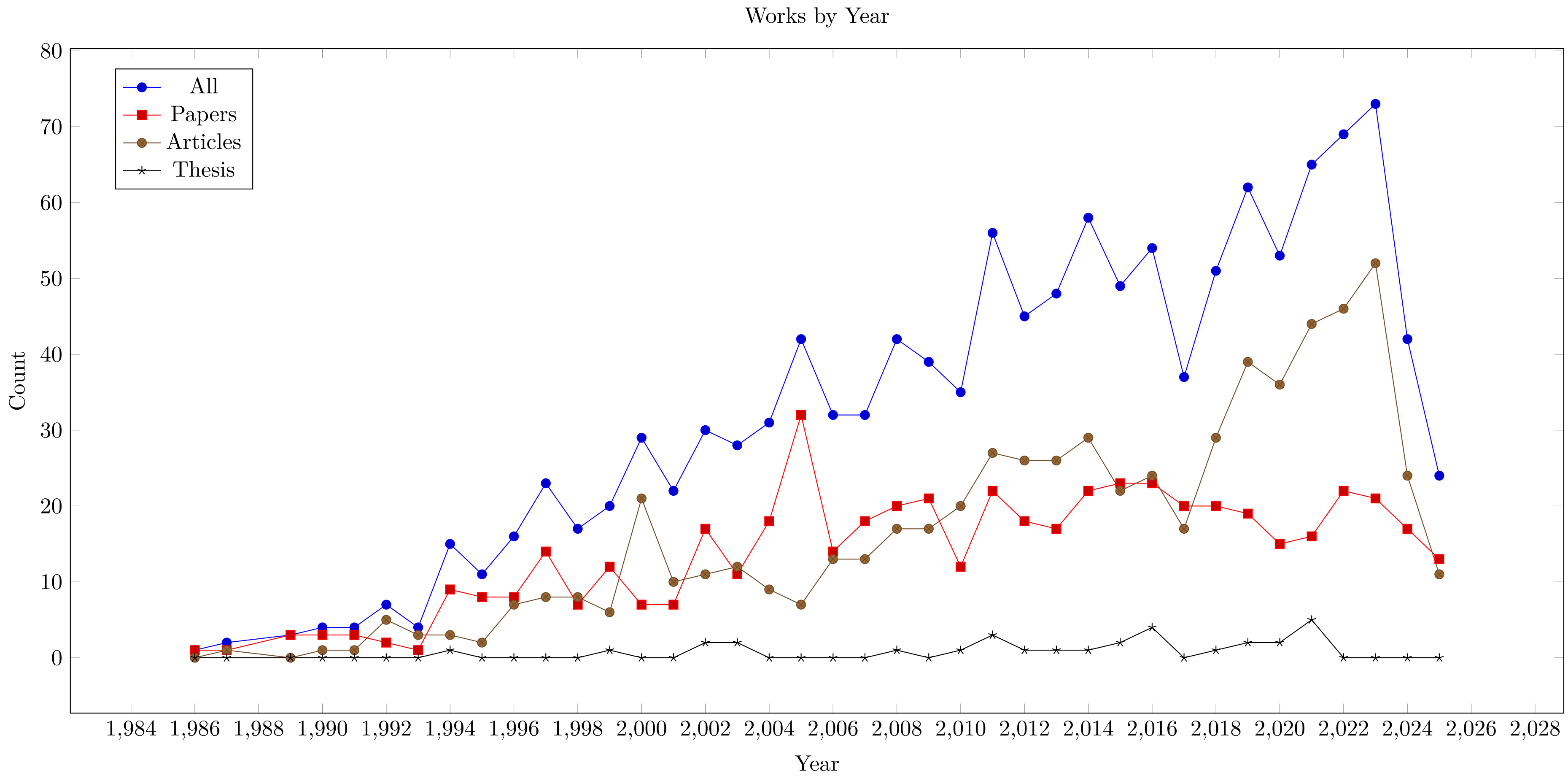
2,018
2,020
2,022
2,024
2,026
2,028

7.2 All Works


Works by Year
All
Papers
Articles
Thesis
Count
Year
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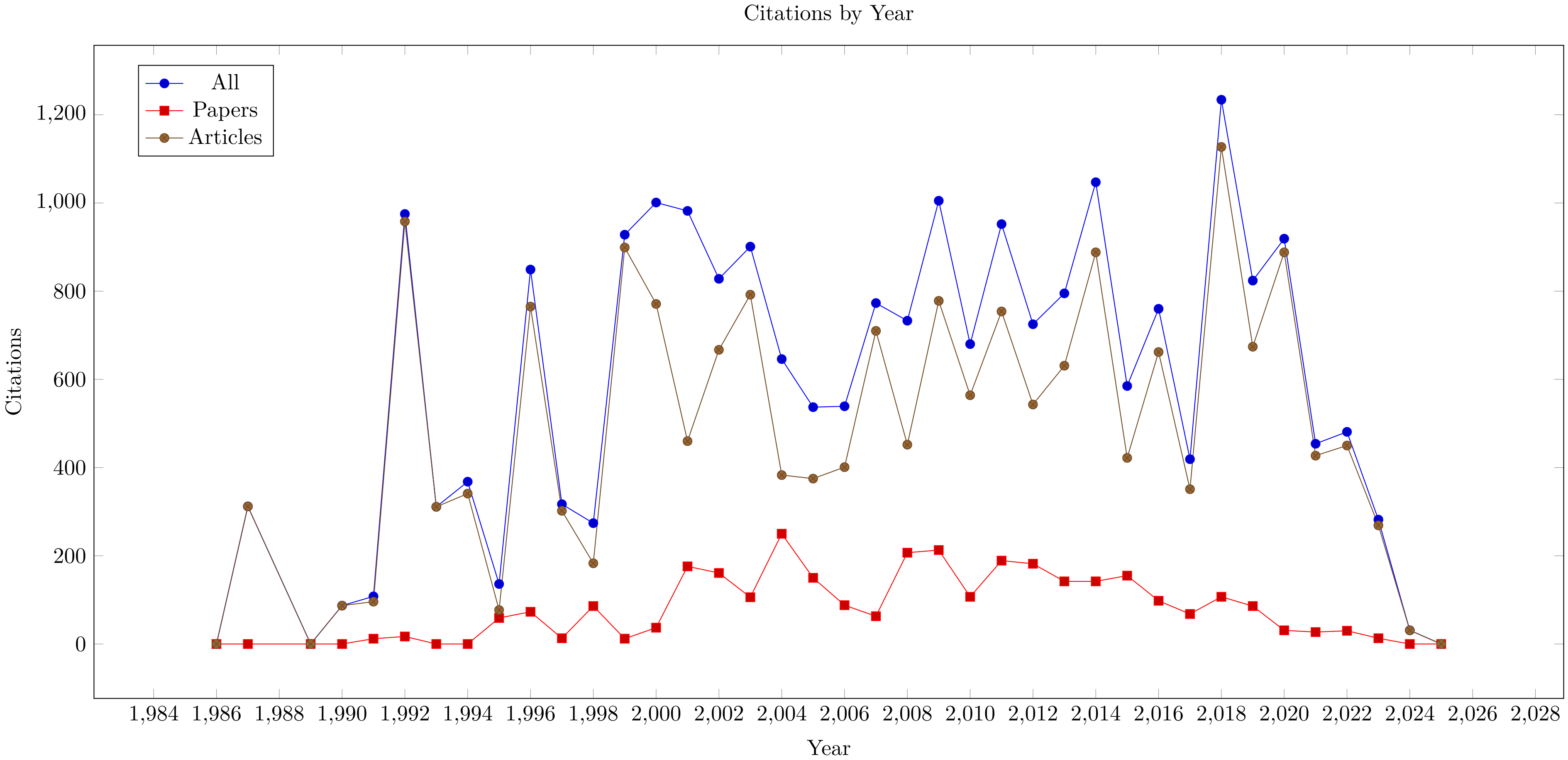
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Articles
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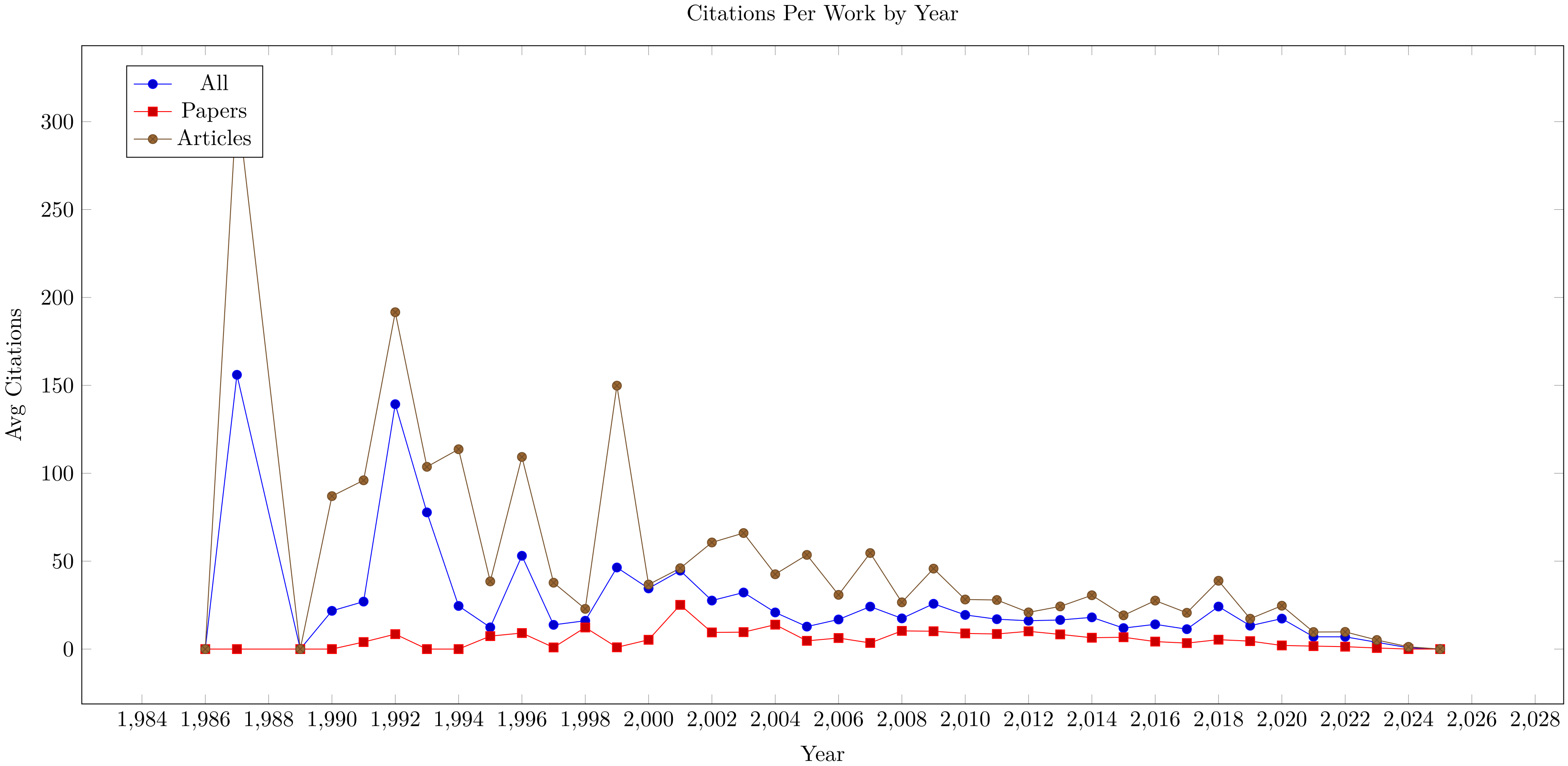
Citations Per Work by Year
All
Papers
Articles
Avg Citations
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200
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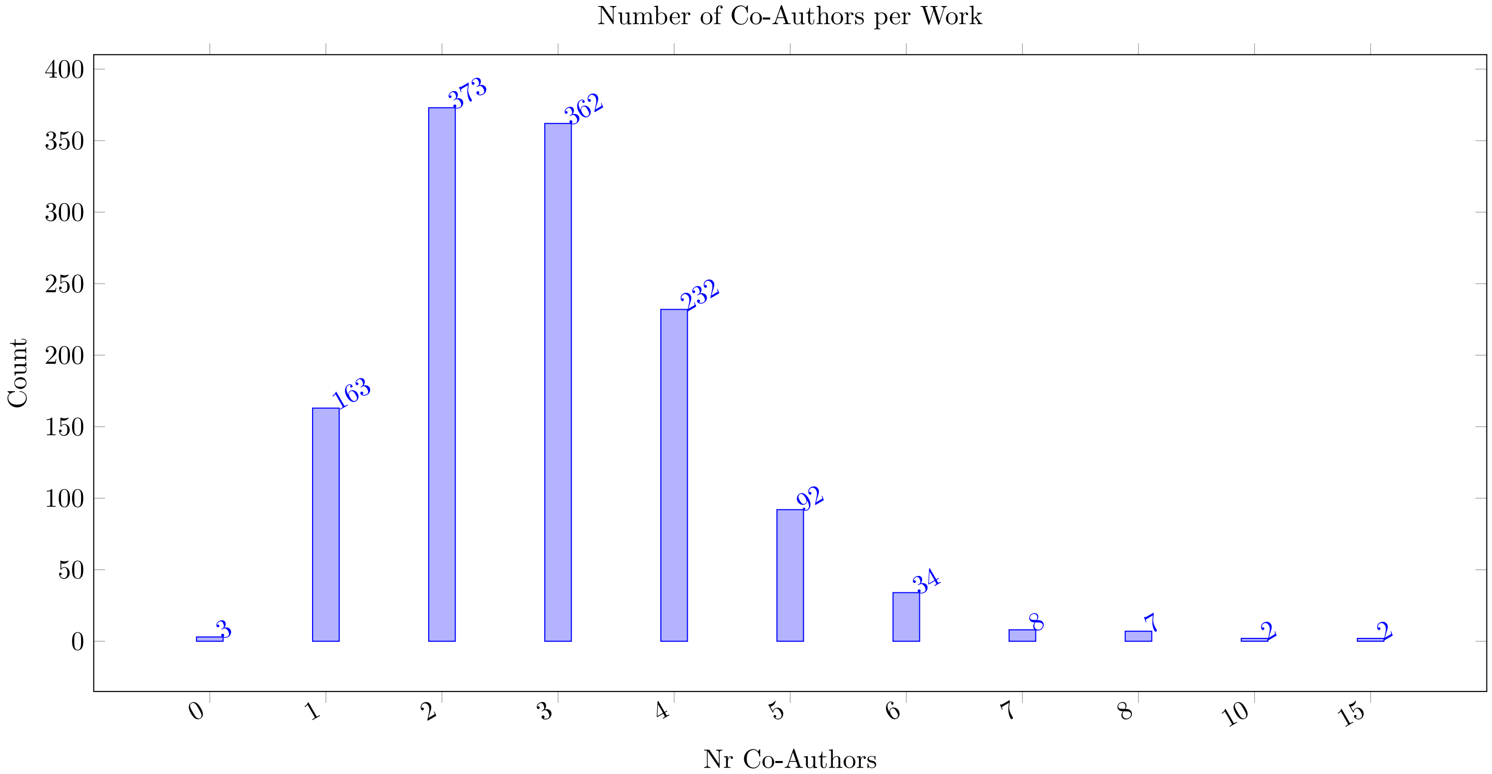
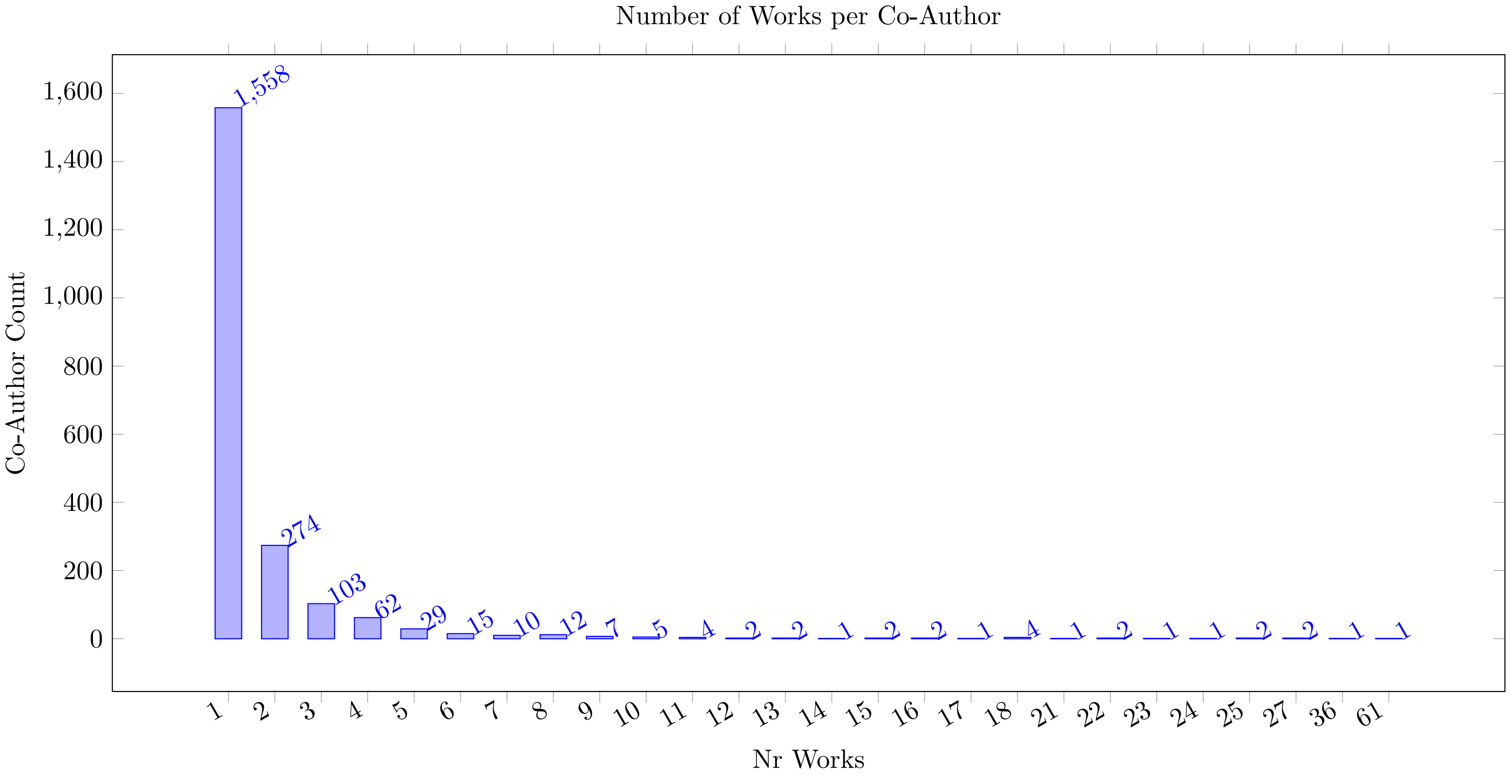
2,020
2,022
2,024
2,026
2,028
Year

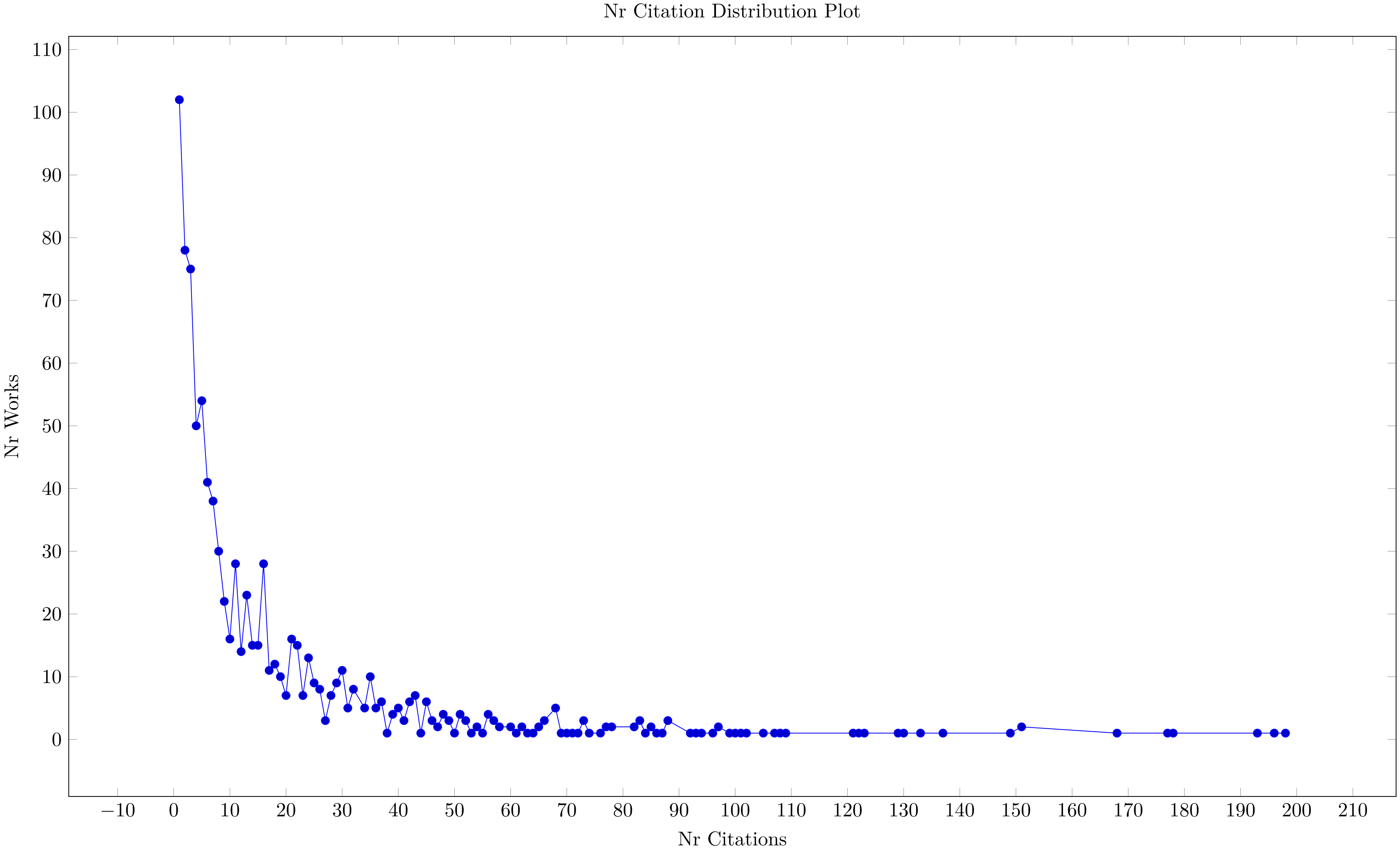
8 Number of Coauthors per Work



9 Number of Works per Author



10 Citation Distribution



11 Similarity Measures

The following distribution plot shows the similarity values between two works based on citations and references counts. If either work does not have citation and reference values, then the similarity is set to NaN. The total similarity count is the average of the similarity for citations and for references. As value we compute the ratio of non-shared references (citations) to the sum of individual references (citations). So both the citation and reference similarity range between zero and one, and the average ranges between zero and one. Low values are very rare, as they require both works to be citing the same papers, and being cited by the same papers. A larger value indicates that items are less similar according to this measure. In the plot we group values into 0.1 wide value bins, so an entry for 0.2 includes values from 0.15 to 0.25.

We observe that low values of this similarity are often found for two works by the same authors that are close in time, where we assumes that the bibliographies in both papers is based on the same literature survey. If neither paper is widely cited, the similarity value is low.

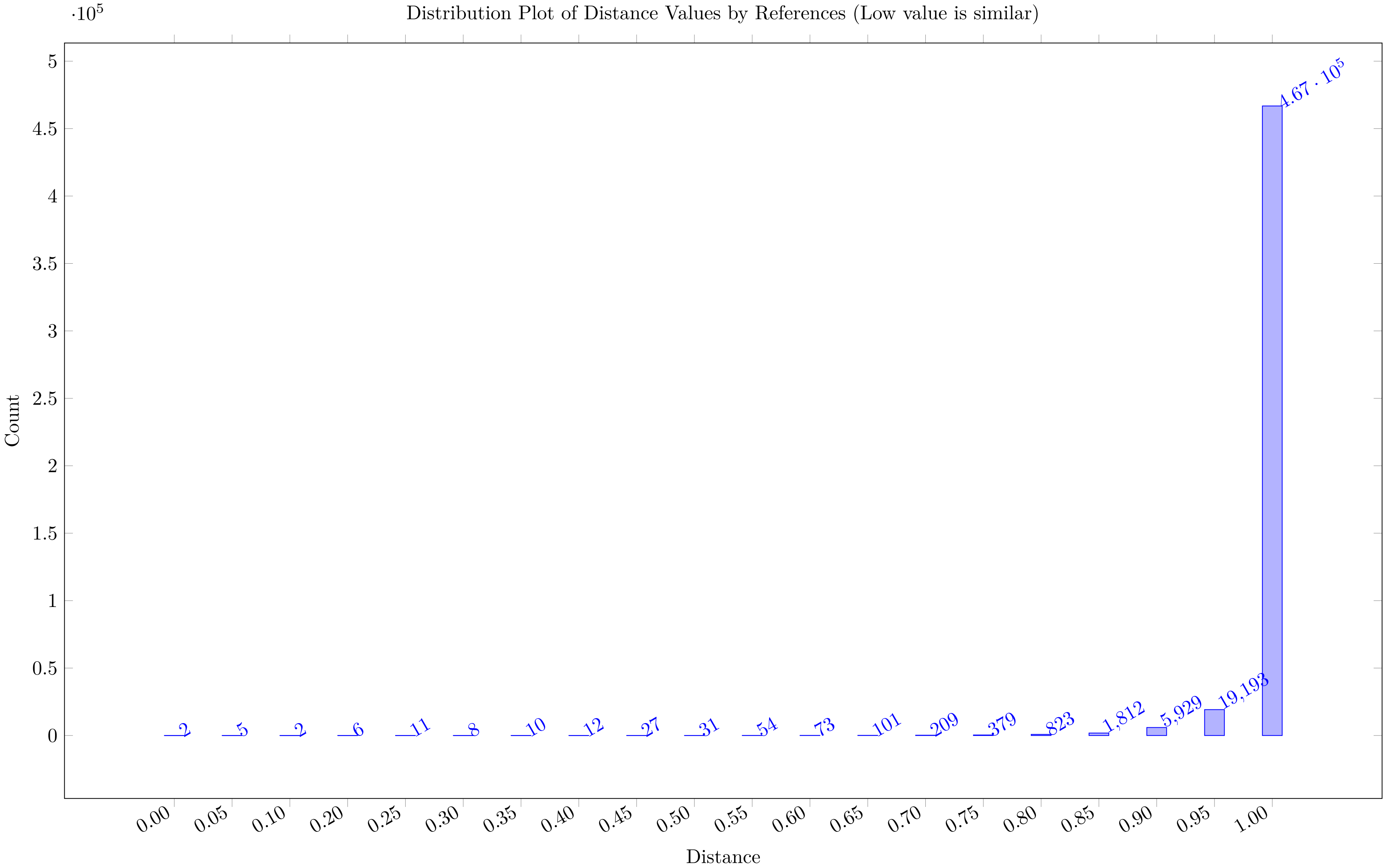
The vast majority of paper pairs has a distance close to one, as their references and citations do not overlap much.

Table 11: Heat Map based on rounded DotProduct Similarity of Concepts (high = similar)

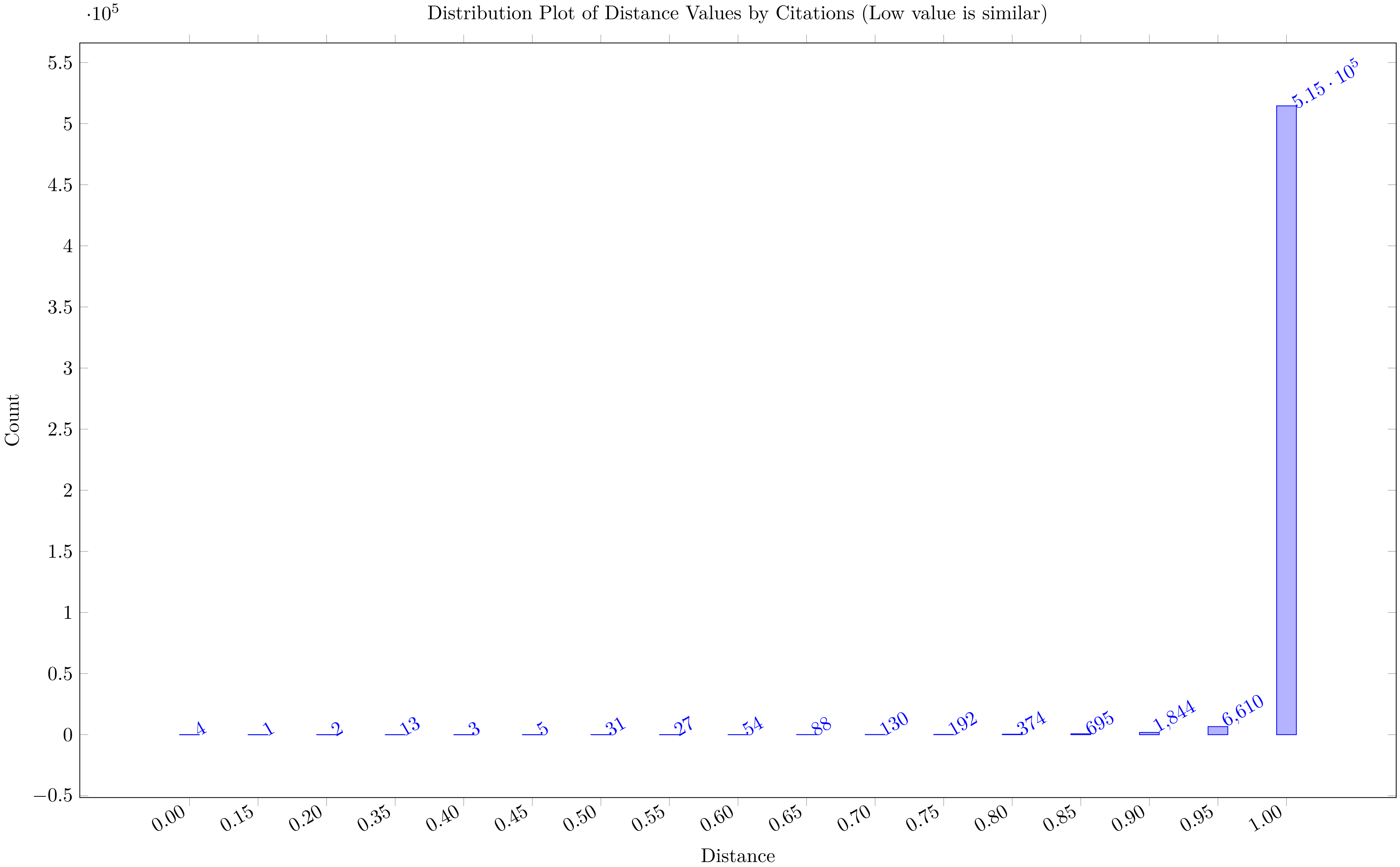
From/To	Total	ZarandiASC20	Tang2020	Ziadlou2024	Schutt11	abs-1902-09244	YunusogluY22	ZhuSZW23	VilimLS15	Siala15a	abs-2211-14492	ZeballosNH11	abs-1911-04766	abs-2402-00459	Zhang2024	ZhangBB22	YuraszeckMCCR23	Sciau2024	ZeballosQH10	
Total		117985.00	95225.00	91440.00	87905.00	87674.00	87418.00	84972.00	84508.00	83311.00	83305.00	80058.00	79964.00	79537.00	78703.00	78484.00	77914.00	76720.00	75604.00	75383.
Baptiste02	111082.00	330.00	260.00	198.00	266.00	173.00	201.00	169.00	172.00	226.00	161.00	172.00	148.00	135.00	128.00	159.00	163.00	185.00	149.00	193.
Astrand21	109824.00	307.00	259.00	187.00	228.00	174.00	200.00	165.00	162.00	191.00	151.00	143.00	157.00	142.00	141.00	153.00	144.00	197.00	132.00	173.
Beck99	100935.00	240.00	208.00	159.00	229.00	154.00	164.00	157.00	155.00	194.00	143.00	147.00	138.00	137.00	140.00	133.00	126.00	181.00	137.00	130.
BartakSR10	94624.00	232.00	190.00	165.00	207.00	134.00	161.00	152.00	143.00	177.00	142.00	140.00	118.00	120.00	131.00	148.00	118.00	145.00	129.00	145.
Dejemeppe16	94561.00	336.00	282.00	214.00	283.00	197.00	219.00	168.00	174.00	251.00	170.00	182.00	171.00	163.00	150.00	160.00	151.00	198.00	157.00	162.
AwadMDMT22	89769.00	226.00	207.00	159.00	168.00	153.00	176.00	134.00	132.00	135.00	133.00	150.00	129.00	120.00	104.00	117.00	130.00	141.00	130.00	132.
BeckDDF98	87227.00	223.00	180.00	158.00	160.00	146.00	158.00	136.00	124.00	161.00	126.00	145.00	111.00	115.00	117.00	116.00	110.00	148.00	127.00	115.
AbreuN22	84937.00	229.00	153.00	162.00	146.00	137.00	167.00	142.00	124.00	128.00	137.00	104.00	108.00	124.00	92.00	116.00	134.00	111.00	111.00	161.
AbreuPNF23	84411.00	232.00	163.00	167.00	155.00	146.00	176.00	135.00	114.00	125.00	124.00	113.00	105.00	123.00	97.00	108.00	125.00	119.00	107.00	161.
AfsarVPG23	84301.00	190.00	151.00	141.00	137.00	138.00	144.00	128.00	121.00	125.00	132.00	108.00	118.00	117.00	108.00	110.00	121.00	139.00	117.00	133.
AstrandJZ20	82936.00	181.00	163.00	123.00	153.00	131.00	131.00	122.00	127.00	150.00	108.00	110.00	109.00	102.00	106.00	122.00	114.00	129.00	101.00	119.
ArtiguesLH13	81571.00	158.00	166.00	128.00	149.00	123.00	152.00	95.00	119.00	125.00	108.00	134.00	109.00	101.00	102.00	97.00	94.00	125.00	113.00	97.
Groleaz21	80476.00	368.00	295.00	226.00	246.00	198.00	227.00	201.00	195.00	234.00	191.00	166.00	171.00	179.00	176.00	181.00	159.00	213.00	144.00	206.
ArmstrongGOS21	79570.00	174.00	161.00	139.00	148.00	121.00	145.00	136.00	119.00	124.00	118.00	99.00	123.00	104.00	129.00	118.00	119.00	122.00	110.00	117.
BartakSR08	78958.00	158.00	148.00	127.00	164.00	108.00	122.00	123.00	121.00	151.00	112.00	105.00	95.00	102.00	108.00	120.00	102.00	113.00	95.00	117.
ColT22	78877.00	248.00	201.00	185.00	162.00	153.00	179.00	158.00	141.00	162.00	147.00	139.00	153.00	138.00	154.00	133.00	142.00	153.00	126.00	158.
Fahimi16	78799.00	239.00	207.00	155.00	257.00	152.00	147.00	159.00	159.00	226.00	140.00	146.00	138.00	118.00	140.00	140.00	131.00	196.00	130.00	136.
Caseau2001	78444.00	196.00	172.00	126.00	197.00	135.00	138.00	126.00	145.00	163.00	125.00	130.00	118.00	122.00	121.00	131.00	136.00	132.00	117.00	125.
Caballero19	77007.00	170.00	170.00	118.00	239.00	129.00	123.00	116.00	158.00	202.00	106.00	109.00	121.00	93.00	118.00	120.00	126.00	159.00	112.00	85.
AbreuNP23	76867.00	203.00	142.00	161.00	131.00	131.00	157.00	137.00	116.00	119.00	124.00	102.00	94.00	107.00	89.00	112.00	126.00	95.00	100.00	150.
Akan2023	76389.00	244.00	161.00	148.00	152.00	142.00	135.00	102.00	111.00	105.00	109.00	109.00	100.00	109.00	79.00	88.00	108.00	139.00	104.00	103.
BlazewiczDP96	76106.00	231.00	184.00	133.00	179.00	126.00	141.00	131.00	123.00	149.00	123.00	114.00	97.00	111.00	106.00	134.00	95.00	129.00	101.00	126.
Godet21a	75517.00	256.00	242.00	184.00	263.00	155.00	166.00	154.00	177.00	248.00	157.00	132.00	159.00	130.00	168.00	149.00	158.00	215.00	130.00	147.
BajestaniB13	75196.00	181.00	179.00	124.00	120.00	115.00	128.00	117.00	110.00	110.00	102.00	114.00	89.00	100.00	94.00	96.00	86.00	137.00	89.00	99.
AbreuAPNM21	74888.00	208.00	143.00	147.00	128.00	121.00	147.00	127.00	103.00	114.00	119.00	89.00	90.00	113.00	85.00	107.00	121.00	96.00	96.00	152.
Ahmeti2024	74554.00	148.00	142.00	98.00	149.00	140.00	123.00	93.00	121.00	117.00	87.00	88.00	123.00	94.00	86.00	93.00	112.00	127.00	88.00	91.
BidotVLB09	74312.00	192.00	155.00	119.00	140.00	134.00	139.00	110.00	122.00	134.00	105.00	126.00	102.00	101.00	110.00	106.00	113.00	132.00	114.00	114.
BeckF98	73691.00	177.00	151.00	111.00	142.00	115.00	122.00	122.00	113.00	142.00	109.00	100.00	102.00	108.00	106.00	97.00	100.00	122.00	97.00	96.
AlfieriGPS23	73324.00	190.00	142.00	140.00	118.00	132.00	141.00	116.00	99.00	107.00	116.00	102.00	82.00	114.00	86.00	109.00	110.00	90.00	90.00	130.
ArkhipovBL19	73076.00	137.00	133.00	93.00	166.00	104.00	105.00	97.00	135.00	121.00	92.00	89.00	97.00	87.00	86.00	103.00	108.00	121.00	87.00	80.
Artigues2011	73022.00	136.00	130.00	100.00	156.00	102.00	98.00	109.00	127.00	146.00	99.00	91.00	75.00	89.00	91.00	122.00	100.00	104.00	84.00	95.
Banaszak2014	72864.00	155.00	115.00	114.00	124.00	96.00	97.00	109.00	105.00	123.00	97.00	105.00	87.00	81.00	79.00	93.00	88.00	114.00	97.00	94.
BosiM2001	71988.00	156.00	162.00	124.00	162.00	110.00	131.00	111.00	137.00	158.00	115.00	114.00	100.00	105.00	105.00	126.00	112.00	106.00	105.00	106.
Brucker2002	71895.00	196.00	190.00	129.00	180.00	132.00	138.00	130.00	152.00	156.00	119.00	120.00	100.00	106.00	95.00	129.00	129.00	140.00	101.00	115.
BeckF00	71683.00	146.00	130.00	106.00	167.00	108.00	105.00	110.00	117.00	156.00	99.00	106.00	93.00	84.00	96.00	107.00	91.00	123.00	92.00	92.
Braune2022	70941.00	146.00	153.00	121.00	165.00	125.00	133.00	128.00	131.00	136.00	115.00	116.00	112.00	104.00	112.00	118.00	118.00	114.00	105.00	100.
Bleukx2025	70742.00	150.00	140.00	136.00	144.00	99.00	106.00	106.00	122.00	145.00	110.00	93.00	98.00	108.00	120.00	106.00	102.00	141.00	95.00	92.
BonninMNE24	70729.00	172.00	156.00	122.00	159.00	113.00	122.00	112.00	119.00	130.00	104.00	98.00	110.00	99.00	101.00	112.00	106.00	118.00	99.00	131.
BeckR03	69682.00	156.00	149.00	127.00	132.00	131.00	127.00	115.00	119.00	134.00	111.00	129.00	93.00	100.00	94.00	103.00	90.00	110.00	107.00	97.
Bit-Monnot23	69484.00	135.00	126.00	105.00	154.00	106.00	99.00	120.00	129.00	177.00	105.00	94.00	103.00	98.00	111.00	121.00	105.00	112.00	86.00	105.
BeckPS03	69439.00	151.00	135.00	116.00	138.00	122.00	120.00	106.00	116.00	131.00	104.00	115.00	91.00	92.00	93.00	99.00	105.00	99.00	101.00	96.
ChenGPSH10	69401.00	163.00	158.00	115.00	164.00	101.00	109.00	121.00	127.00	149.00	110.00	120.00	96.00	96.00	101.00	113.00	97.00	127.00	102.00	106.
Other		109519.00	87971.00	85530.00	80678.00	82112.00	81399.00	79567.00	78972.00	76854.00	78200.00	75050.00	75231.00	74846.00	74048.00	73459.00	72989.00	71003.00	70990.00	70201.

Table 12: Heat Map based on 100*Cosine Similarity of Concepts (high = similar)

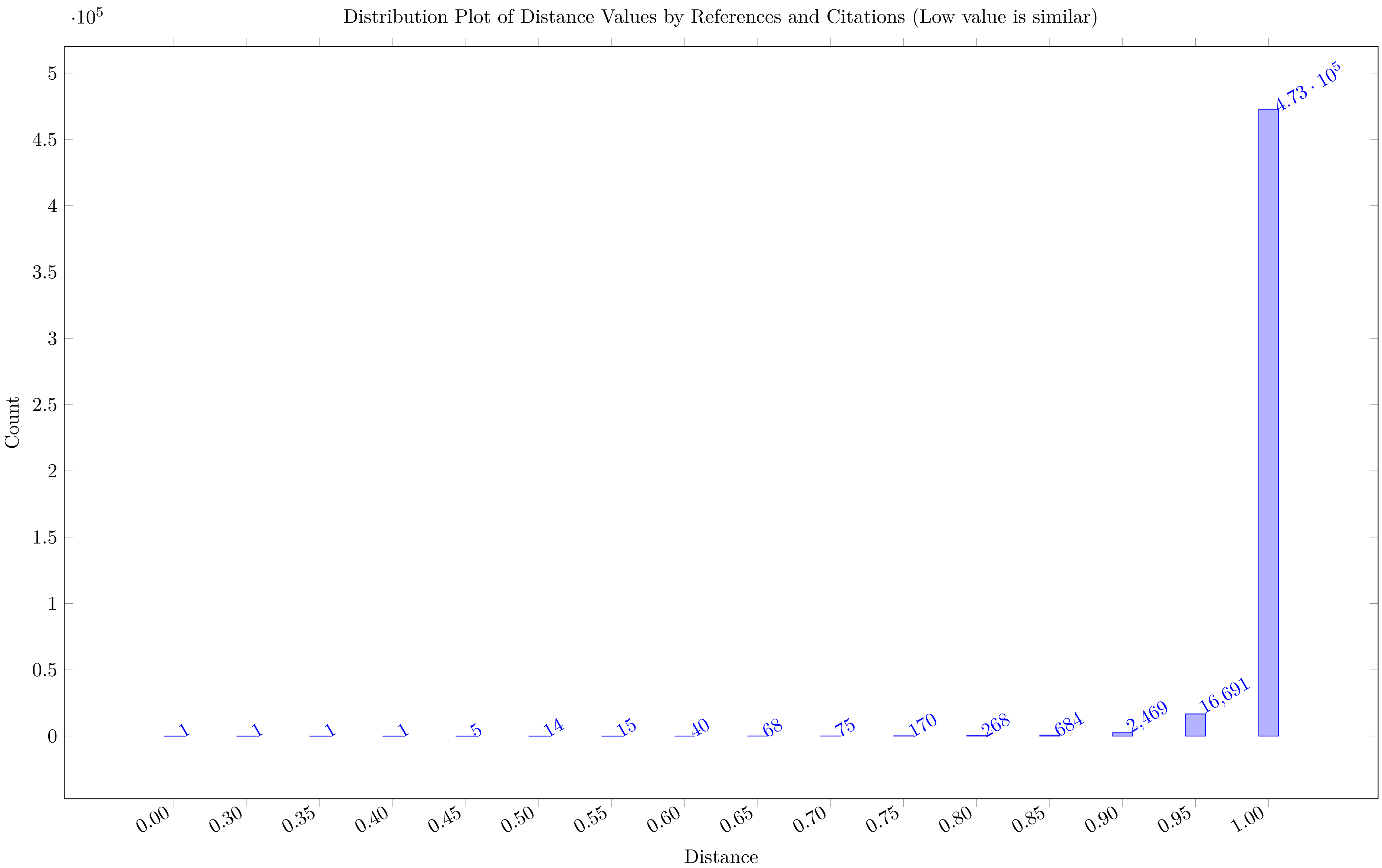
From/To	Total	ZeballosM09	ZeballosH05	abs-2312-13682	abs-1901-07914	VilimLS15	ZhangYW21	Xujun2009	ZeballosQH10	Zhang2005	abs-1902-09244	abs-2211-14492	Wolf05	ZhangBB22	abs-2306-05747	abs-2305-19888	ZeballosNH11	WatsonB08	ZibranR11a	ZhouGL15	WikarekS19	ZouZ20	abs-2402-00459	Other
Total		573.68	572.98	560.54	560.32	555.34	552.66	551.97	549.87	547.43	546.51	542.68	540.66	540.15	538.98	538.75	538.05	537.68	537.14	531.81	530.92	530.70	529.82	
Banaszak2014	572.90	0.77	0.76	0.59	0.67	0.65	0.62	0.76	0.66	0.66	0.56	0.59	0.68	0.60	0.57	0.57	0.66	0.64	0.60	0.65	0.65	0.60	0.51	558.87
Banaszak2008	569.21	0.80	0.77	0.60	0.70	0.64	0.59	0.80	0.70	0.67	0.56	0.62	0.71	0.59	0.59	0.56	0.66	0.67	0.62	0.63	0.64	0.56	0.50	555.02
Bartak02a	565.34	0.62	0.66	0.60	0.65	0.70	0.61	0.68	0.62	0.70	0.57	0.59	0.66	0.61	0.57	0.53	0.59	0.63	0.50	0.56	0.67	0.56	0.59	551.86
BartakSR08	563.98	0.60	0.66	0.57	0.60	0.68	0.62	0.70	0.59	0.80	0.58	0.63	0.70	0.71	0.59	0.56	0.60	0.67	0.52	0.65	0.70	0.54	0.59	550.11
AfsarVPG23	553.21	0.62	0.65	0.54	0.53	0.63	0.74	0.56	0.68	0.52	0.69	0.69	0.62	0.61	0.67	0.59	0.58	0.69	0.50	0.60	0.55	0.57	0.62	539.78
BeckPS03	552.42	0.73	0.78	0.57	0.64	0.73	0.68	0.72	0.70	0.70	0.73	0.65	0.74	0.66	0.64	0.60	0.74	0.73	0.49	0.64	0.72	0.53	0.59	537.71
AstrandJZ20	551.95	0.61	0.63	0.61	0.56	0.67	0.66	0.67	0.59	0.57	0.66	0.57	0.62	0.68	0.64	0.62	0.59	0.67	0.52	0.67	0.63	0.50	0.55	538.47
ArkhipovBL19	543.56	0.54	0.54	0.51	0.53	0.80	0.64	0.55	0.57	0.63	0.59	0.54	0.71	0.64	0.55	0.59	0.54	0.60	0.41	0.54	0.72	0.47	0.52	530.85
AstrandJZ18	543.06	0.67	0.66	0.69	0.69	0.62	0.52	0.69	0.64	0.54	0.52	0.51	0.60	0.53	0.51	0.55	0.53	0.56	0.64	0.56	0.54	0.54	0.48	530.29
Benedetti2008	540.21	0.66	0.69	0.60	0.61	0.68	0.63	0.71	0.65	0.75	0.60	0.60	0.78	0.61	0.61	0.60	0.64	0.64	0.50	0.64	0.66	0.57	0.55	526.22
AngelsmarkJ00	535.51	0.63	0.53	0.48	0.60	0.58	0.48	0.64	0.49	0.68	0.39	0.52	0.60	0.51	0.47	0.44	0.56	0.56	0.54	0.54	0.57	0.48	0.48	523.73
Astrand0F21	534.71	0.57	0.64	0.64	0.64	0.65	0.68	0.59	0.59	0.50	0.56	0.55	0.59	0.66	0.63	0.59	0.51	0.64	0.46	0.54	0.63	0.47	0.58	521.77
ArtiguesLH13	534.50	0.68	0.71	0.58	0.52	0.62	0.48	0.66	0.65	0.68	0.61	0.56	0.63	0.53	0.49	0.58	0.71	0.52	0.59	0.57	0.53	0.54	0.53	521.52
Beck07	534.22	0.64	0.64	0.56	0.56	0.73	0.70	0.73	0.58	0.64	0.64	0.67	0.68	0.73	0.68	0.52	0.61	0.85	0.60	0.60	0.65	0.52	0.59	520.11
AbidinK20	528.64	0.55	0.55	0.56	0.50	0.52	0.58	0.42	0.60	0.41	0.56	0.49	0.45	0.51	0.46	0.51	0.56	0.51	0.64	0.46	0.49	0.68	0.63	517.03
AalianPG23	527.76	0.62	0.59	0.73	0.59	0.59	0.54	0.63	0.59	0.47	0.58	0.44	0.56	0.52	0.56	0.61	0.48	0.55	0.61	0.50	0.49	0.54	0.45	515.52
BeckDDF98	527.09	0.72	0.73	0.57	0.63	0.59	0.62	0.67	0.67	0.73	0.67	0.60	0.61	0.58	0.57	0.52	0.71	0.62	0.56	0.58	0.65	0.53	0.56	513.41
Astrand2020	526.52	0.57	0.70	0.64	0.62	0.59	0.67	0.66	0.62	0.54	0.57	0.51	0.64	0.61	0.58	0.59	0.54	0.61	0.53	0.53	0.63	0.51	0.53	513.54
Ahmeti2024	524.12	0.53	0.56	0.62	0.51	0.68	0.65	0.57	0.54	0.51	0.75	0.48	0.57	0.55	0.57	0.58	0.50	0.62	0.56	0.50	0.53	0.56	0.53	511.67
BeckFW11	523.30	0.61	0.62	0.63	0.60	0.74	0.72	0.67	0.63	0.59	0.62	0.72	0.70	0.75	0.77	0.62	0.55	0.93	0.53	0.62	0.69	0.46	0.66	508.88
Alaka21	522.71	0.53	0.60	0.56	0.57	0.52	0.62	0.53	0.60	0.47	0.53	0.49	0.51	0.53	0.48	0.55	0.46	0.48	0.52	0.54	0.50	0.66	0.59	510.88
Bartak02	522.22	0.67	0.59	0.53	0.59	0.60	0.47	0.69	0.53	0.70	0.45	0.49	0.59	0.51	0.44	0.44	0.60	0.51	0.56	0.51	0.66	0.53	0.46	510.10
Balduccini2017	519.24	0.58	0.55	0.52	0.58	0.61	0.48	0.62	0.47	0.61	0.47	0.54	0.54	0.49	0.50	0.46	0.54	0.58	0.50	0.55	0.60	0.52	0.48	507.43
AwadMDMT22	518.73	0.61	0.66	0.53	0.47	0.61	0.64	0.55	0.66	0.59	0.68	0.61	0.68	0.57	0.58	0.66	0.71	0.58	0.51	0.57	0.56	0.56	0.56	505.58
AkramNHRSA23	516.45	0.54	0.56	0.57	0.63	0.55	0.61	0.56	0.55	0.52	0.47	0.63	0.57	0.50	0.58	0.59	0.43	0.59	0.57	0.58	0.43	0.54	0.64	504.25
BartakSR10	514.54	0.58	0.64	0.47	0.60	0.62	0.60	0.58	0.62	0.77	0.56	0.61	0.70	0.68	0.54	0.55	0.62	0.59	0.44	0.63	0.68	0.49	0.53	501.44
Astrand21	514.00	0.54	0.57	0.53	0.49	0.61	0.61	0.54	0.55	0.52	0.62	0.56	0.55	0.60	0.57	0.62	0.55	0.57	0.47	0.66	0.55	0.49	0.54	501.68
Artigues2011	513.78	0.53	0.54	0.44	0.51	0.71	0.62	0.57	0.52	0.62	0.55	0.55	0.63	0.72	0.53	0.47	0.52	0.63	0.38	0.54	0.70	0.41	0.51	501.59
AlakaP23	511.08	0.52	0.55	0.51	0.61	0.51	0.62	0.53	0.58	0.47	0.51	0.45	0.45	0.50	0.43	0.51	0.49	0.45	0.52	0.49	0.53	0.65	0.53	499.67
Adelgren2023	511.00	0.52	0.53	0.56	0.54	0.54	0.57	0.55	0.51	0.54	0.48	0.53	0.60	0.57	0.53	0.65	0.52	0.56	0.41	0.56	0.55	0.44	0.47	499.27
AlesioBNG15	510.69	0.55	0.53	0.51	0.56	0.54	0.60	0.50	0.56	0.54	0.48	0.53	0.58	0.51	0.52	0.56	0.54	0.55	0.64	0.56	0.49	0.64	0.53	498.68
BockmayrP06	509.63	0.70	0.65	0.73	0.70	0.65	0.48	0.62	0.64	0.63	0.55	0.54	0.60	0.54	0.55	0.60	0.58	0.59	0.61	0.60	0.58	0.54	0.58	496.37
Bocewicz2009	509.01	0.70	0.72	0.54	0.65	0.64	0.59	0.71	0.61	0.67	0.54	0.55	0.74	0.60	0.56	0.54	0.66	0.64	0.59	0.58	0.67	0.56	0.50	495.43
Beck99	508.55	0.60	0.62	0.51	0.57	0.62	0.63	0.57	0.60	0.71	0.59	0.57	0.65	0.56	0.56	0.55	0.60	0.63	0.50	0.55	0.60	0.53	0.56	495.66
Beck06	505.03	0.61	0.65	0.57	0.59	0.64	0.63	0.70	0.56	0.61	0.61	0.67	0.65	0.67	0.66	0.49	0.60	0.84	0.47	0.64	0.68	0.38	0.59	491.51
BeckF98	504.08	0.55	0.62	0.46	0.60	0.61	0.68	0.57	0.58	0.70	0.59	0.59	0.66	0.55	0.58	0.55	0.55	0.60	0.42	0.51	0.63	0.49	0.59	491.39
BeckF00a	504.02	0.66	0.61	0.49	0.56	0.64	0.54	0.69	0.56	0.73	0.49	0.52	0.70	0.56	0.51	0.46	0.60	0.63	0.52	0.55	0.64	0.50	0.45	491.41
BeniniLMR11	502.08	0.65	0.66	0.62	0.61	0.65	0.56	0.61	0.64	0.58	0.57	0.52	0.62	0.56	0.53	0.58	0.63	0.56	0.58	0.49	0.56	0.62	0.53	489.15
BenderWS21	502.04	0.68	0.69	0.70	0.68	0.65	0.66	0.74	0.65	0.56	0.63	0.52	0.65	0.53	0.59	0.70	0.57	0.63	0.63	0.56	0.60	0.59	0.48	488.34
CarchraeB09	500.27	0.69	0.68	0.71	0.67	0.80	0.70	0.71	0.64	0.63	0.65	0.71	0.73	0.75	0.76	0.61	0.65	0.85	0.54	0.62	0.67	0.51	0.68	485.30
Alesio2013	499.34	0.57	0.56	0.52	0.57	0.52	0.55	0.50	0.58	0.58	0.45	0.52	0.64	0.51	0.53	0.57	0.53	0.55	0.61	0.56	0.49	0.55	0.51	487.38
AbreuN22	498.54	0.48	0.58	0.56	0.48	0.59	0.66	0.45	0.58	0.38	0.62	0.64	0.54	0.58	0.62	0.70	0.50	0.62	0.41	0.65	0.55	0.48	0.59	486.28
Other		547.91	546.55	536.53	535.54	528.80	527.08	525.80	524.76	522.04	522.40	518.81	514.23	515.38	515.07	514.97	513.74	511.54	514.82	507.74	505.67	508.32	506.86	



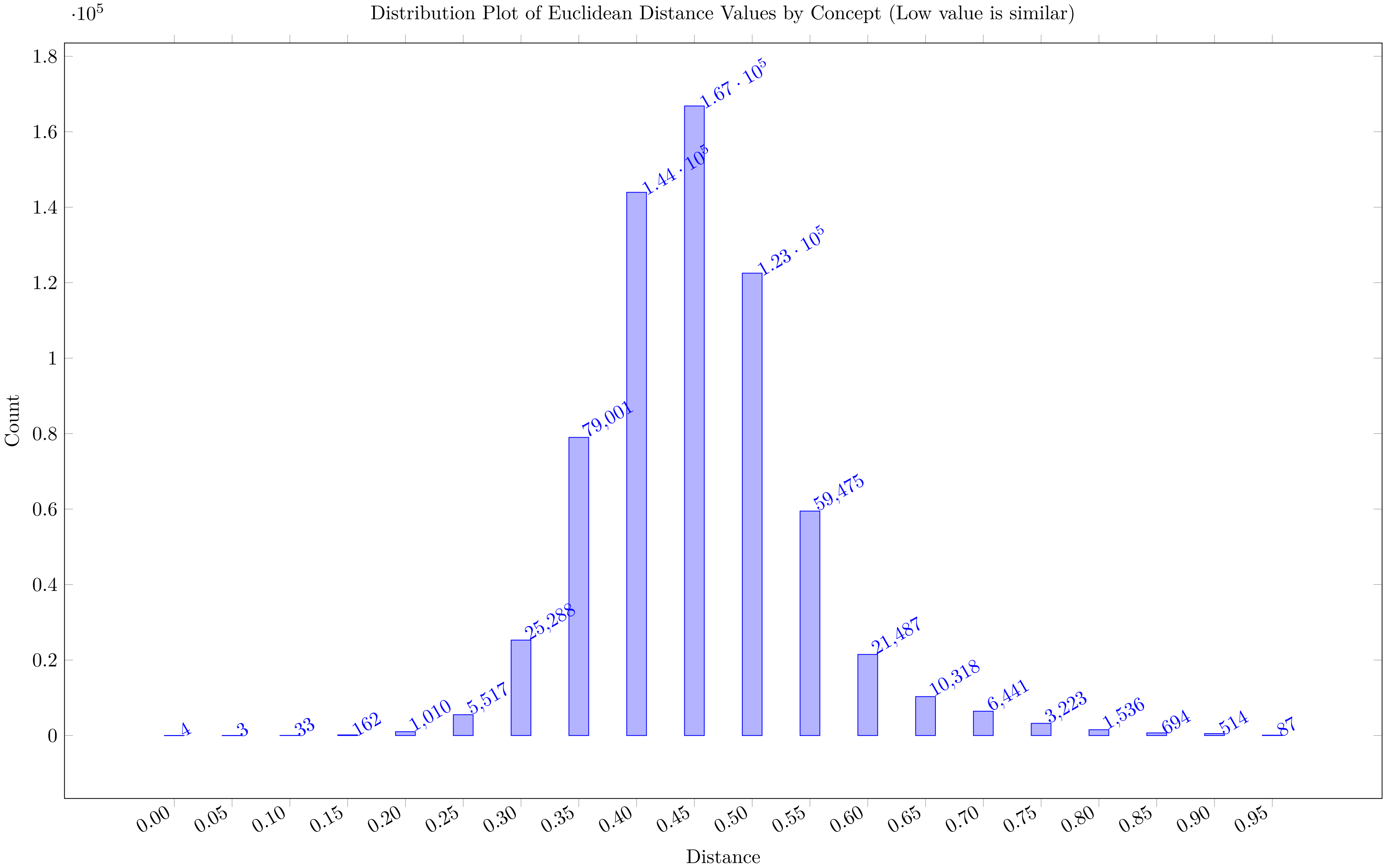
·10^5
Distribution Plot of Distance Values by References (Low value is similar)
Count
Distance
5
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
0.00
0.05
0.10
0.20
0.25
0.30
0.35
0.40
0.45
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
0.95
1.00
2
5
2
6
11
8
10
12
27
31
54
73
101
209
379
823
1,812
5,929
19,193
4.67 · 10^5



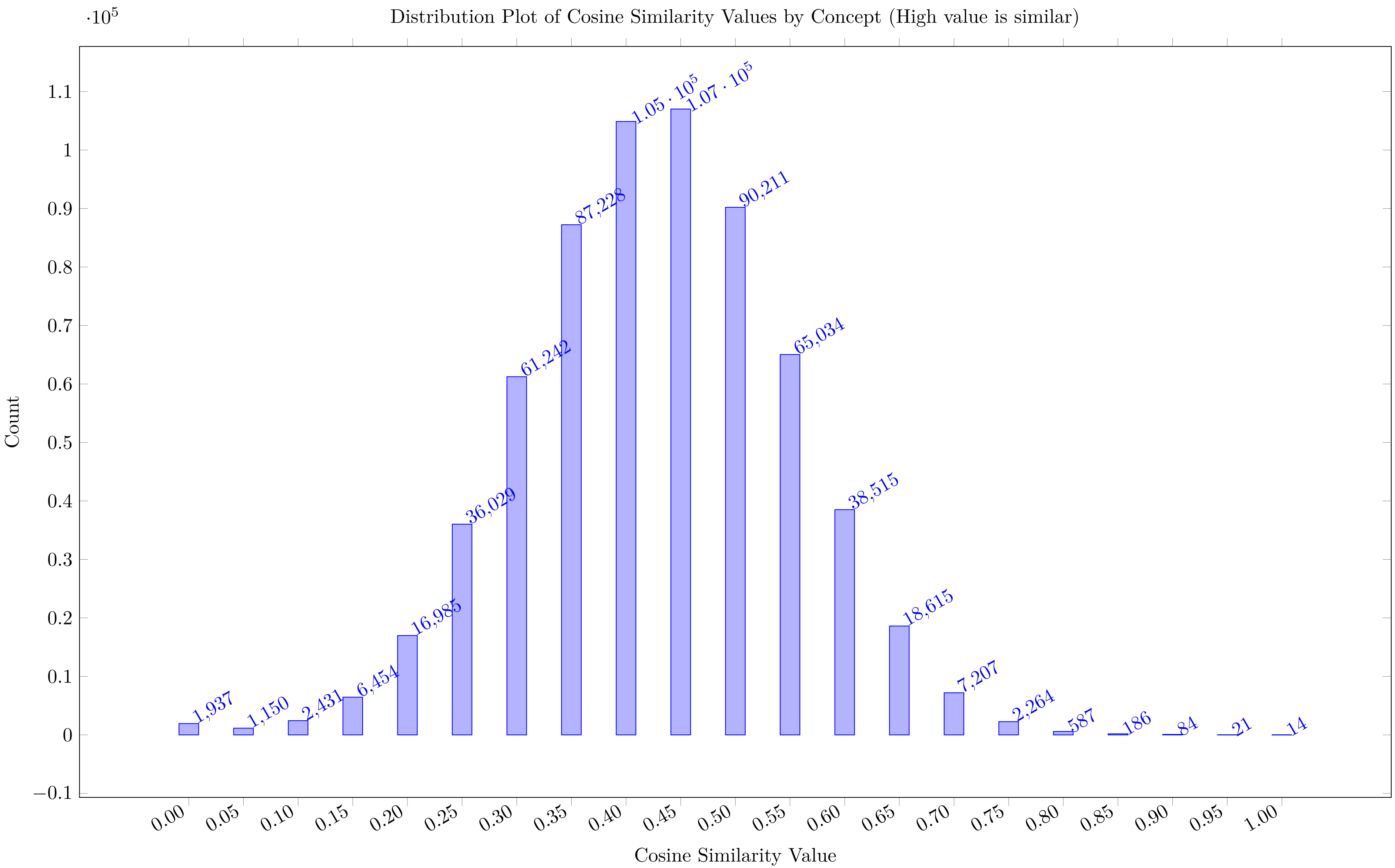
Distribution Plot of Distance Values by Citations (Low value is similar)
$\cdot 10^5$
Count
Distance
5.5
5
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
−0.5
0.00
0.15
0.20
0.35
0.40
0.45
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
0.95
1.00
4
1
2
13
3
5
31
27
54
88
130
192
374
695
1,844
6,610
$5.15 \cdot 10^5$

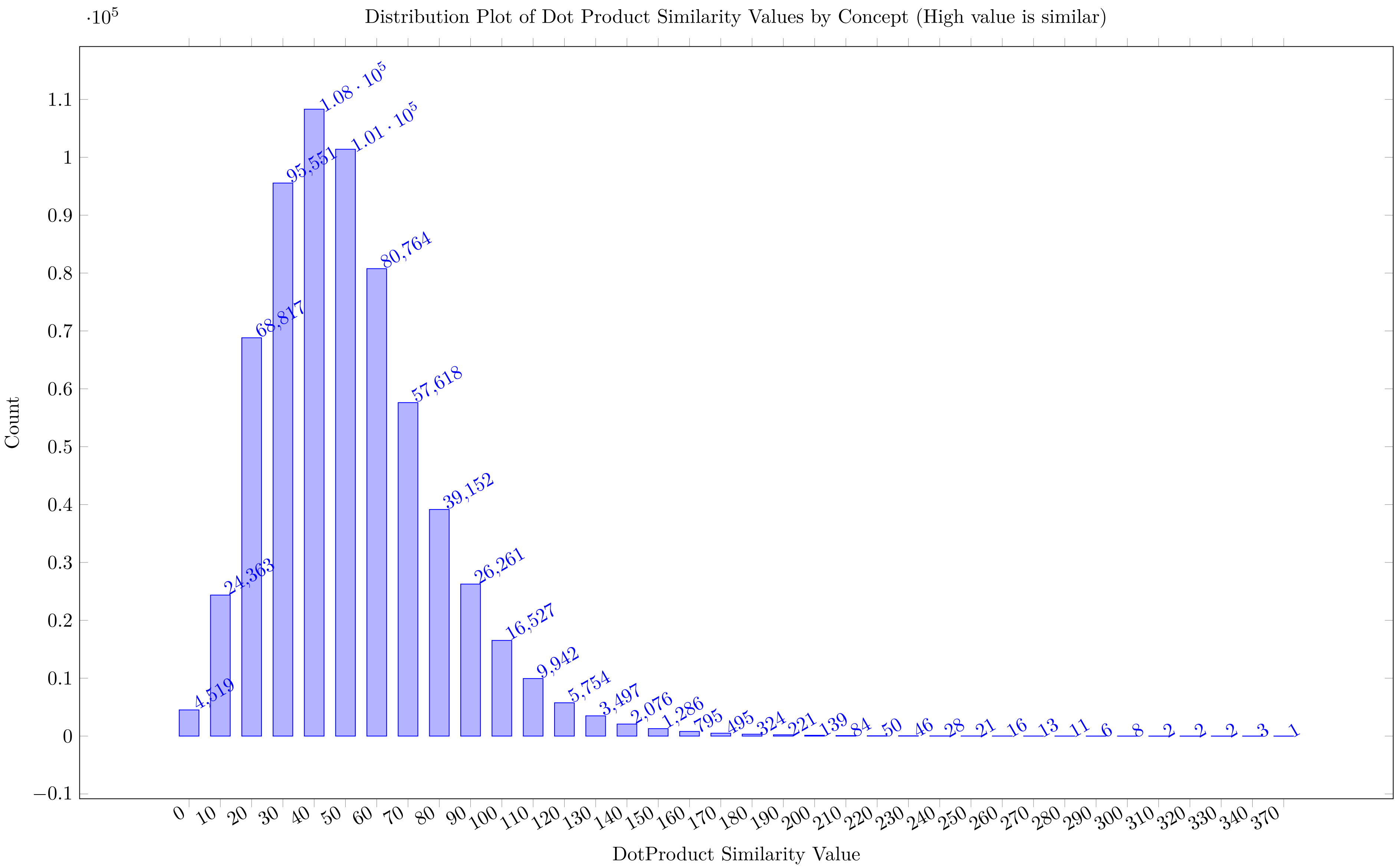


The similarity by concept uses the Euclidean distance between the feature vectors for two works. We translate the MatchLevel for each Concept into a linear scale, and then calculate the distances as the square root of the sum of squared differences for each feature. The distribution plot below rounds the distances to integer values. Similarity values of this type are only calculated when both works have a local copy, from which we extract the features. If either work does not have a local copy, the similarity is set to be NaN.



Distribution Plot of Euclidean Distance Values by Concept (Low value is similar)
$\cdot 10^5$
Count
Distance
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.45
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
0.95
4
3
33
162
1,010
5,517
25,288
79,001
$1.44 \cdot 10^5$
$1.67 \cdot 10^5$
$1.23 \cdot 10^5$
59,475
21,487
10,318
6,441
3,223
1,536
694
514
87


Distribution Plot of Cosine Similarity Values by Concept (High value is similar)
·10^5
Count
Cosine Similarity Value
1.1
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
−0.1
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.45
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
0.95
1.00
1,937
1,150
2,431
6,454
16,985
36,029
61,242
87,228
1.05 · 10^5
1.07 · 10^5
90,211
65,034
38,515
18,615
7,207
2,264
587
186
84
21
14

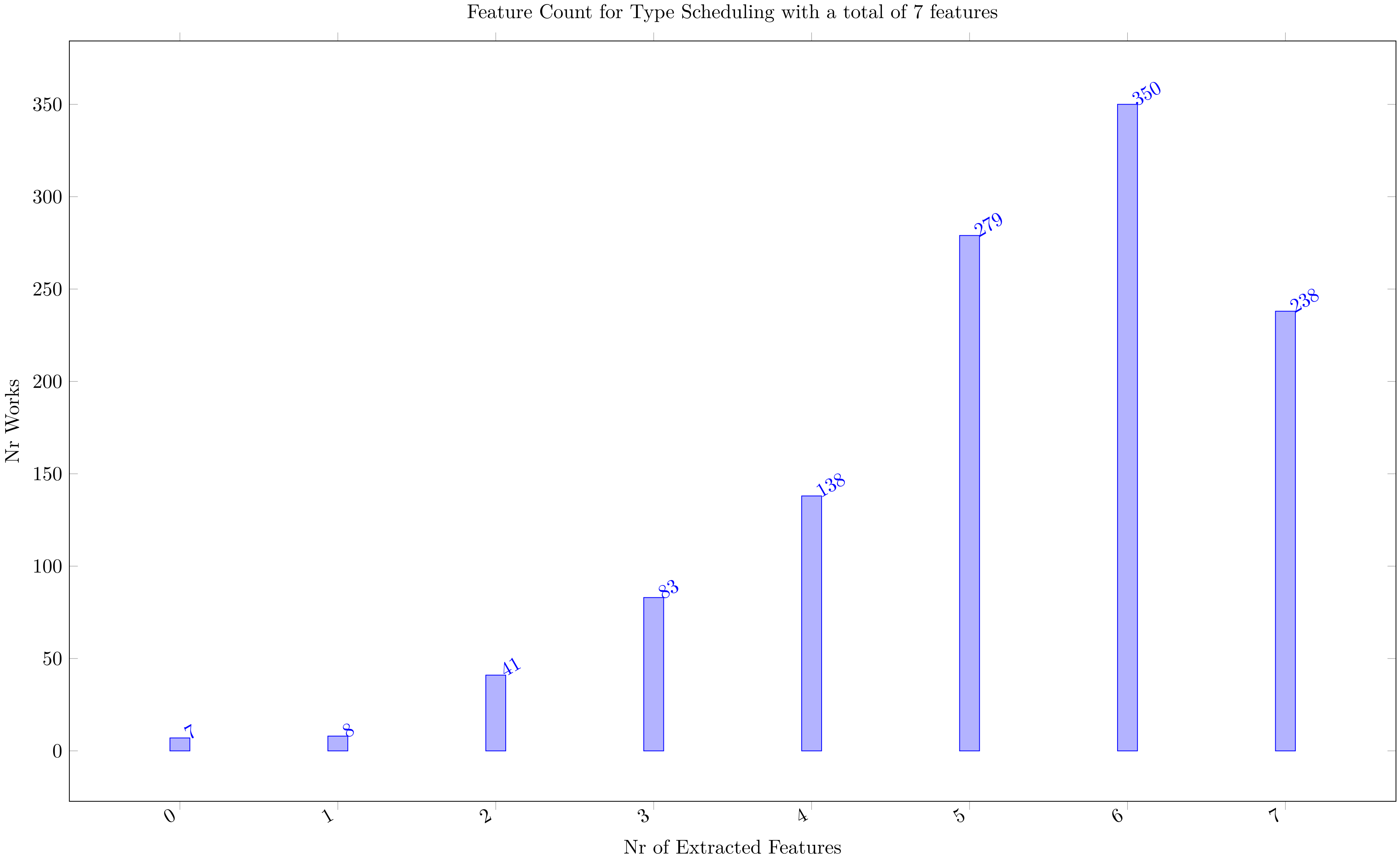


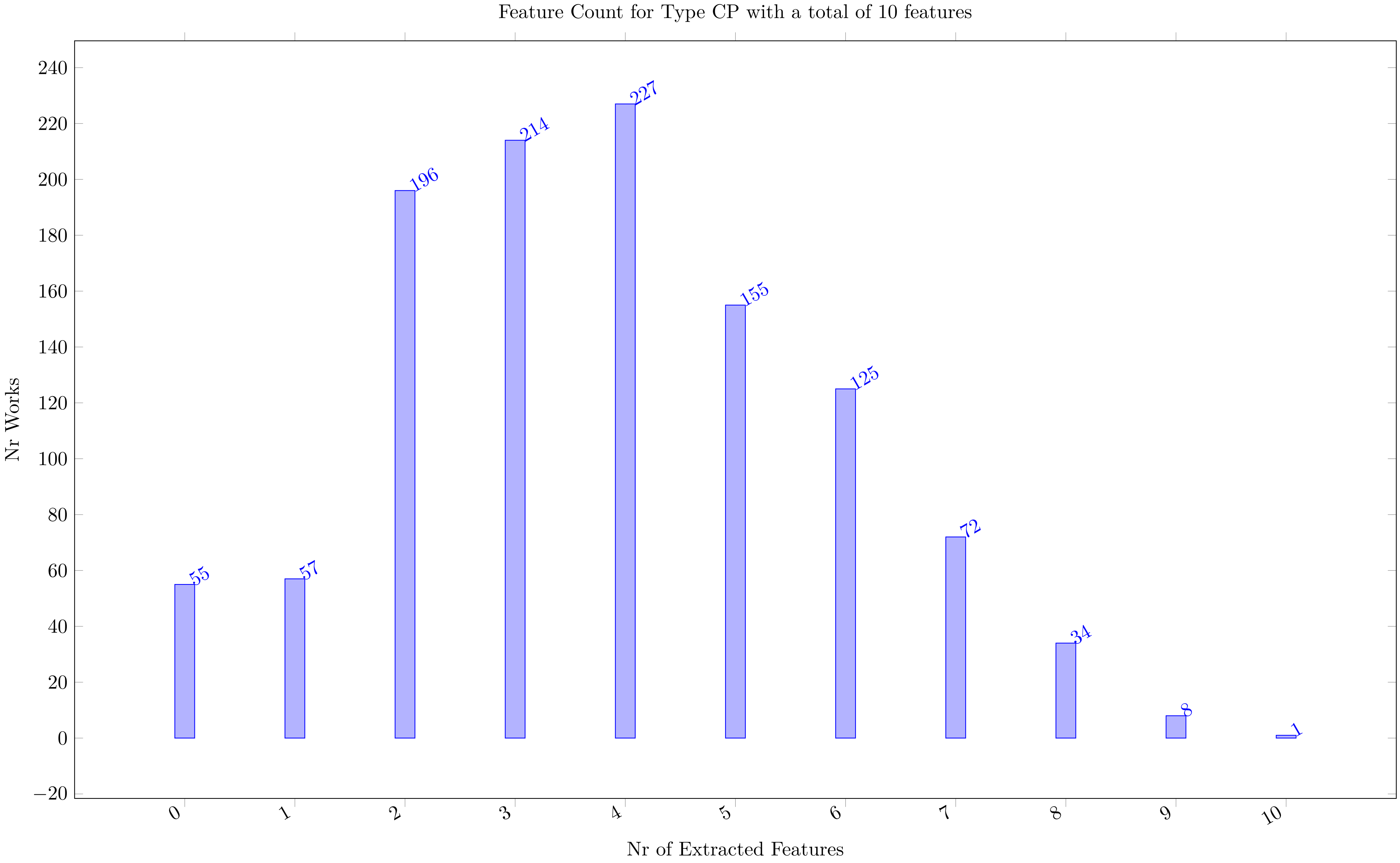
Distribution Plot of Dot Product Similarity Values by Concept (High value is similar)
·10^5
Count
DotProduct Similarity Value
4,519
24,363
68,817
95,551
1.08 · 10^5
1.01 · 10^5
80,764
57,618
39,152
26,261
16,527
9,942
5,754
3,497
2,076
1,286
795
495
324
221
139
84
50
46
28
21
16
13
11
6
8
2
2
2
3
1

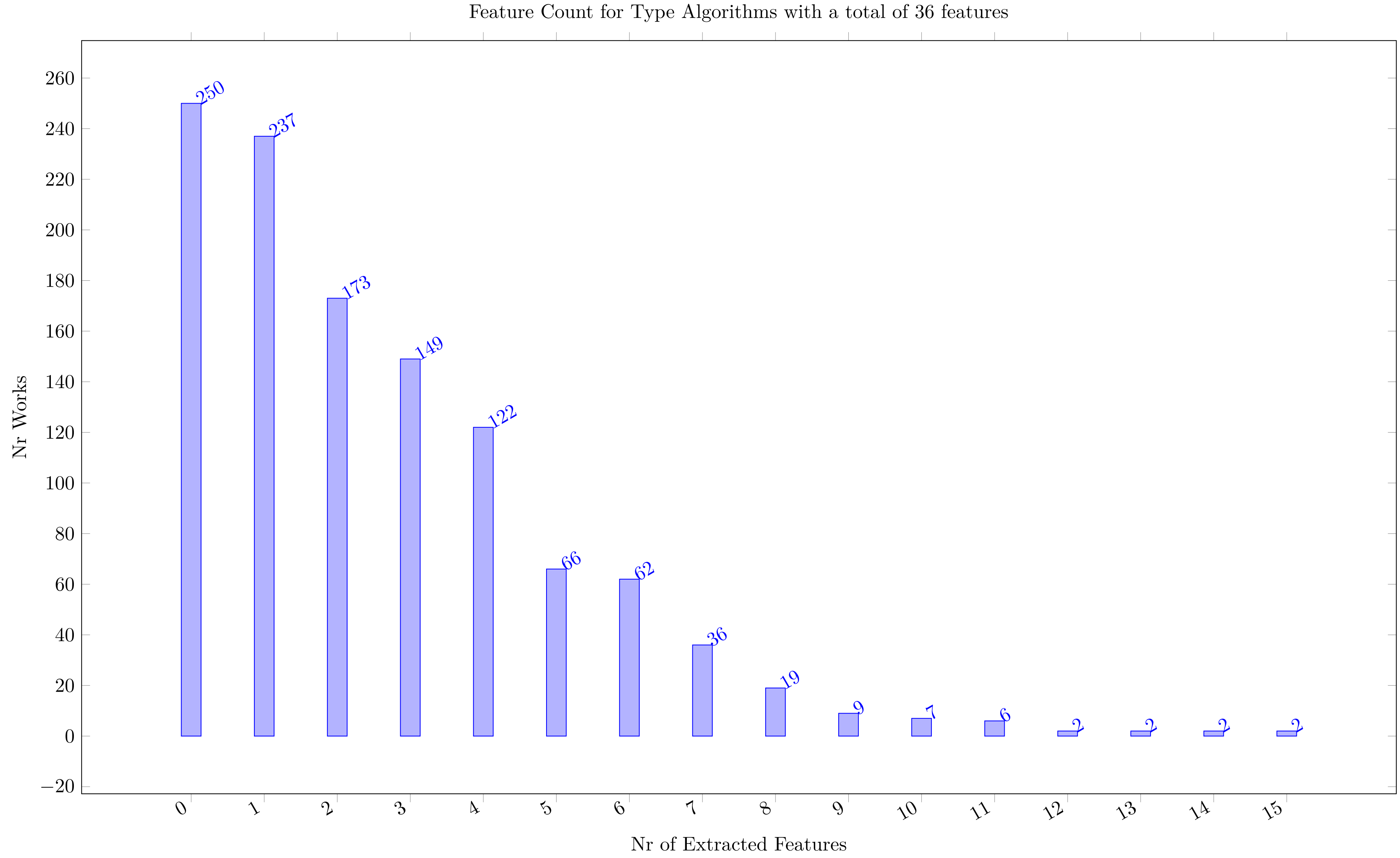
12 Concept Distribution

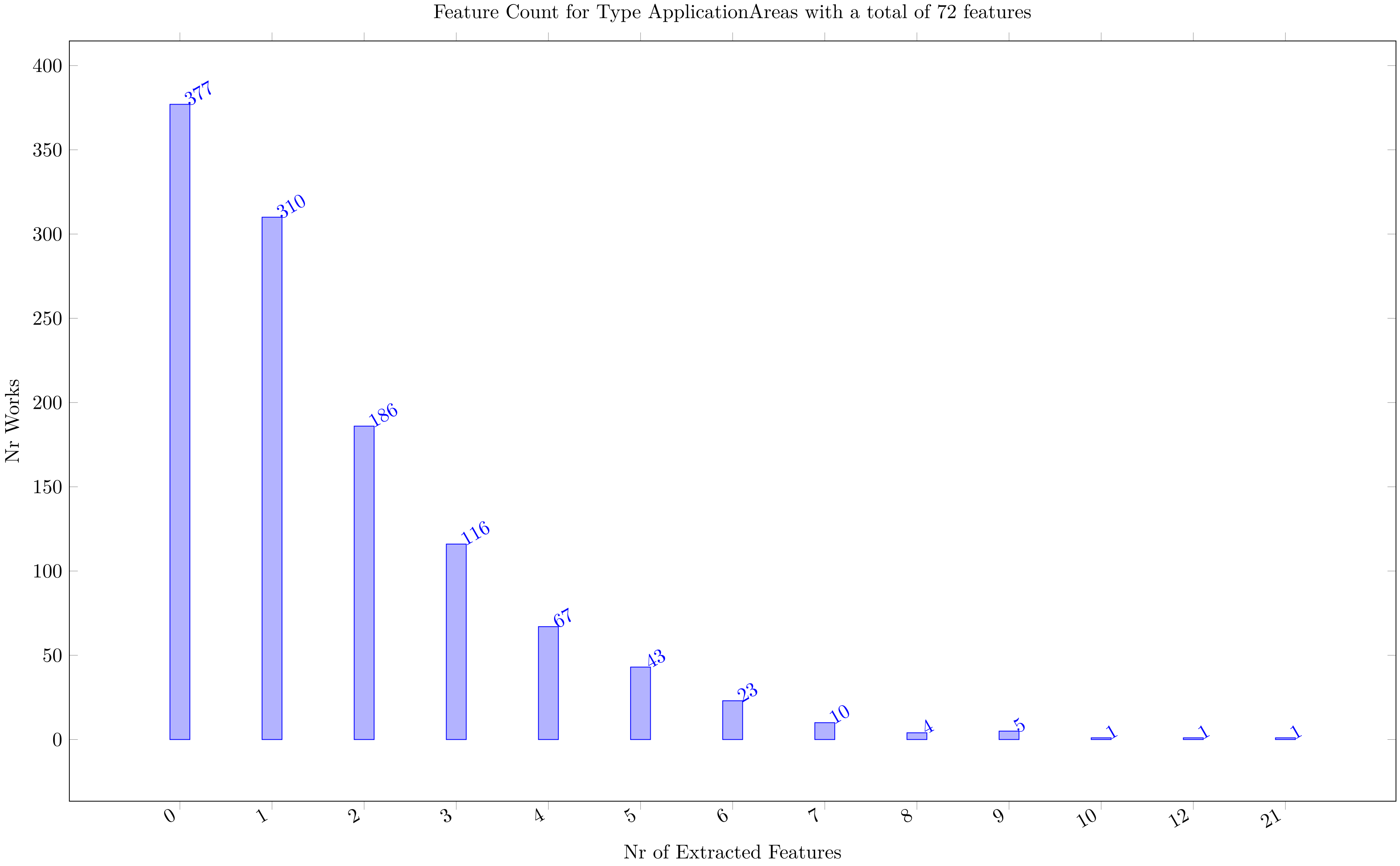
For each concept type, we count how many features are extracted by the individual works that do have a local copy, e.g. for which we can extract features. We can compare the number of features extracted to the number of concepts of a given type, which is stated in the title of the diagram.

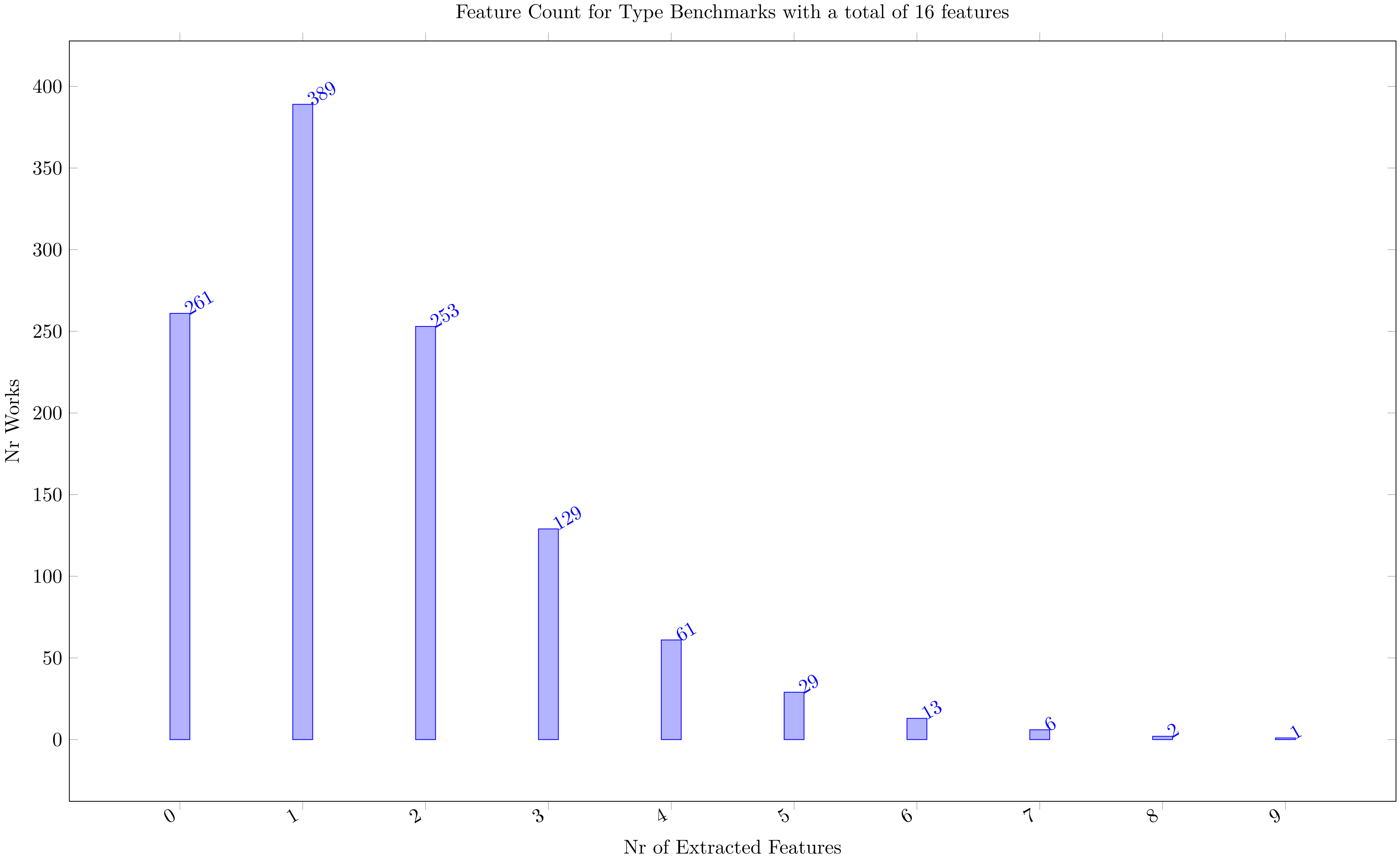
A high count indicates that a work covers many of the concepts of the given type, a low count might mean that our ontology does not have relevant concepts for that work.

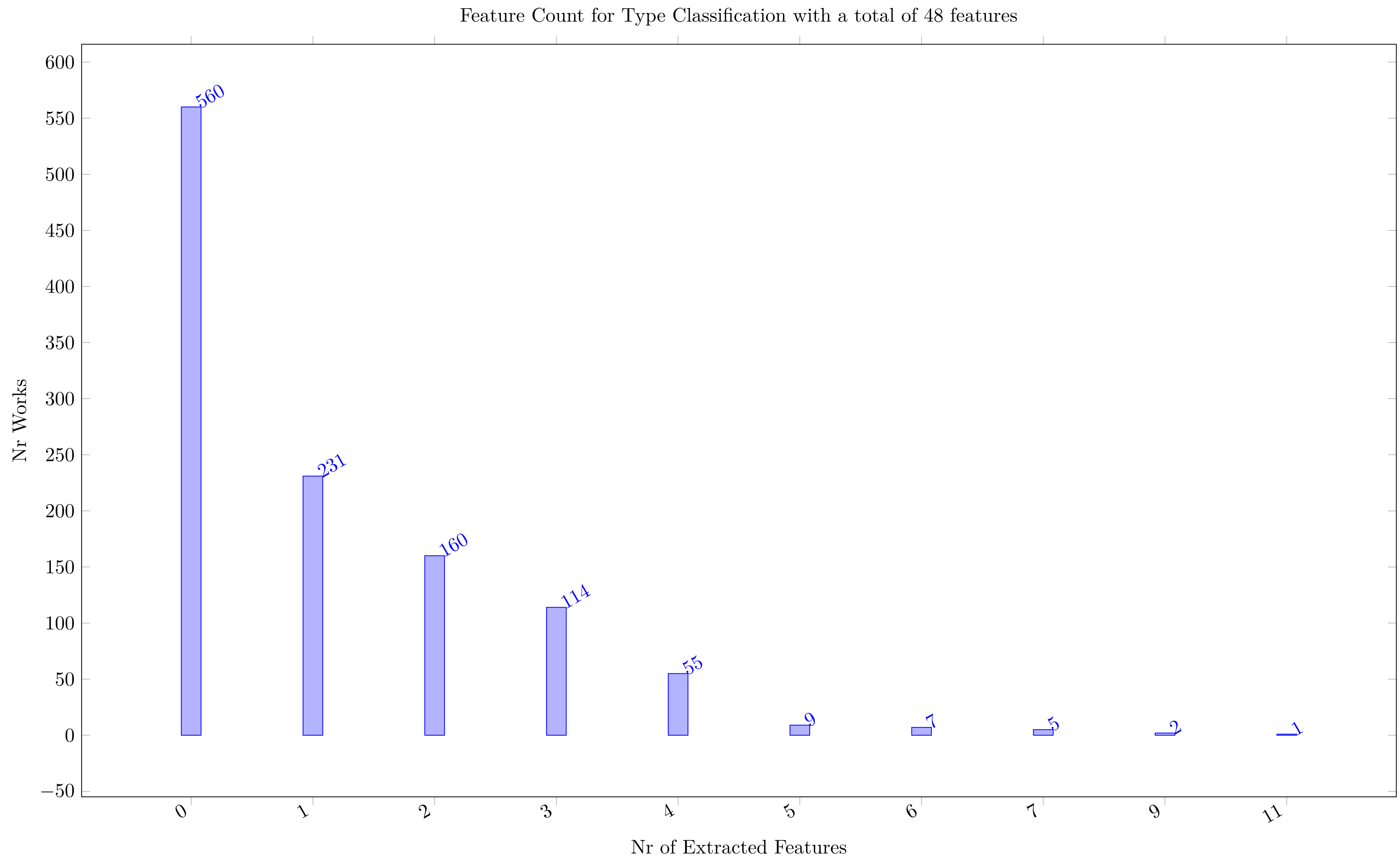

Feature Count for Type Scheduling with a total of 7 features
Nr Works
Nr of Extracted Features
0
50
100
150
200
250
300
350
0
1
2
3
4
5
6
7
7
8
41
83
138
279
350
238

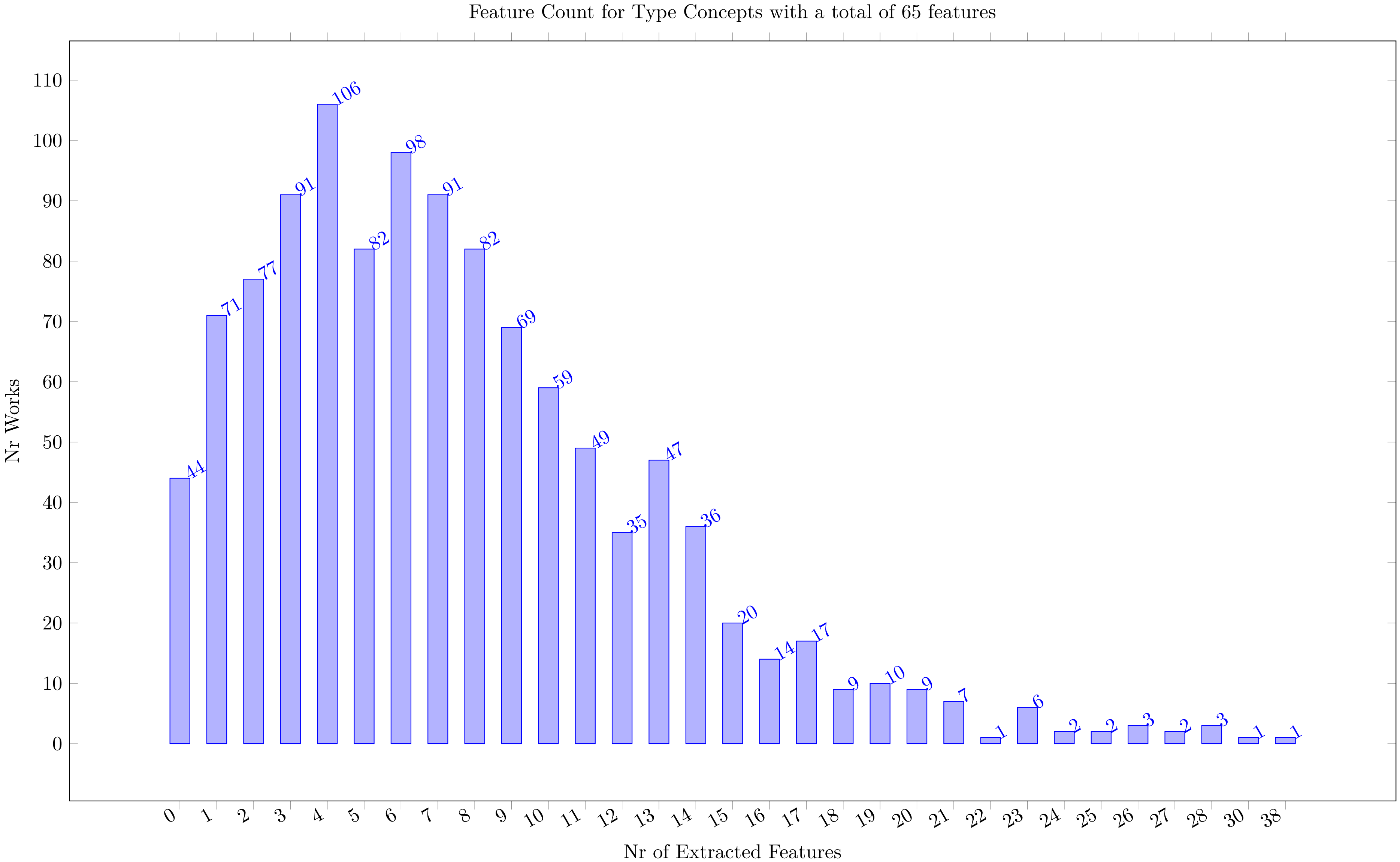

Feature Count for Type CP with a total of 10 features
Nr Works
Nr of Extracted Features
240
220
200
180
160
140
120
100
80
60
40
20
0
−20
0
1
2
3
4
5
6
7
8
9
10
55
57
196
214
227
155
125
72
34
8
1


Feature Count for Type Algorithms with a total of 36 features
Nr Works
260
240
220
200
180
160
140
120
100
80
60
40
20
0
−20
250
237
173
149
122
66
62
36
19
9
7
6
2
2
2
2
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
Nr of Extracted Features

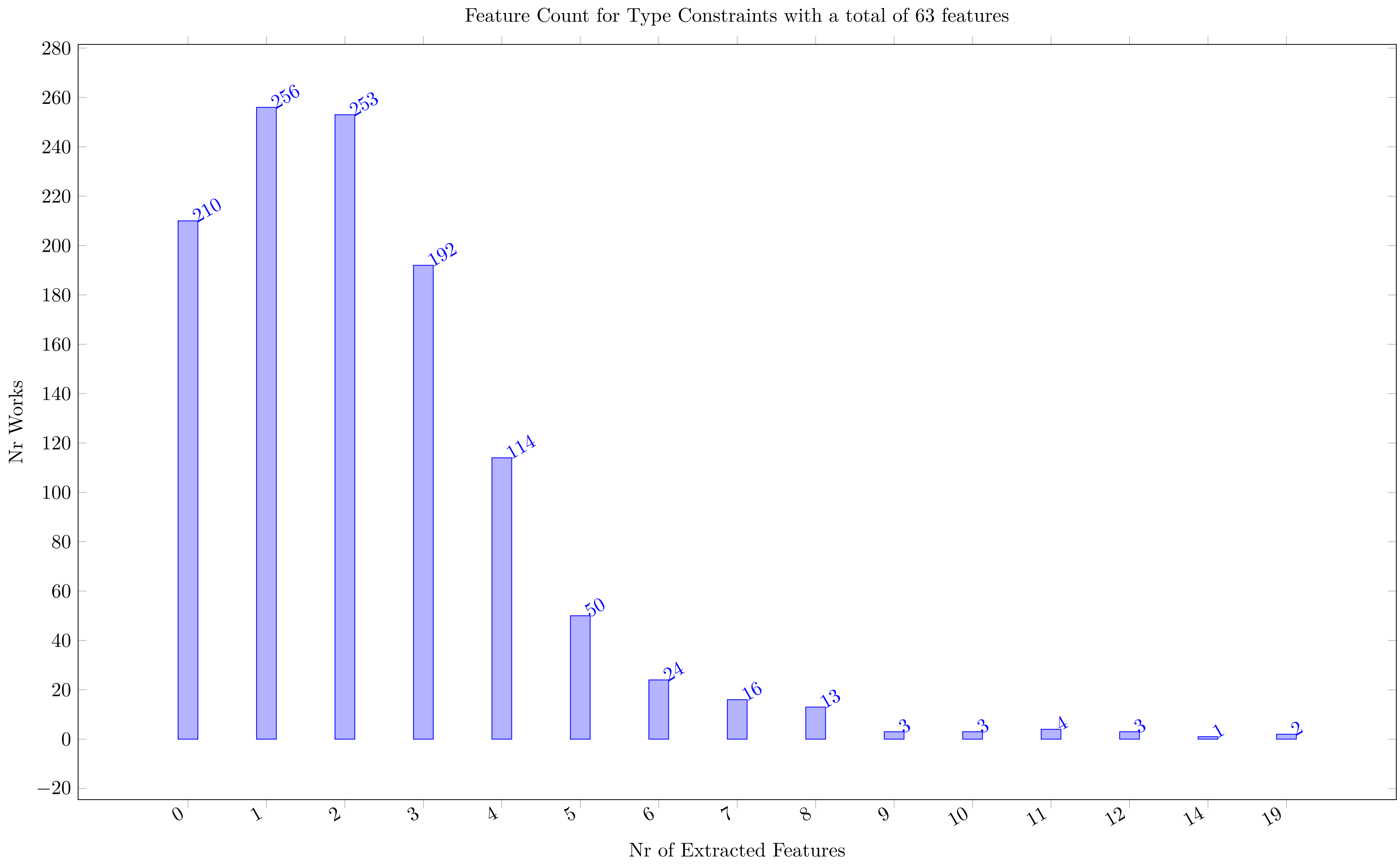

Feature Count for Type ApplicationAreas with a total of 72 features
Nr Works
Nr of Extracted Features
400
350
300
250
200
150
100
50
0
377
310
186
116
67
43
23
10
4
5
1
1
1
0
1
2
3
4
5
6
7
8
9
10
12
21

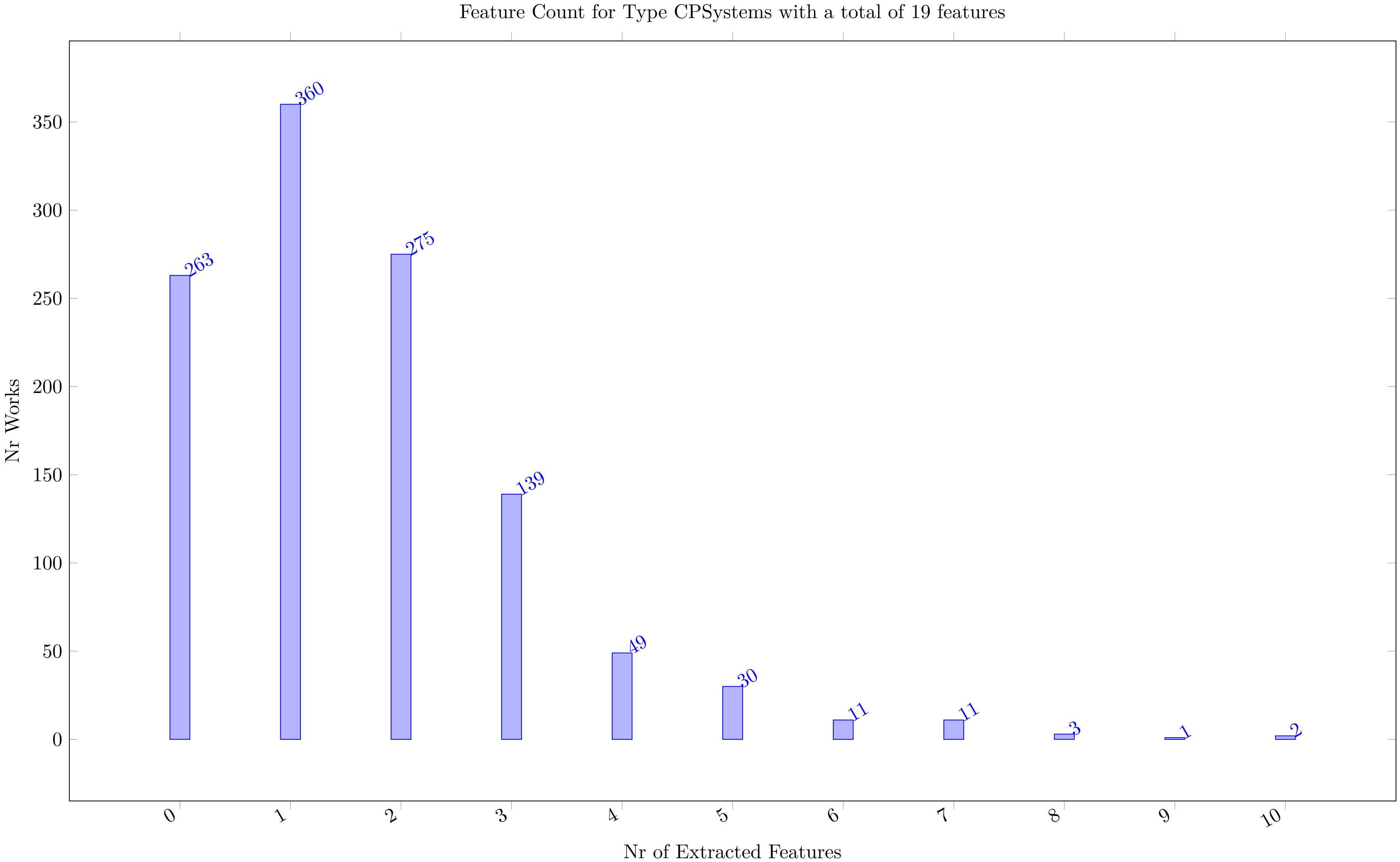

Feature Count for Type Benchmarks with a total of 16 features
Nr Works
Nr of Extracted Features
400
350
300
250
200
150
100
50
0
261
389
253
129
61
29
13
6
2
1
0
1
2
3
4
5
6
7
8
9

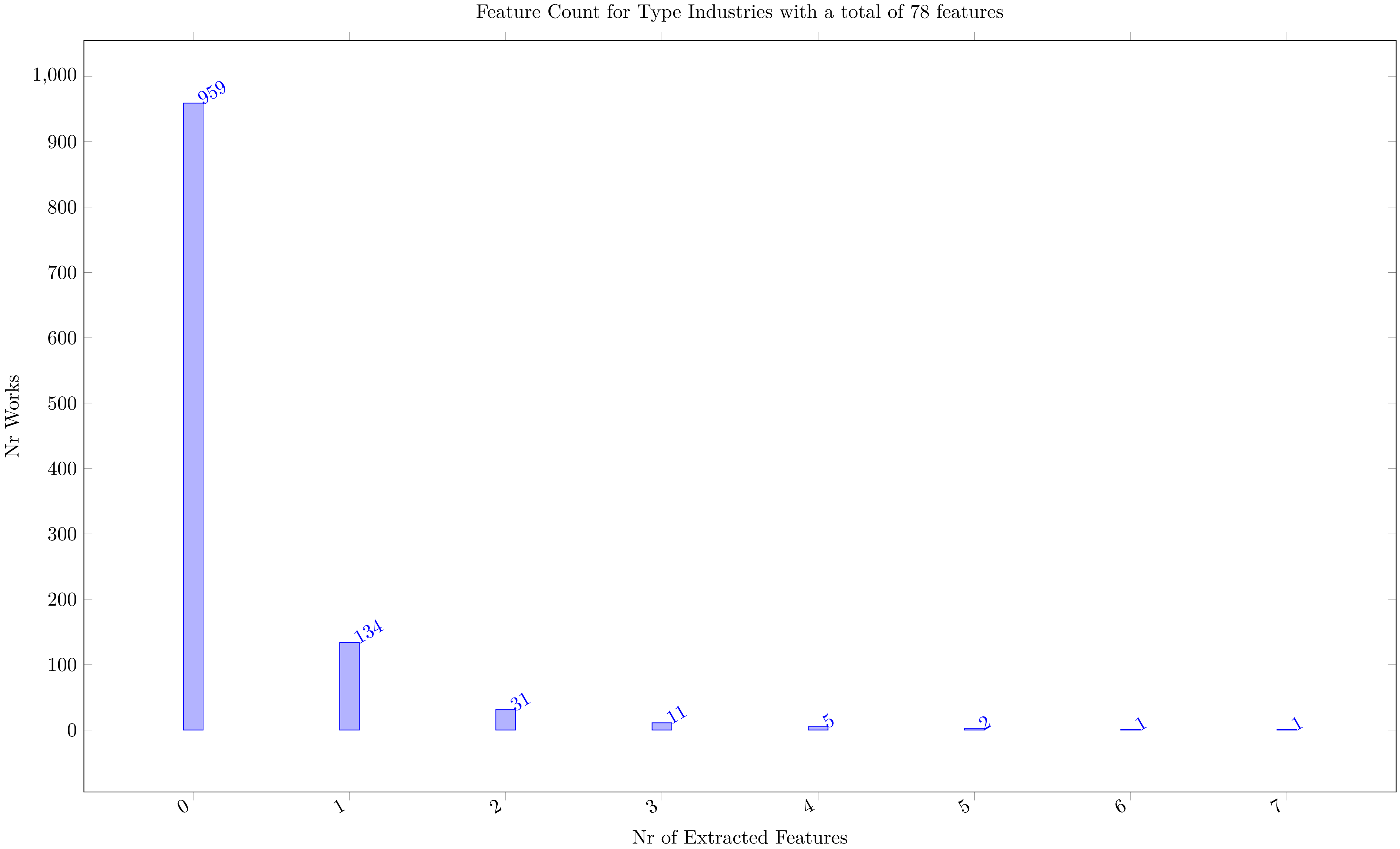

Feature Count for Type Classification with a total of 48 features
Nr Works
Nr of Extracted Features
560
231
160
114
55
9
7
5
2
1
0
1
2
3
4
5
6
7
9
11
600
550
500
450
400
350
300
250
200
150
100
50
0
−50



Feature Count for Type Concepts with a total of 65 features
Nr Works
Nr of Extracted Features
0 10 20 30 40 50 60 70 80 90 100 110
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 30 38
44 71 77 91 106 82 98 91 82 69 59 49 35 47 36 20 14 17 9 10 9 7 1 6 2 2 3 2 3 1 1


Feature Count for Type Constraints with a total of 63 features
Nr Works
Nr of Extracted Features
280
260
240
220
200
180
160
140
120
100
80
60
40
20
0
−20
0
1
2
3
4
5
6
7
8
9
10
11
12
14
19
210
256
253
192
114
50
24
16
13
3
3
4
3
1
2


Feature Count for Type CPSystems with a total of 19 features
Nr Works
Nr of Extracted Features
0
50
100
150
200
250
300
350
0
1
2
3
4
5
6
7
8
9
10
263
360
275
139
49
30
11
11
3
1
2

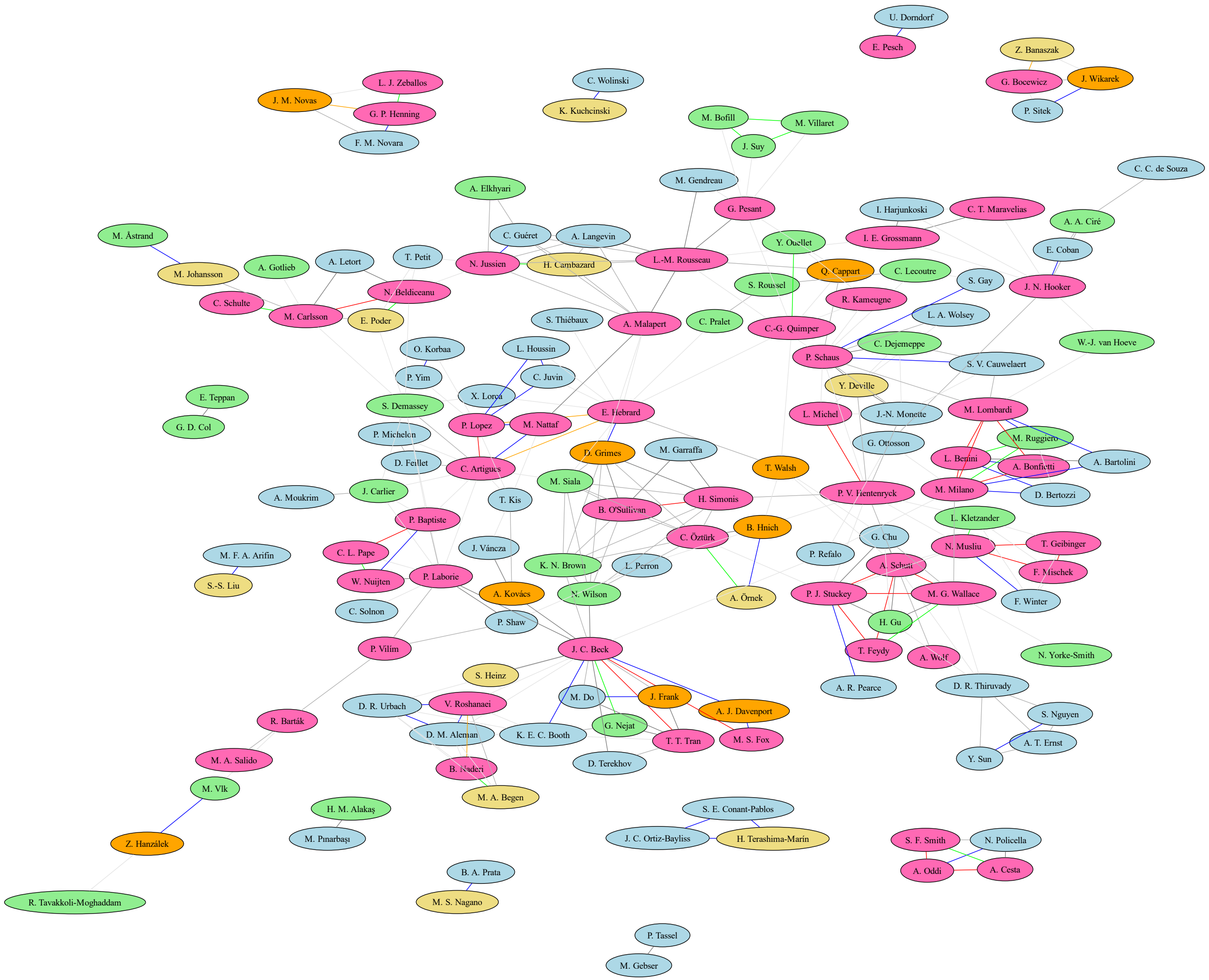

Feature Count for Type Industries with a total of 78 features
Nr Works
Nr of Extracted Features
1,000
900
800
700
600
500
400
300
200
100
0
959
134
31
11
5
2
1
1
0
1
2
3
4
5
6
7

13 Coauthor graph

The coauthor plot is created by graphviz, and is based on the coauthor relations extracted from the author fields of the works. Authors with few works (less than coauthorLimit) are not shown, to avoid a cluttered view. Note that this analysis depends on the use of canonical forms of author names. If bib entries come from many different sources, we will need to check this manually. DBLP seems to be using ORCID values and typically identifies the authors of a work with a canonical representation of their name. Accents and umlauts are other sources of having multiple forms of the name of the same author. Note that the risk of two different authors using the same name should be low for very specific literature surveys, but cannot be checked automatically with the data sources currently used.

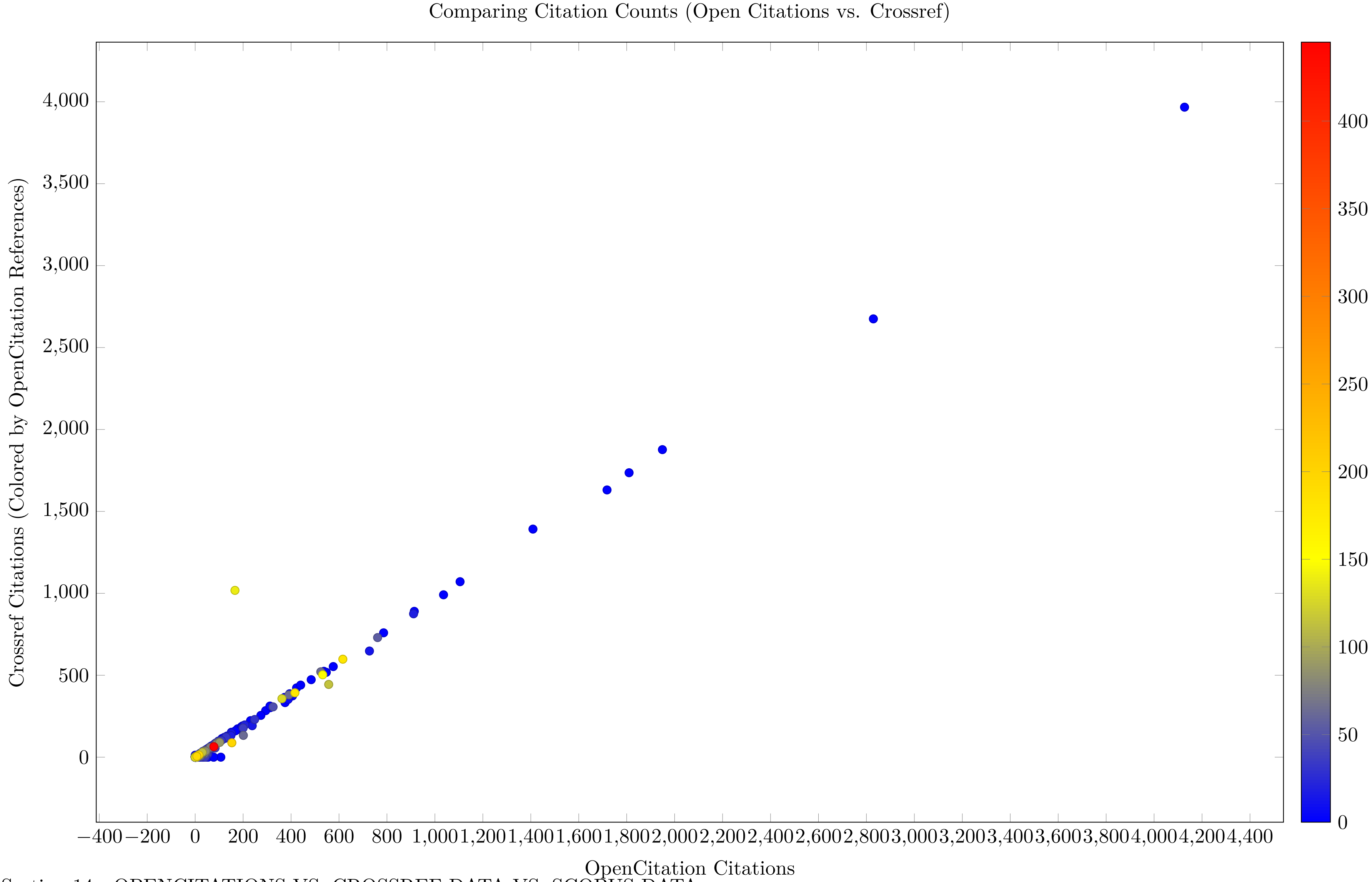
The plots can be made with different layout tools in graphviz, it seems that fdp produces the most consistent visually attractive plots for this type of display. This probably needs more work on parameter settings to be fully automated.

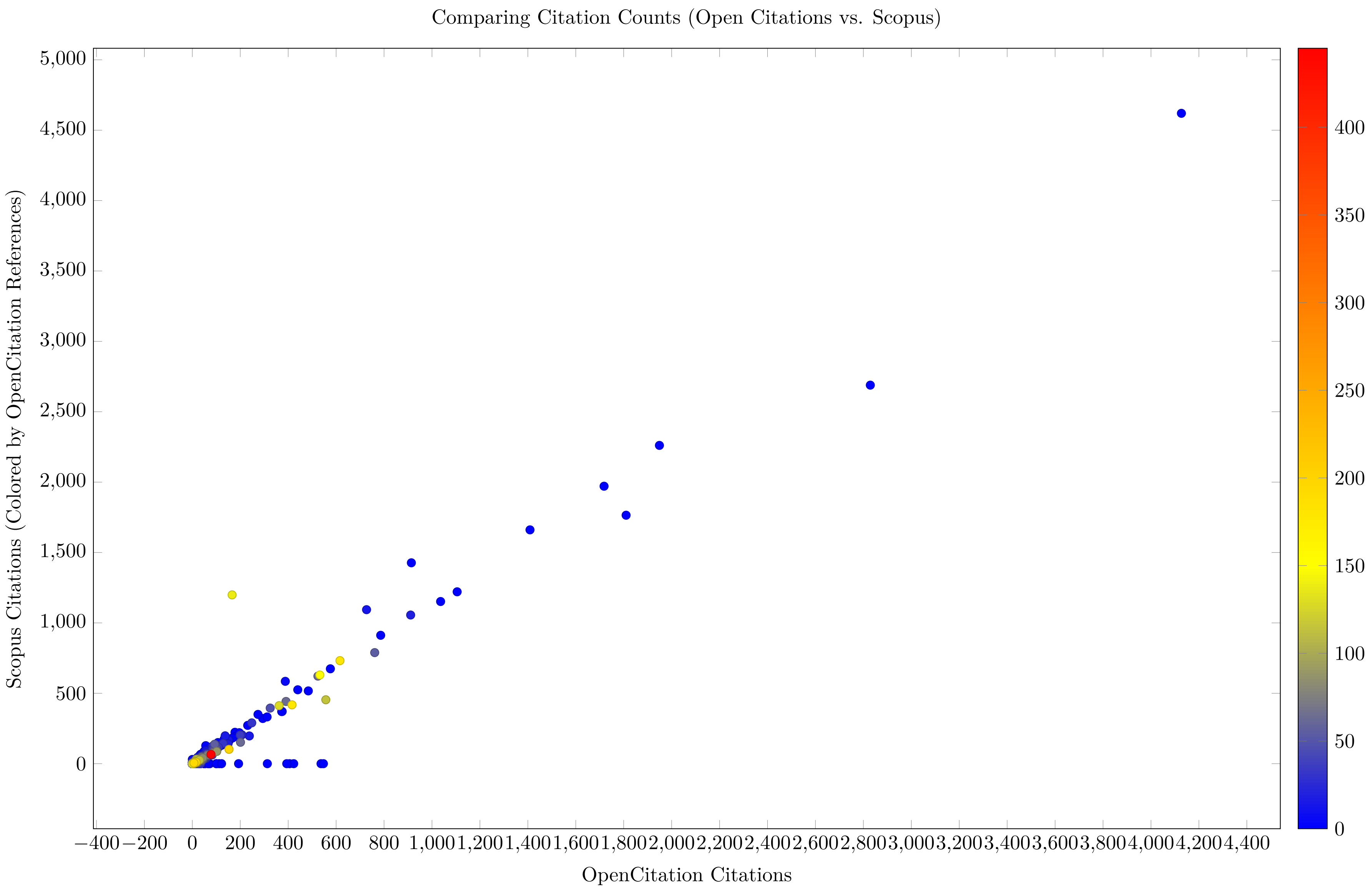
Figure 1: Coauthor Graph Drawn with fdp (Graphviz, CoauthorLimit = 4)

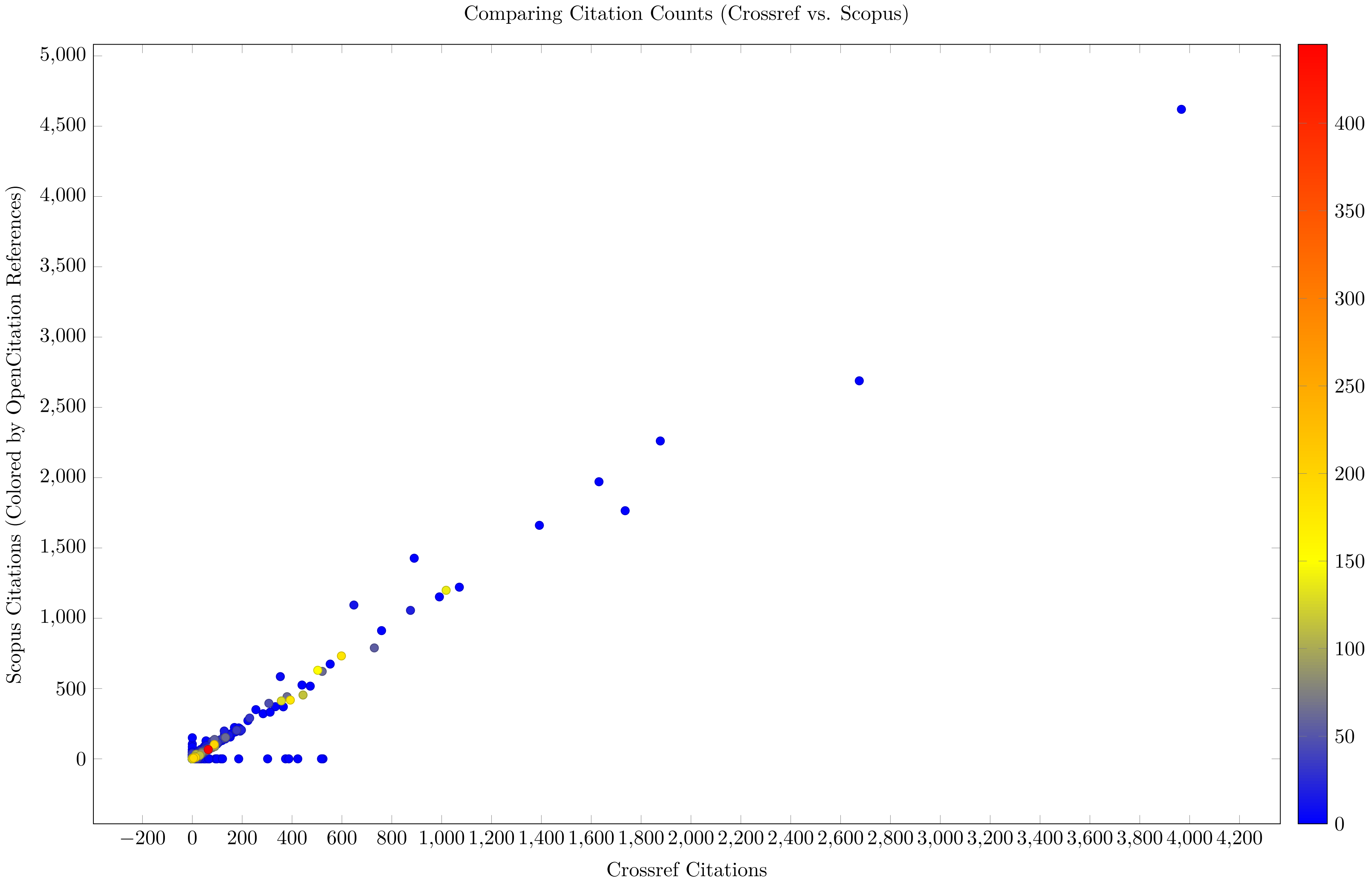


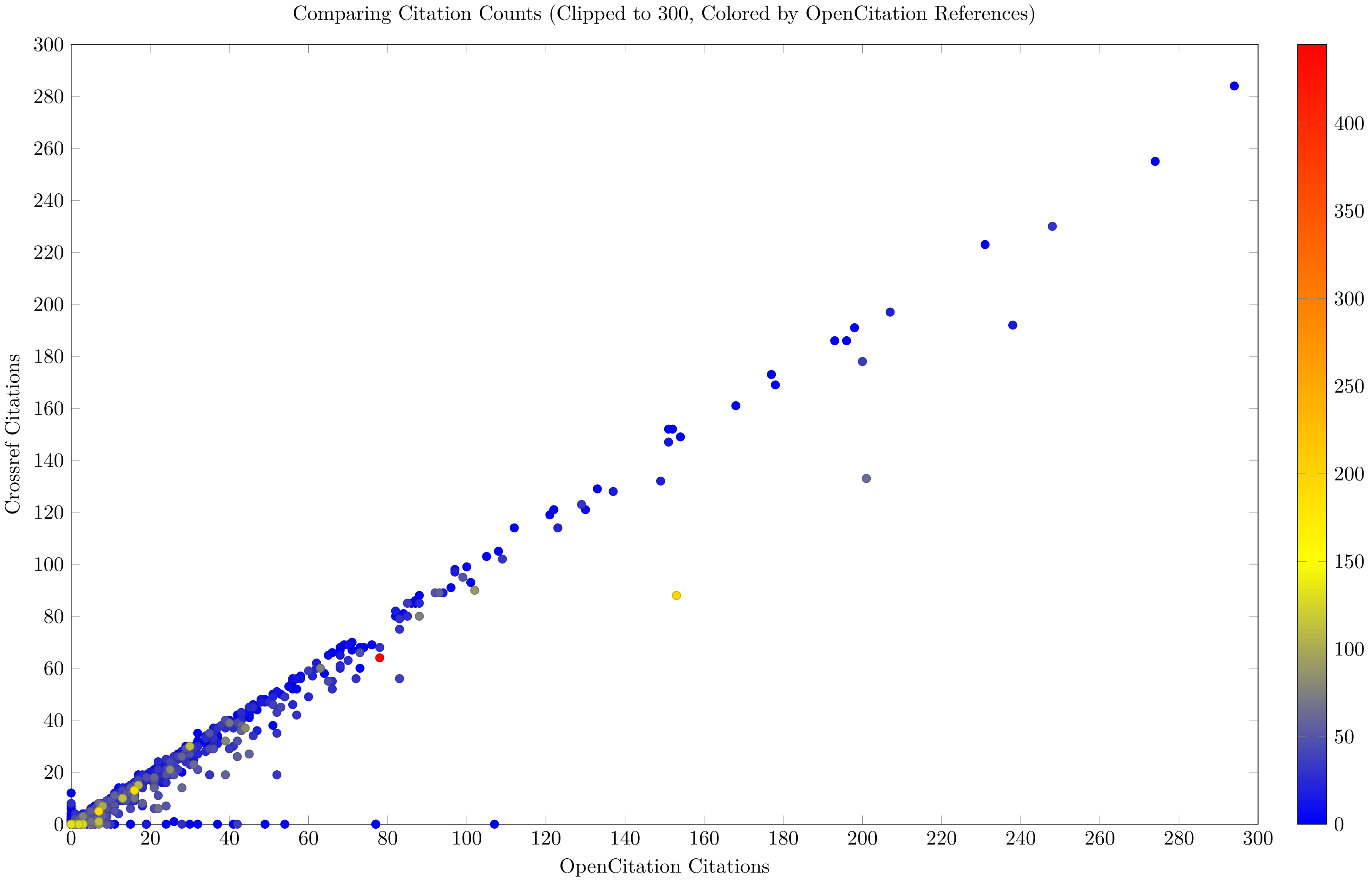
14 OpenCitations vs. Crossref Data vs. Scopus Data

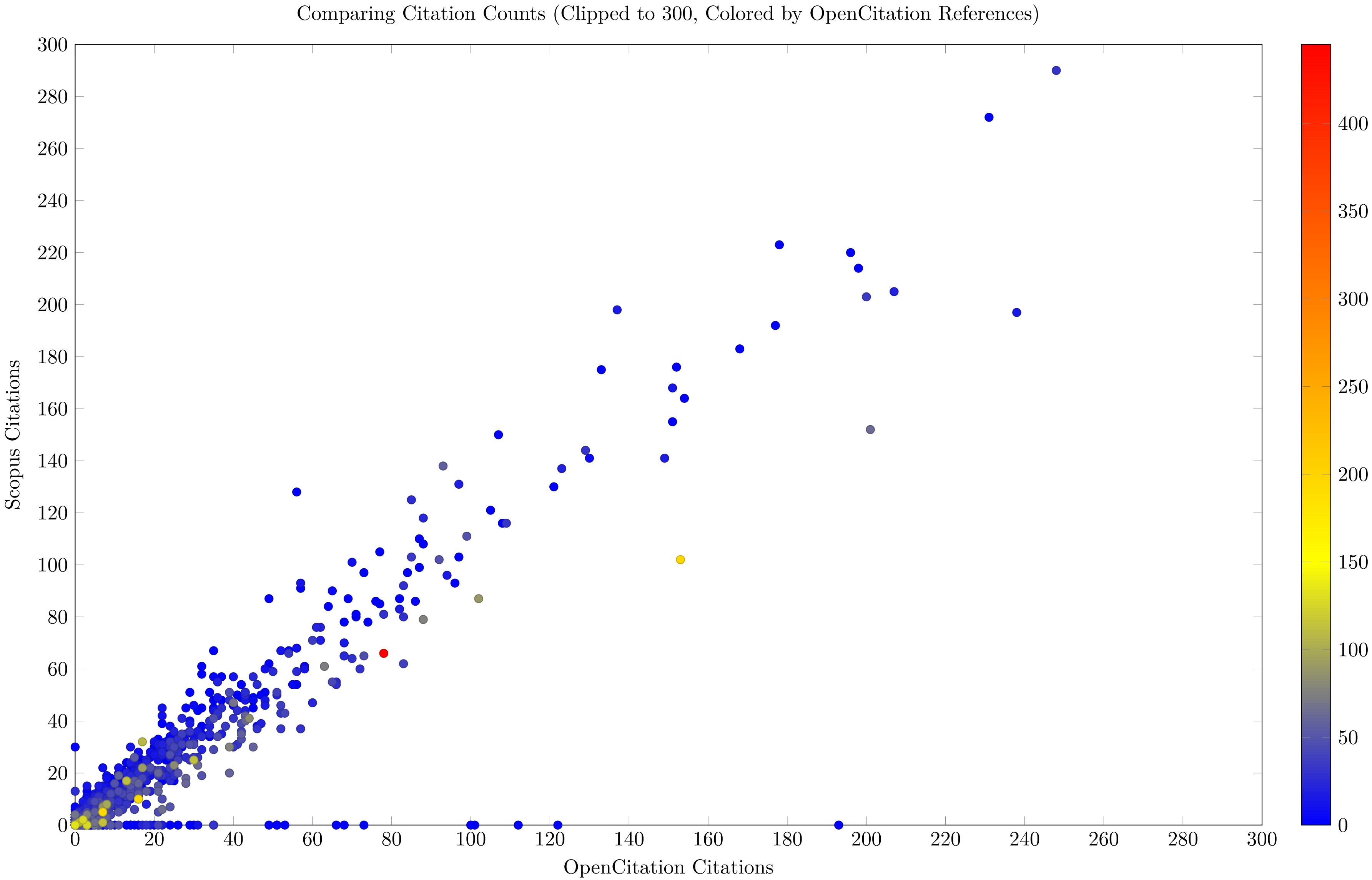
14.1 Citation Comparison

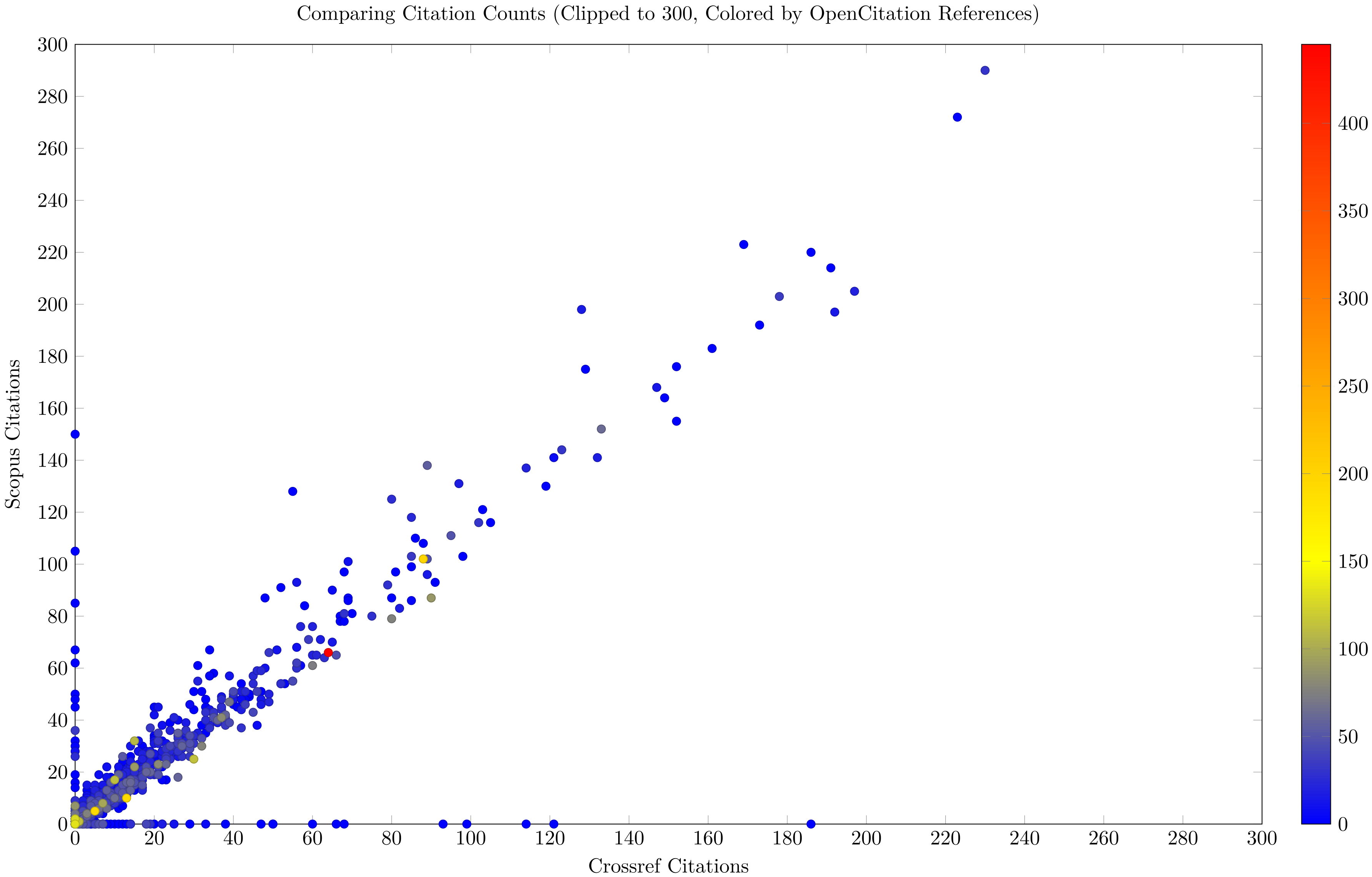



Comparing Citation Counts (Open Citations vs. Scopus)
Scopus Citations (Colored by OpenCitation References)
OpenCitation Citations
5,000
4,500
4,000
3,500
3,000
2,500
2,000
1,500
1,000
500
0
−400
−200
0
200
400
600
800
1,000
1,200
1,400
1,600
1,800
2,000
2,200
2,400
2,600
2,800
3,000
3,200
3,400
3,600
3,800
4,000
4,200
4,400
400
350
300
250
200
150
100
50
0

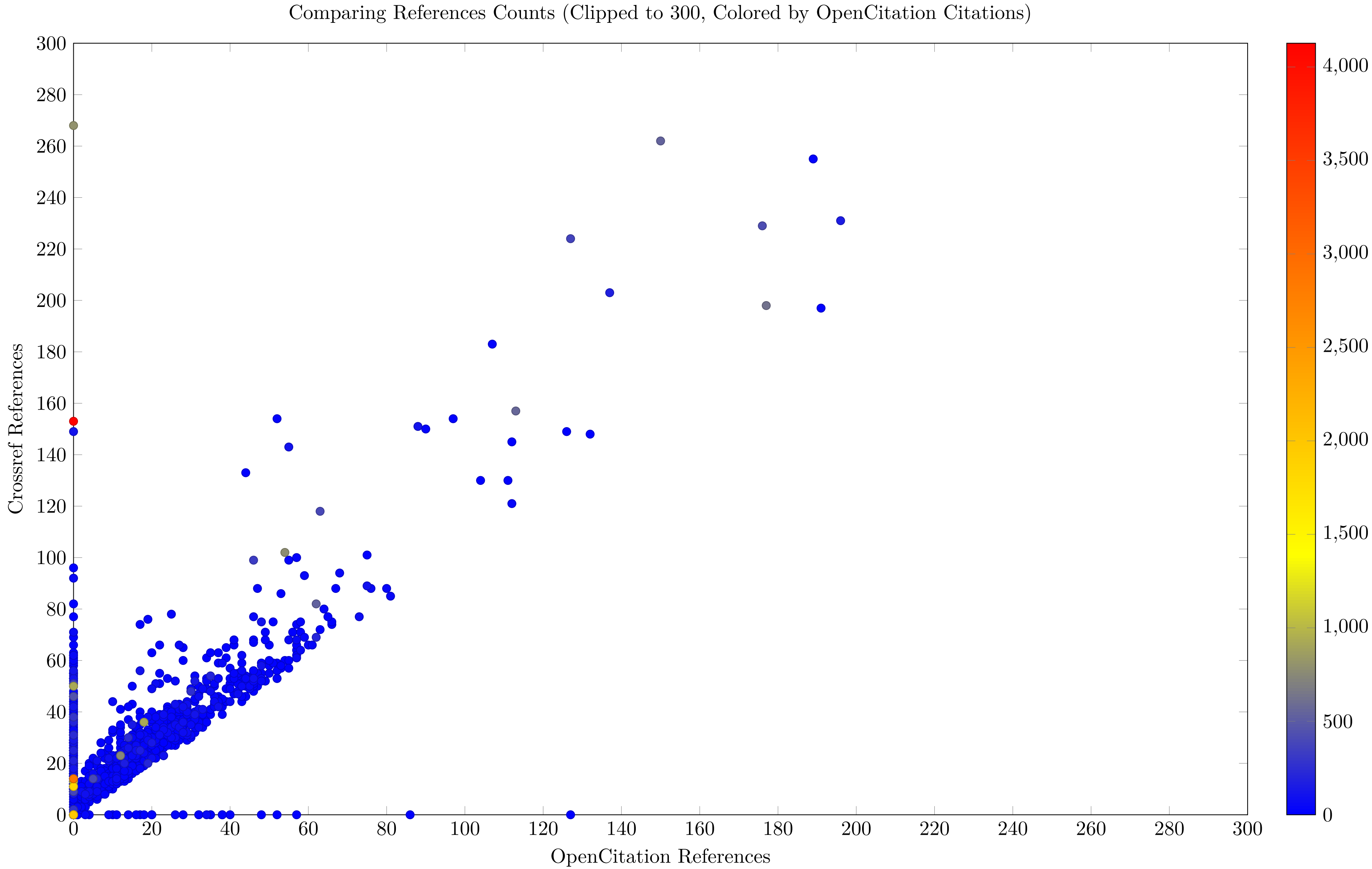

Comparing Citation Counts (Crossref vs. Scopus)
Scopus Citations (Colored by OpenCitation References)
Crossref Citations
5,000
4,500
4,000
3,500
3,000
2,500
2,000
1,500
1,000
500
0
−200
200
400
600
800
1,000
1,200
1,400
1,600
1,800
2,000
2,200
2,400
2,600
2,800
3,000
3,200
3,400
3,600
3,800
4,000
4,200
400
350
300
250
200
150
100
50


Comparing Citation Counts (Clipped to 300, Colored by OpenCitation References)
Crossref Citations
OpenCitation Citations
0
20
40
60
80
100
120
140
160
180
200
220
240
260
280
300
0
50
100
150
200
250
300
350
400

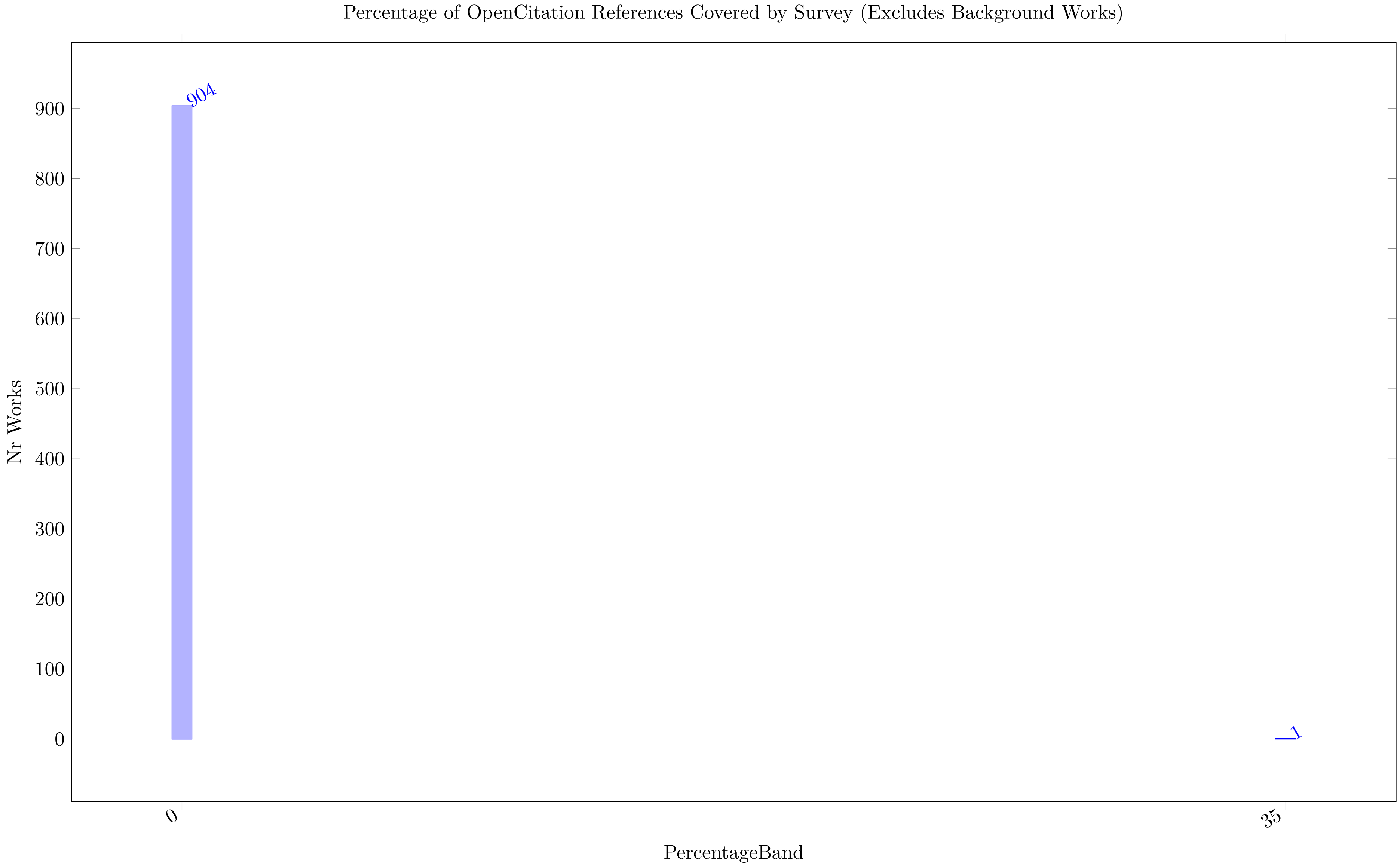

Comparing Citation Counts (Clipped to 300, Colored by OpenCitation References)
Scopus Citations
OpenCitation Citations
0
20
40
60
80
100
120
140
160
180
200
220
240
260
280
300
0
50
100
150
200
250
300
350
400

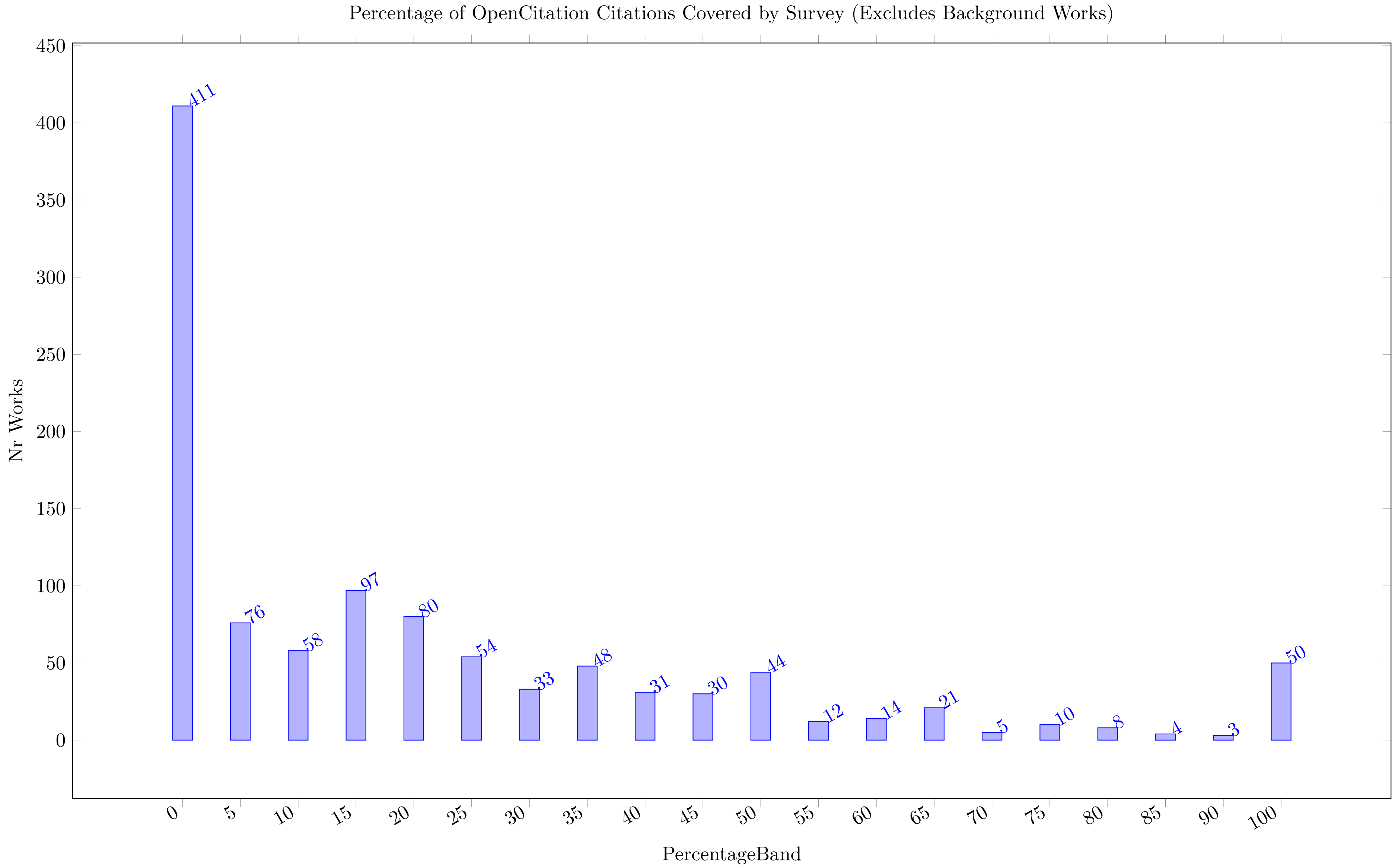

Comparing Citation Counts (Clipped to 300, Colored by OpenCitation References)
Scopus Citations
Crossref Citations
0
20
40
60
80
100
120
140
160
180
200
220
240
260
280
300
0
50
100
150
200
250
300
350
400

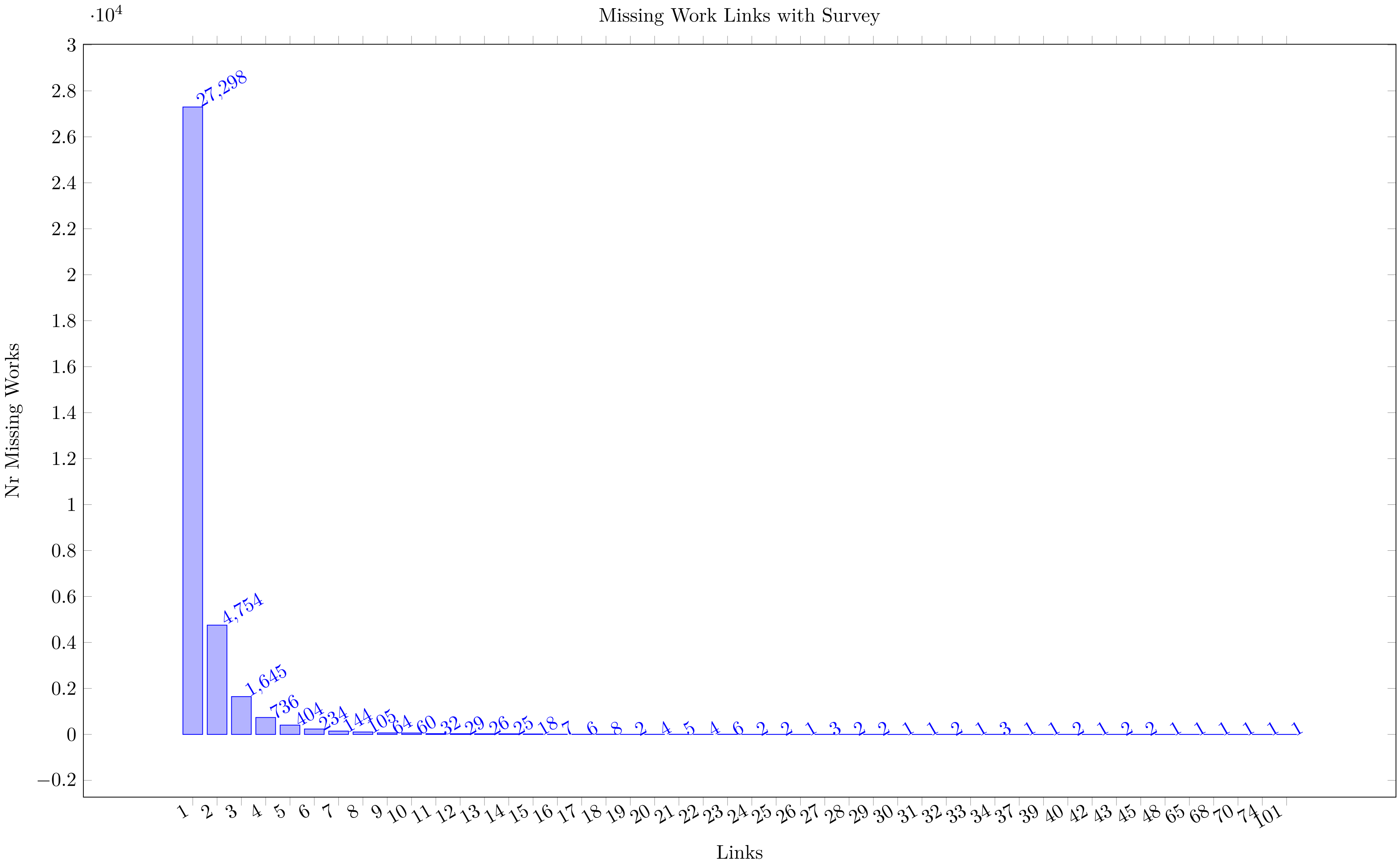
14.2 References Comparison



14.3 Percentage Cover




Percentage of OpenCitation Citations Covered by Survey (Excludes Background Works)
Nr Works
PercentageBand
0
50
100
150
200
250
300
350
400
450
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
80
85
90
100
411
76
58
97
80
54
33
48
31
30
44
12
14
21
5
10
8
4
3
50


Missing Work Links with Survey
Nr Missing Works
Links
·10^4
27,298
4,754
1,645
736
404
234
144
105
64
60
32
29
26
25
18
7
6
8
2
4
5
4
6
2
2
1
3
2
2
1
1
2
1
3
1
1
2
1
2
2
1
1
1
1
1
1

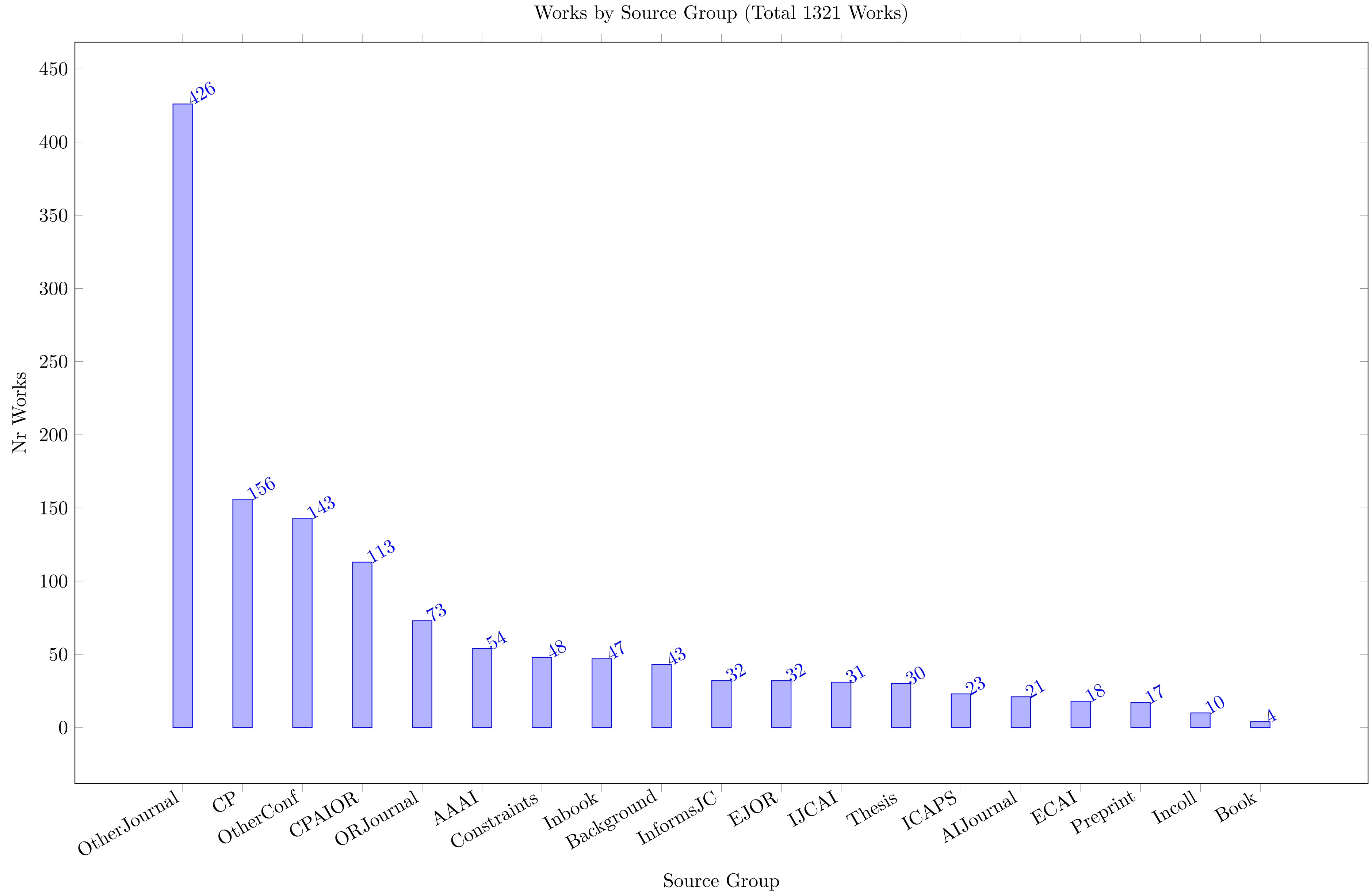
15 Citations by Year and Source Group

We have defined a number of source groups to group publications of a given type together, without using the full conference series and journal distinctions for grouping. The following table lists all defined source groups for this survey. Adding groups requires updates to the source code.

Table 13: Source Groups

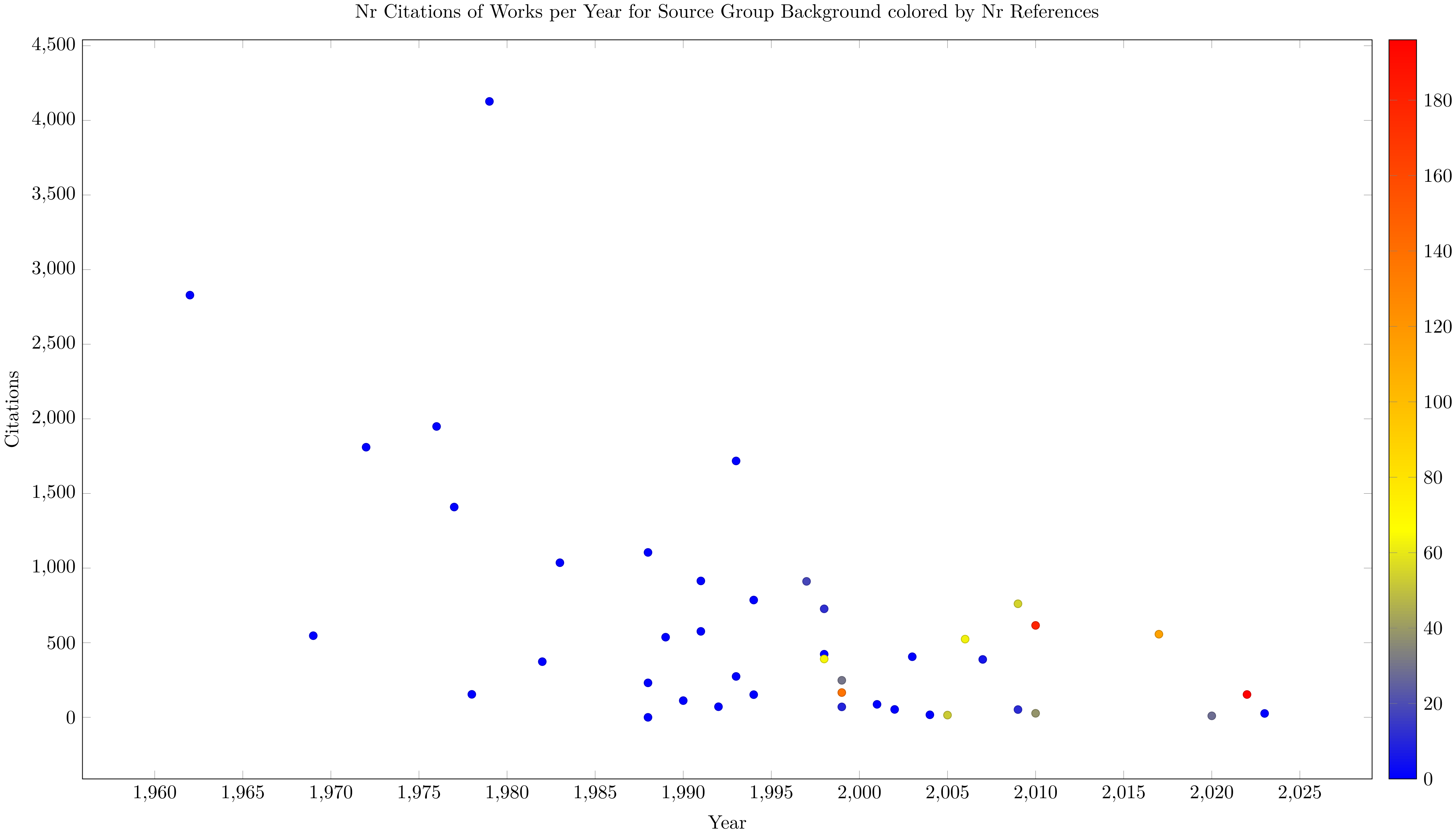
Name	Description
Background	Background material
CP	The CP conference (from 1995)
CPAIOR	The CPAIOR conference (starting 2004)
ICAPS	The ICAPS conference
AAAI	AAAI conference
IJCAI	IJCAI Conference
ECAI	ECAI Conference
OtherConf	Any other conference
Constraints	The Constraint Journal
EJOR	The European Journal on Operations Research
InformsJC	The Informs Journal on Computing
AIJournal	Other AI Journals
ORJournal	Other OR Journals
JoPR	Journal of Peace Research
JoCR	Journal of Conflict Resolution
CMPS	Conflict Management and Piece Science
Preprint	A non reviewed preprint
OtherJournal	Any other Journal
Book	A book
Inbook	Chapter in a Book
Incoll	Chapter in a Collection
Thesis	A thesis
Other	Any other published work

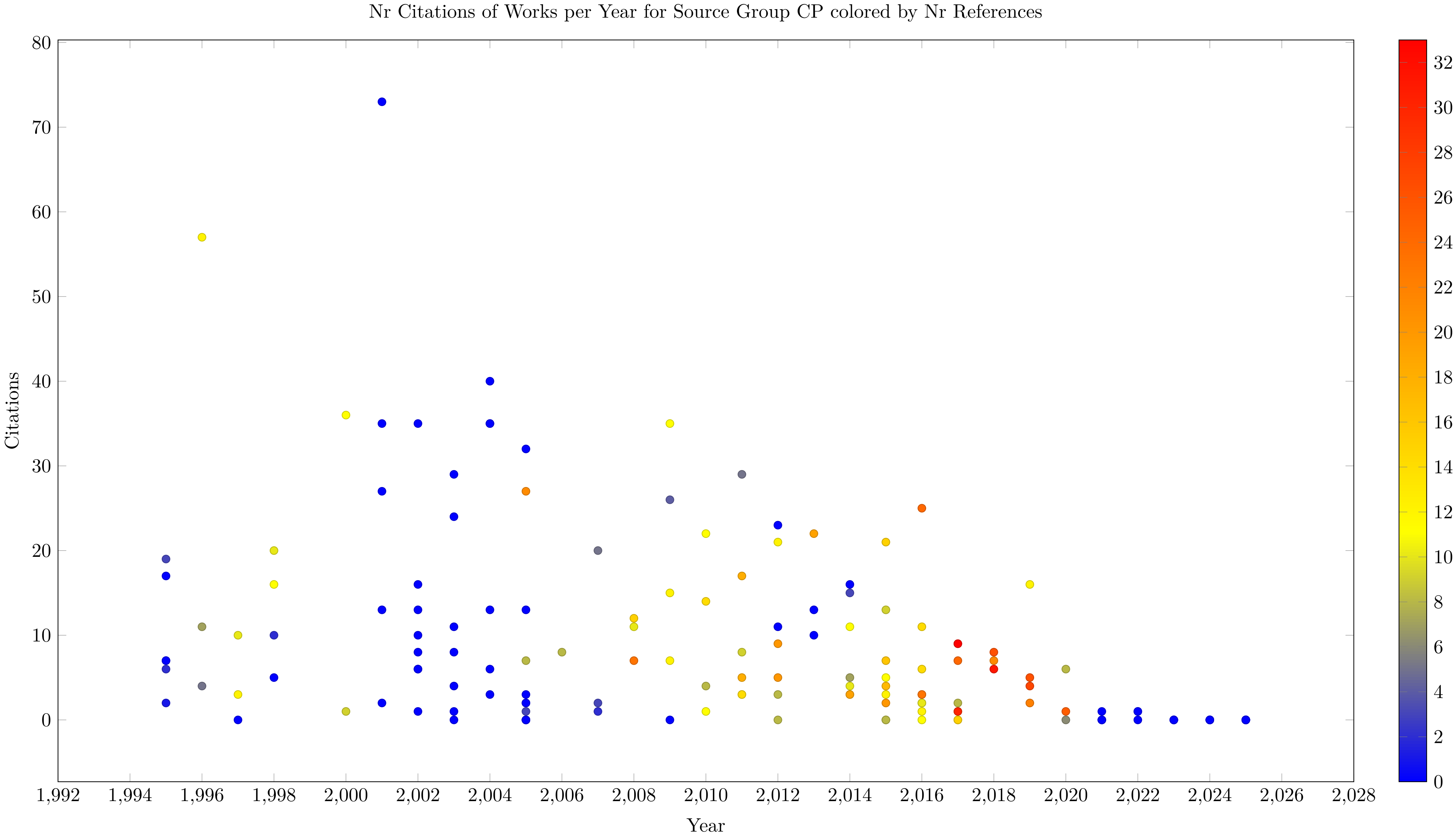
The first plot in this section shows how many works in each source group have been published. This considers the complete time period of the survey.

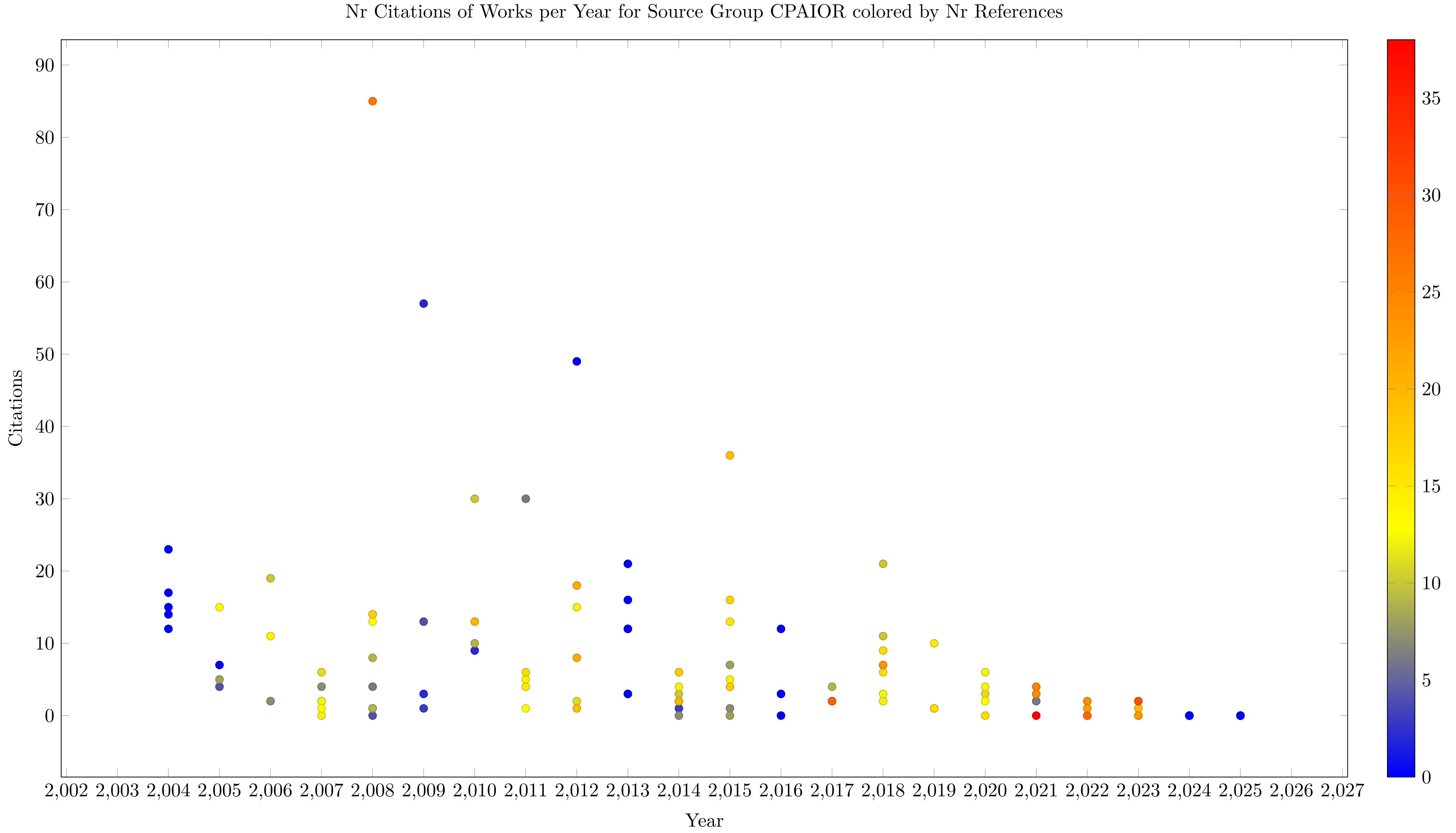

Works by Source Group (Total 1321 Works)
Nr Works
Source Group
0
50
100
150
200
250
300
350
400
450
426
156
143
113
73
54
48
47
43
32
32
31
30
23
21
18
17
10
4
OtherJournal
CP
OtherConf
CPAIOR
ORJournal
AAAI
Constraints
Inbook
Background
InformsJC
EJOR
IJCAI
Thesis
ICAPS
AIJournal
ECAI
Preprint
Incoll
Book

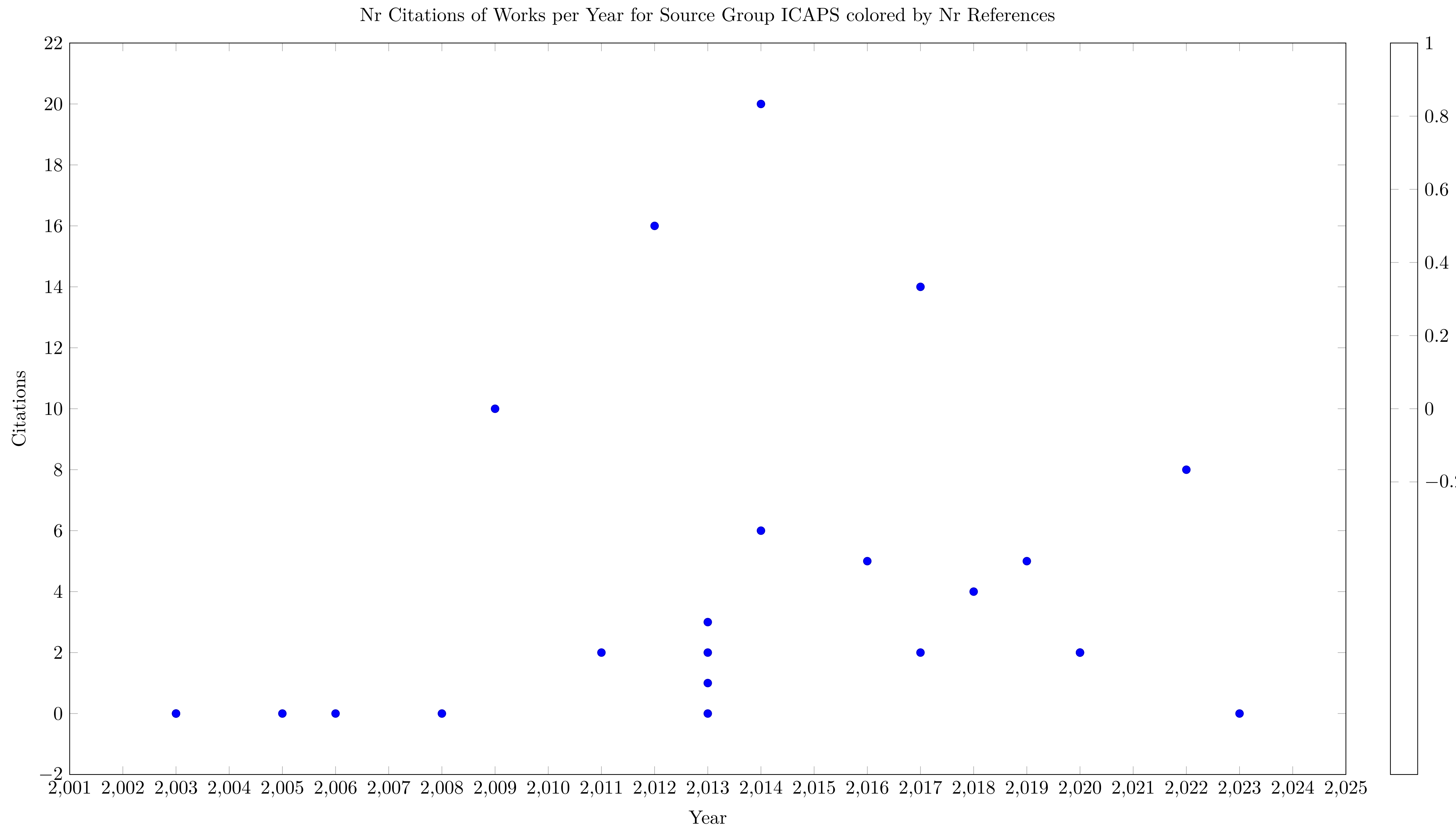
15.1 Source Group Citations by Year

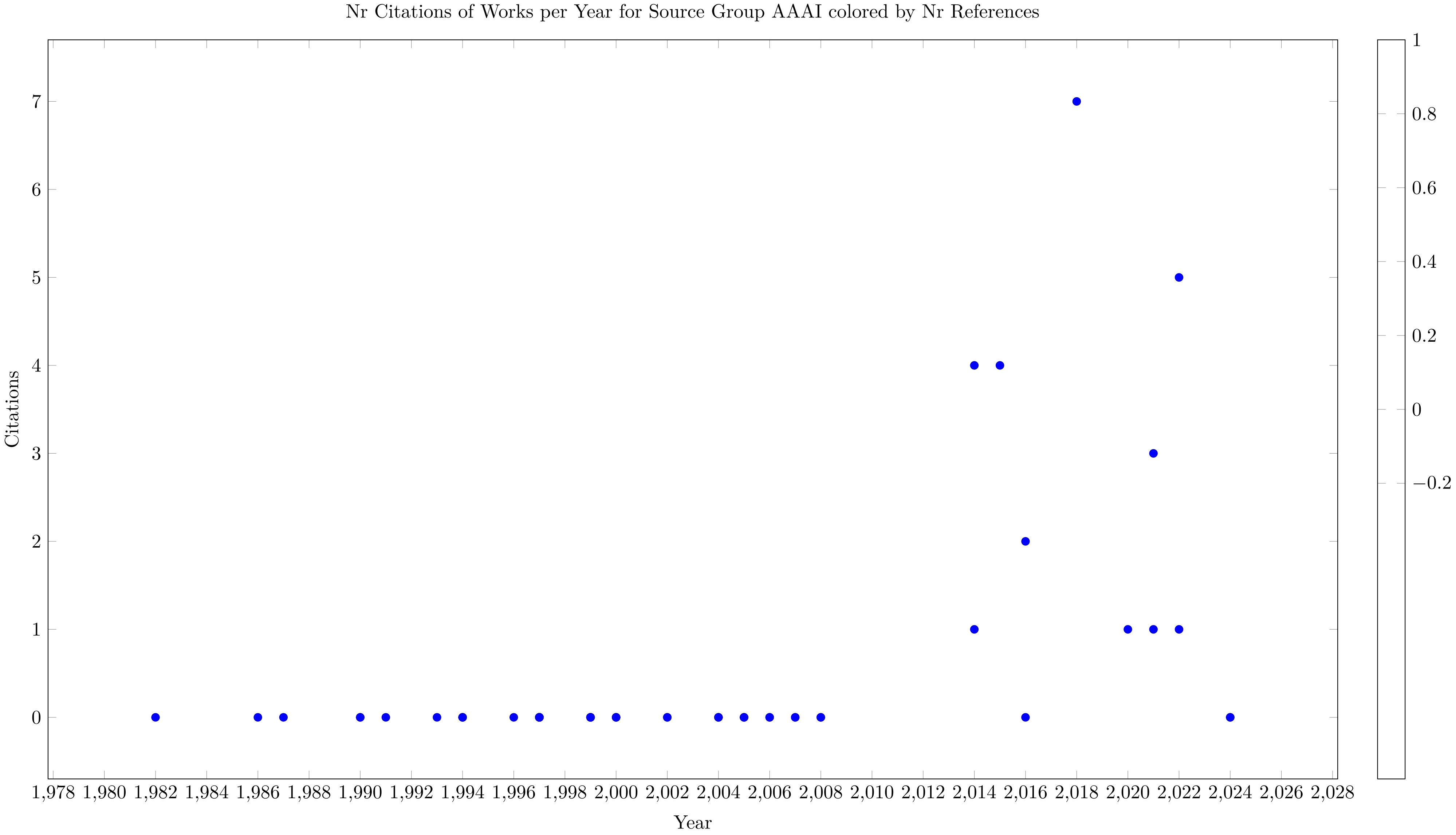
We plot for each source group the number of citations obtained by papers published in a given year. This plot gives both an indication in which period the source group was active, and how significant the works in the source are. It is of course natural that more recent papers have fewer citations than papers published many tears ago.

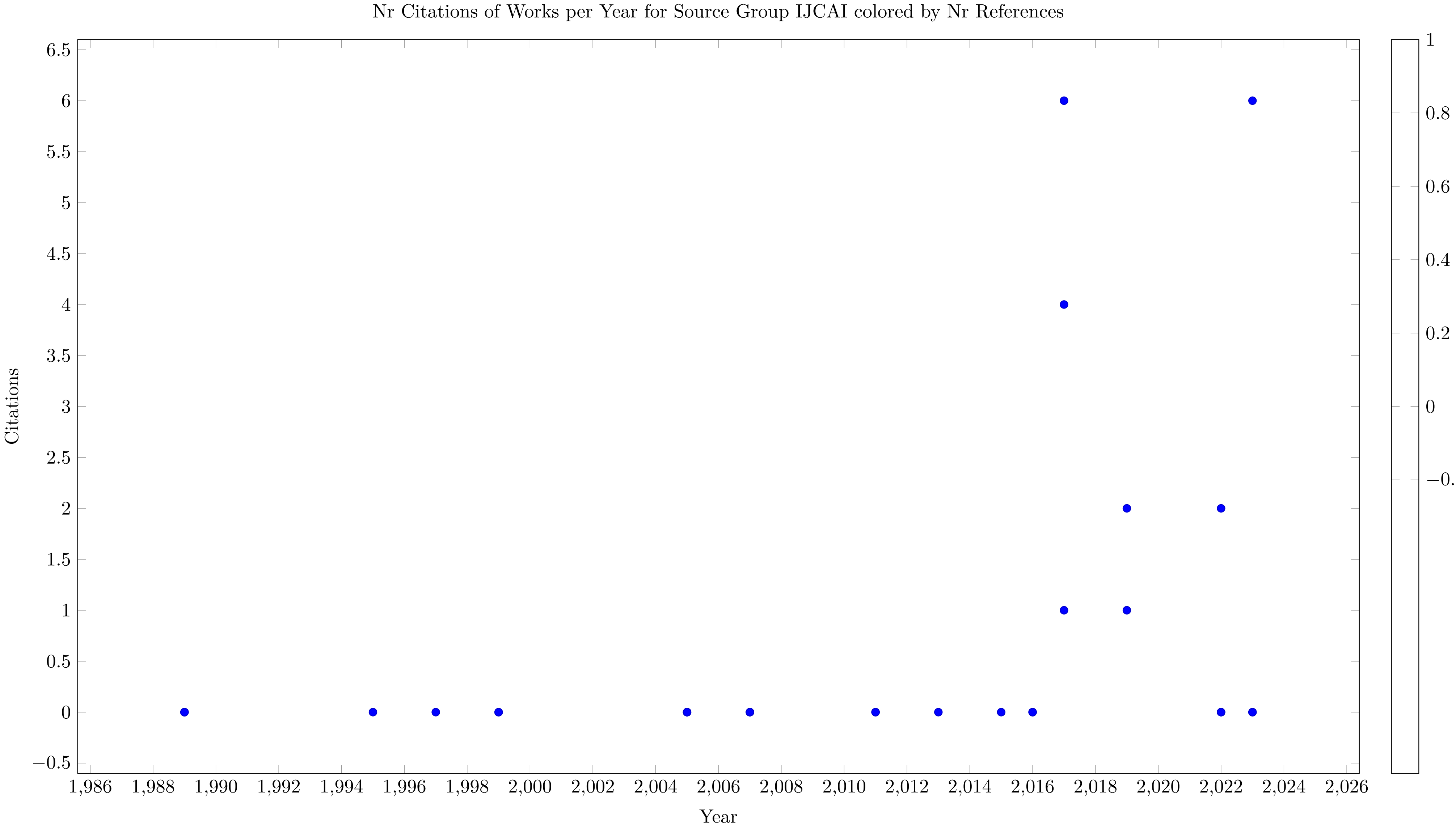

Nr Citations of Works per Year for Source Group Background colored by Nr References
Citations
Year
4,500
4,000
3,500
3,000
2,500
2,000
1,500
1,000
500
0
1,960
1,965
1,970
1,975
1,980
1,985
1,990
1,995
2,000
2,005
2,010
2,015
2,020
2,025
180
160
140
120
100
80
60
40
20
0

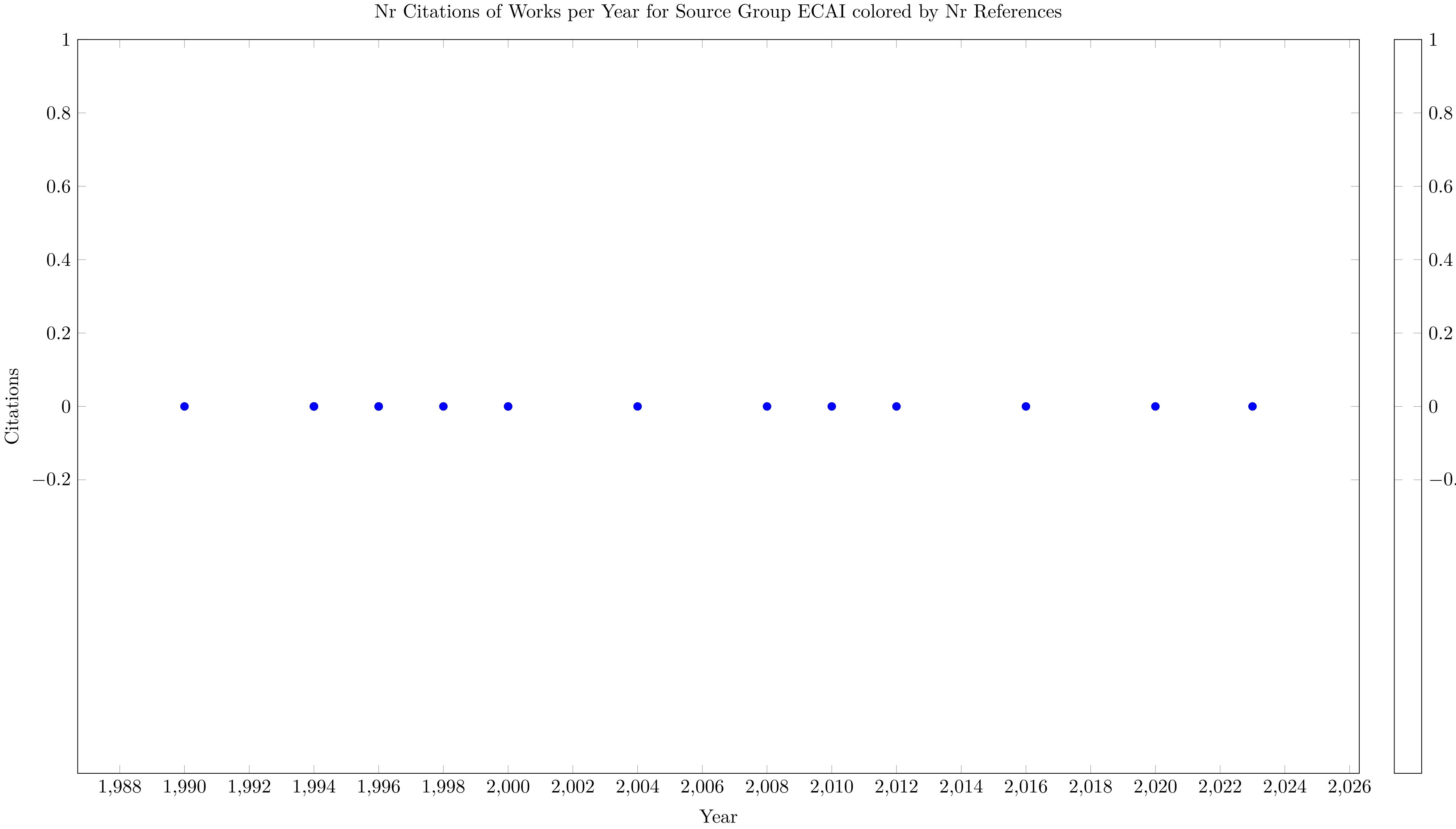

Nr Citations of Works per Year for Source Group CP colored by Nr References
Citations
Year
0
10
20
30
40
50
60
70
80
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026
2,028
0
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32

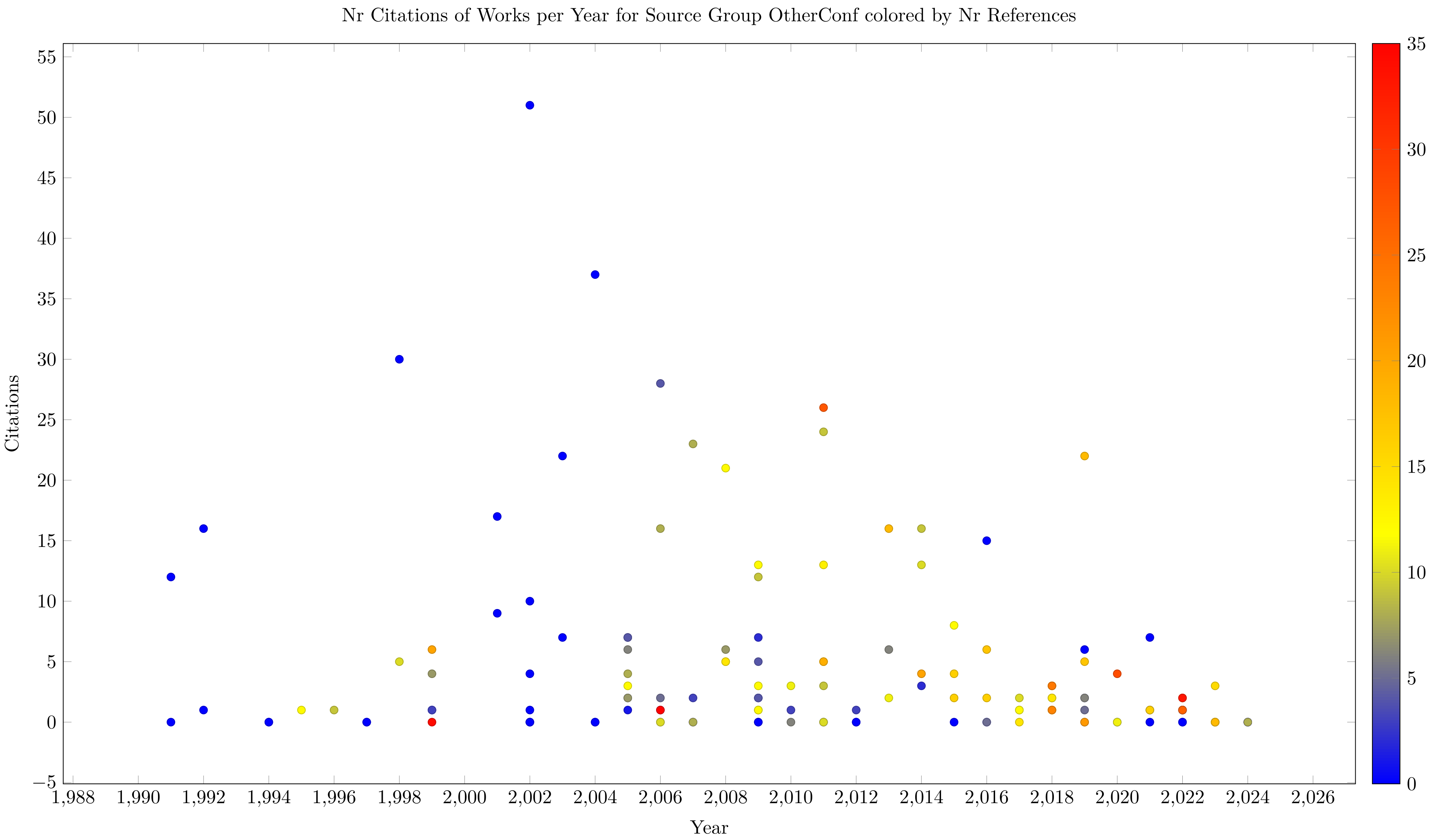

Nr Citations of Works per Year for Source Group CPAIOR colored by Nr References
Citations
Year
90
80
70
60
50
40
30
20
10
0
2,002 2,003 2,004 2,005 2,006 2,007 2,008 2,009 2,010 2,011 2,012 2,013 2,014 2,015 2,016 2,017 2,018 2,019 2,020 2,021 2,022 2,023 2,024 2,025 2,026 2,027
35
30
25
20
15
10
5
0

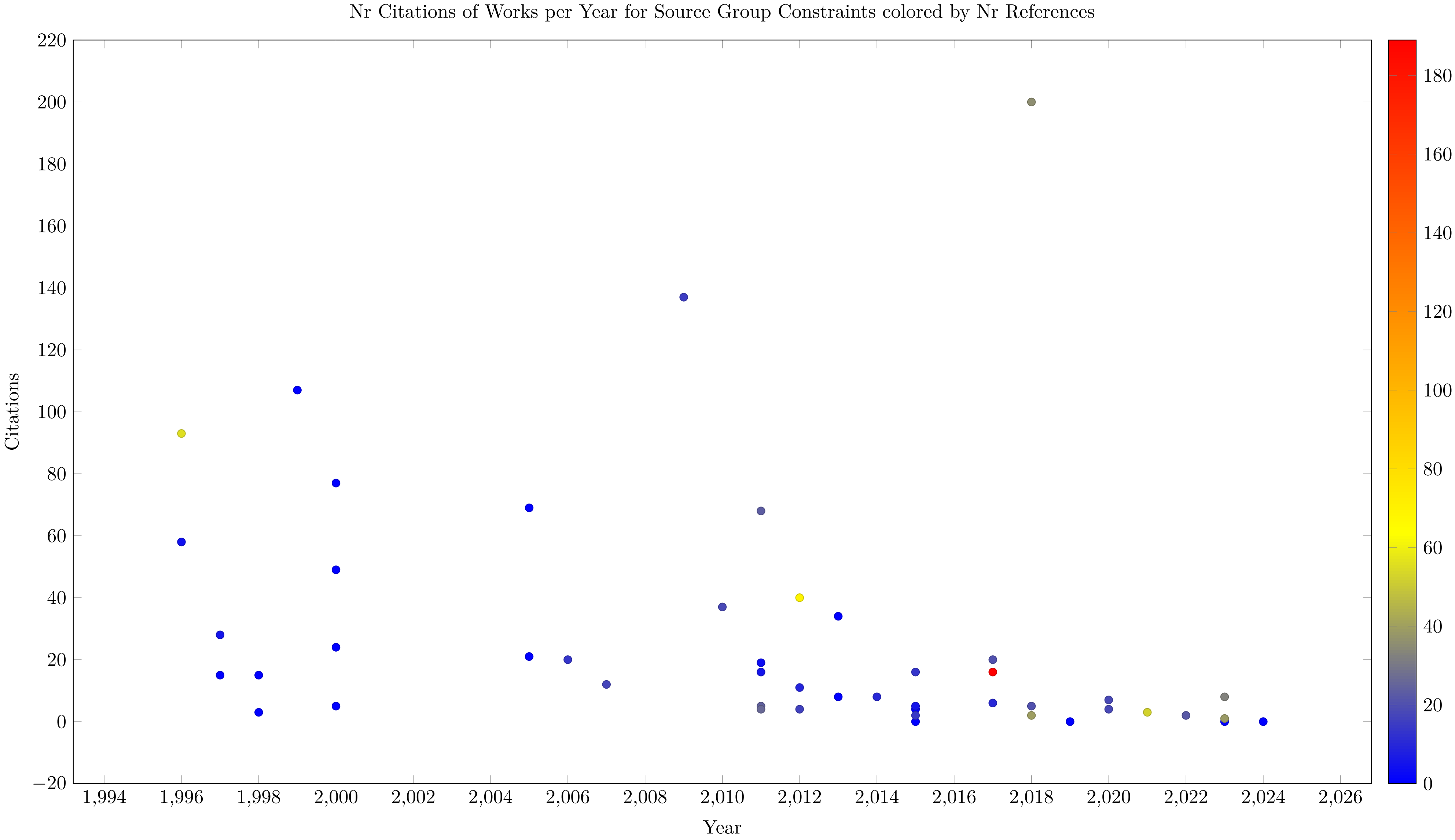

Nr Citations of Works per Year for Source Group ICAPS colored by Nr References
Citations
Year
22
20
18
16
14
12
10
8
6
4
2
0
−2
2,001 2,002 2,003 2,004 2,005 2,006 2,007 2,008 2,009 2,010 2,011 2,012 2,013 2,014 2,015 2,016 2,017 2,018 2,019 2,020 2,021 2,022 2,023 2,024 2,025
1
0.8
0.6
0.4
0.2
0
−0.2

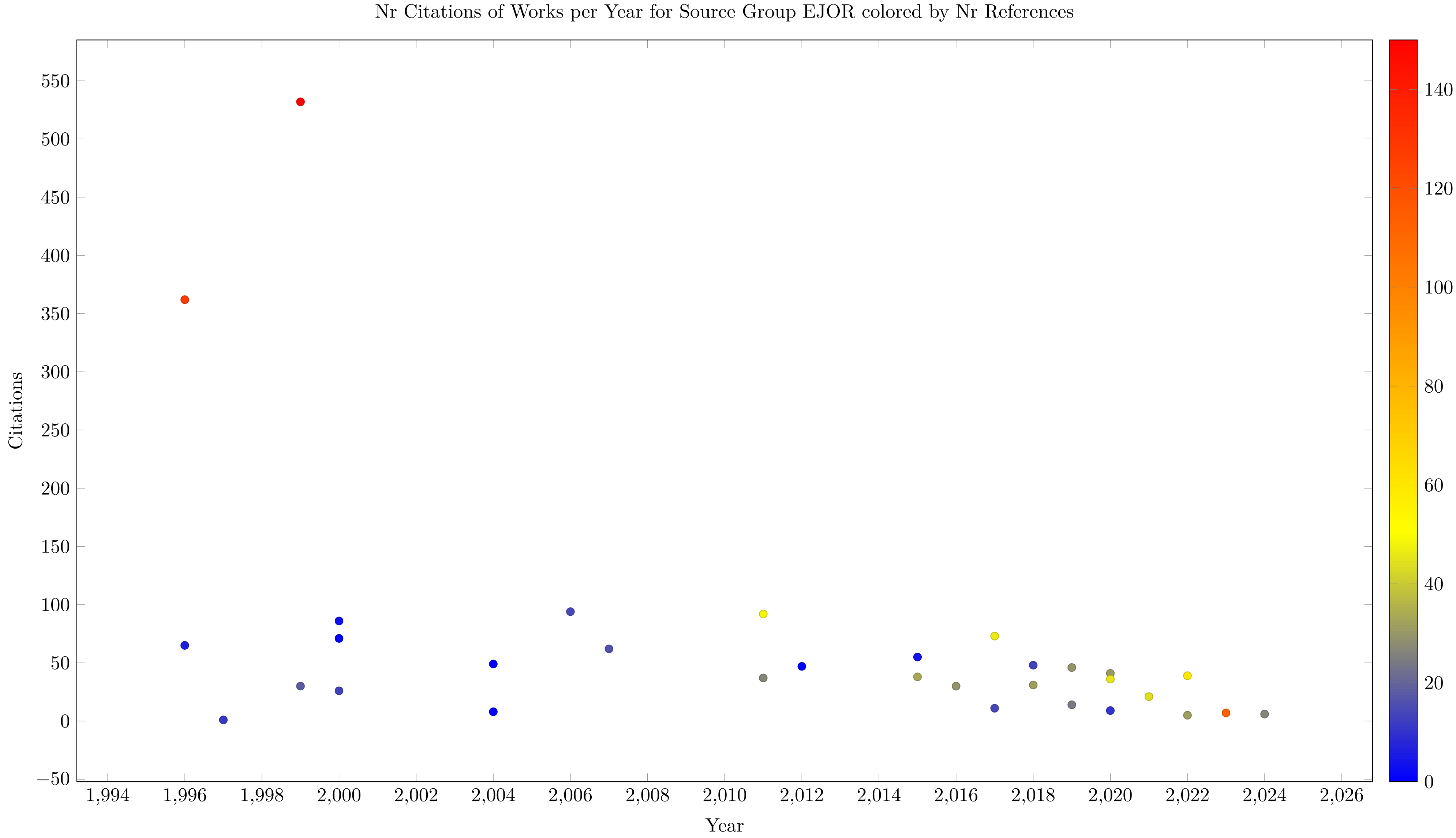

Nr Citations of Works per Year for Source Group AAAI colored by Nr References
Citations
Year
0
1
2
3
4
5
6
7
1,978
1,980
1,982
1,984
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026
2,028
1
0.8
0.6
0.4
0.2
0
−0.2

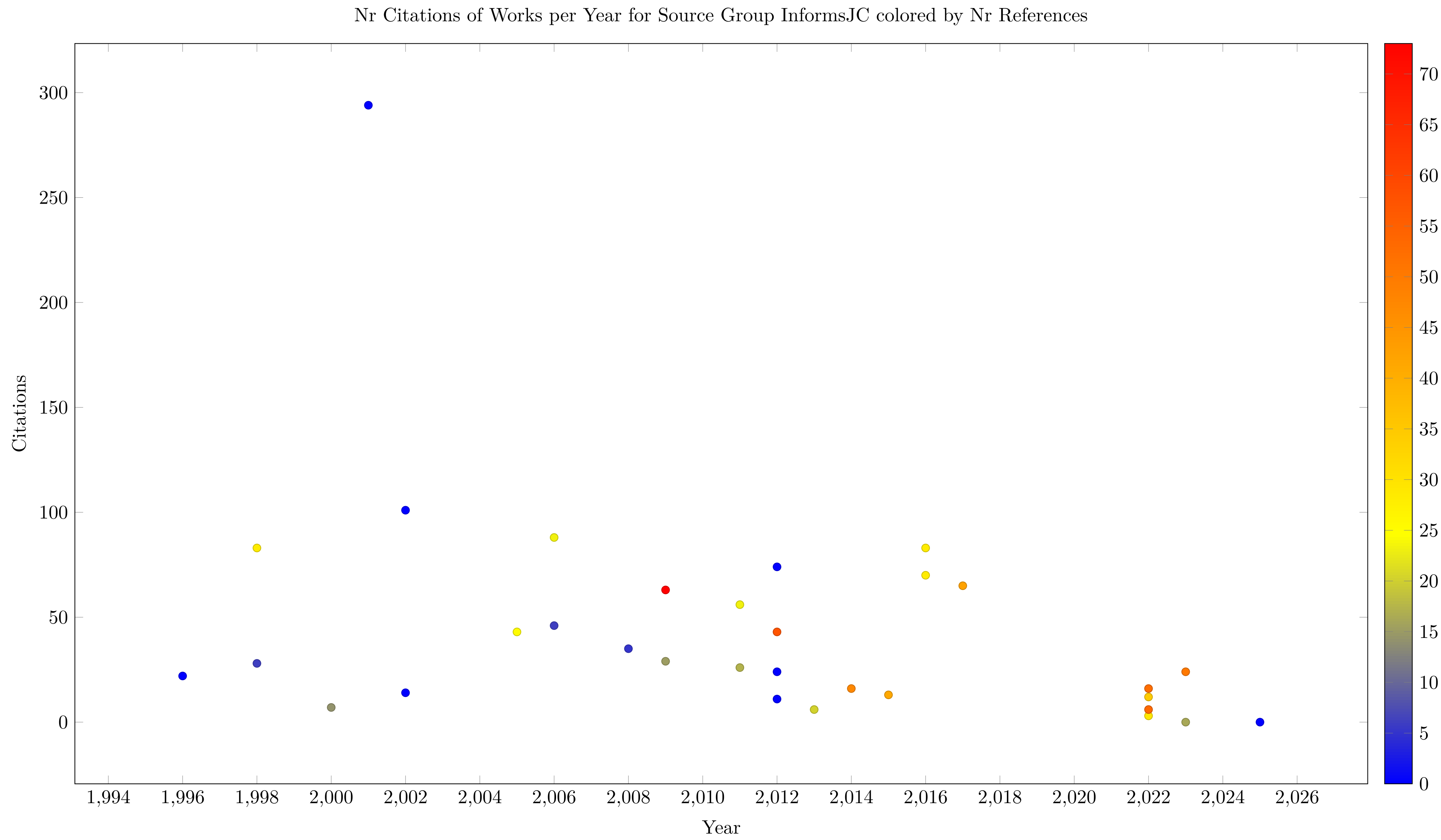

Nr Citations of Works per Year for Source Group IJCAI colored by Nr References
Citations
Year
6.5
6
5.5
5
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
−0.5
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026
1
0.8
0.6
0.4
0.2
0

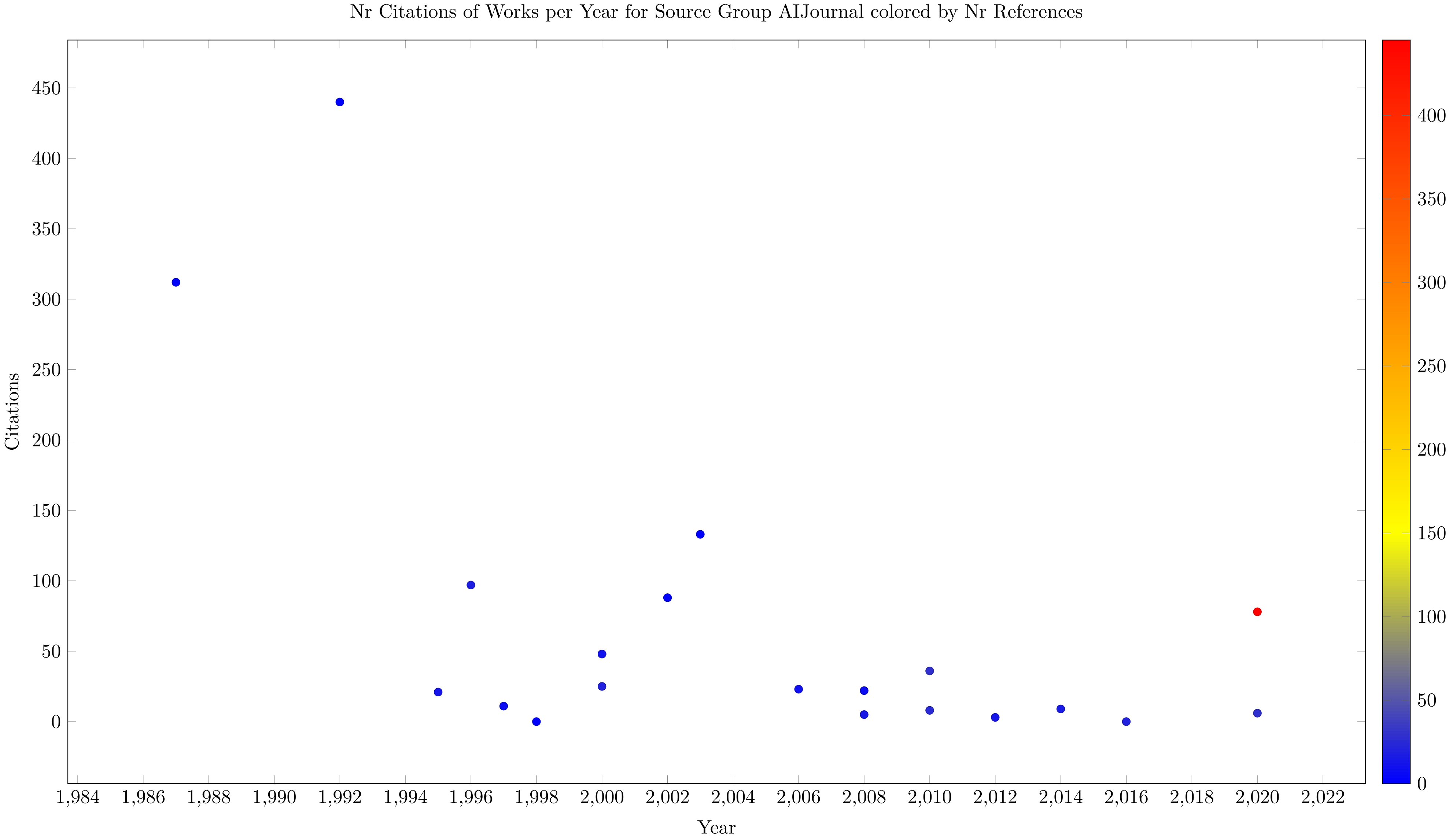

Nr Citations of Works per Year for Source Group ECAI colored by Nr References
Citations
Year
1
0.8
0.6
0.4
0.2
0
−0.2
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026

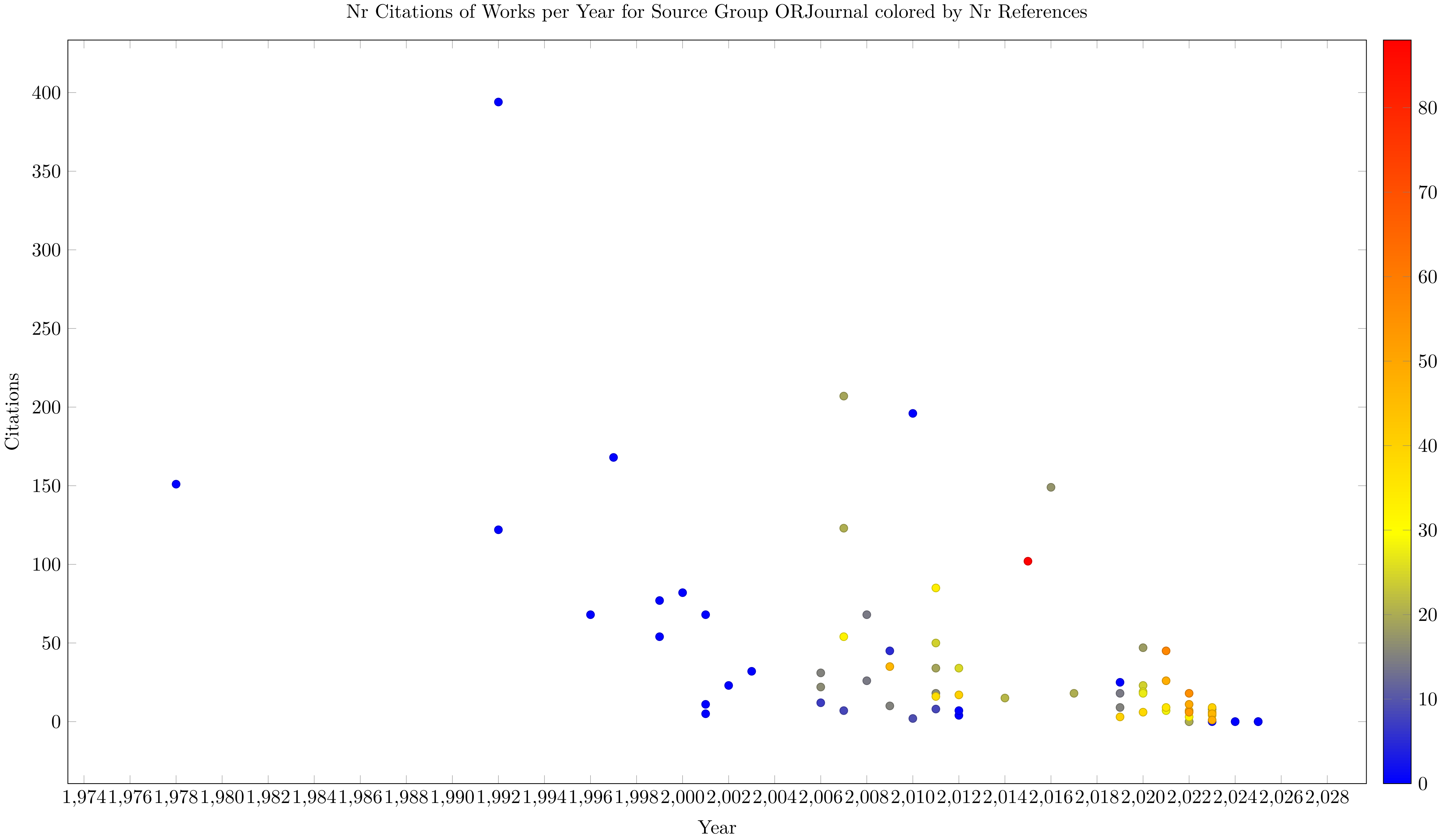

Nr Citations of Works per Year for Source Group OtherConf colored by Nr References
Citations
Year
−5
0
5
10
15
20
25
30
35
40
45
50
55
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026
0
5
10
15
20
25
30
35

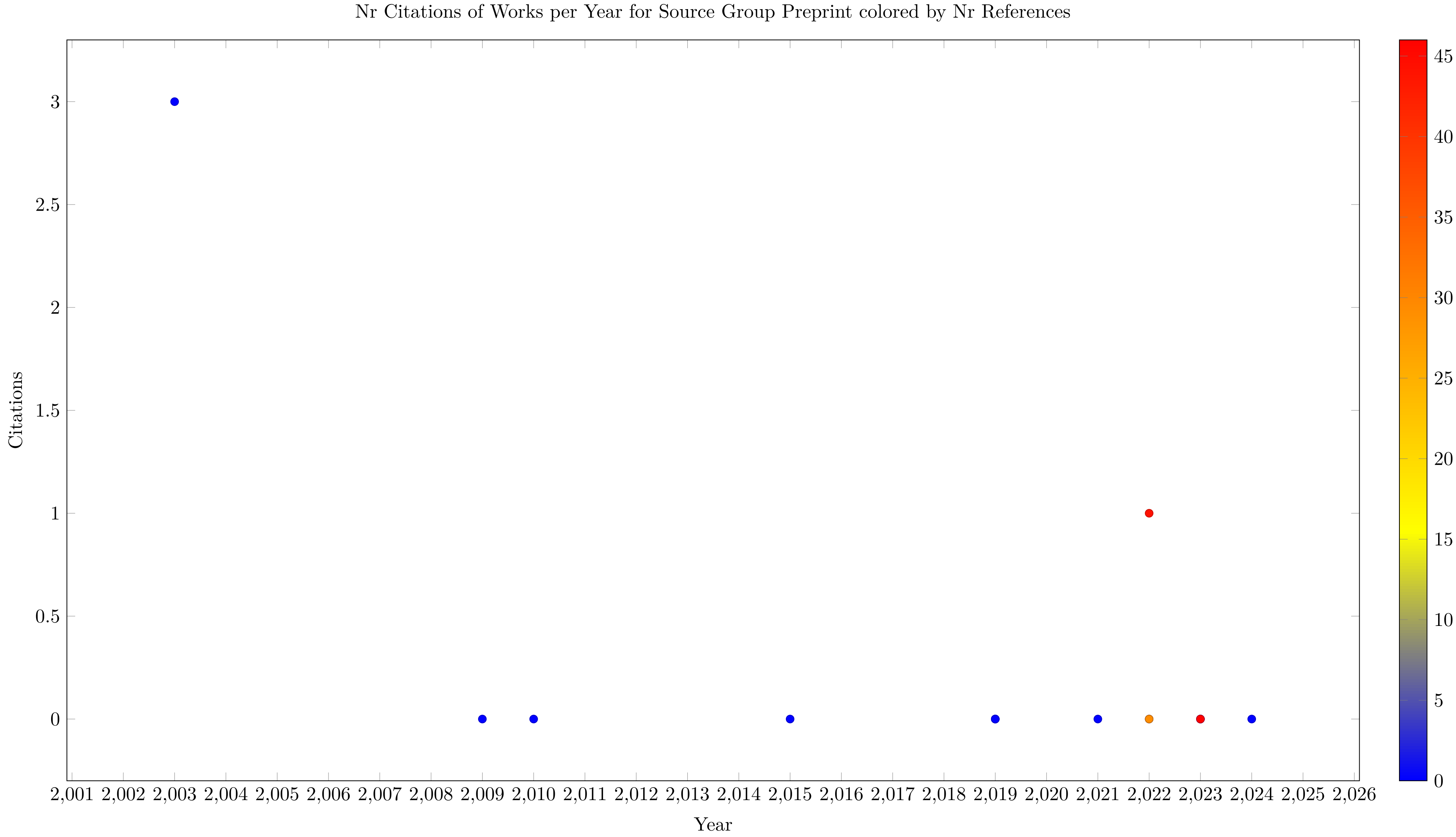

Nr Citations of Works per Year for Source Group Constraints colored by Nr References
Citations
Year
220
200
180
160
140
120
100
80
60
40
20
0
−20
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026
180
160
140
120
100
80
60
40
20
0

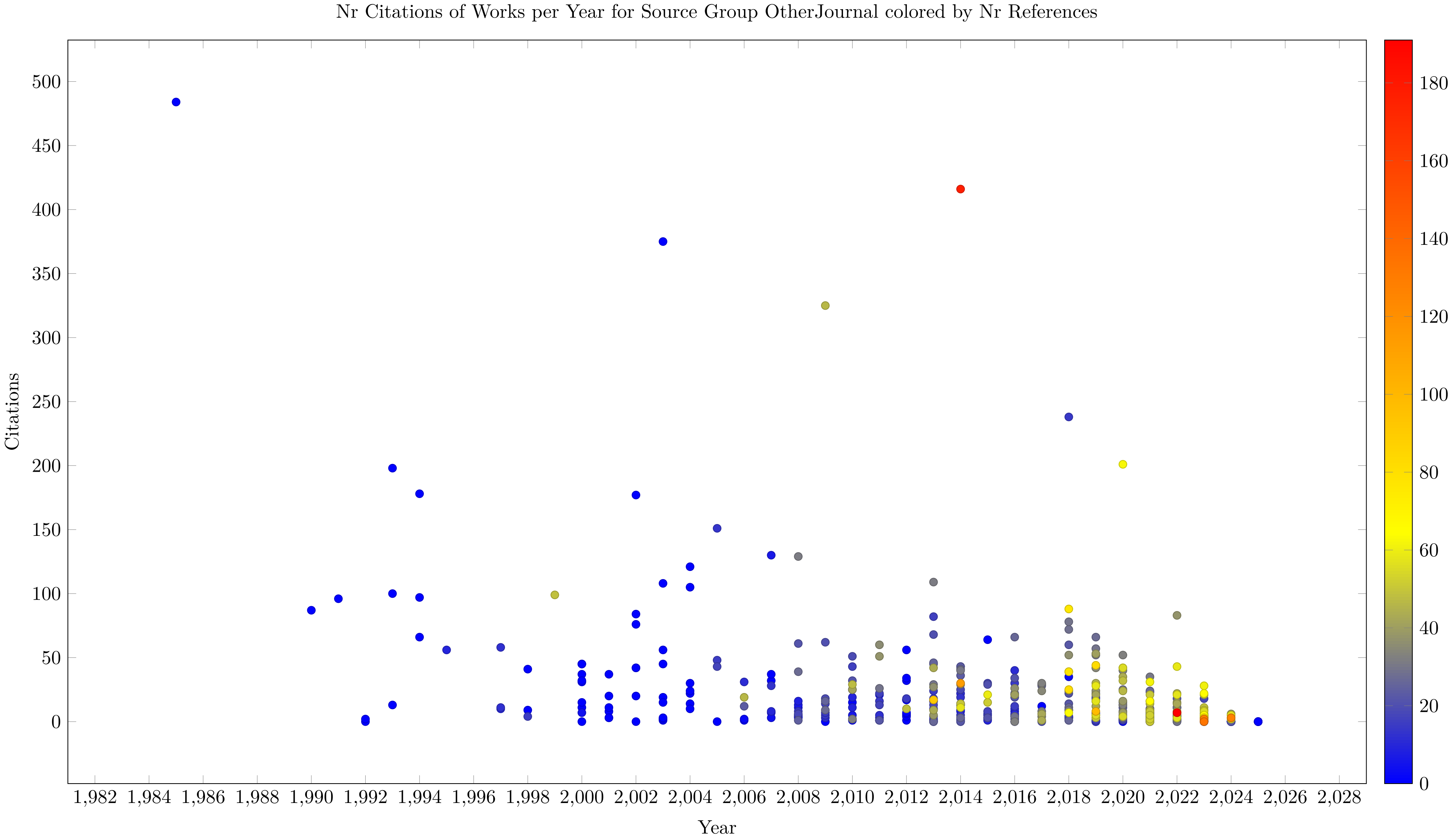

Nr Citations of Works per Year for Source Group EJOR colored by Nr References
Citations
Year
550
500
450
400
350
300
250
200
150
100
50
0
−50
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026
140
120
100
80
60
40
20
0


Nr Citations of Works per Year for Source Group InformsJC colored by Nr References
Citations
Year
0
50
100
150
200
250
300
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70


Nr Citations of Works per Year for Source Group AIJournal colored by Nr References
Citations
Year
450
400
350
300
250
200
150
100
50
0
1,984
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
400
350
300
250
200
150
100
50
0


Nr Citations of Works per Year for Source Group ORJournal colored by Nr References
Citations
Year
0
50
100
150
200
250
300
350
400
1,974
1,976
1,978
1,980
1,982
1,984
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026
2,028
0
10
20
30
40
50
60
70
80


Nr Citations of Works per Year for Source Group Preprint colored by Nr References
Citations
Year
3
2.5
2
1.5
1
0.5
0
2,001 2,002 2,003 2,004 2,005 2,006 2,007 2,008 2,009 2,010 2,011 2,012 2,013 2,014 2,015 2,016 2,017 2,018 2,019 2,020 2,021 2,022 2,023 2,024 2,025 2,026
45
40
35
30
25
20
15
10
5
0


Nr Citations of Works per Year for Source Group OtherJournal colored by Nr References
Citations
Year
0
50
100
150
200
250
300
350
400
450
500
1,982
1,984
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026
2,028
0
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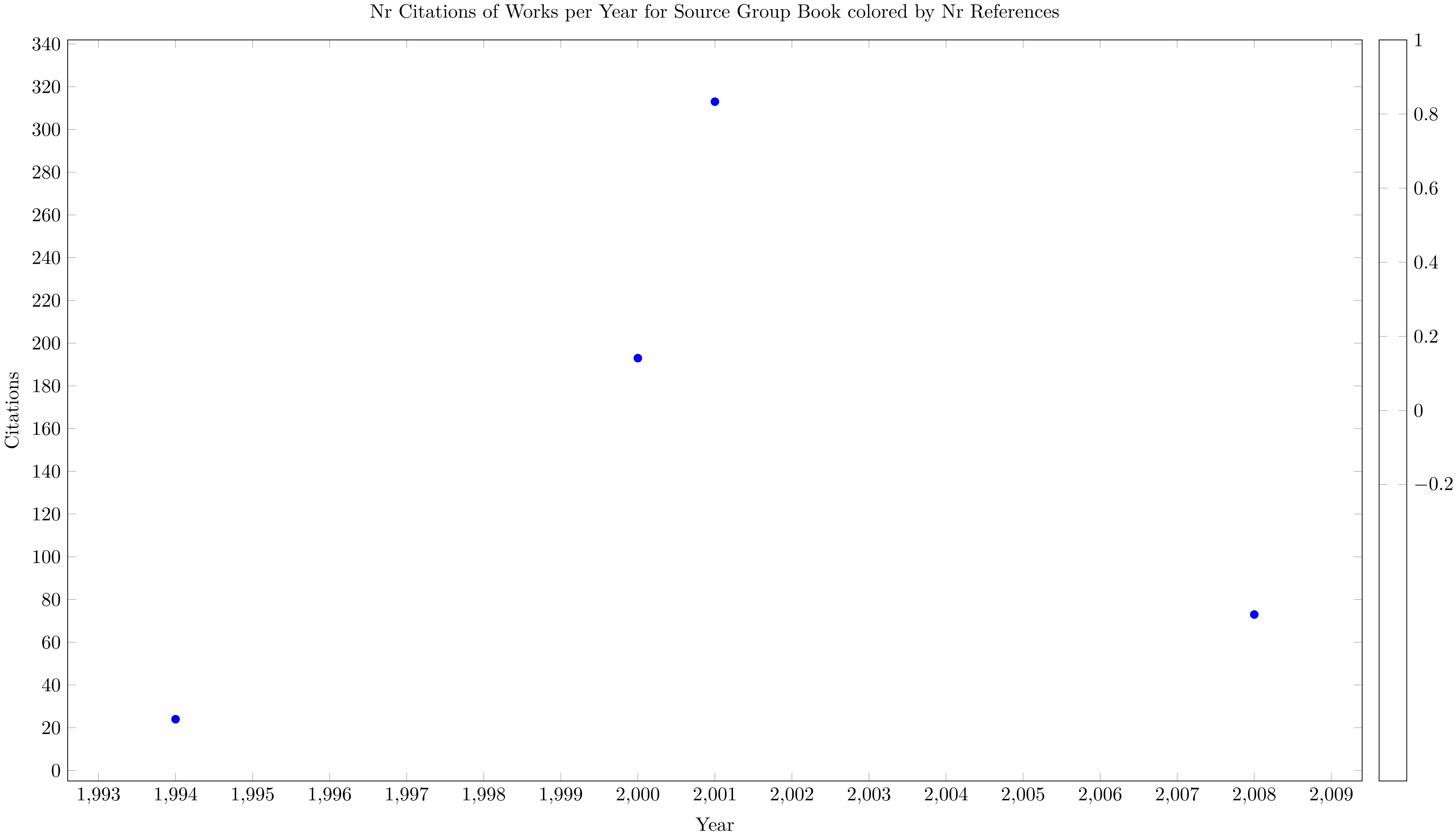
40
60
80
100
120
140
160
180


Nr Citations of Works per Year for Source Group Book colored by Nr References
Citations
340
320
300
280
260
240
220
200
180
160
140
120
100
80
60
40
20
0
1,993
1,994
1,995
1,996
1,997
1,998
1,999
2,000
2,001
2,002
2,003
2,004
2,005
2,006
2,007
2,008
2,009
Year
1
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0.6
0.4
0.2
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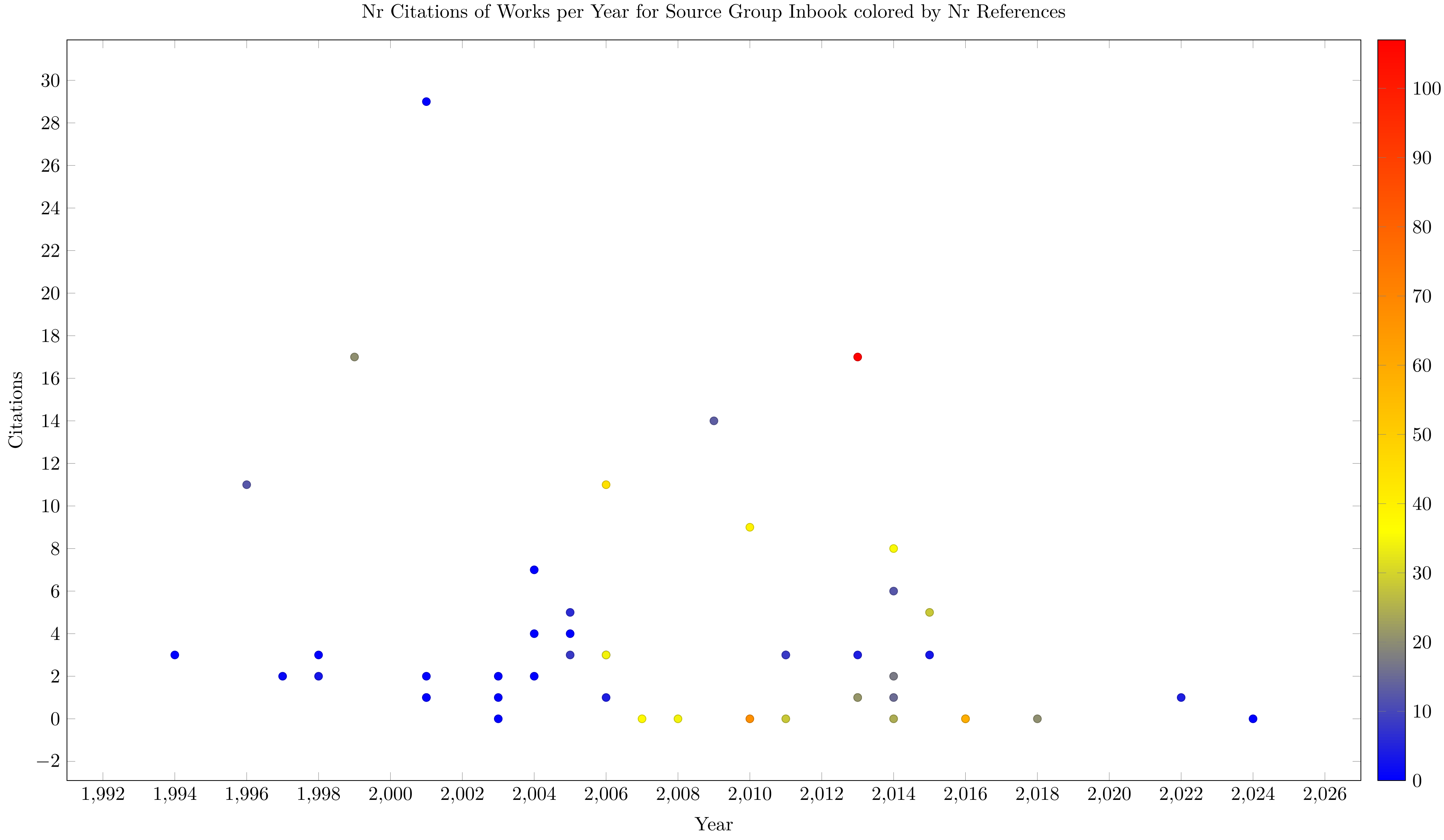
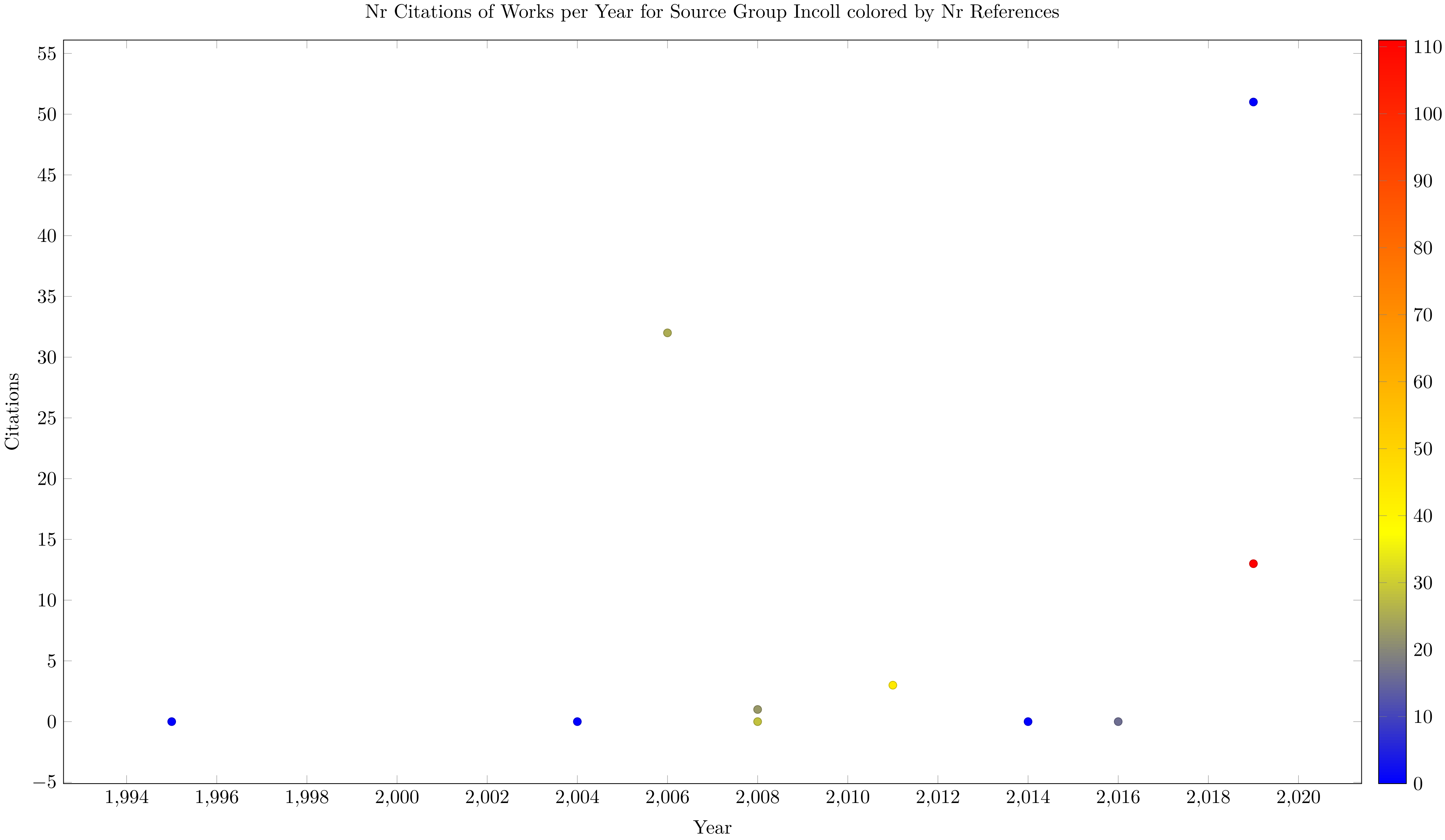
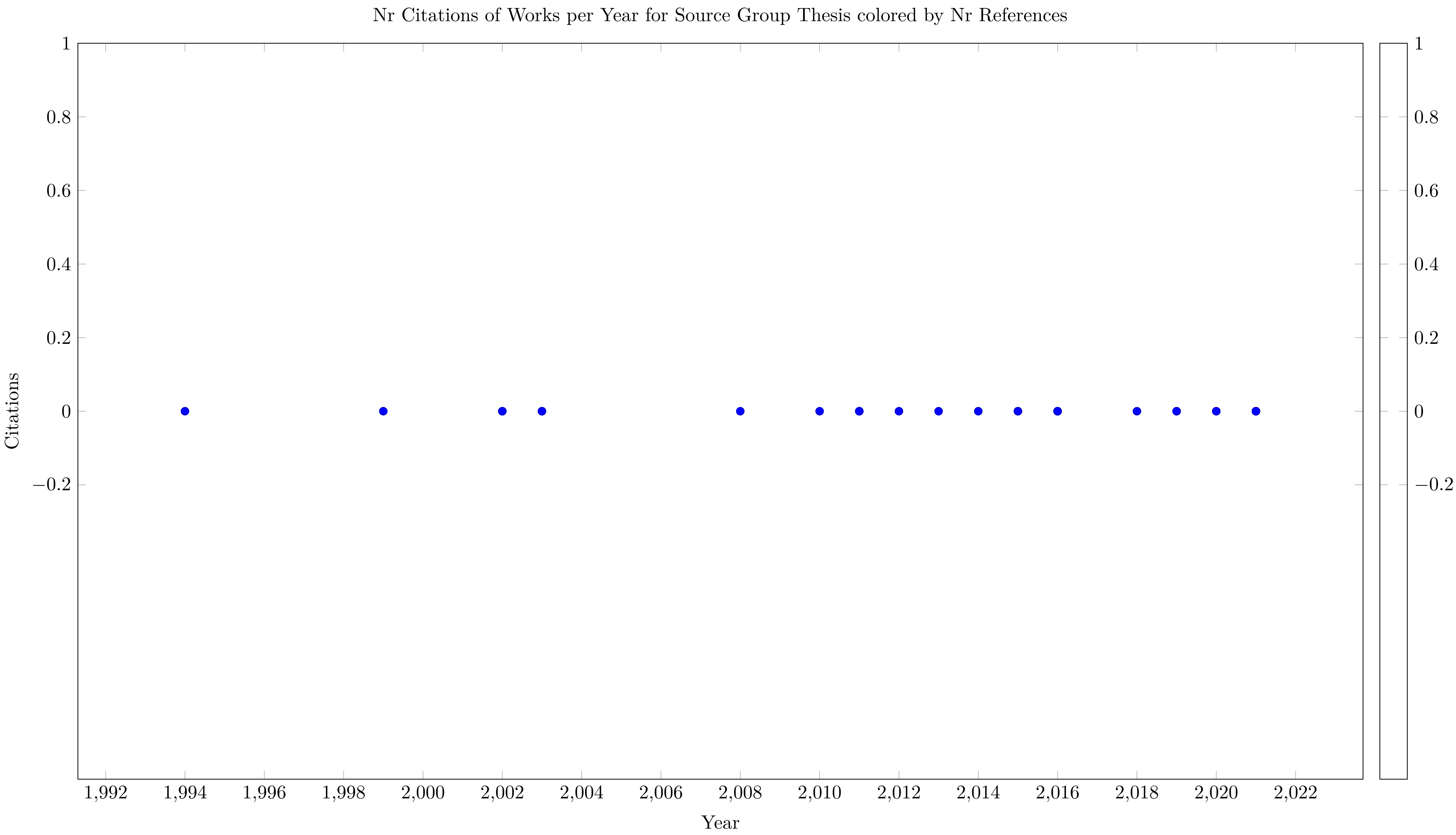
−0.2


Nr Citations of Works per Year for Source Group Inbook colored by Nr References
Citations
Year
30
28
26
24
22
20
18
16
14
12
10
8
6
4
2
0
−2
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026
100
90
80
70
60
50
40
30
20
10
0


Nr Citations of Works per Year for Source Group Incoll colored by Nr References
Citations
Year
55
50
45
40
35
30
25
20
15
10
5
0
−5
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
110
100
90
80
70
60
50
40
30
20
10
0


Nr Citations of Works per Year for Source Group Thesis colored by Nr References
Citations
Year
1
0.8
0.6
0.4
0.2
0
−0.2
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022

15.2 Reference Flows

The following table looks at references between source groups that are contained in the survey, i.e. where bot the citing and the cited work is included in the survey. We show how many papers referred to in the group on the left belong to the group in the column.

Table 14: Reference Flows

	CPAIOR
OtherJournal	1

The entries in the previous table are not directly comparable, without knowing how many works are in group. The next table presents a normalized view, where we divide the flow count by the product of the group sizes. This produces a likelihood of a paper in the source group citing a paper in the target group, given as a percentage from 0 to 100. We can see that the likelihood does not depend on the prestige of the target, e.g. papers at AAAI are cited much less than papers in CP.

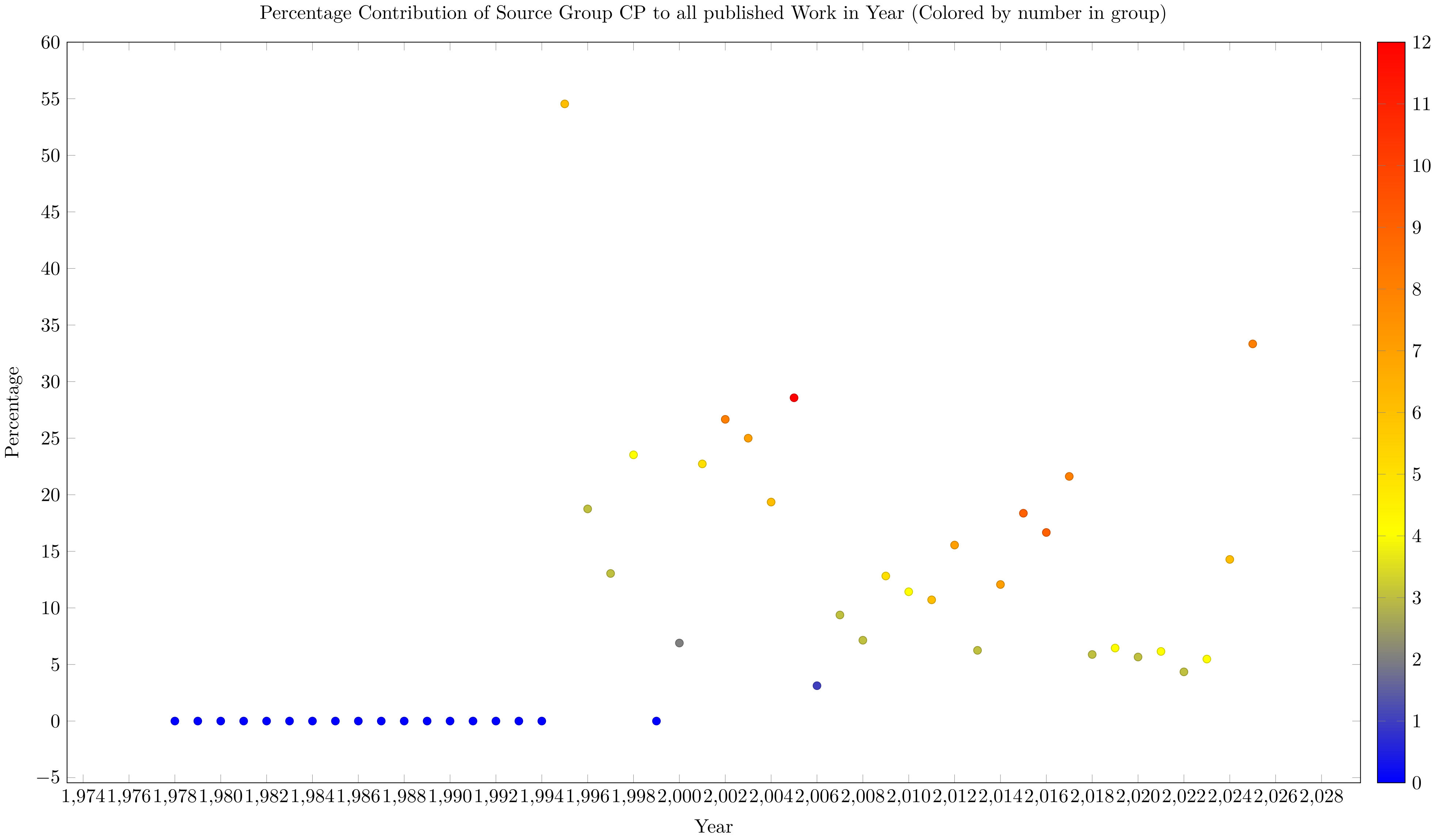
Note that the numbers are derived from the flows contained in the survey, which are based on the OpenCitation reference links. If such links are missing, or we are missing works in some group, then the results will be affected.

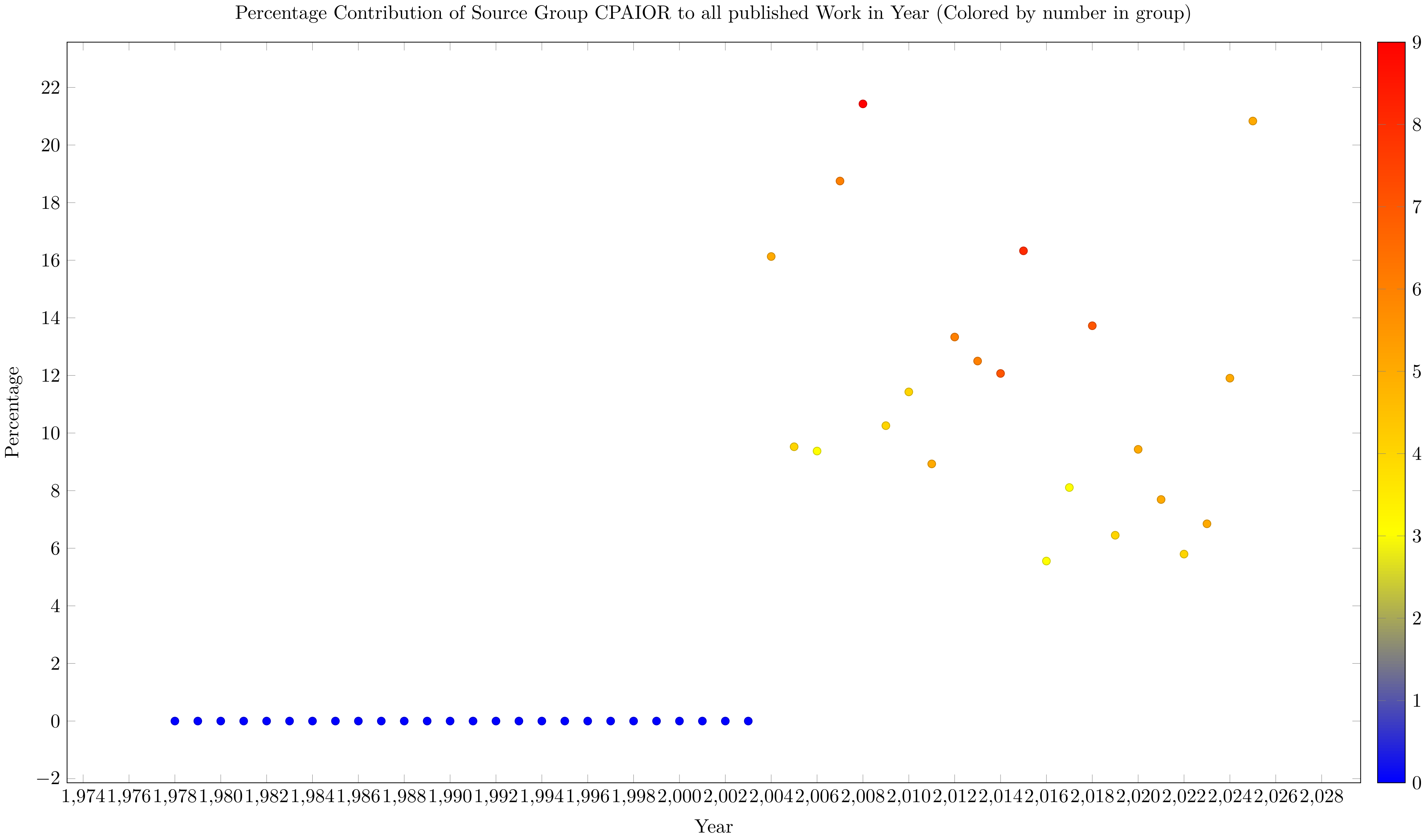
Table 15: Reference Flows Normalized

	CPAIOR
OtherJournal	0.00

16 Contribution of Source Group to Total Works per Year

The following plots show the percentage of works published in a year belonging to a specific source group. This plot helps to understand how important that group is to the field over time


Percentage Contribution of Source Group CP to all published Work in Year (Colored by number in group)
Percentage
Year
60
55
50
45
40
35
30
25
20
15
10
5
0
−5
1,974
1,976
1,978
1,980
1,982
1,984
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026
2,028
12
11
10
9
8
7
6
5
4
3
2
1
0


Percentage Contribution of Source Group CPAIOR to all published Work in Year (Colored by number in group)
Percentage
Year
−2
0
2
4
6
8
10
12
14
16
18
20
22
1,974 1,976 1,978 1,980 1,982 1,984 1,986 1,988 1,990 1,992 1,994 1,996 1,998 2,000 2,002 2,004 2,006 2,008 2,010 2,012 2,014 2,016 2,018 2,020 2,022 2,024 2,026 2,028
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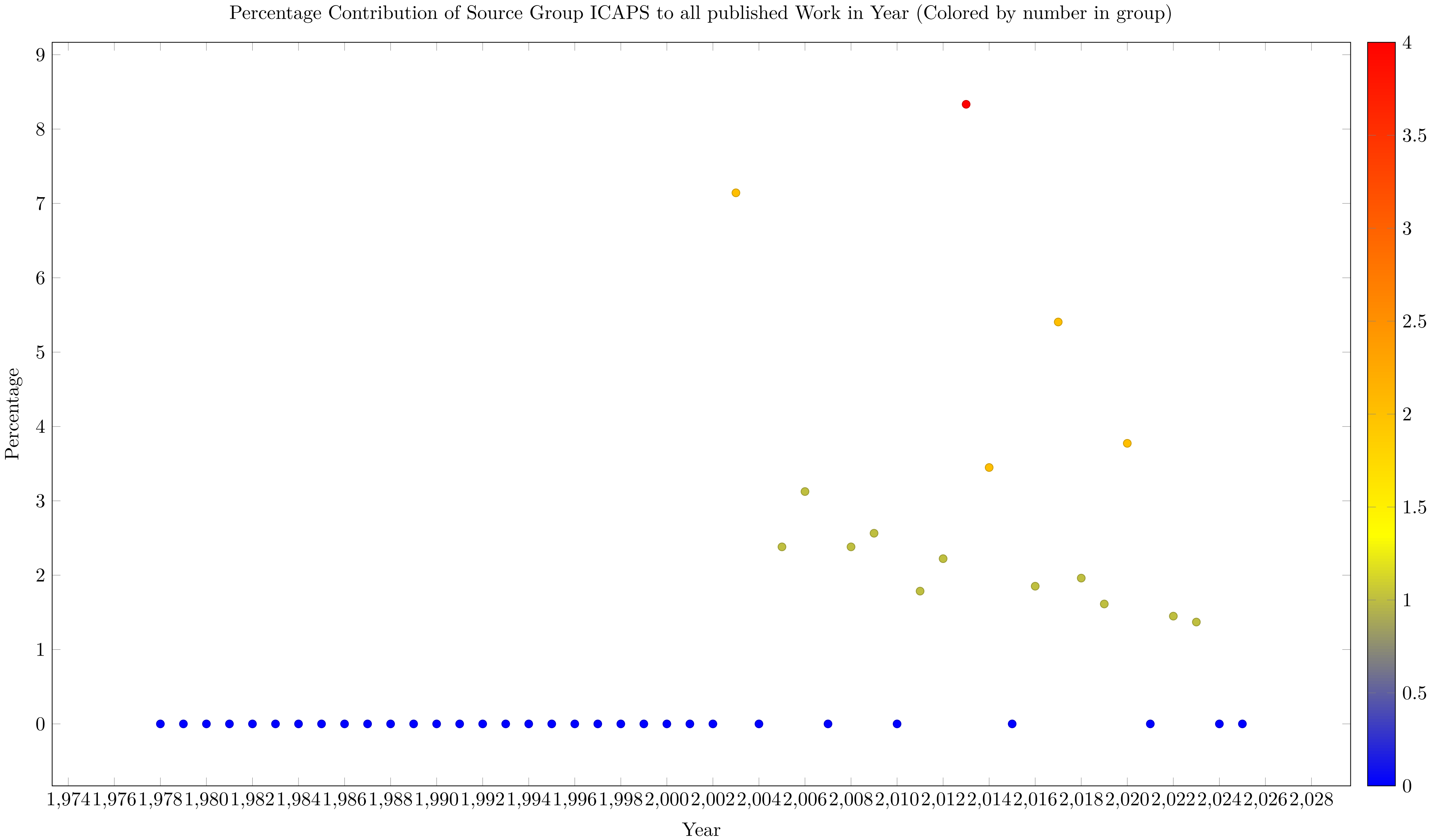
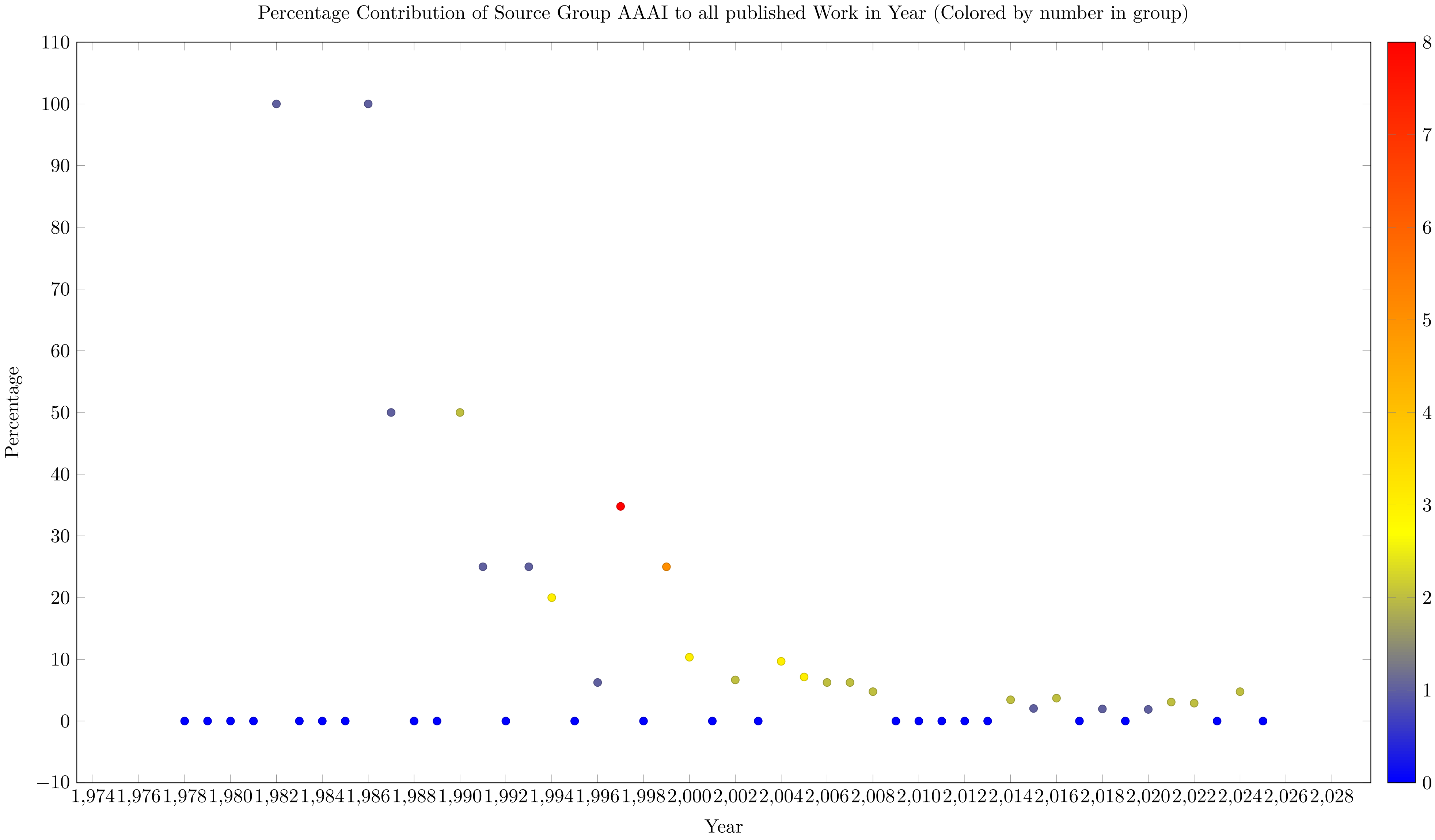
5
6
7
8
9


Percentage Contribution of Source Group ICAPS to all published Work in Year (Colored by number in group)
Percentage
Year
0
1
2
3
4
5
6
7
8
9
1,974
1,976
1,978
1,980
1,982
1,984
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026
2,028
0
0.5
1
1.5
2
2.5
3
3.5
4


Percentage Contribution of Source Group AAAI to all published Work in Year (Colored by number in group)
Percentage
Year
110
100
90
80
70
60
50
40
30
20
10
0
−10
1,974
1,976
1,978
1,980
1,982
1,984
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
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2,022
2,024
2,026
2,028
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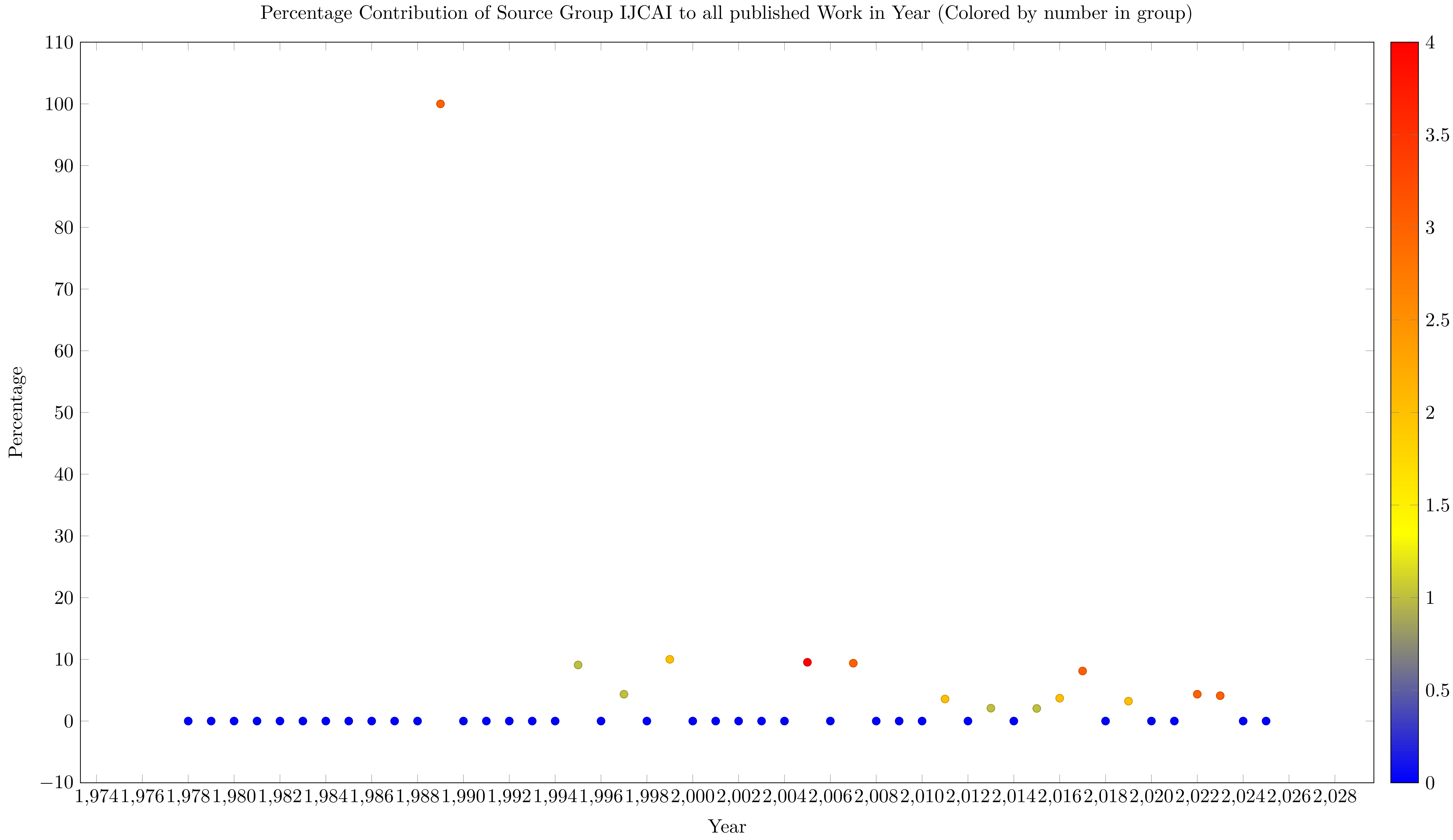
7
6
5
4
3
2
1
0


Percentage Contribution of Source Group IJCAI to all published Work in Year (Colored by number in group)
Percentage
Year
110
100
90
80
70
60
50
40
30
20
10
0
−10
1,974 1,976 1,978 1,980 1,982 1,984 1,986 1,988 1,990 1,992 1,994 1,996 1,998 2,000 2,002 2,004 2,006 2,008 2,010 2,012 2,014 2,016 2,018 2,020 2,022 2,024 2,026 2,028
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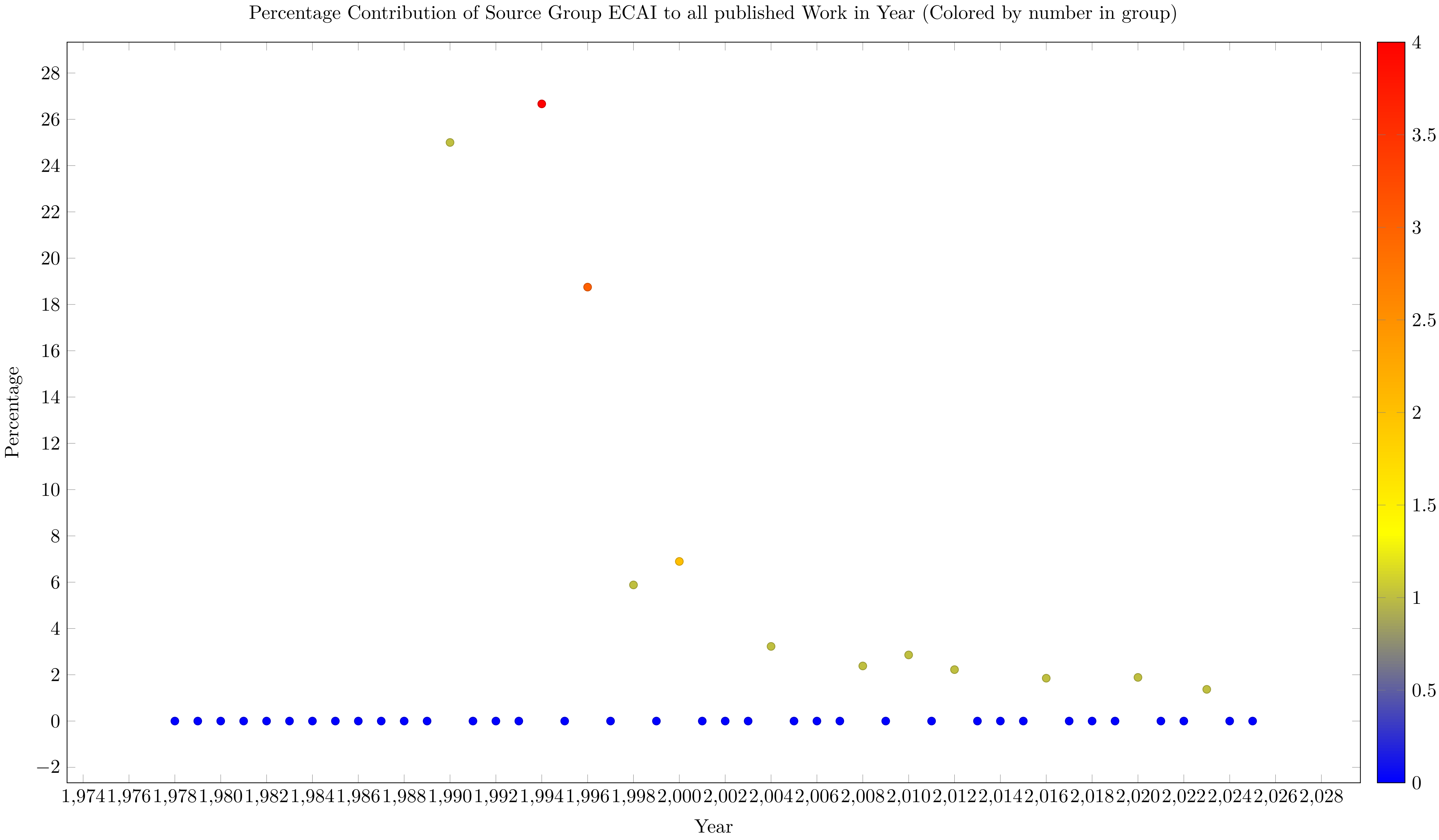
2.5
2
1.5
1
0.5
0


Percentage Contribution of Source Group ECAI to all published Work in Year (Colored by number in group)
Percentage
Year
−2
0
2
4
6
8
10
12
14
16
18
20
22
24
26
28
1,974 1,976 1,978 1,980 1,982 1,984 1,986 1,988 1,990 1,992 1,994 1,996 1,998 2,000 2,002 2,004 2,006 2,008 2,010 2,012 2,014 2,016 2,018 2,020 2,022 2,024 2,026 2,028
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0.5
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1.5
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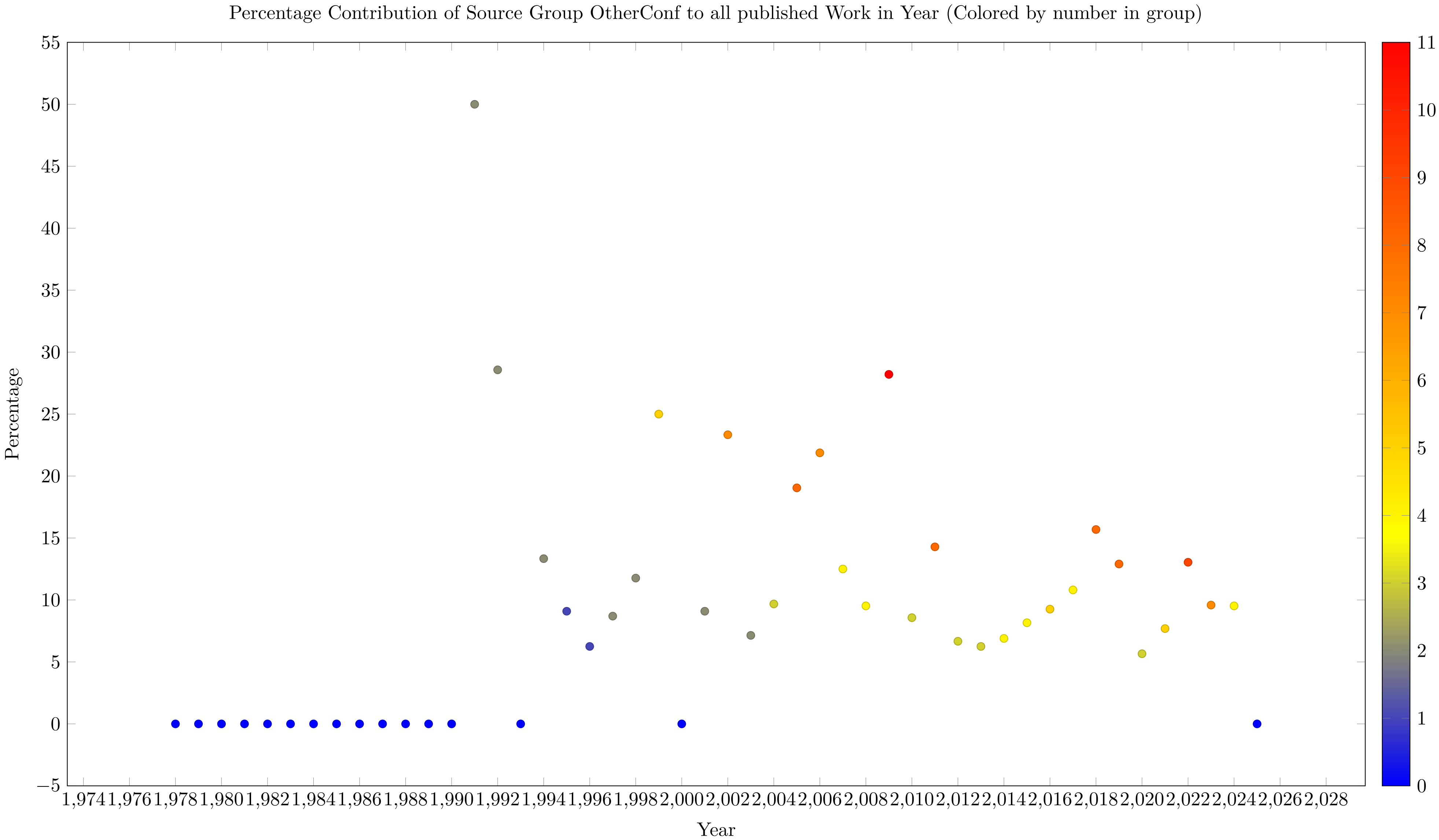
3.5
4


Percentage Contribution of Source Group OtherConf to all published Work in Year (Colored by number in group)
Percentage
Year
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50
45
40
35
30
25
20
15
10
5
0
−5
1,974
1,976
1,978
1,980
1,982
1,984
1,986
1,988
1,990
1,992
1,994
1,996
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2,018
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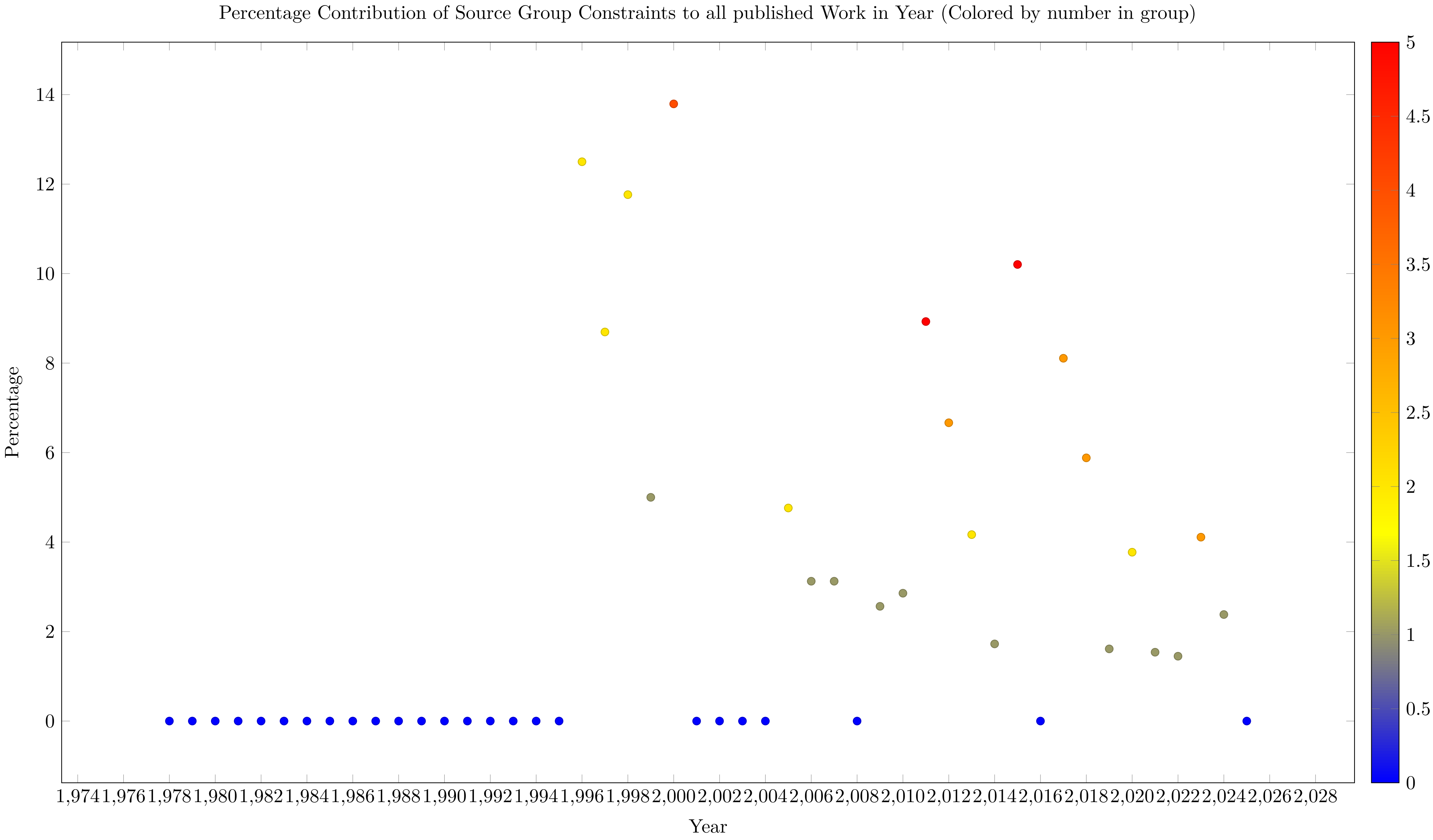
Percentage Contribution of Source Group Constraints to all published Work in Year (Colored by number in group)
Percentage
Year
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12
14
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2,008
2,010
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2,016
2,018
2,020
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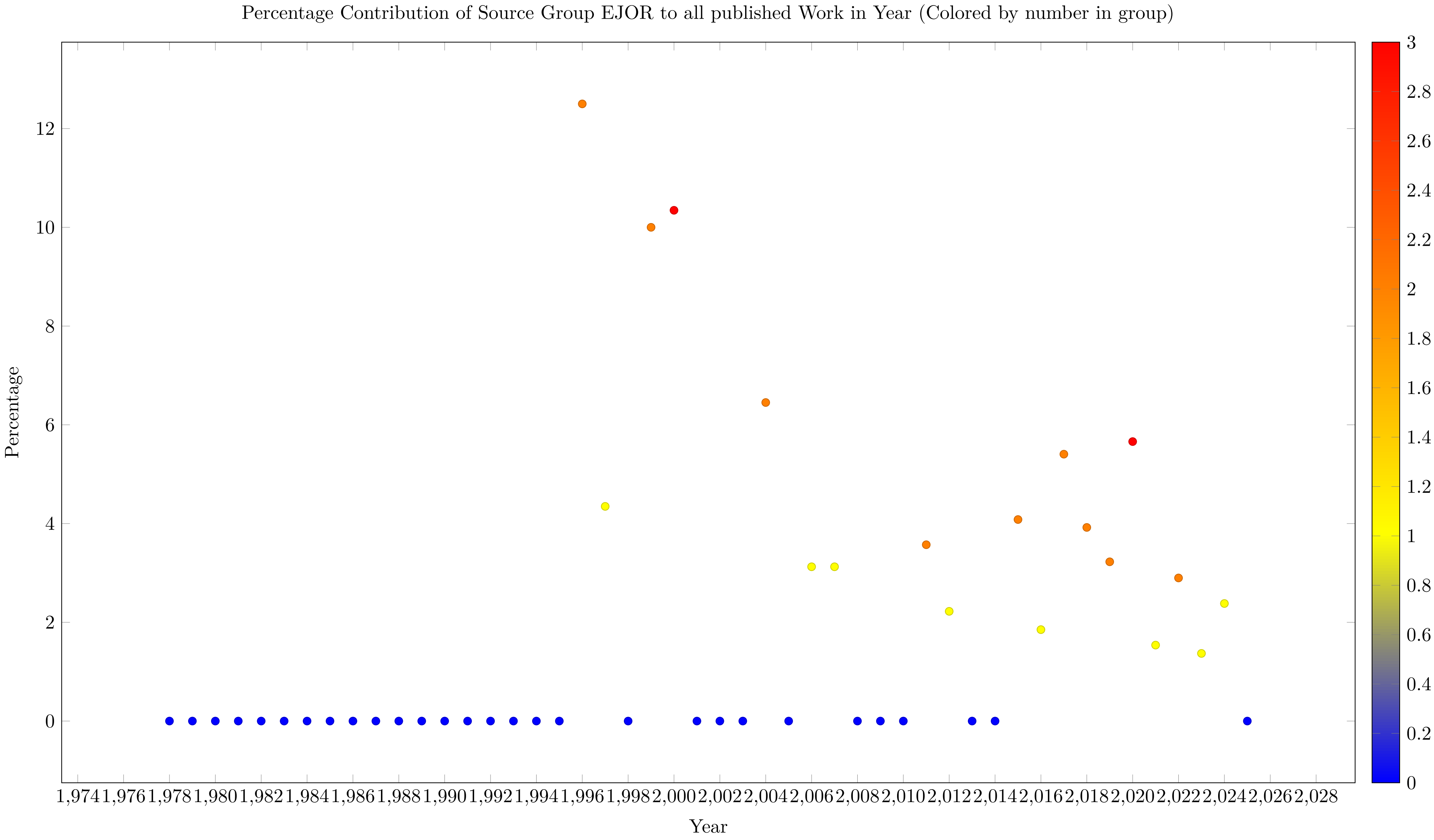
4.5
5


Percentage Contribution of Source Group EJOR to all published Work in Year (Colored by number in group)
Percentage
Year
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2
4
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8
10
12
1,974
1,976
1,978
1,980
1,982
1,984
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026
2,028
0
0.2
0.4
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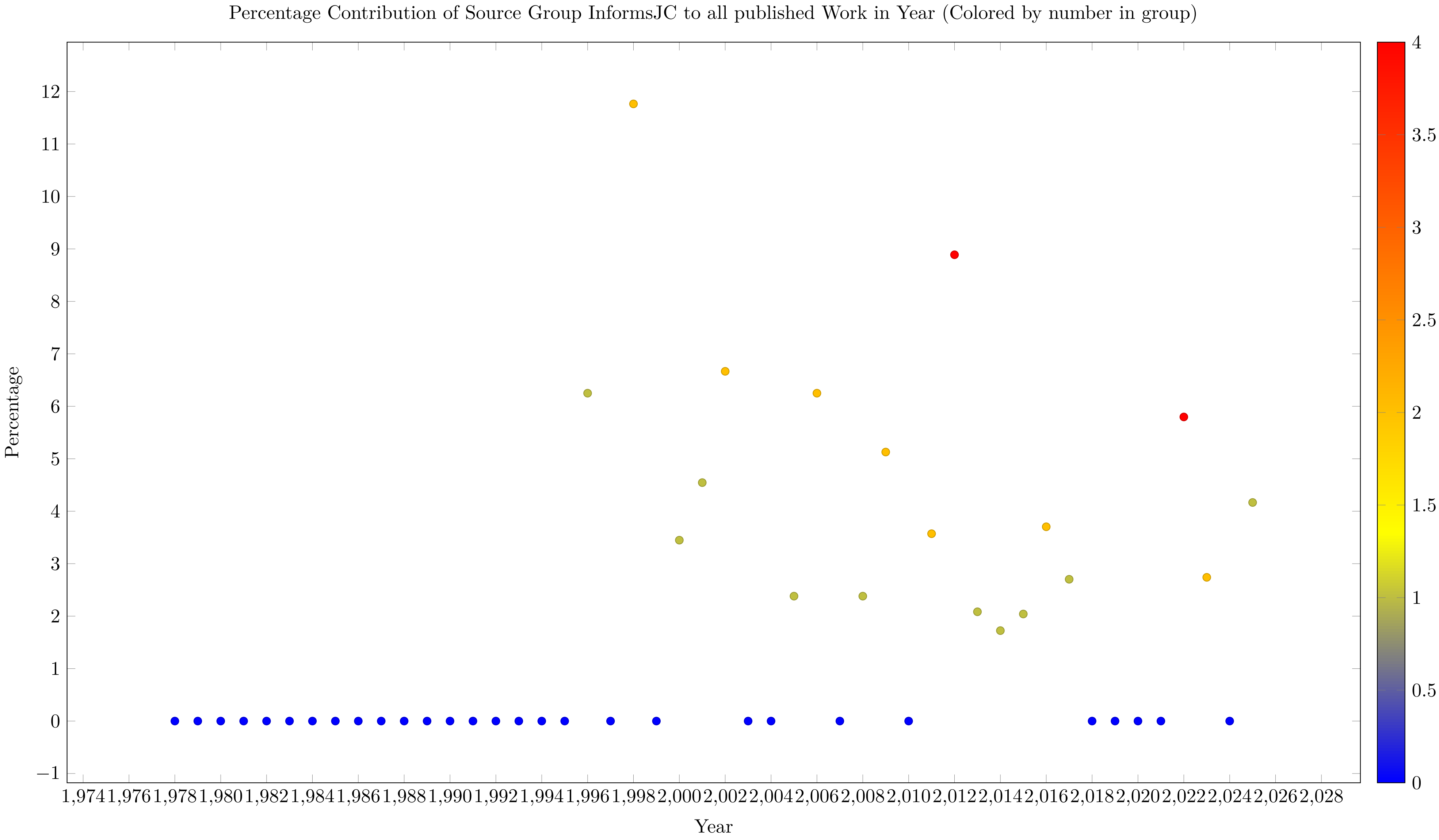
1.2
1.4
1.6
1.8
2
2.2
2.4
2.6
2.8
3


Percentage Contribution of Source Group InformsJC to all published Work in Year (Colored by number in group)
Percentage
Year
−1
0
1
2
3
4
5
6
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12
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1,994
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2,008
2,010
2,012
2,014
2,016
2,018
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2,026
2,028
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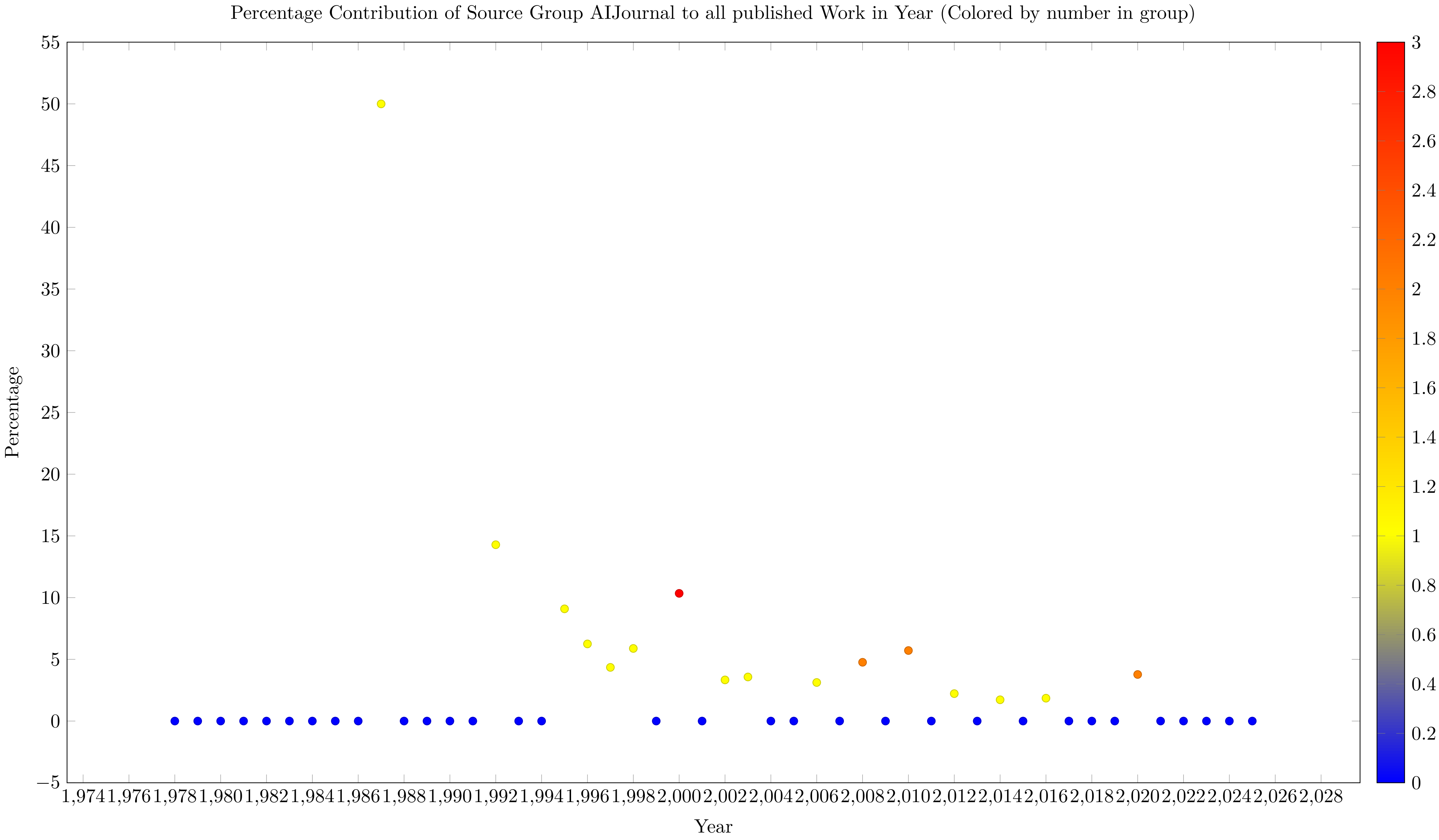
Percentage Contribution of Source Group AIJournal to all published Work in Year (Colored by number in group)
Percentage
Year
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50
45
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35
30
25
20
15
10
5
0
−5
1,974 1,976 1,978 1,980 1,982 1,984 1,986 1,988 1,990 1,992 1,994 1,996 1,998 2,000 2,002 2,004 2,006 2,008 2,010 2,012 2,014 2,016 2,018 2,020 2,022 2,024 2,026 2,028
3
2.8
2.6
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0.8
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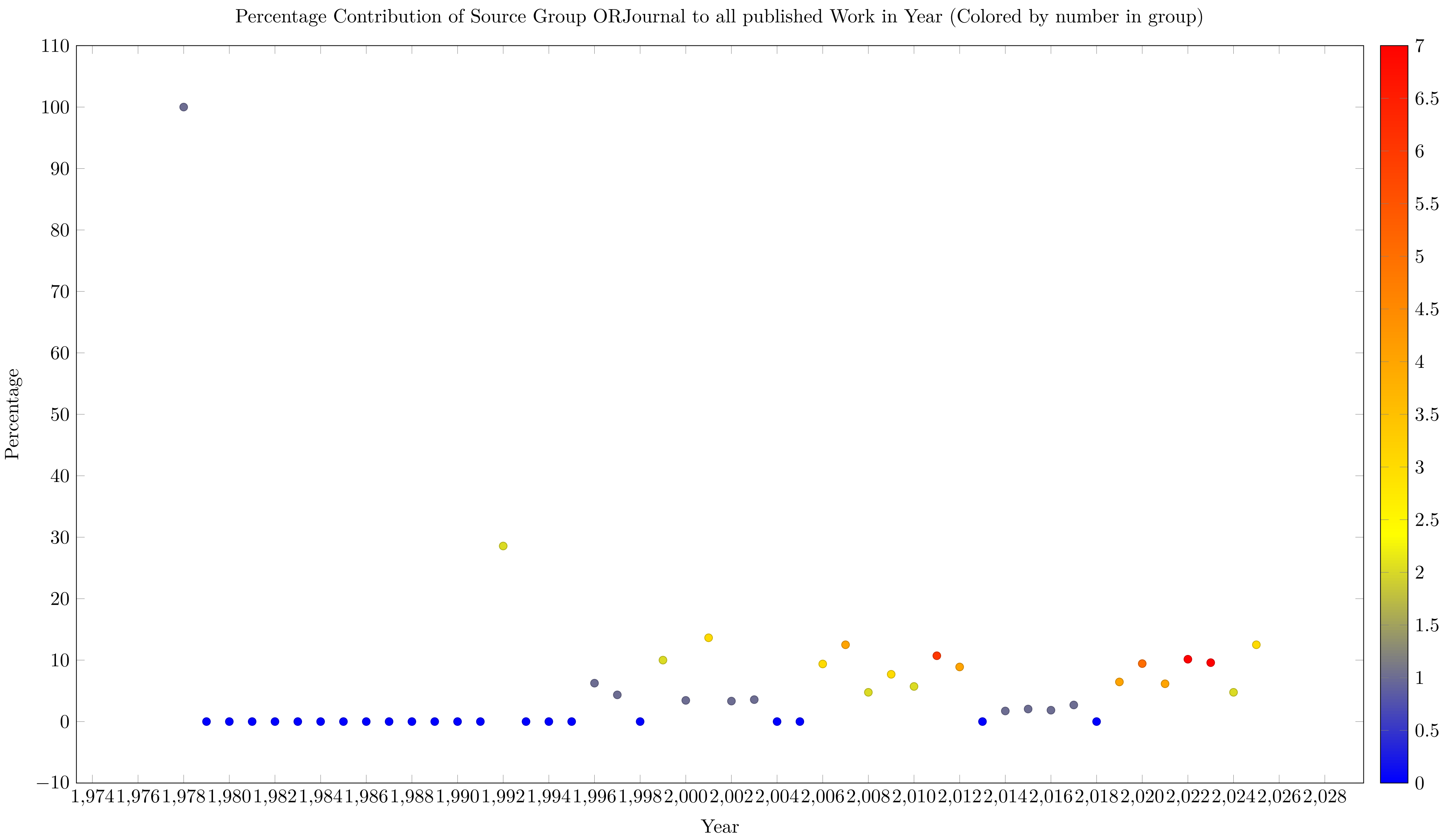
0.4
0.2
0


Percentage Contribution of Source Group ORJournal to all published Work in Year (Colored by number in group)
Percentage
Year
110
100
90
80
70
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20
10
0
−10
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1,976
1,978
1,980
1,982
1,984
1,986
1,988
1,990
1,992
1,994
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2,026
2,028
7
6.5
6
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4.5
4
3.5
3
2.5
2
1.5
1
0.5
0


Percentage Contribution of Source Group Preprint to all published Work in Year (Colored by number in group)
Percentage
Year
0
1
2
3
4
5
6
7
1,974 1,976 1,978 1,980 1,982 1,984 1,986 1,988 1,990 1,992 1,994 1,996 1,998 2,000 2,002 2,004 2,006 2,008 2,010 2,012 2,014 2,016 2,018 2,020 2,022 2,024 2,026 2,028
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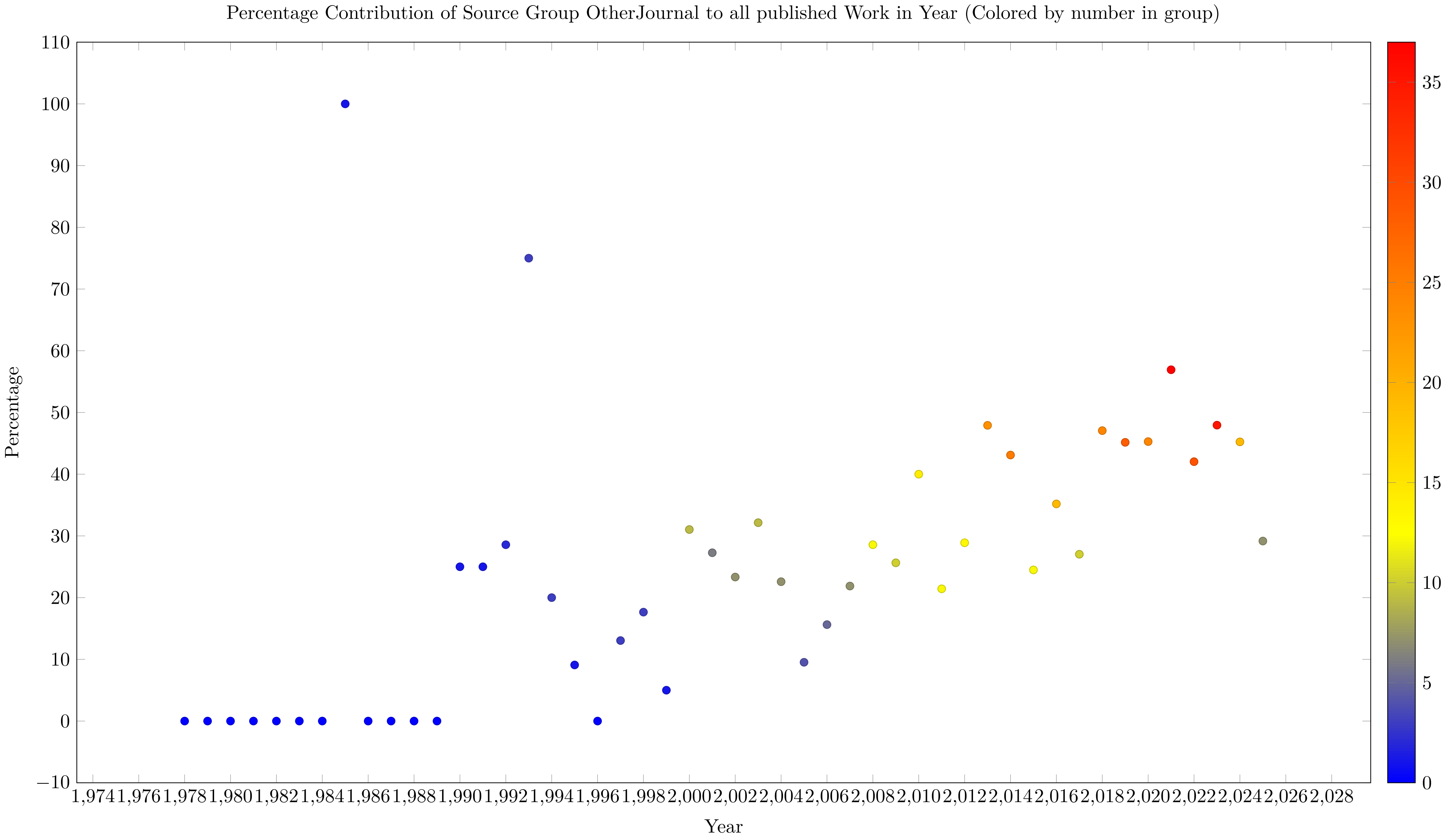
Percentage Contribution of Source Group OtherJournal to all published Work in Year (Colored by number in group)
Percentage
Year
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100
90
80
70
60
50
40
30
20
10
0
−10
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1,976
1,978
1,980
1,982
1,984
1,986
1,988
1,990
1,992
1,994
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2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026
2,028
35
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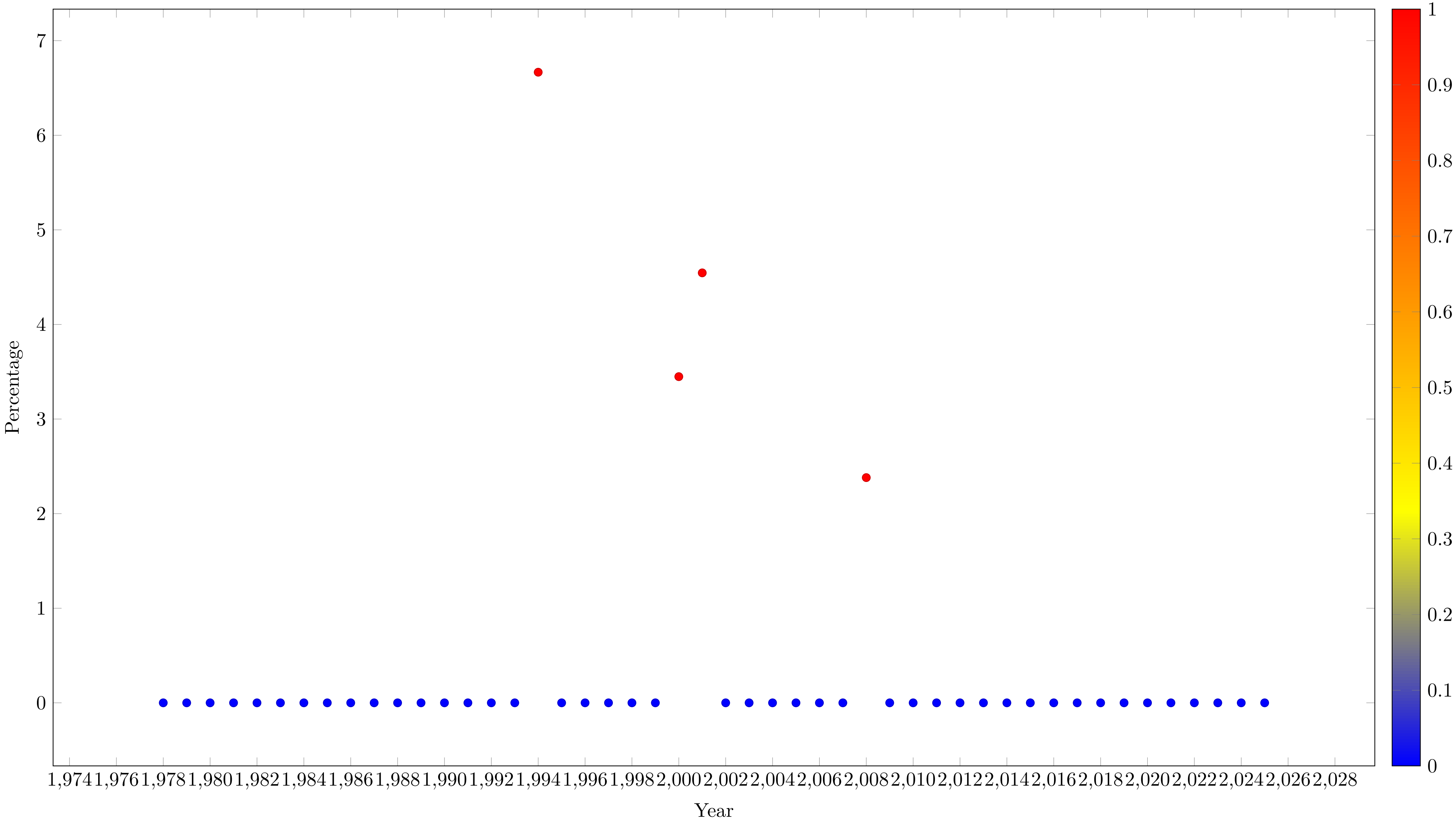
25
20
15
10
5
0

Percentage Contribution of Source Group Book to all published Work in Year (Colored by number in group)




Percentage Contribution of Source Group Inbook to all published Work in Year (Colored by number in group)
Percentage
Year
0
2
4
6
8
10
12
14
16
18
20
1,974
1,976
1,978
1,980
1,982
1,984
1,986
1,988
1,990
1,992
1,994
1,996
1,998
2,000
2,002
2,004
2,006
2,008
2,010
2,012
2,014
2,016
2,018
2,020
2,022
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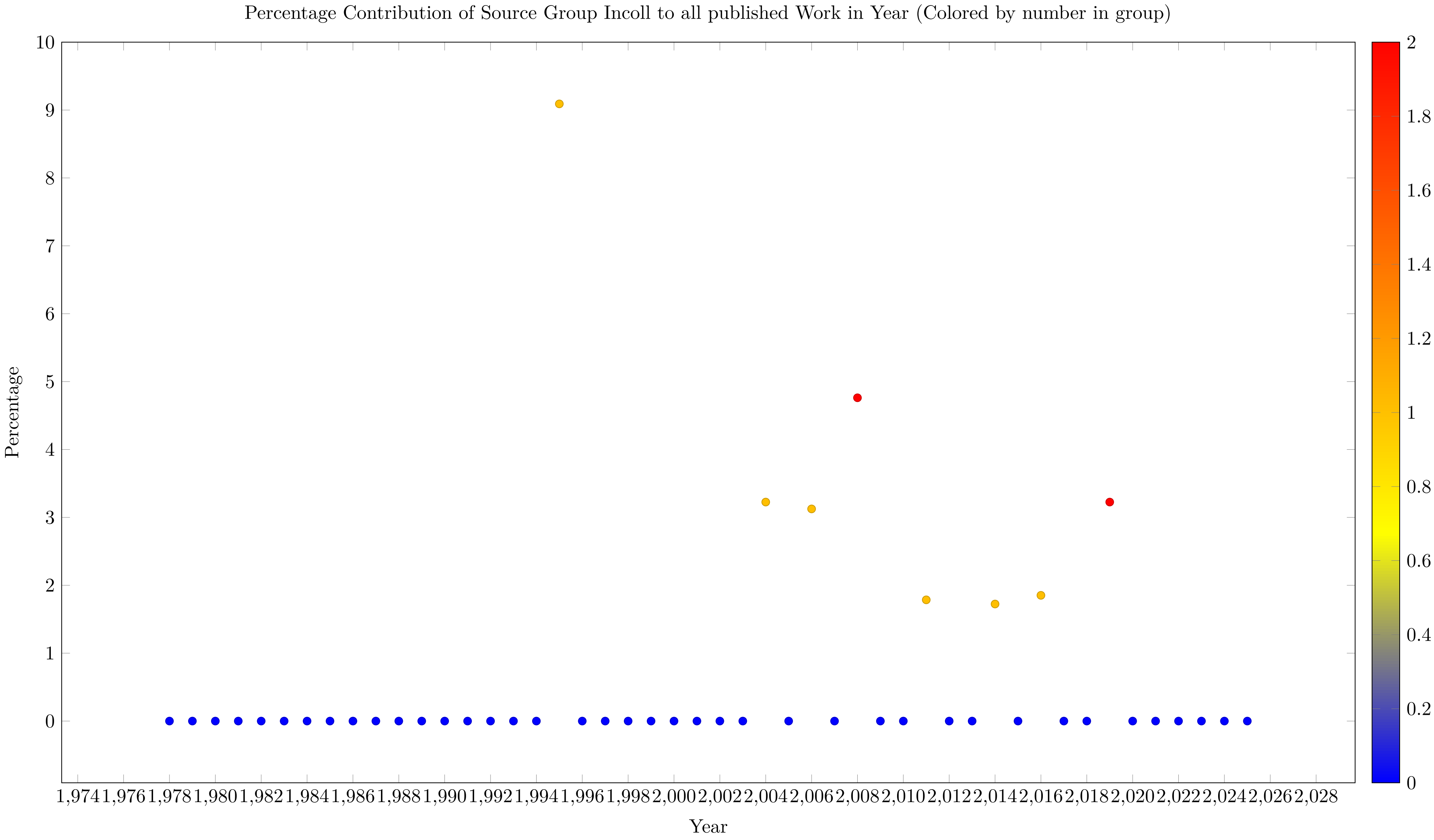
2.5
3
3.5
4
4.5
5


Percentage Contribution of Source Group Incoll to all published Work in Year (Colored by number in group)
Percentage
Year
0
1
2
3
4
5
6
7
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9
10
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1,978
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2,006
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2,010
2,012
2,014
2,016
2,018
2,020
2,022
2,024
2,026
2,028
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0.6
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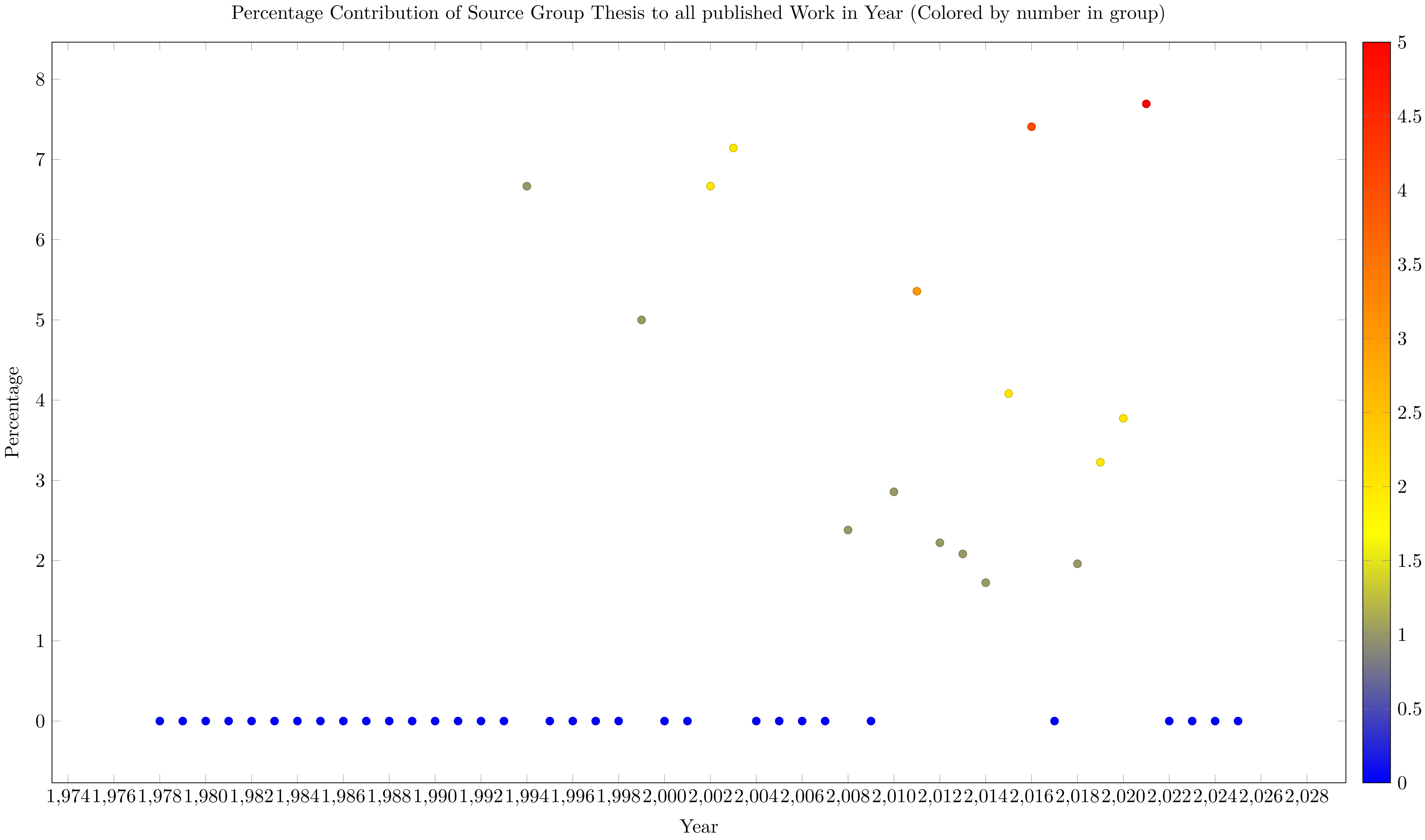
1
1.2
1.4
1.6
1.8
2


Percentage Contribution of Source Group Thesis to all published Work in Year (Colored by number in group)
Percentage
Year
8
7
6
5
4
3
2
1
0
1,974 1,976 1,978 1,980 1,982 1,984 1,986 1,988 1,990 1,992 1,994 1,996 1,998 2,000 2,002 2,004 2,006 2,008 2,010 2,012 2,014 2,016 2,018 2,020 2,022 2,024 2,026 2,028
5
4.5
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3.5
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2.5
2
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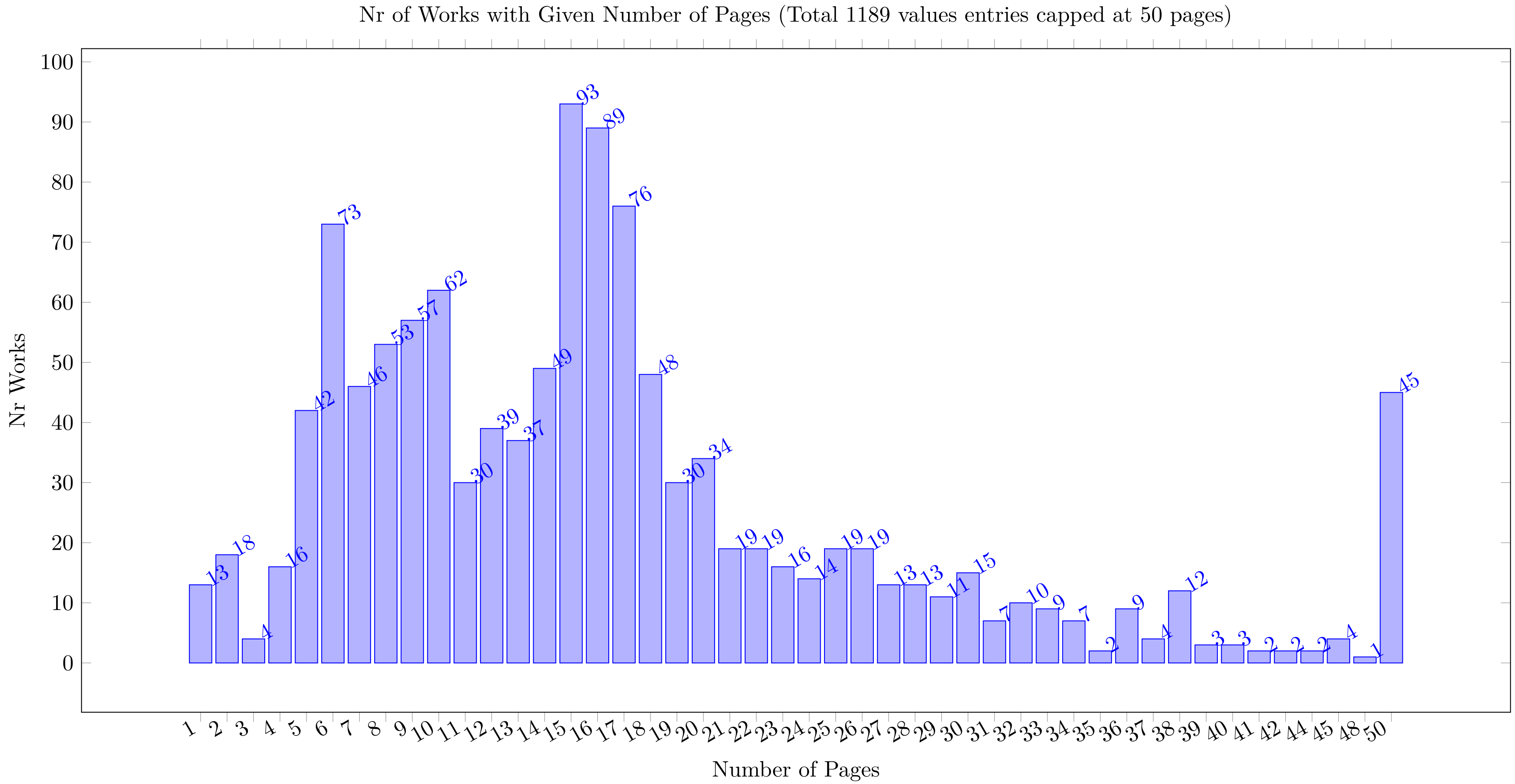
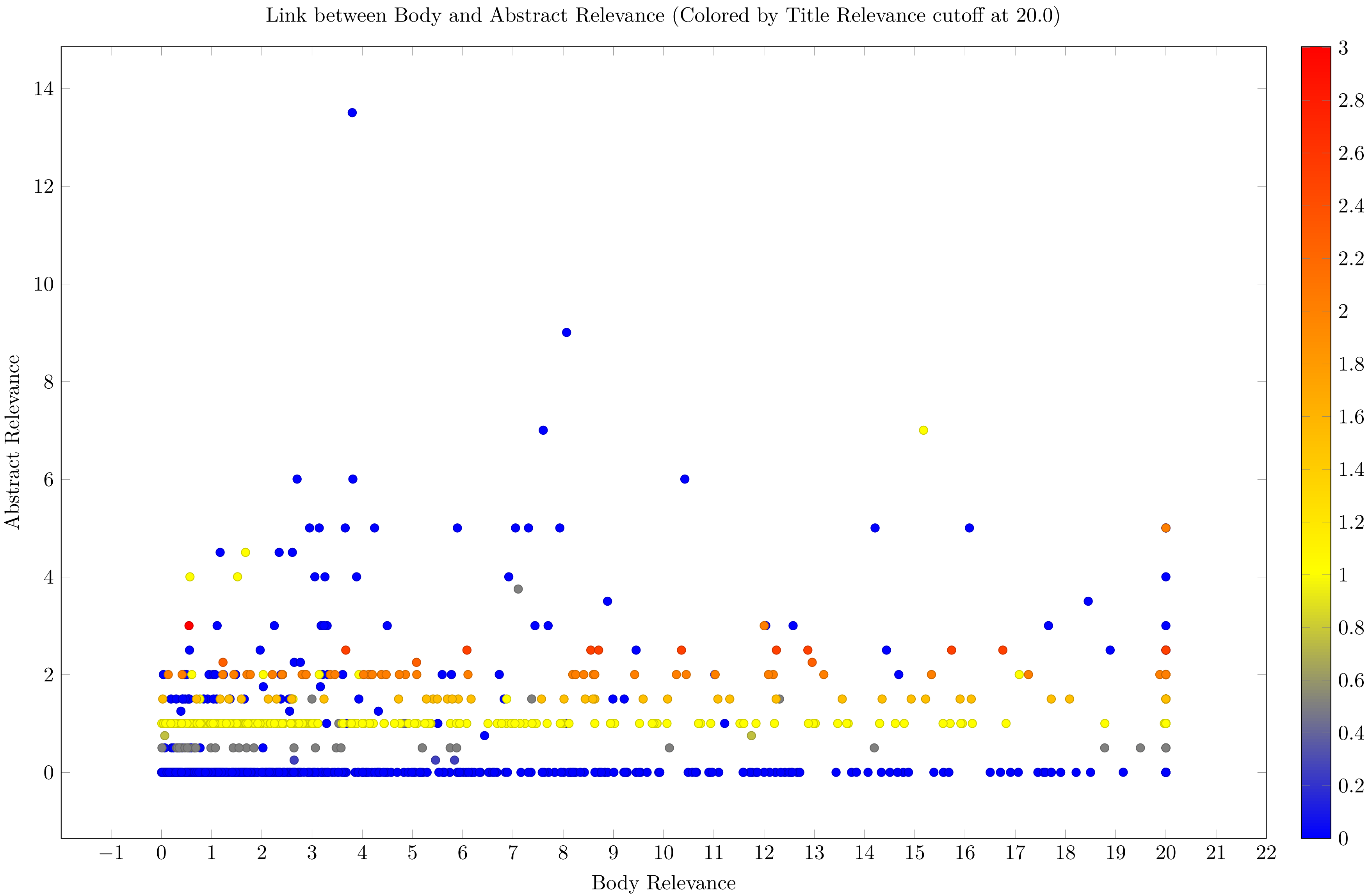
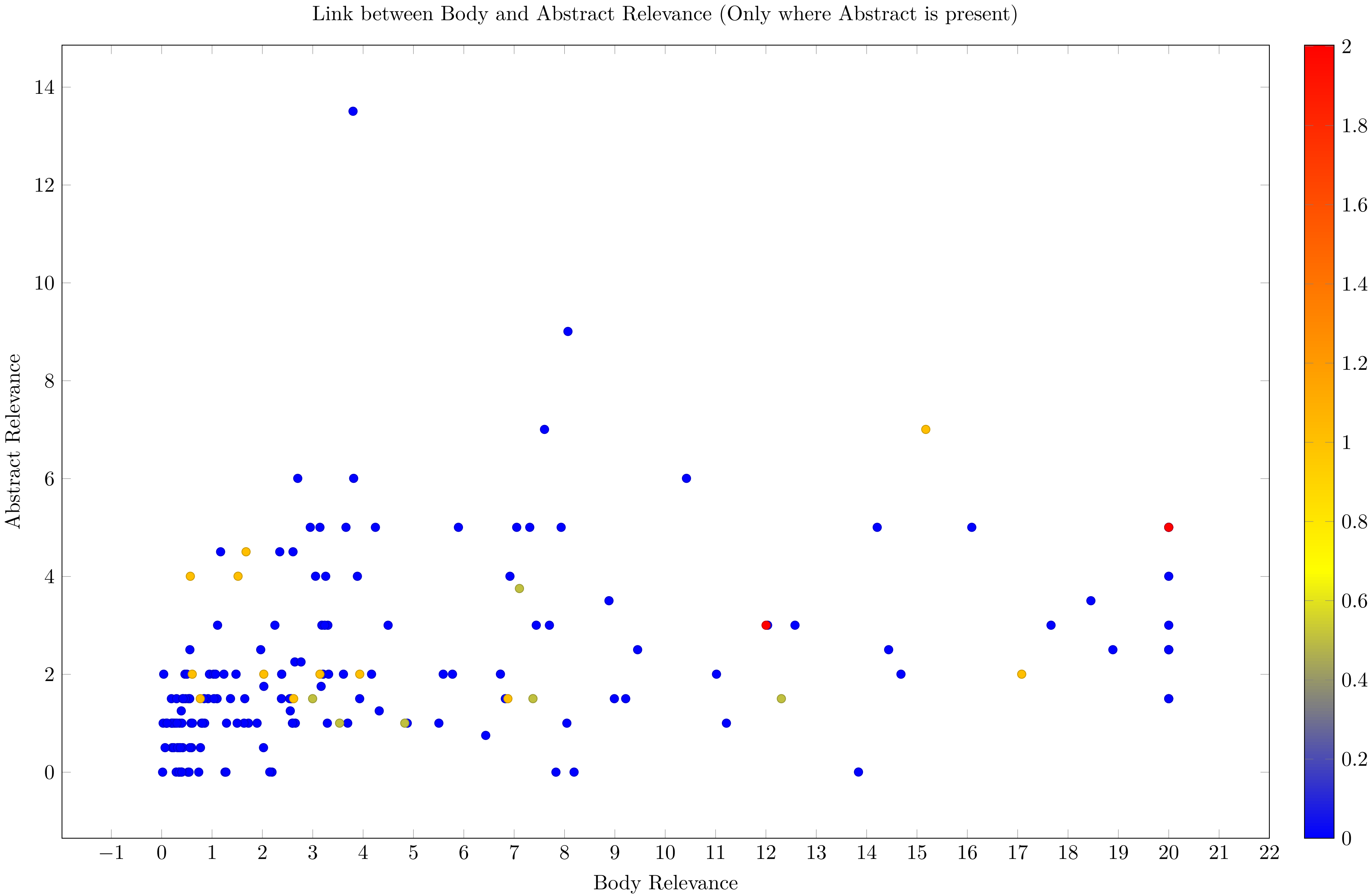
17 Page Length Distribution



18 Relevance Distribution




Link between Body and Abstract Relevance (Only where Abstract is present)
Abstract Relevance
Body Relevance
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2
3
4
5
6
7
8
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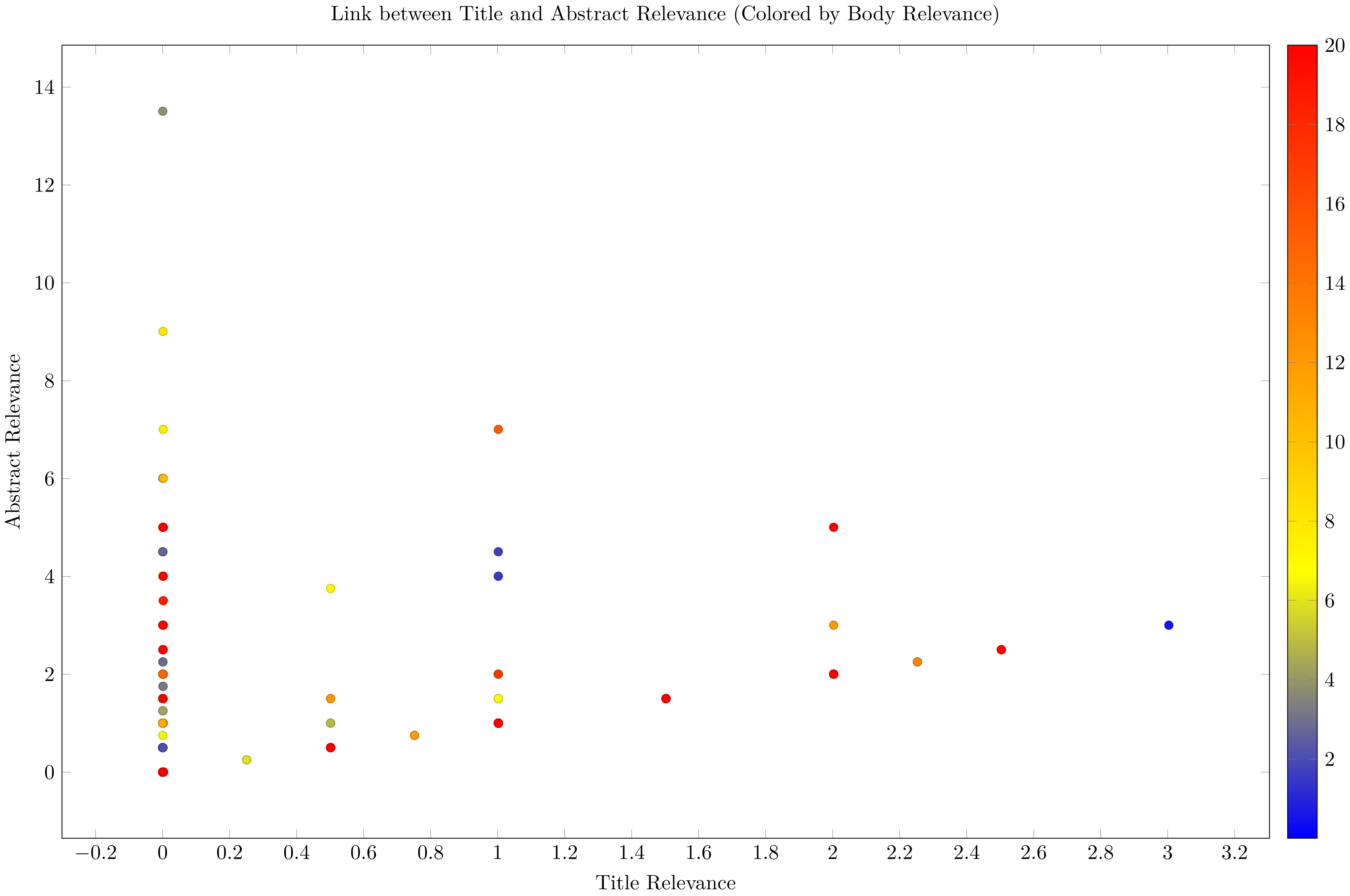
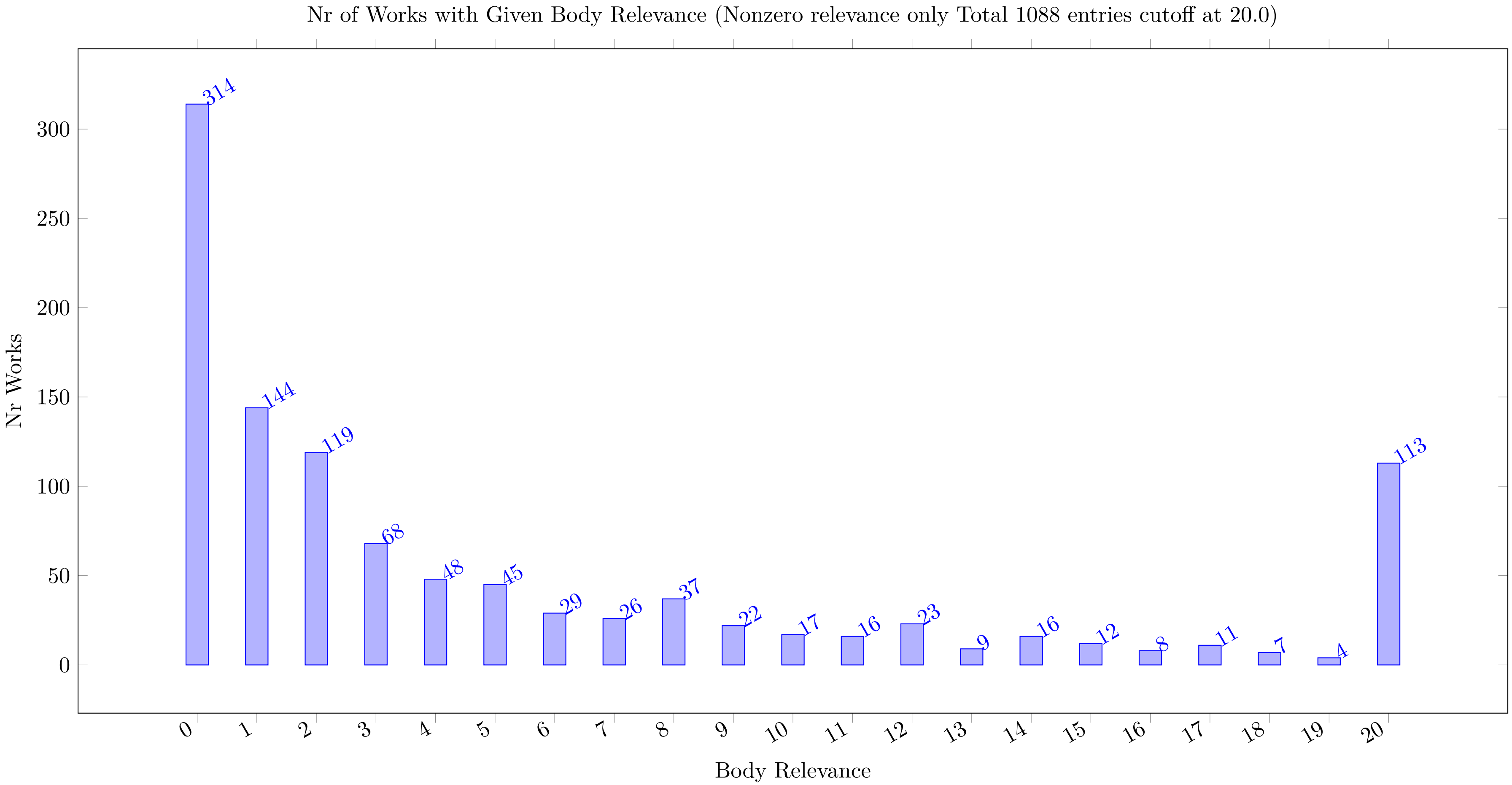
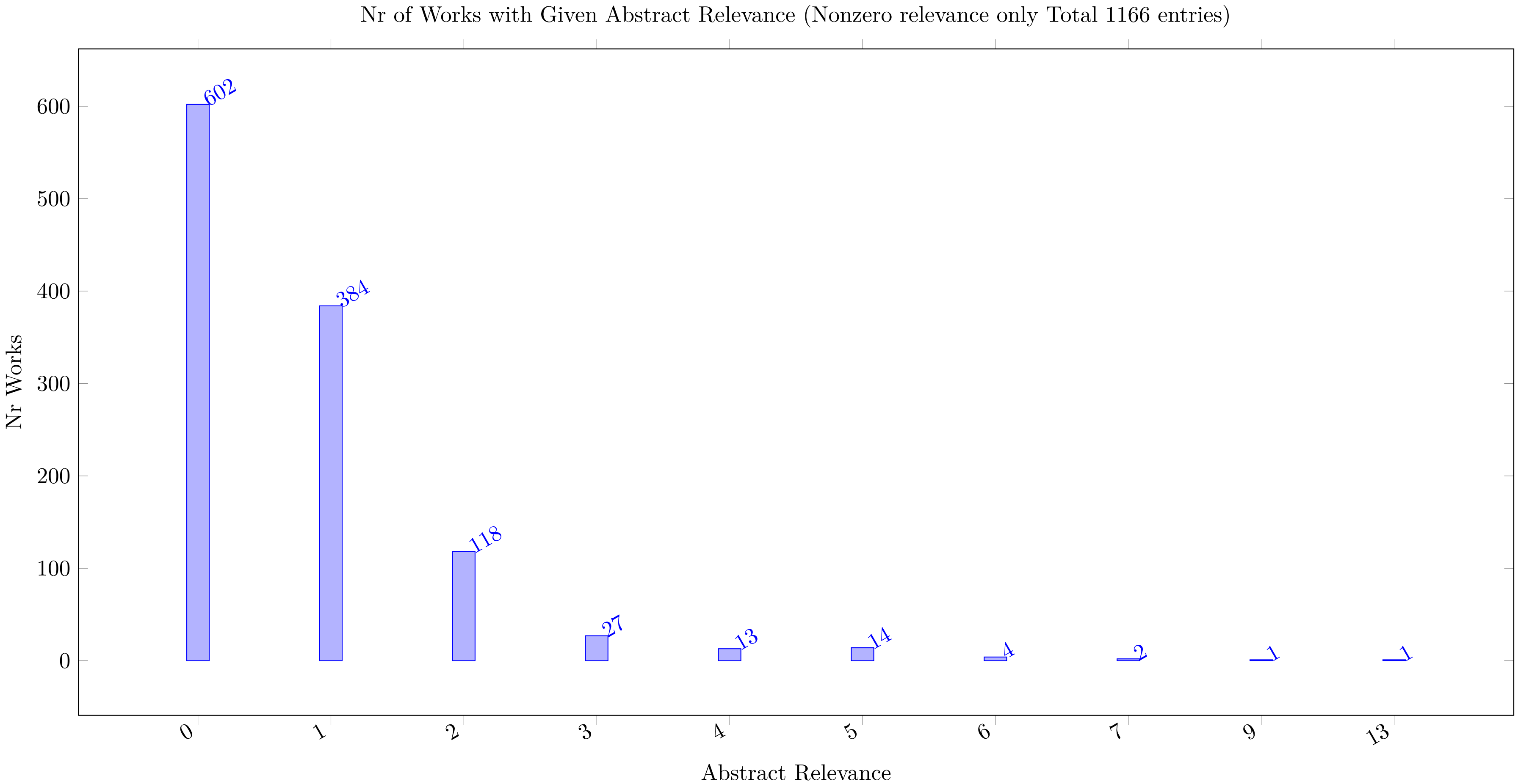
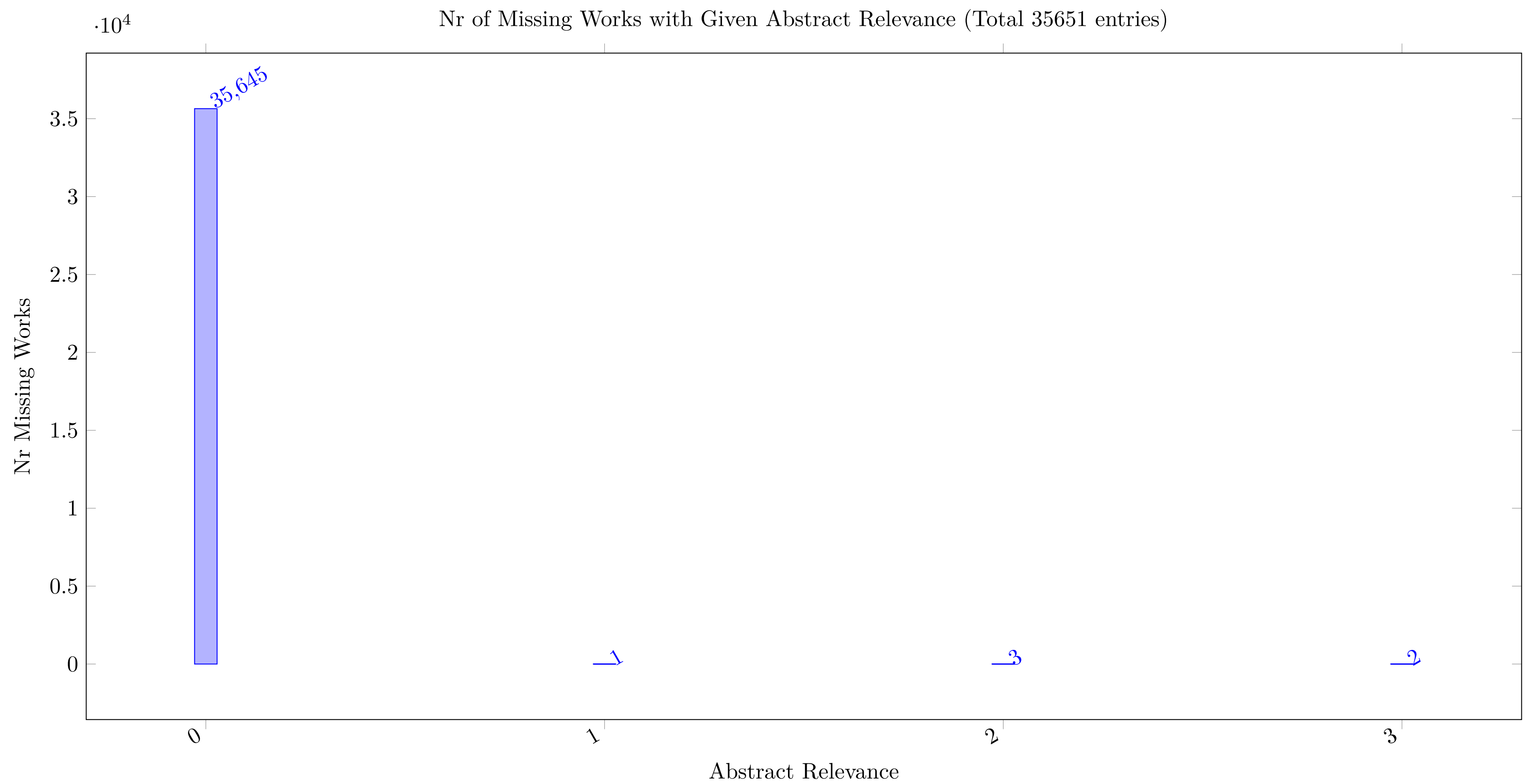
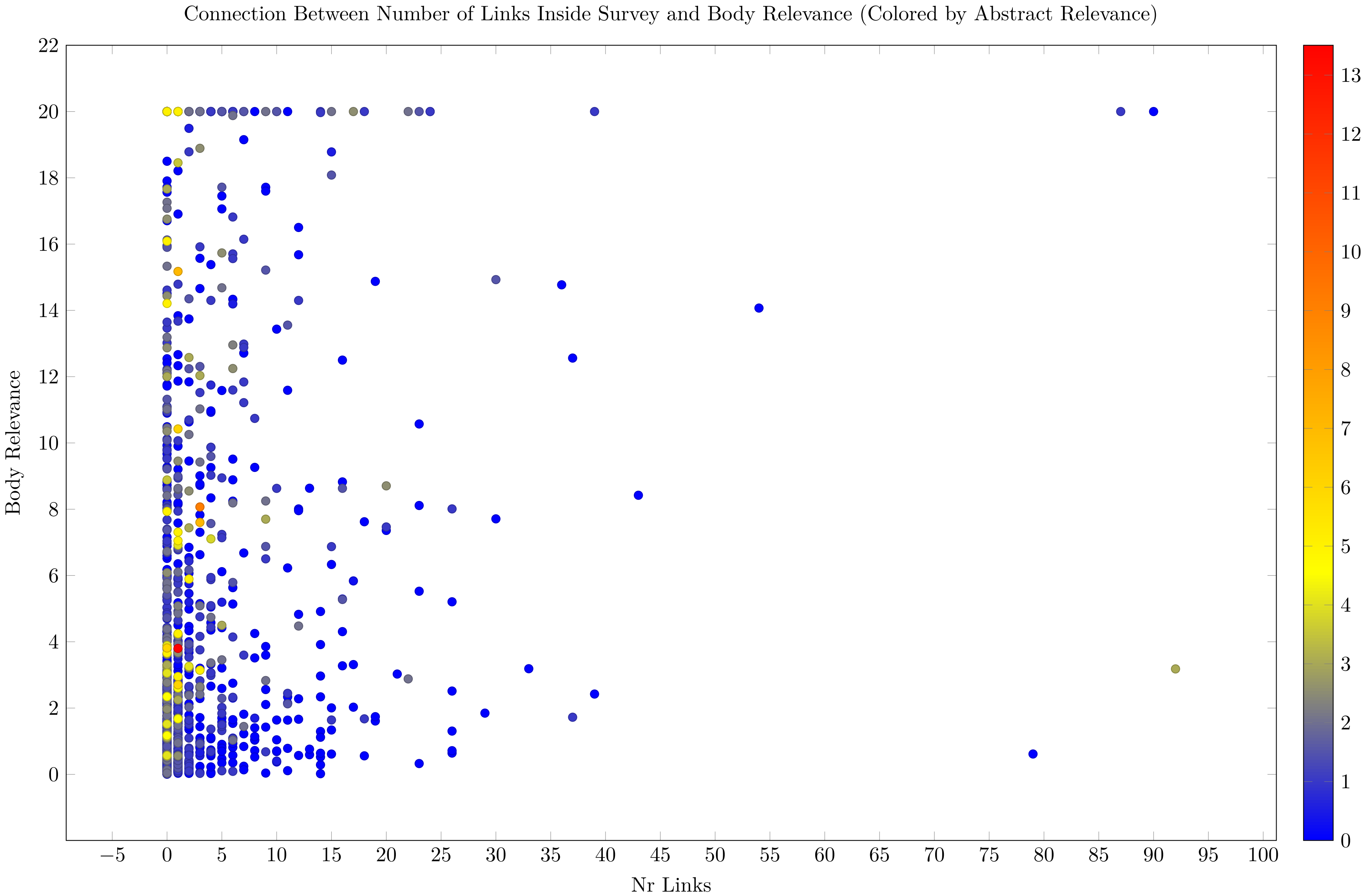
11
12
13
14
15
16
17
18
19
20
21
22
0
2
4
6
8
10
12
14
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

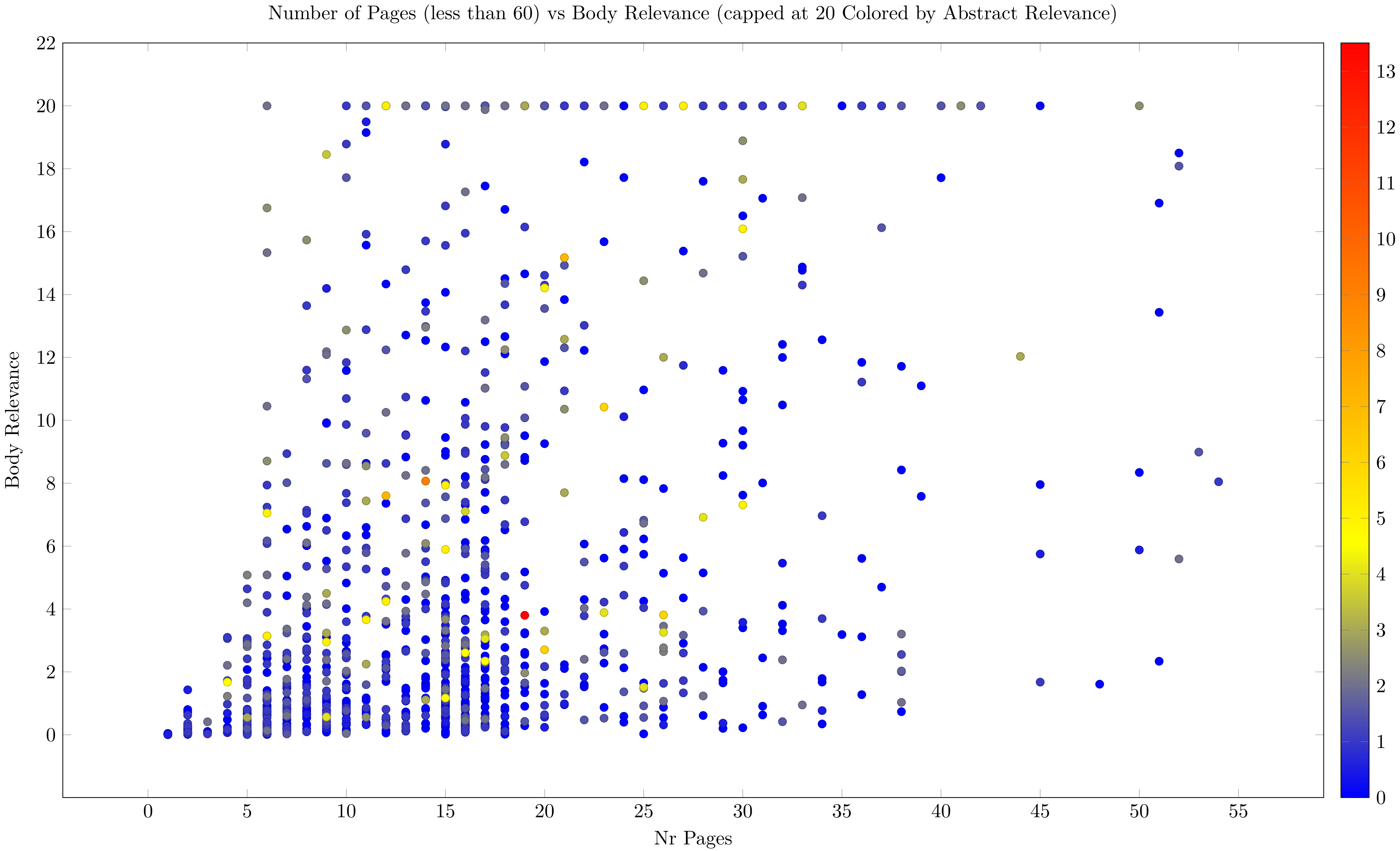

Link between Title and Abstract Relevance (Colored by Body Relevance)
Abstract Relevance
Title Relevance
14
12
10
8
6
4
2
0
−0.2
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
2.2
2.4
2.6
2.8
3
3.2
20
18
16
14
12
10
8
6
4
2


Nr of Works with Given Body Relevance (Nonzero relevance only Total 1088 entries cutoff at 20.0)
Nr Works
Body Relevance
0
50
100
150
200
250
300
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
314
144
119
68
48
45
29
26
37
22
17
16
23
9
16
12
8
11
7
4
113

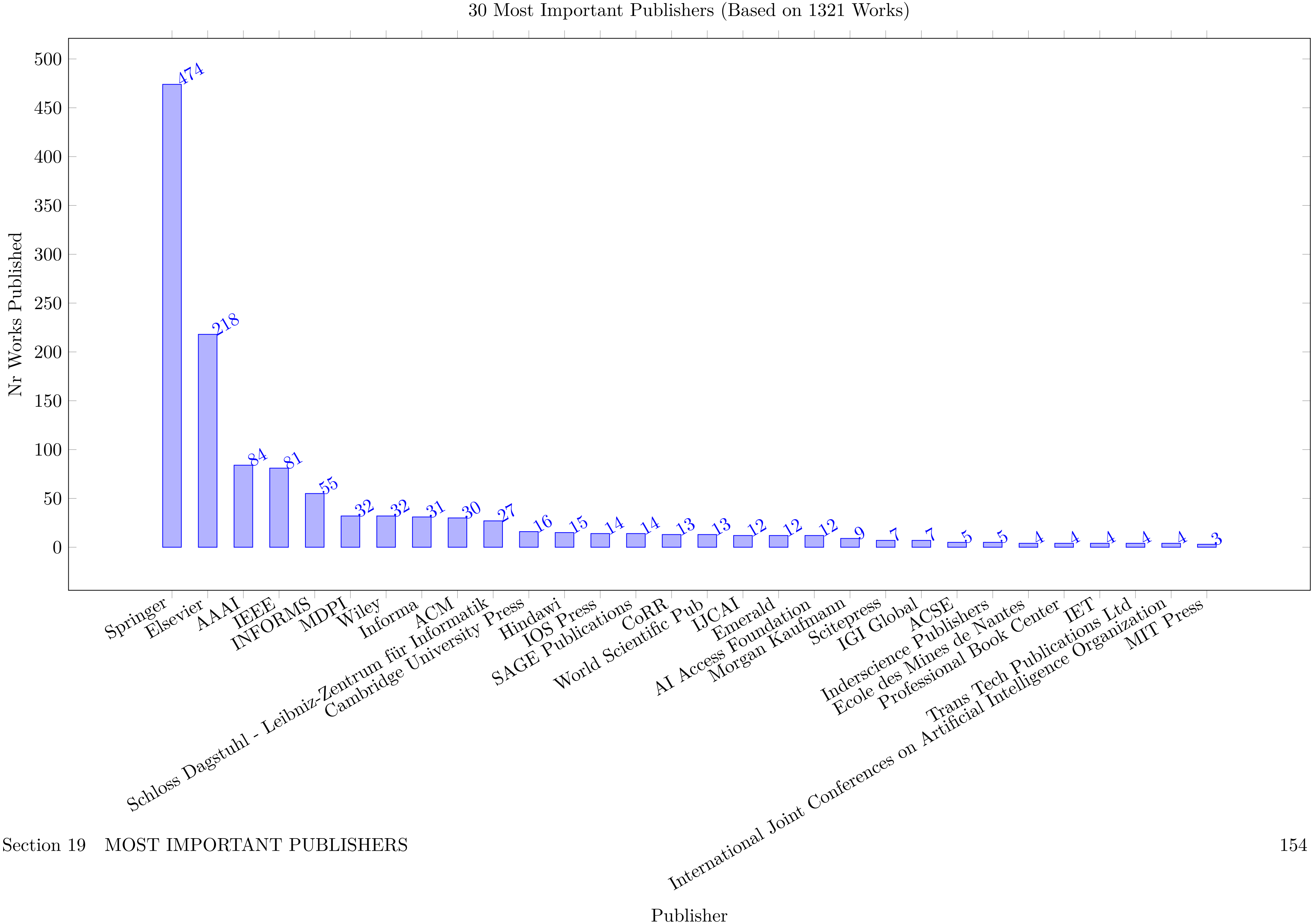

Nr of Works with Given Abstract Relevance (Nonzero relevance only Total 1166 entries)
Nr Works
Abstract Relevance
0
100
200
300
400
500
600
0
1
2
3
4
5
6
7
9
13
602
384
118
27
13
14
4
2
1
1

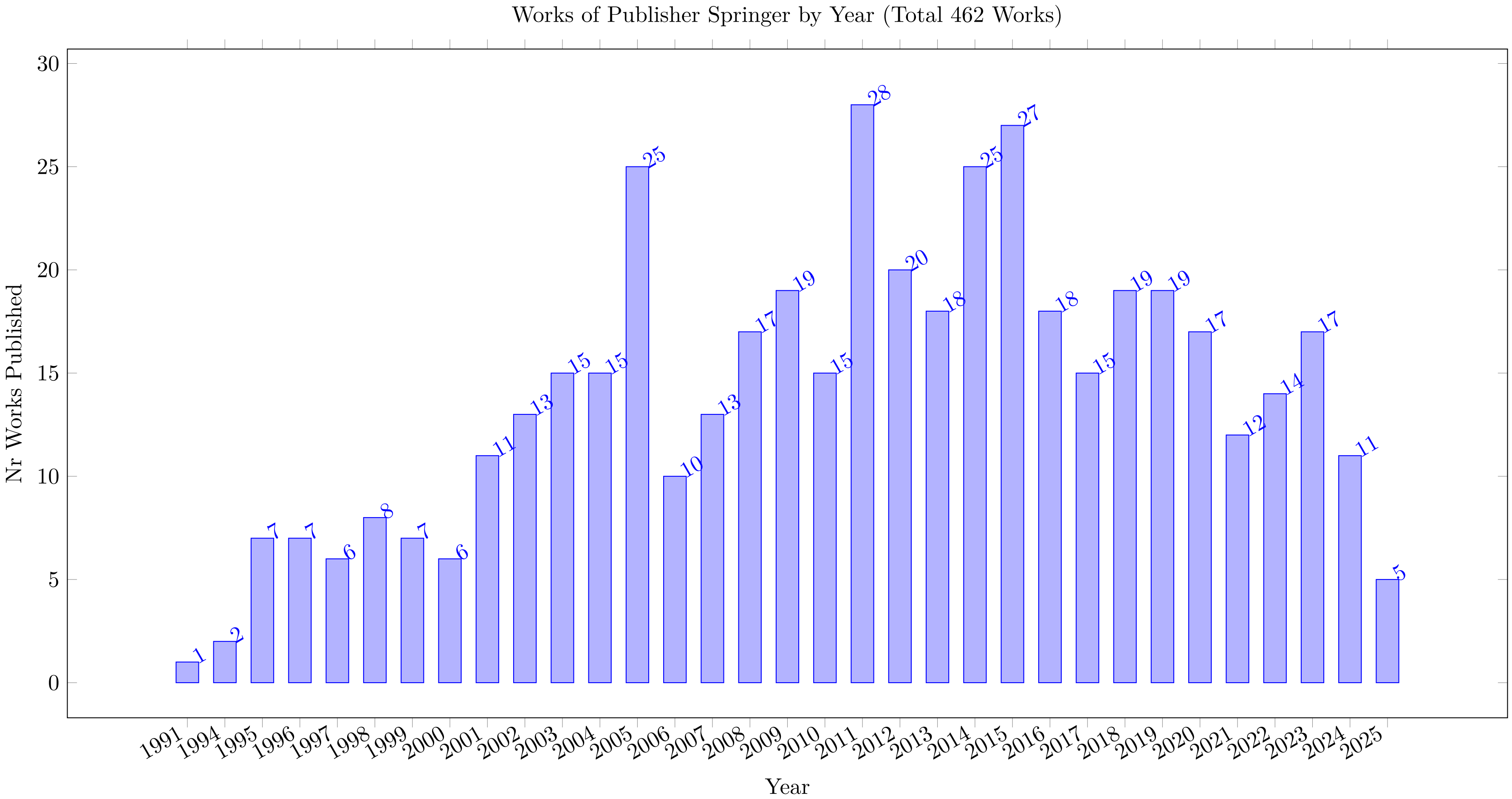

Nr of Missing Works with Given Abstract Relevance (Total 35651 entries)
$\cdot 10^4$
35,645
1
3
2
3.5
3
2.5
2
1.5
1
0.5
0
Nr Missing Works
0
1
2
3
Abstract Relevance


Connection Between Number of Links Inside Survey and Body Relevance (Colored by Abstract Relevance)
Body Relevance
Nr Links
22
20
18
16
14
12
10
8
6
4
2
0
−5
0
5
10
15
20
25
30
35
40
45
50
55
60
65
70
75
80
85
90
95
100
13
12
11
10
9
8
7
6
5
4
3
2
1
0


Number of Pages (less than 60) vs Body Relevance (capped at 20 Colored by Abstract Relevance)
Body Relevance
Nr Pages
0
2
4
6
8
10
12
14
16
18
20
22
5
15
20
25
30
35
40
45
50
55
1
3
7
9
11
13

19 Most Important Publishers




Works of Publisher Springer by Year (Total 462 Works)
Nr Works Published
Year
0
5
10
15
20
25
30
1991
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
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2007
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2011
2012
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2014
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2018
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2020
2021
2022
2023
2024
2025
1
2
7
7
6
8
7
6
11
13
15
15
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28
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18
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19
19
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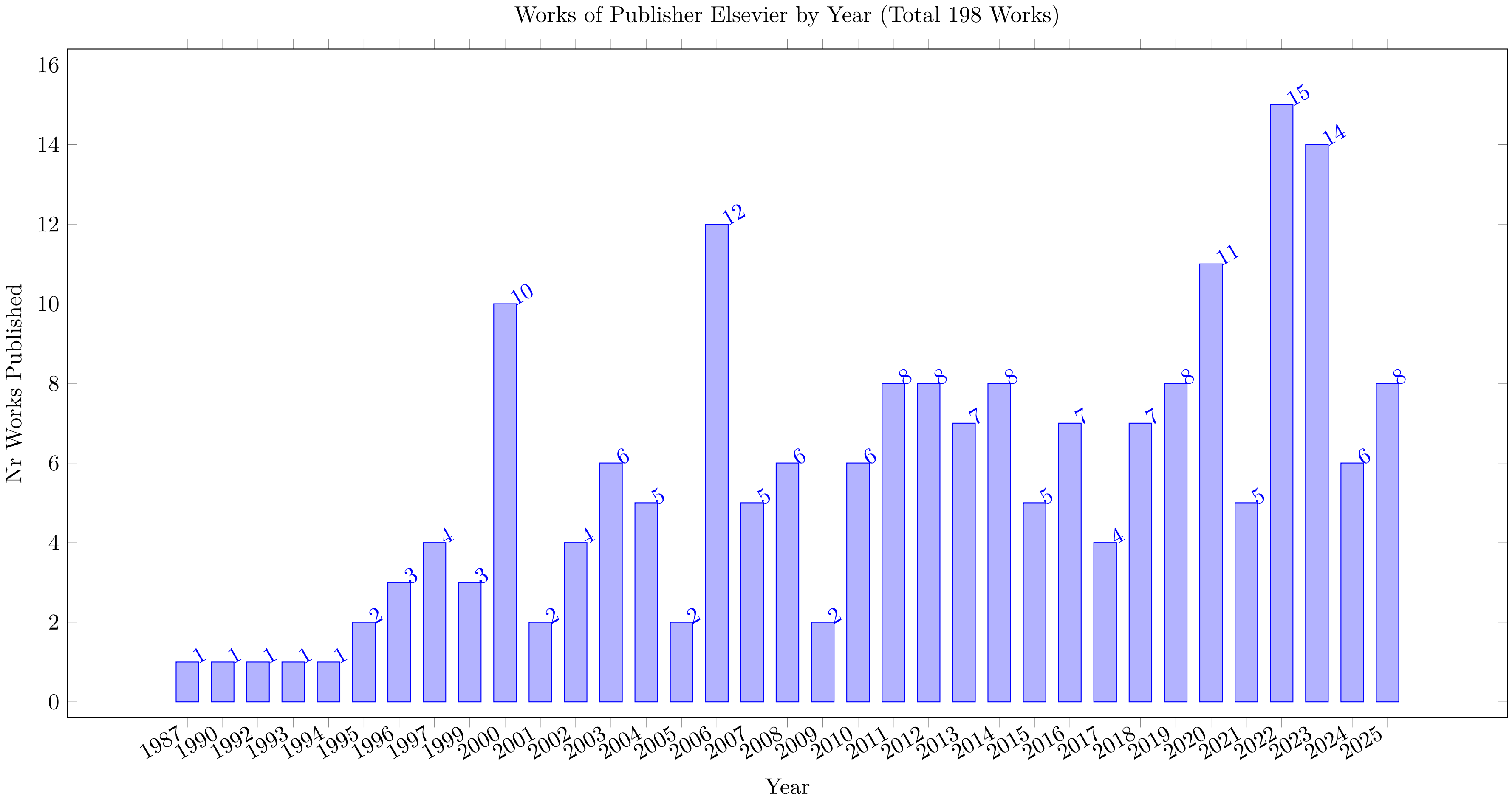
14
17
11
5


Works of Publisher Elsevier by Year (Total 198 Works)
Nr Works Published
Year
0
2
4
6
8
10
12
14
16
1987
1990
1992
1993
1994
1995
1996
1997
1999
2000
2001
2002
2003
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2019
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2021
2022
2023
2024
2025
1
1
1
1
1
2
3
4
3
10
2
4
6
5
2
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8
5
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7
8
11
5
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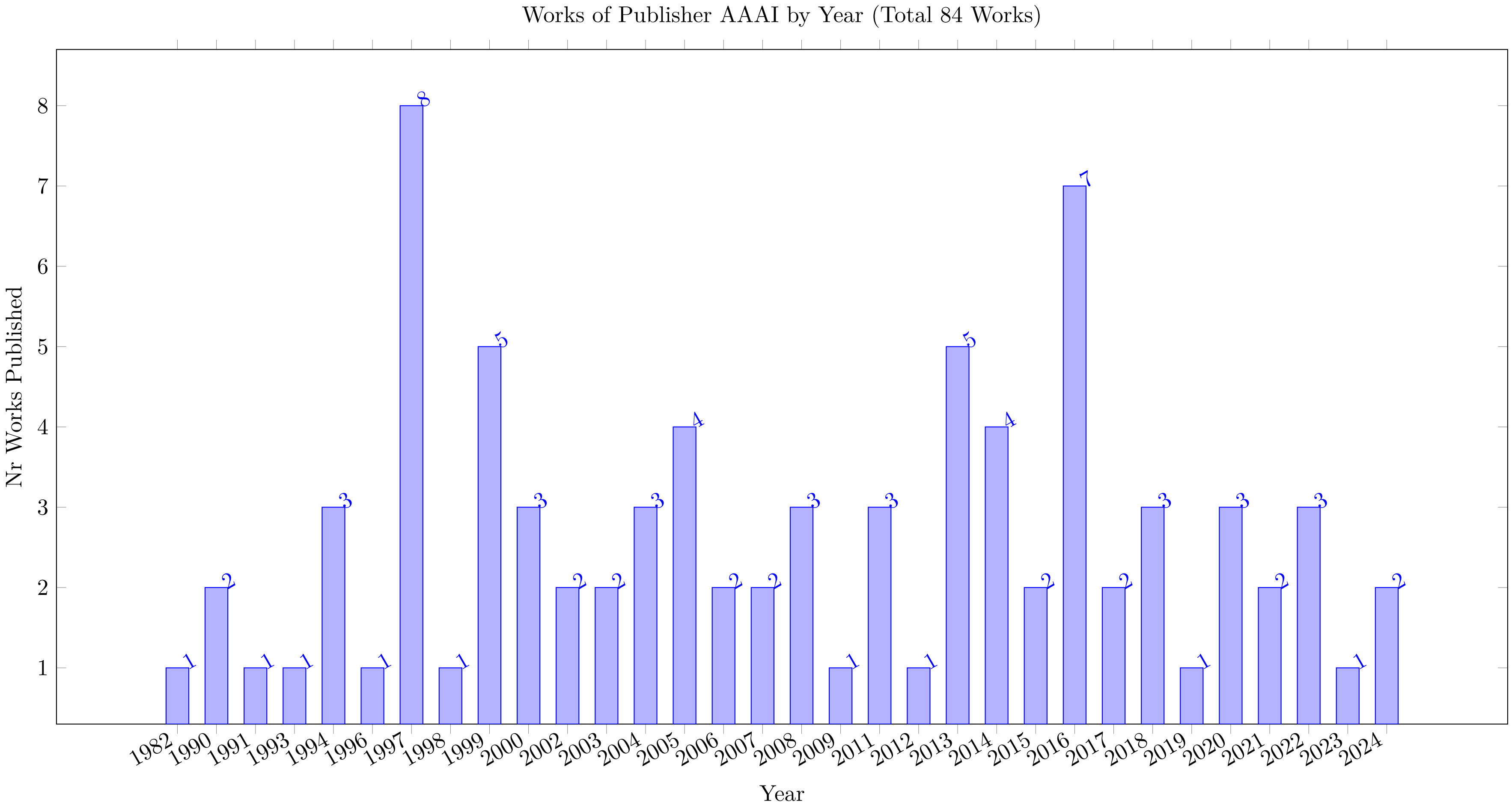
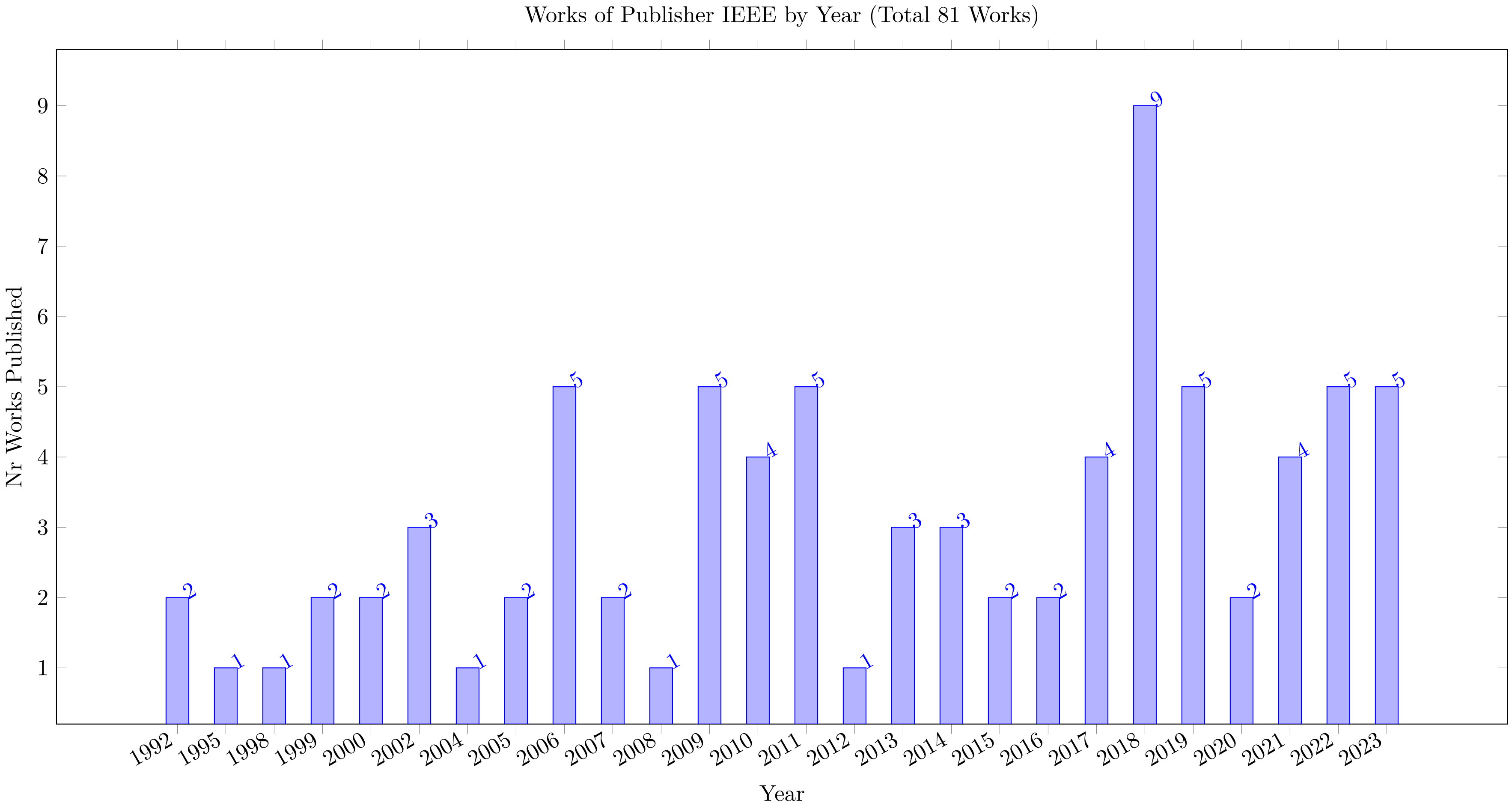
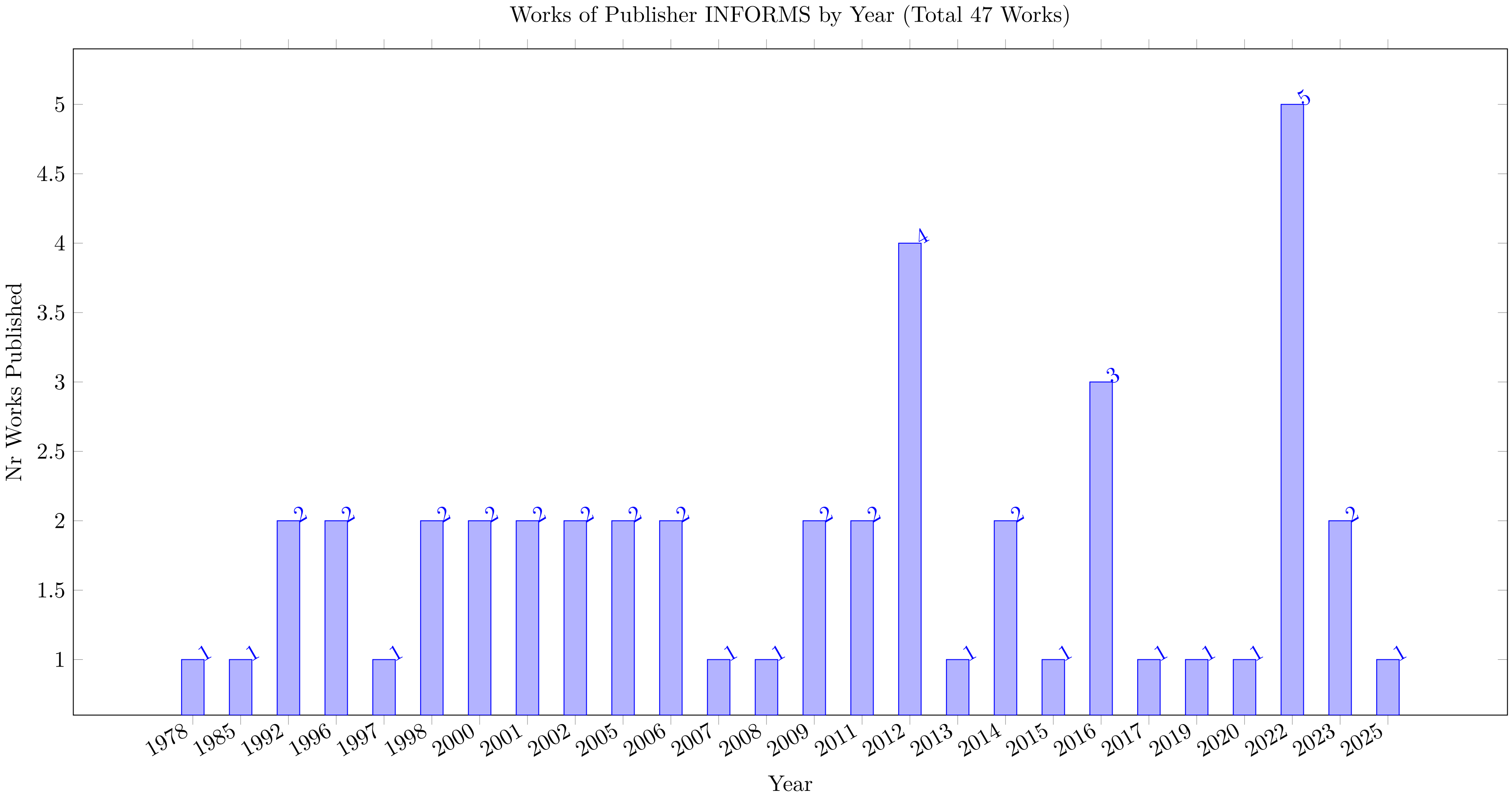
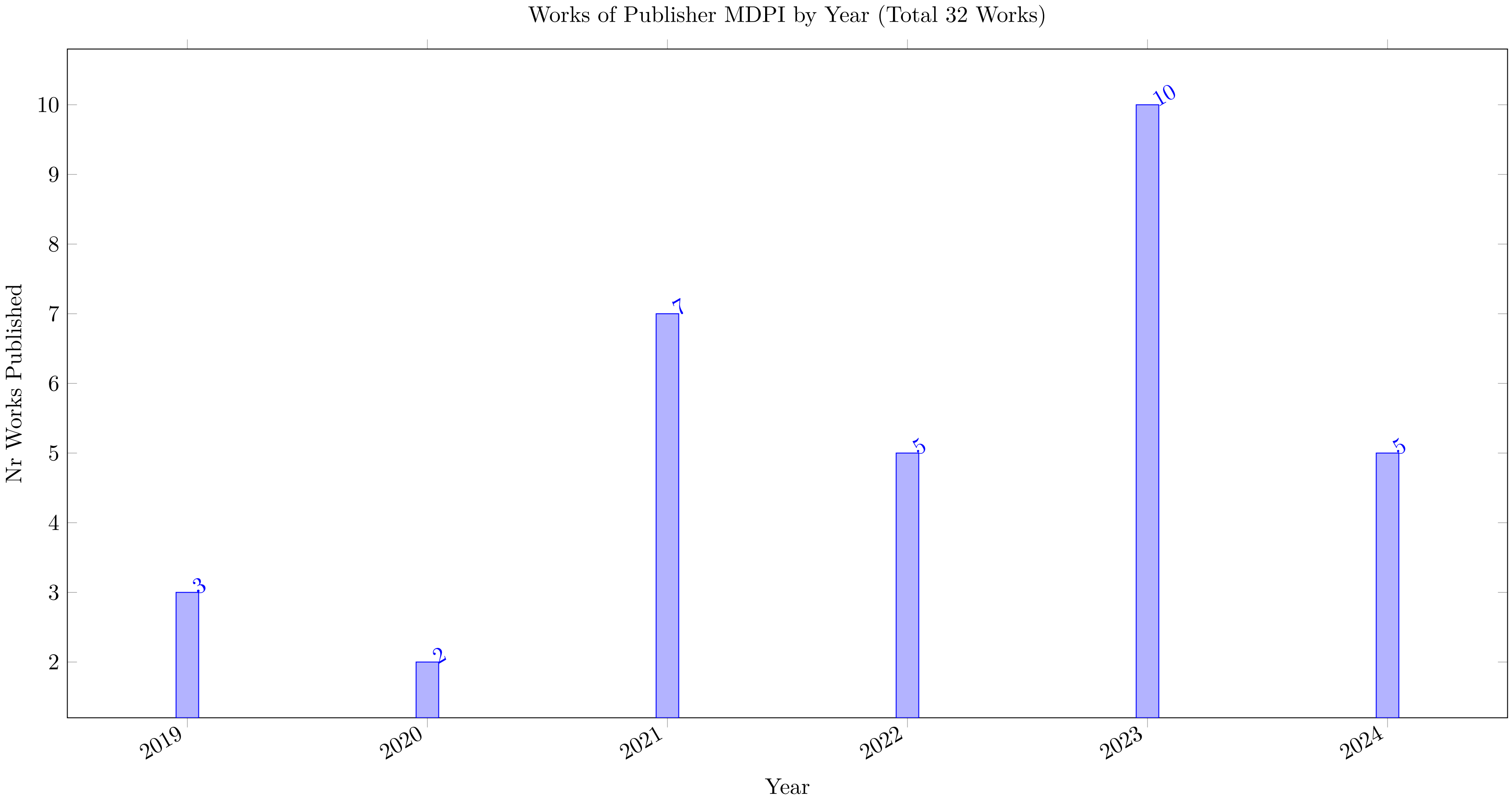
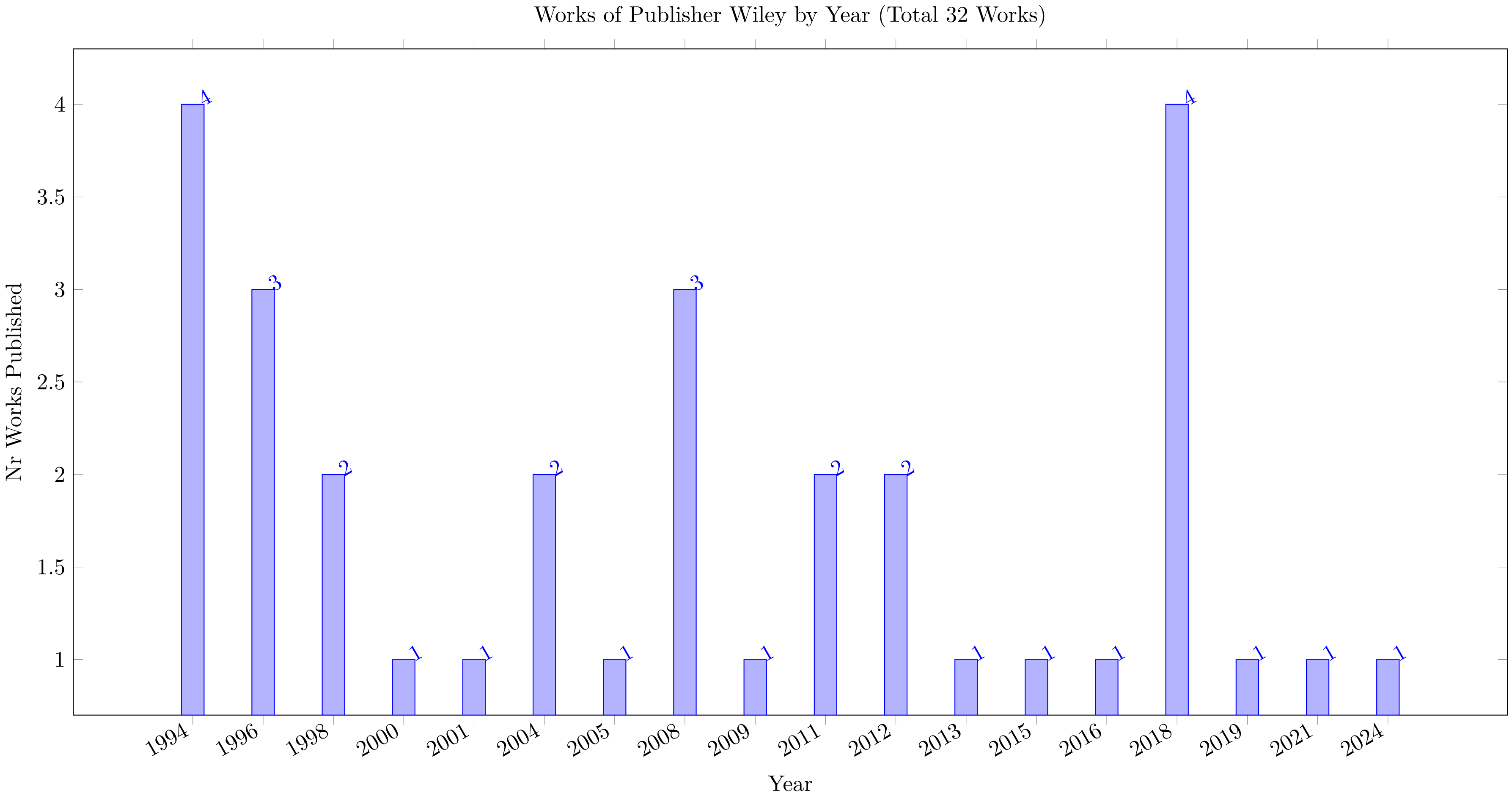
14
6
8

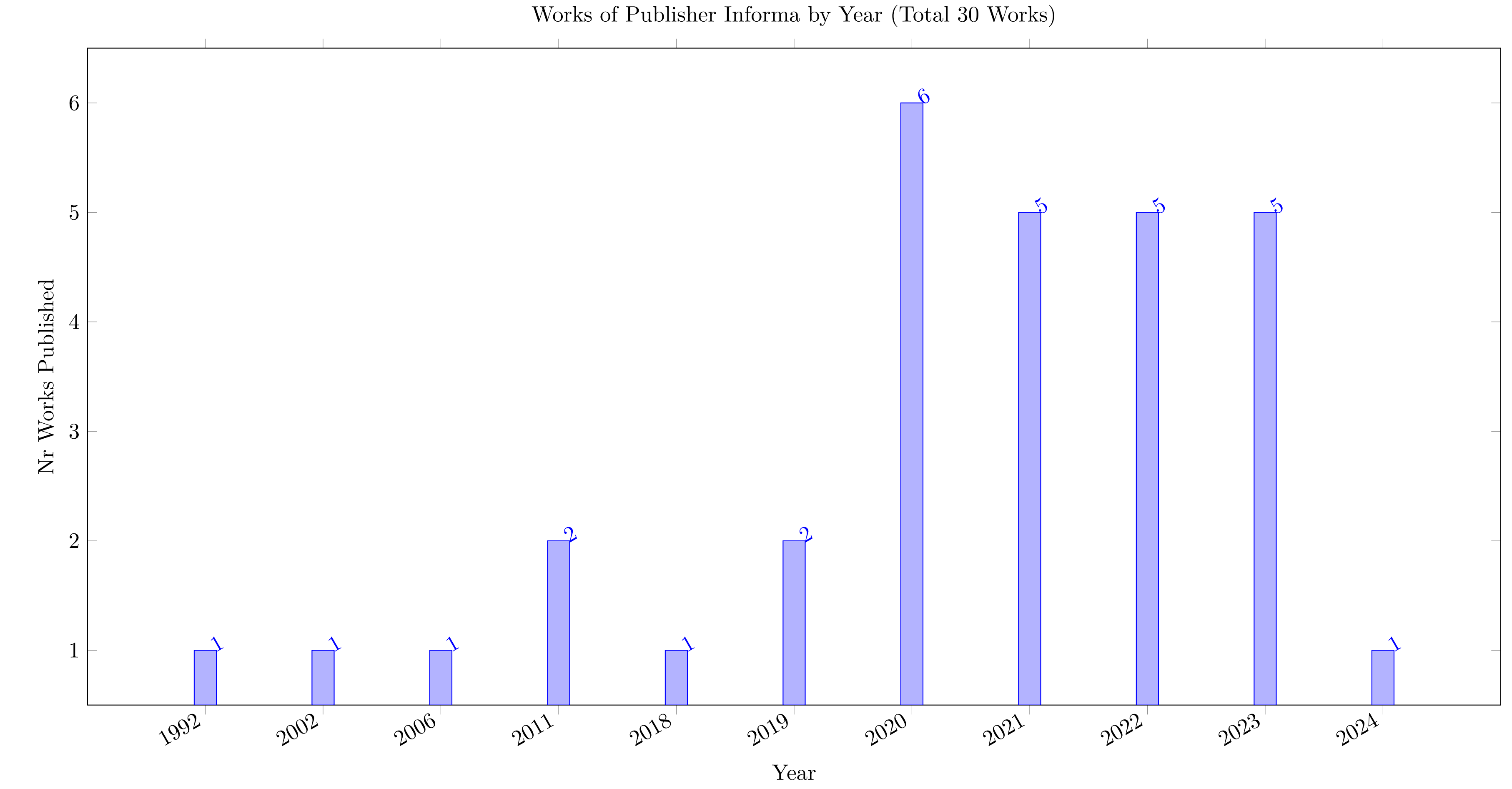

Works of Publisher AAAI by Year (Total 84 Works)
Nr Works Published
Year
1982 1990 1991 1993 1994 1996 1997 1998 1999 2000 2002 2003 2004 2005 2006 2007 2008 2009 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024
1 2 1 1 3 1 8 1 5 3 2 2 3 4 2 2 3 1 3 1 5 4 2 7 2 3 1 3 2 3 1 2

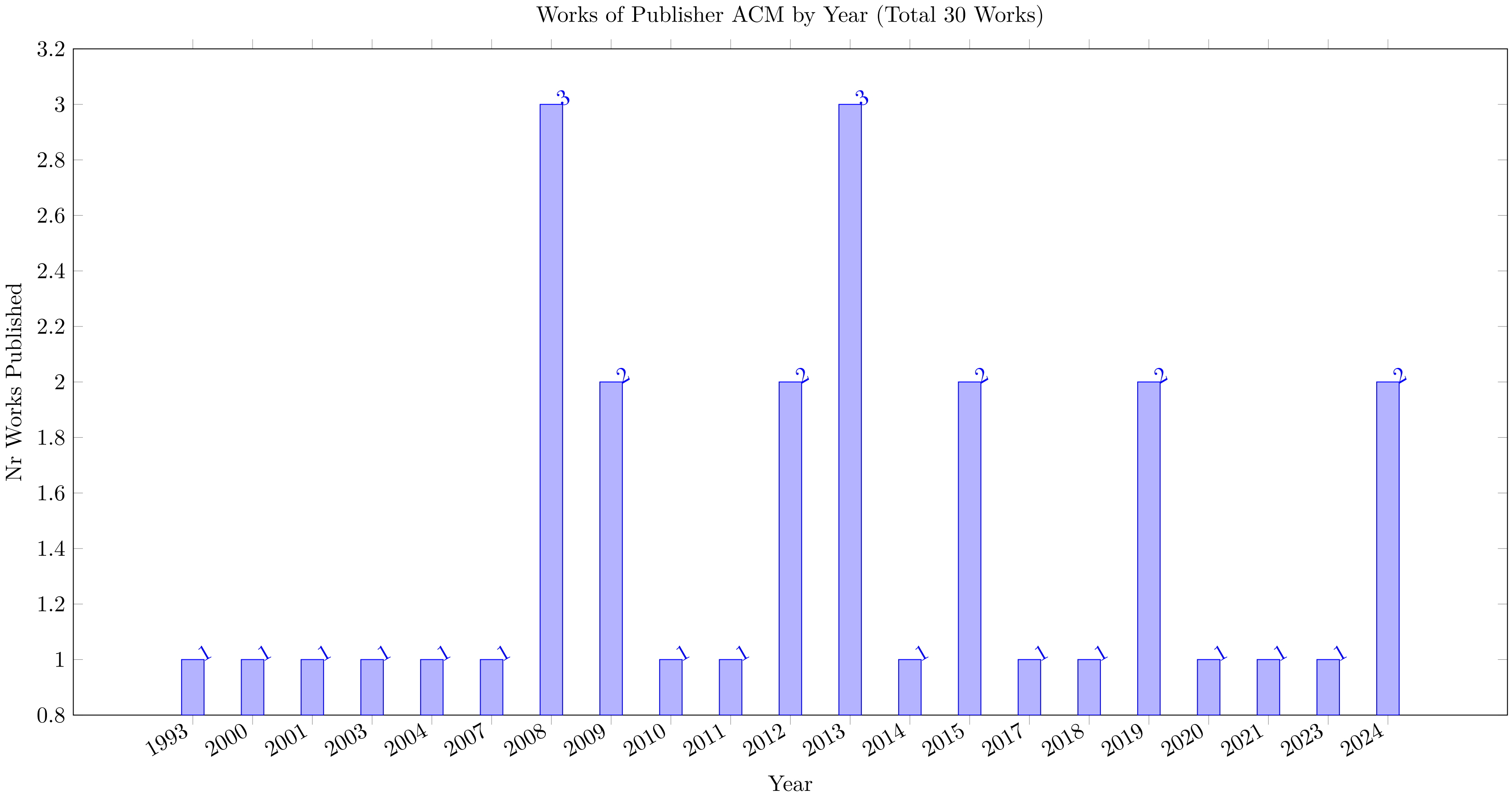

Works of Publisher IEEE by Year (Total 81 Works)
Nr Works Published
Year
1992 1995 1998 1999 2000 2002 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023
2 1 1 2 2 3 1 2 5 2 1 5 4 5 1 3 3 2 2 4 9 5 2 4 5 5

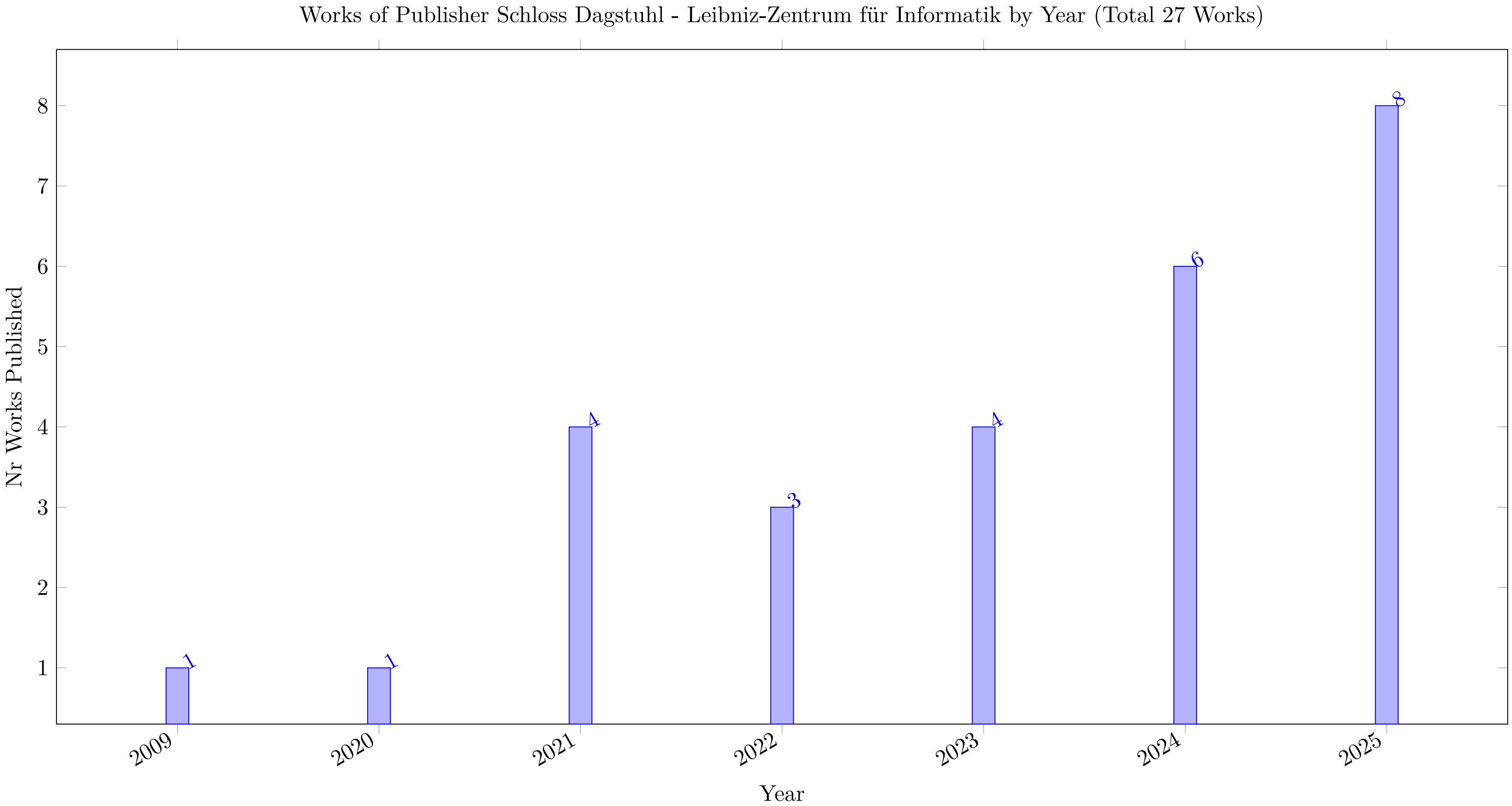

Works of Publisher INFORMS by Year (Total 47 Works)
Nr Works Published
Year
1978
1985
1992
1996
1997
1998
2000
2001
2002
2005
2006
2007
2008
2009
2011
2012
2013
2014
2015
2016
2017
2019
2020
2022
2023
2025
1
1
2
2
1
2
2
2
2
2
2
1
1
2
2
4
1
2
1
3
1
1
1
5
2
1

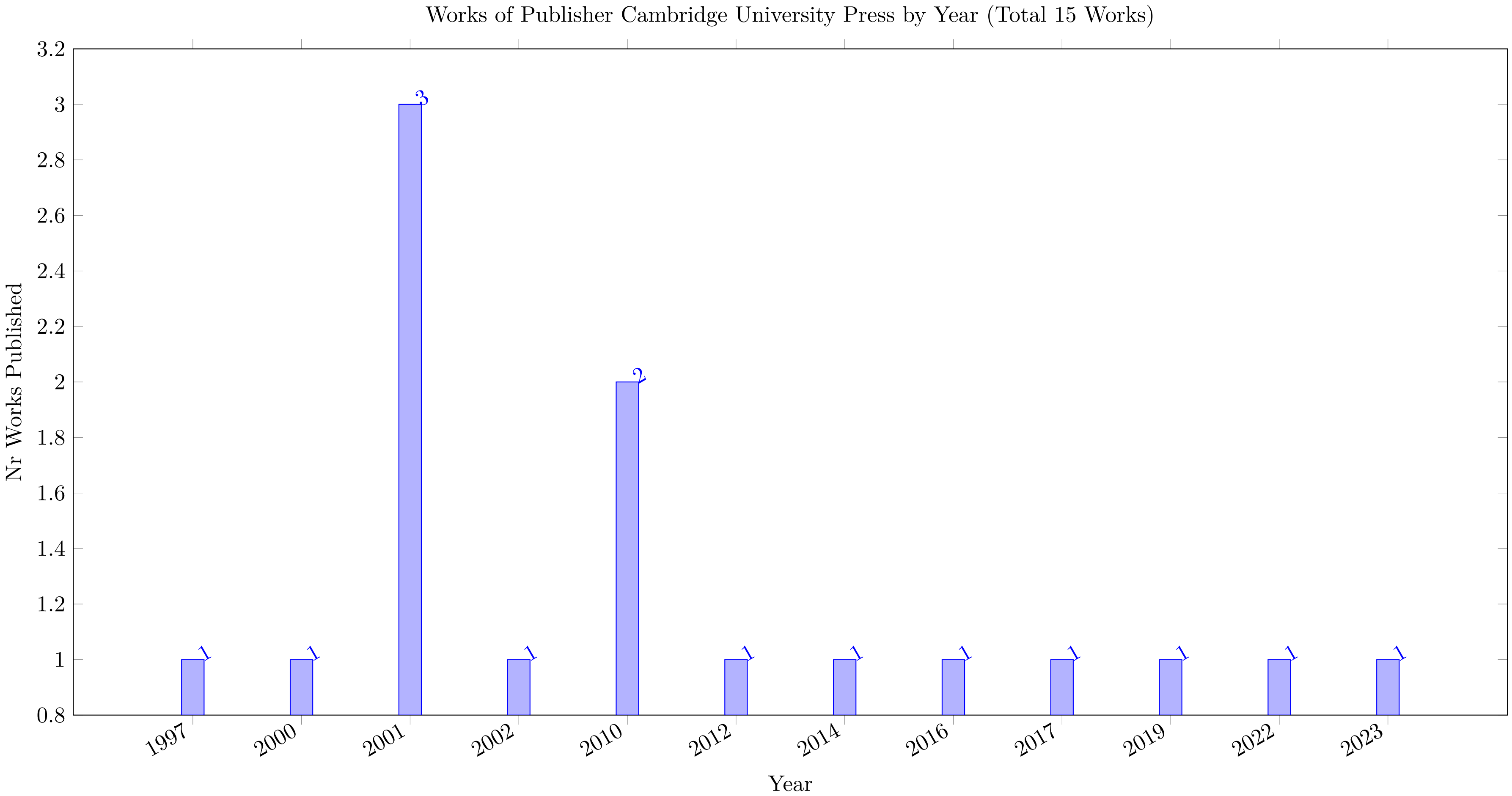

Works of Publisher MDPI by Year (Total 32 Works)
Nr Works Published
Year
2019
2020
2021
2022
2023
2024
3
2
7
5
10
5

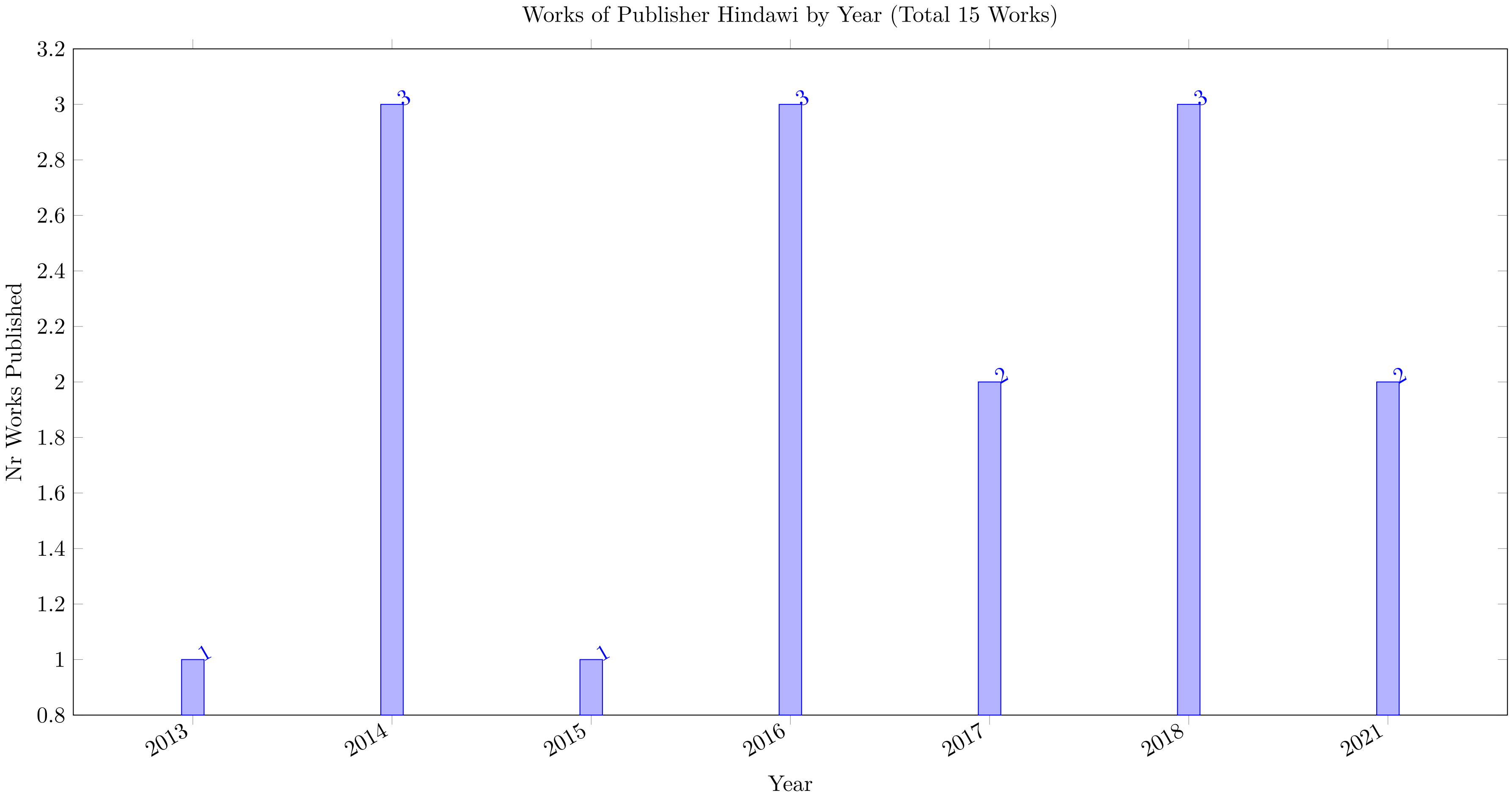

Works of Publisher Wiley by Year (Total 32 Works)
Nr Works Published
Year
4
3.5
3
2.5
2
1.5
1
1994
1996
1998
2000
2001
2004
2005
2008
2009
2011
2012
2013
2015
2016
2018
2019
2021
2024
4
3
2
1
1
2
1
3
1
2
2
1
1
1
4
1
1
1

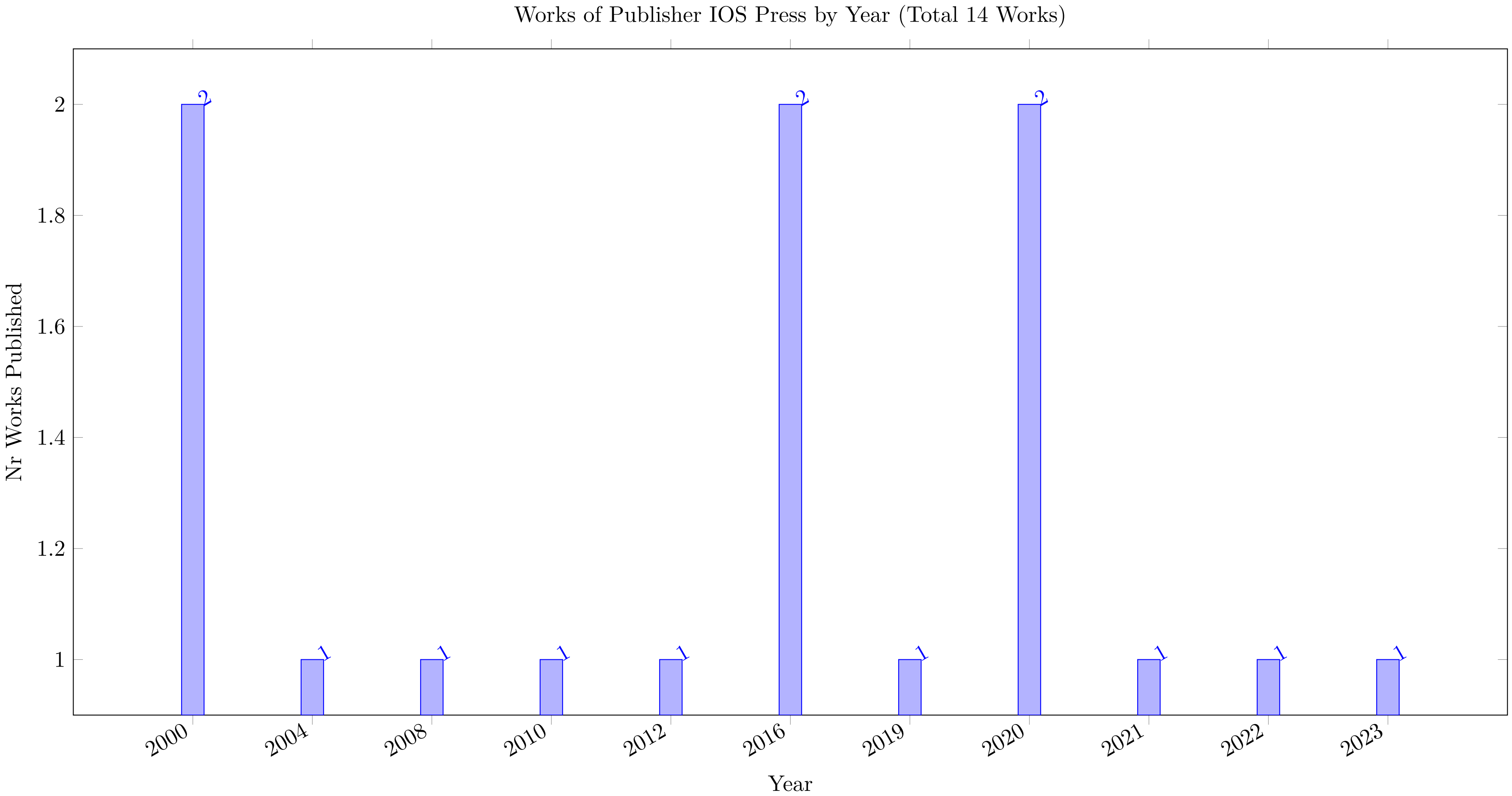

Works of Publisher Informa by Year (Total 30 Works)
Nr Works Published
Year
1992
2002
2006
2011
2018
2019
2020
2021
2022
2023
2024
1
1
1
2
1
2
6
5
5
5
1


Works of Publisher ACM by Year (Total 30 Works)
Nr Works Published
Year
3.2
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0.8
1993
2000
2001
2003
2004
2007
2008
2009
2010
2011
2012
2013
2014
2015
2017
2018
2019
2020
2021
2023
2024


Works of Publisher Schloss Dagstuhl - Leibniz-Zentrum für Informatik by Year (Total 27 Works)
Nr Works Published
Year
2009
2020
2021
2022
2023
2024
2025
1
1
4
3
4
6
8


Works of Publisher Cambridge University Press by Year (Total 15 Works)
Nr Works Published
Year
3.2
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0.8
1997
2000
2001
2002
2010
2012
2014
2016
2017
2019
2022
2023
1
1
3
1
2
1
1
1
1
1
1
1


Works of Publisher Hindawi by Year (Total 15 Works)
Nr Works Published
Year
3.2
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0.8
2013
2014
2015
2016
2017
2018
2021
1
3
1
3
2
3
2


Works of Publisher IOS Press by Year (Total 14 Works)
Nr Works Published
Year
2000
2004
2008
2010
2012
2016
2019
2020
2021
2022
2023
2
1
1
1
1
2
1
2
1
1
1
1
1.2
1.4
1.6
1.8
2


Works of Publisher SAGE Publications by Year (Total 14 Works)
Nr Works Published
3.2
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0.8
2
1
2
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3
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1
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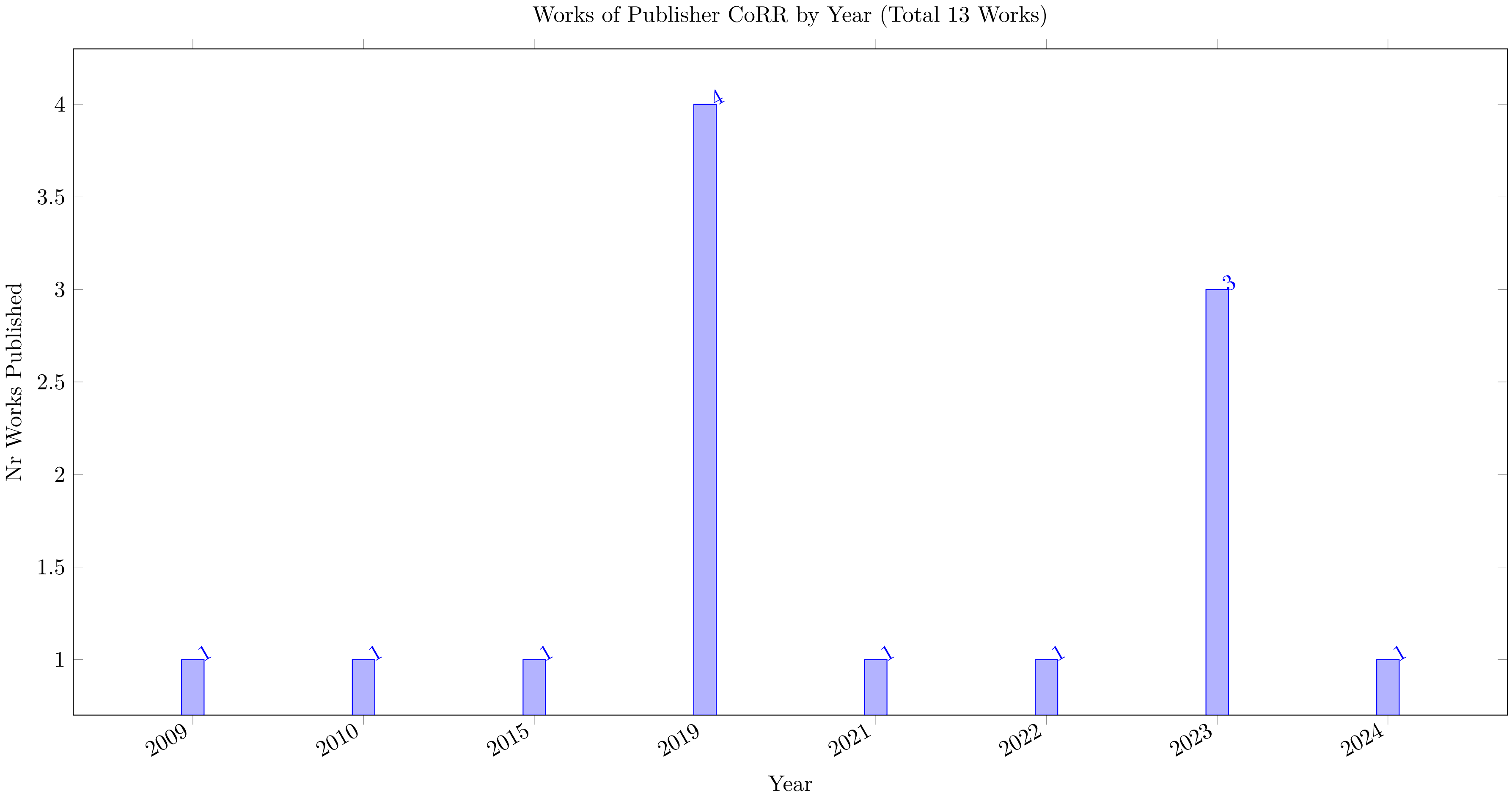
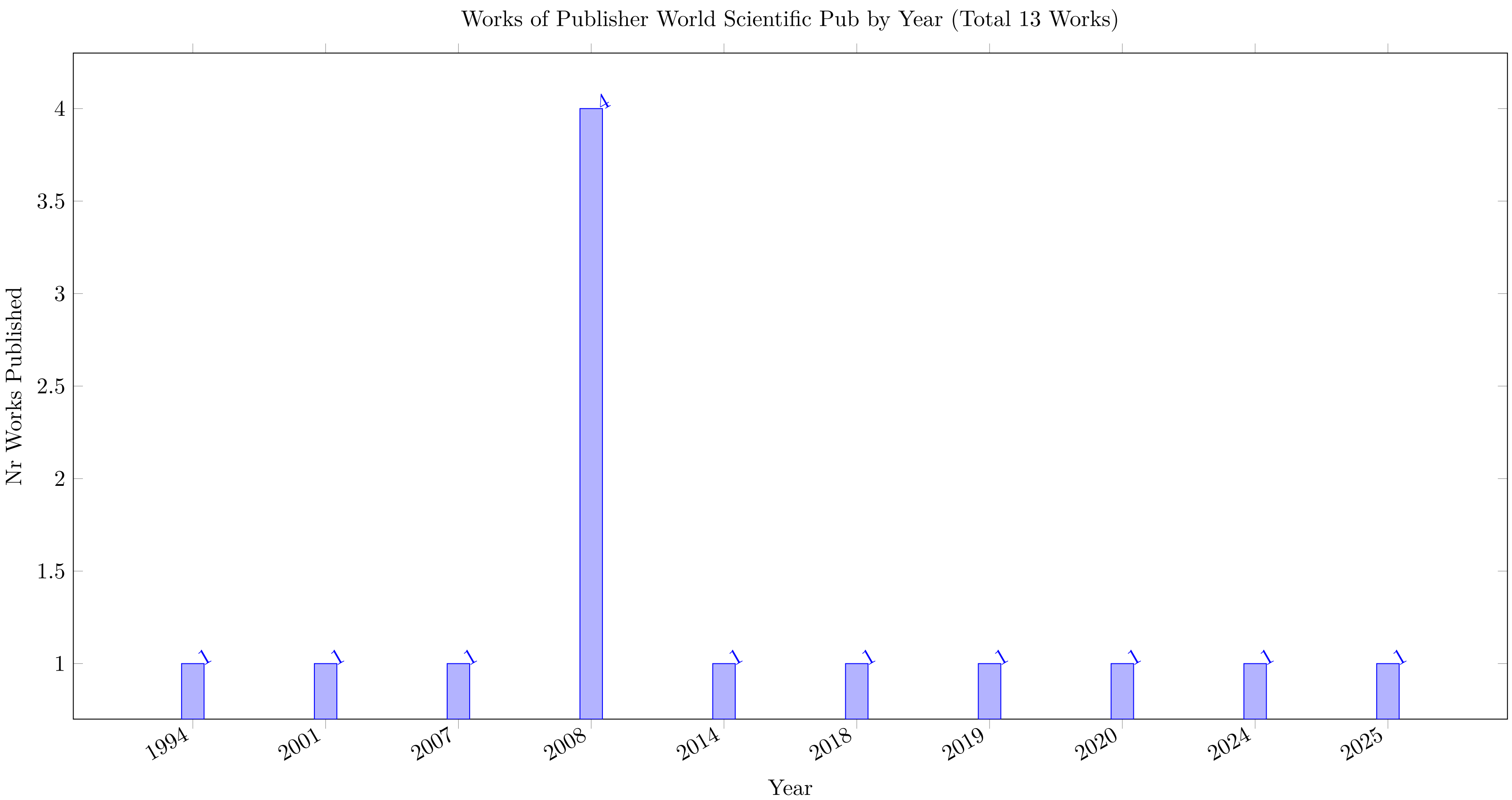
2013
2014
2015
2016
2017
2021
2023
2024
Year


Works of Publisher CoRR by Year (Total 13 Works)
Nr Works Published
Year
4
3.5
3
2.5
2
1.5
1
2009
2010
2015
2019
2021
2022
2023
2024
1
1
1
4
1
1
3
1


Works of Publisher World Scientific Pub by Year (Total 13 Works)
Nr Works Published
Year
1994
2001
2007
2008
2014
2018
2019
2020
2024
2025
4
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1
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1
1
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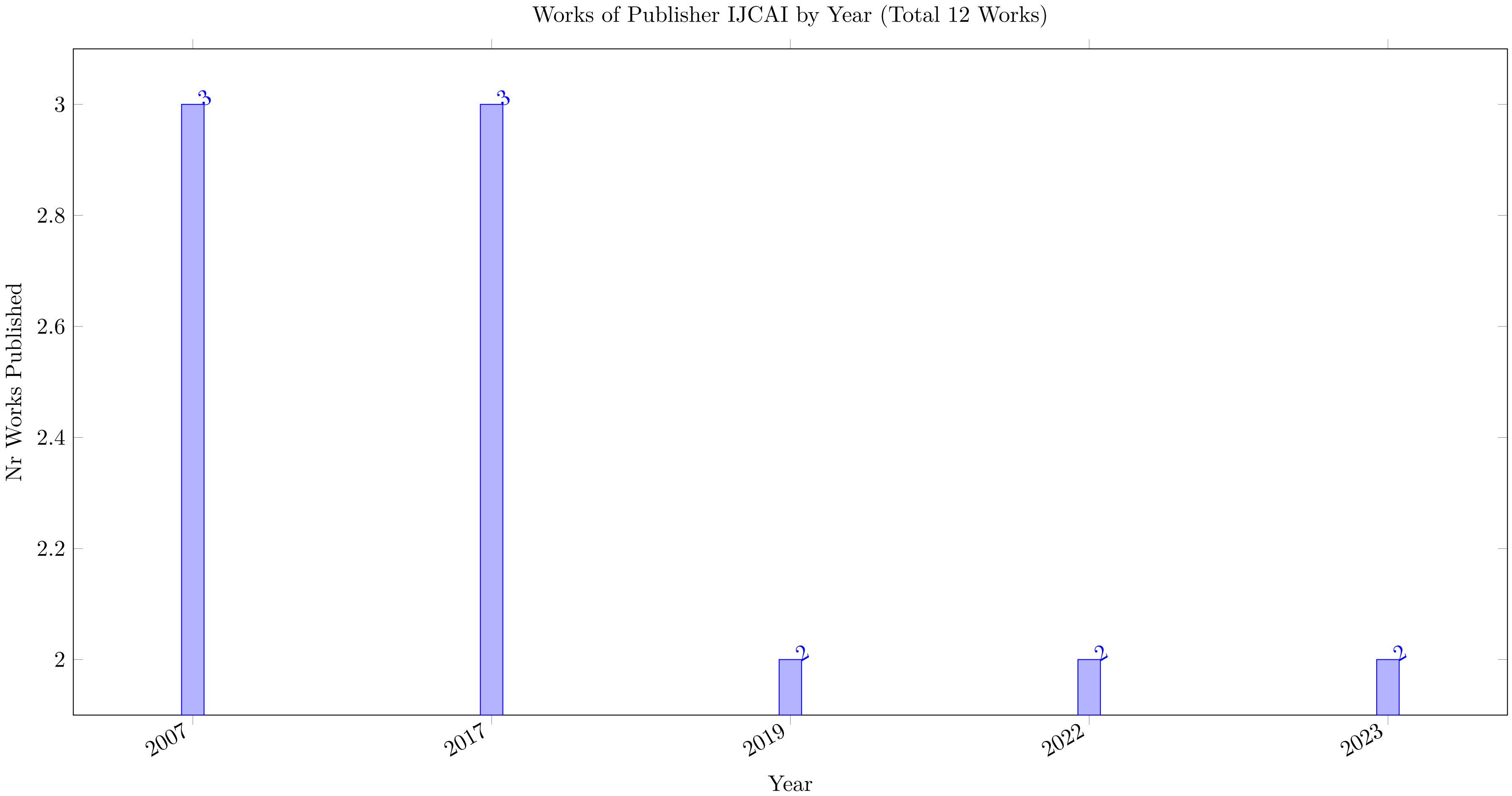
0.5
1
1.5
2
2.5
3
3.5
4


Works of Publisher IJCAI by Year (Total 12 Works)
Nr Works Published
Year
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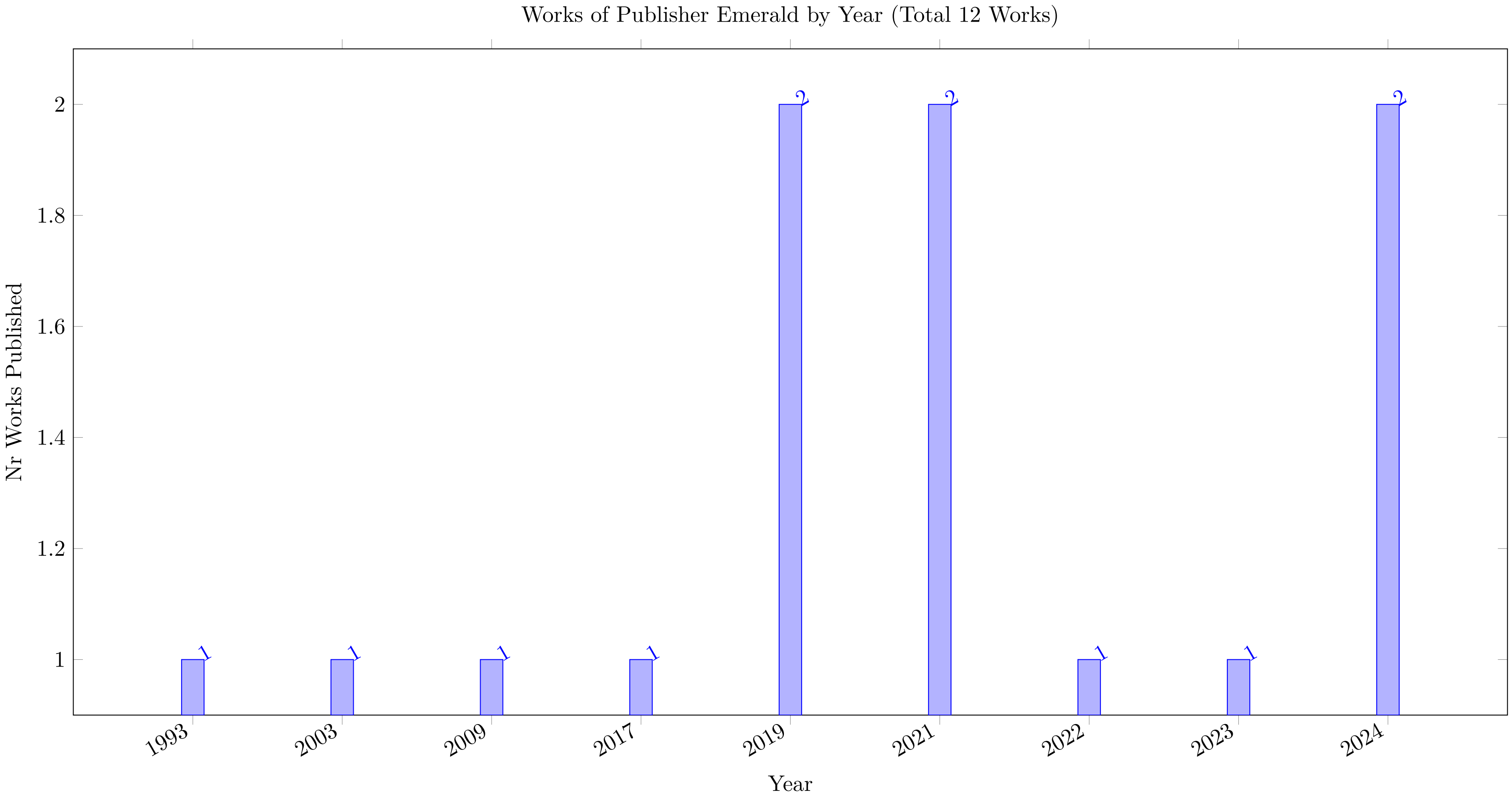
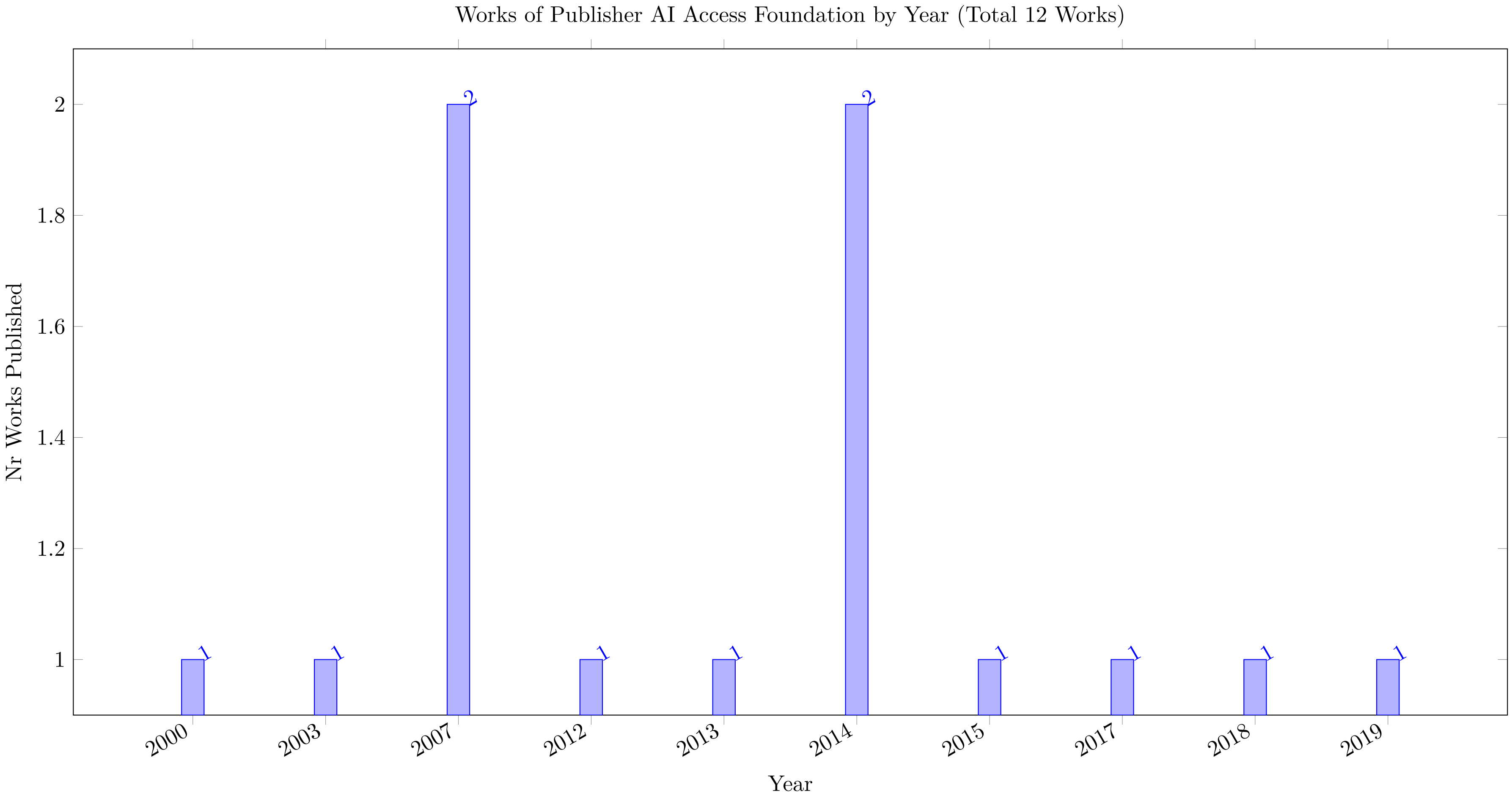
2.8
2.6
2.4
2.2
2
2007
2017
2019
2022
2023
3
3
2
2
2


Works of Publisher Emerald by Year (Total 12 Works)
Nr Works Published
Year
2
1.8
1.6
1.4
1.2
1
1993
2003
2009
2017
2019
2021
2022
2023
2024
1
1
1
1
2
2
1
1
2


Works of Publisher AI Access Foundation by Year (Total 12 Works)
Nr Works Published
Year
2
1.8
1.6
1.4
1.2
1
2000
2003
2007
2012
2013
2014
2015
2017
2018
2019
1
1
2
1
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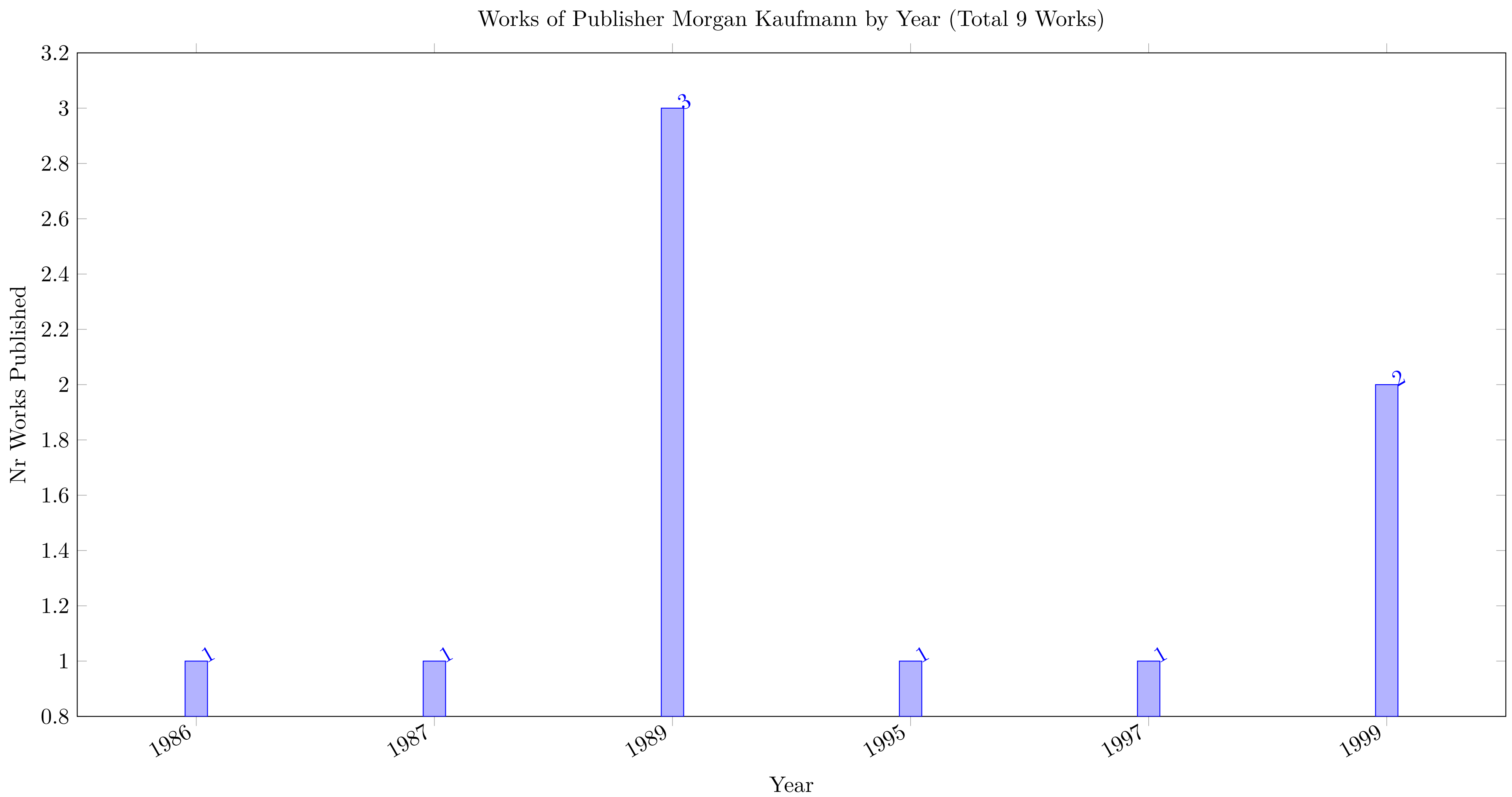
2
1
1
1
1


Works of Publisher Morgan Kaufmann by Year (Total 9 Works)
Nr Works Published
3.2
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0.8
1
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3
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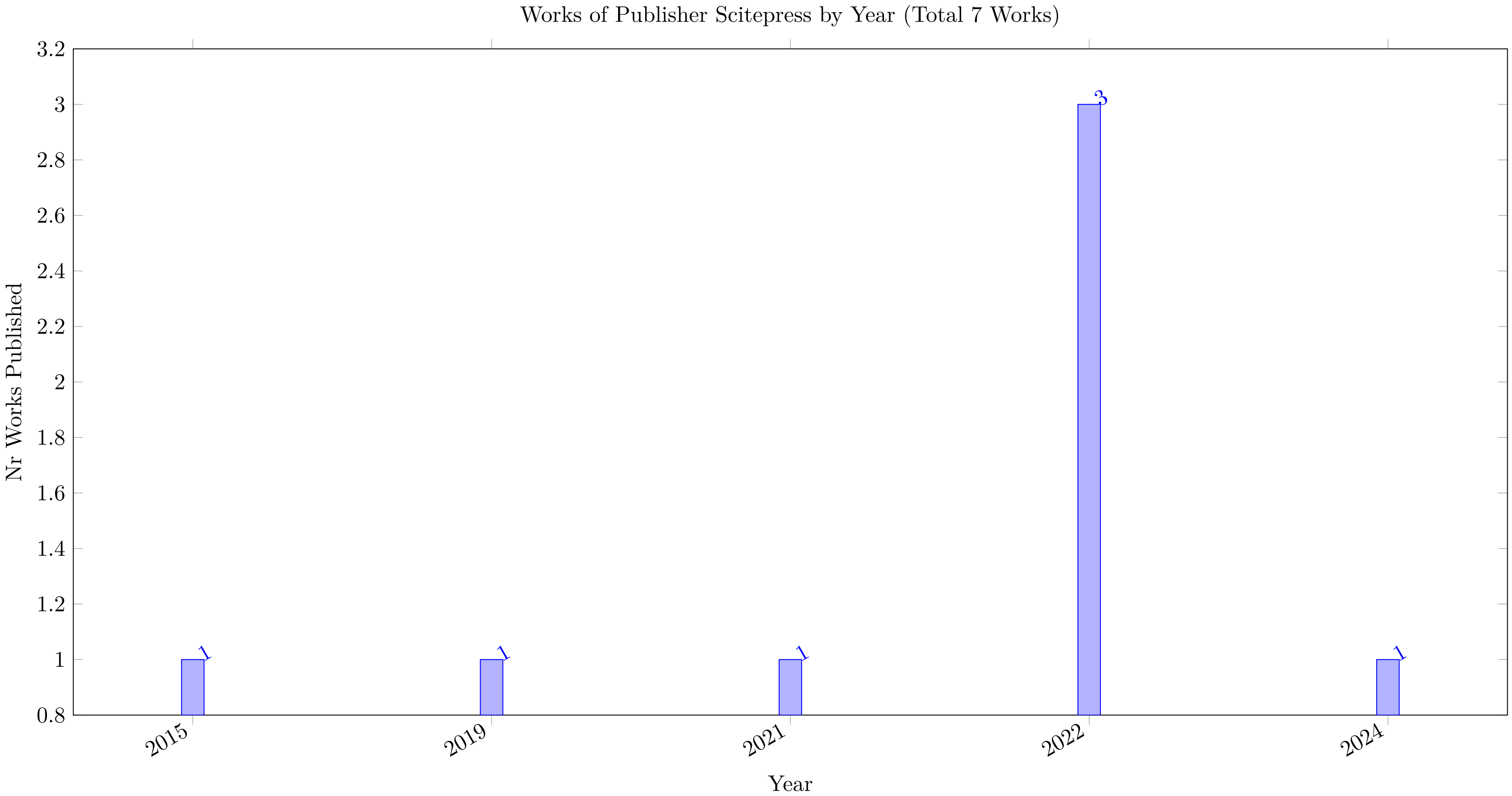
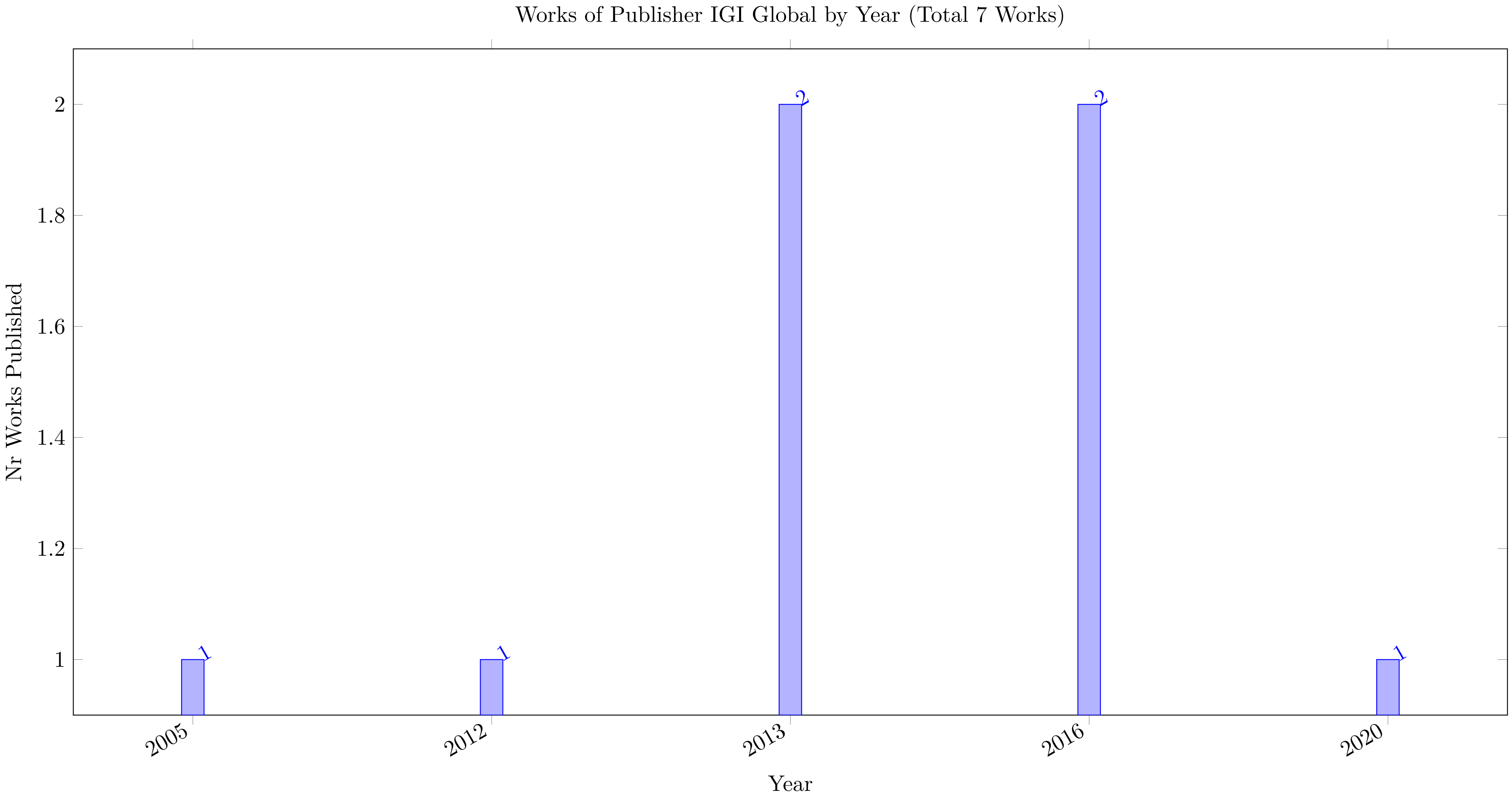
2
1986
1987
1989
1995
1997
1999
Year


Works of Publisher Scitepress by Year (Total 7 Works)
Nr Works Published
Year
3.2
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0.8
2015
2019
2021
2022
2024
1
1
1
3
1


Works of Publisher IGI Global by Year (Total 7 Works)
Nr Works Published
Year
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2012
2013
2016
2020
1
1
2
2
1
1
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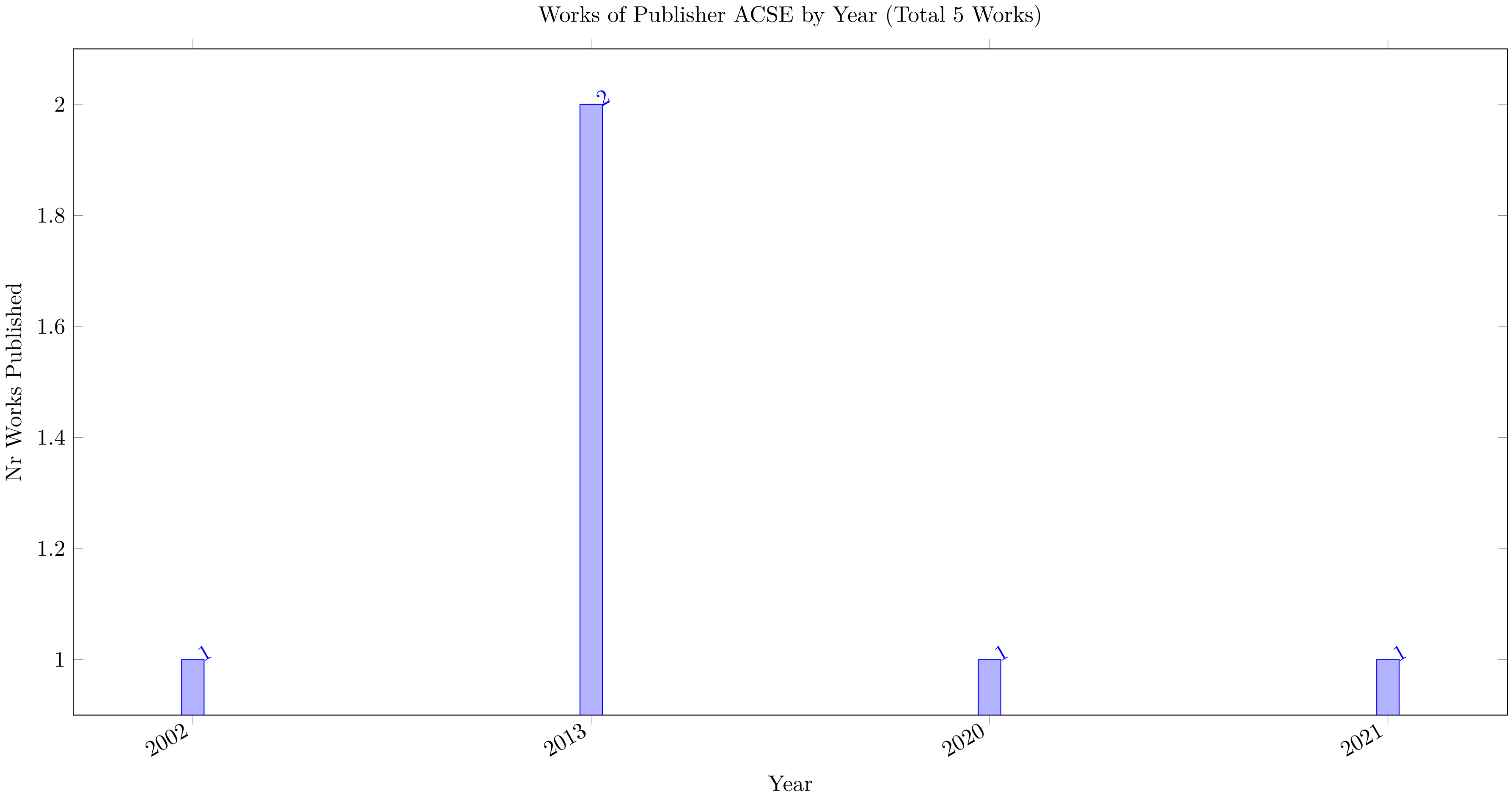
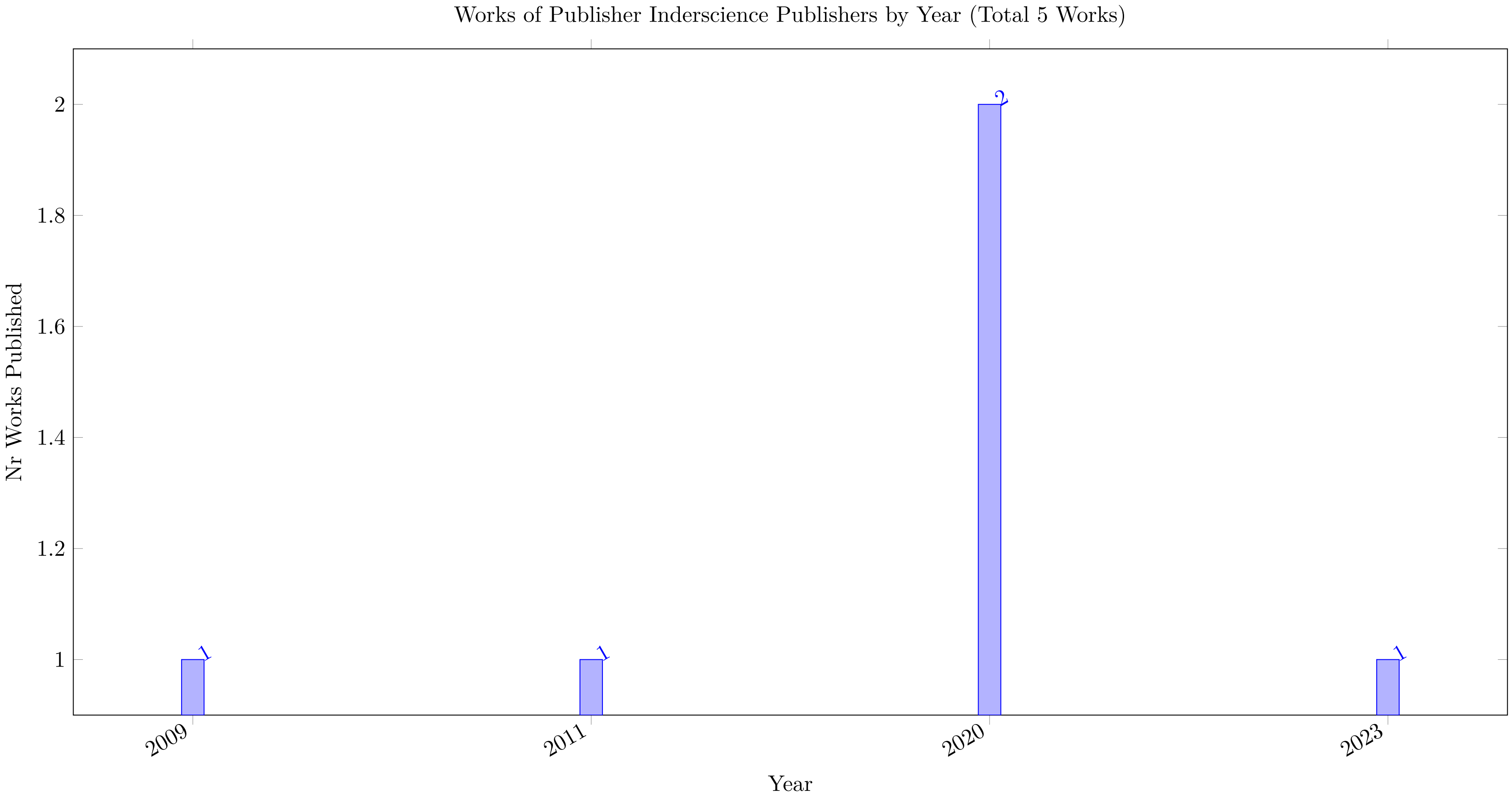
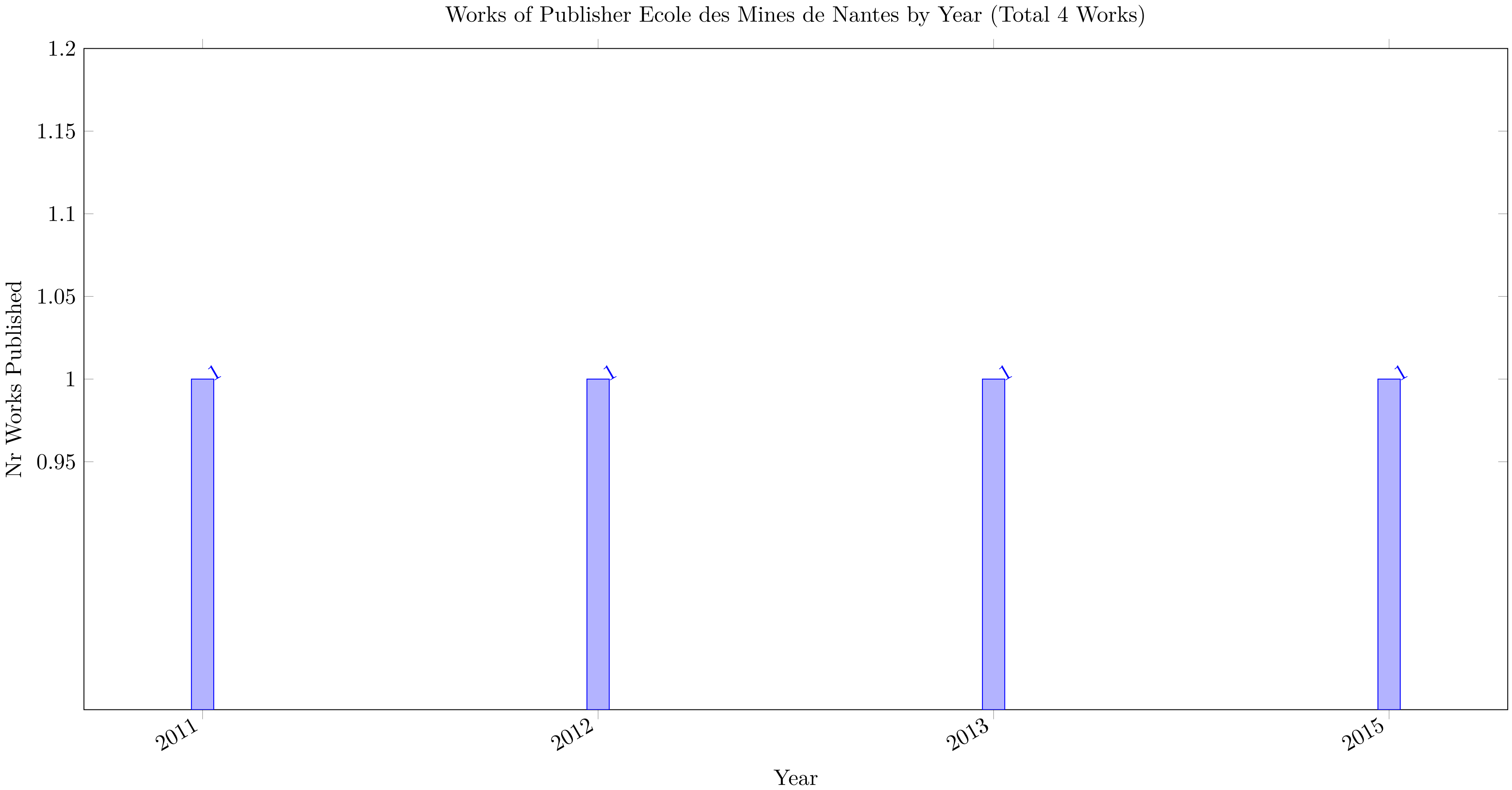
1.4
1.6
1.8
2


Works of Publisher ACSE by Year (Total 5 Works)
Nr Works Published
Year
2002
2013
2020
2021
1
2
1
1
1
1.2
1.4
1.6
1.8
2


Works of Publisher Inderscience Publishers by Year (Total 5 Works)
Nr Works Published
Year
2009
2011
2020
2023
1
1
2
1
1
1.2
1.4
1.6
1.8
2


Works of Publisher Ecole des Mines de Nantes by Year (Total 4 Works)
Nr Works Published
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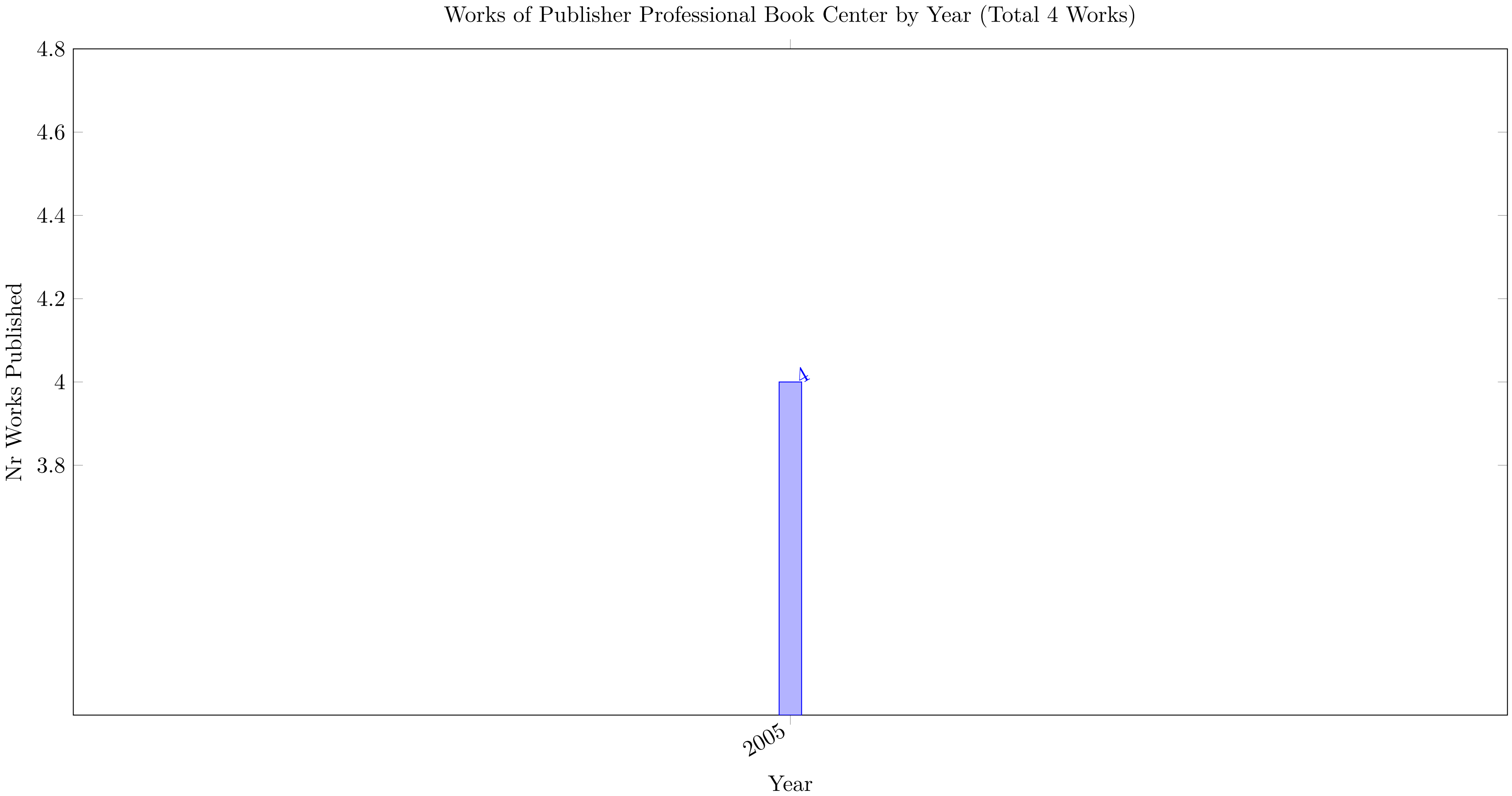
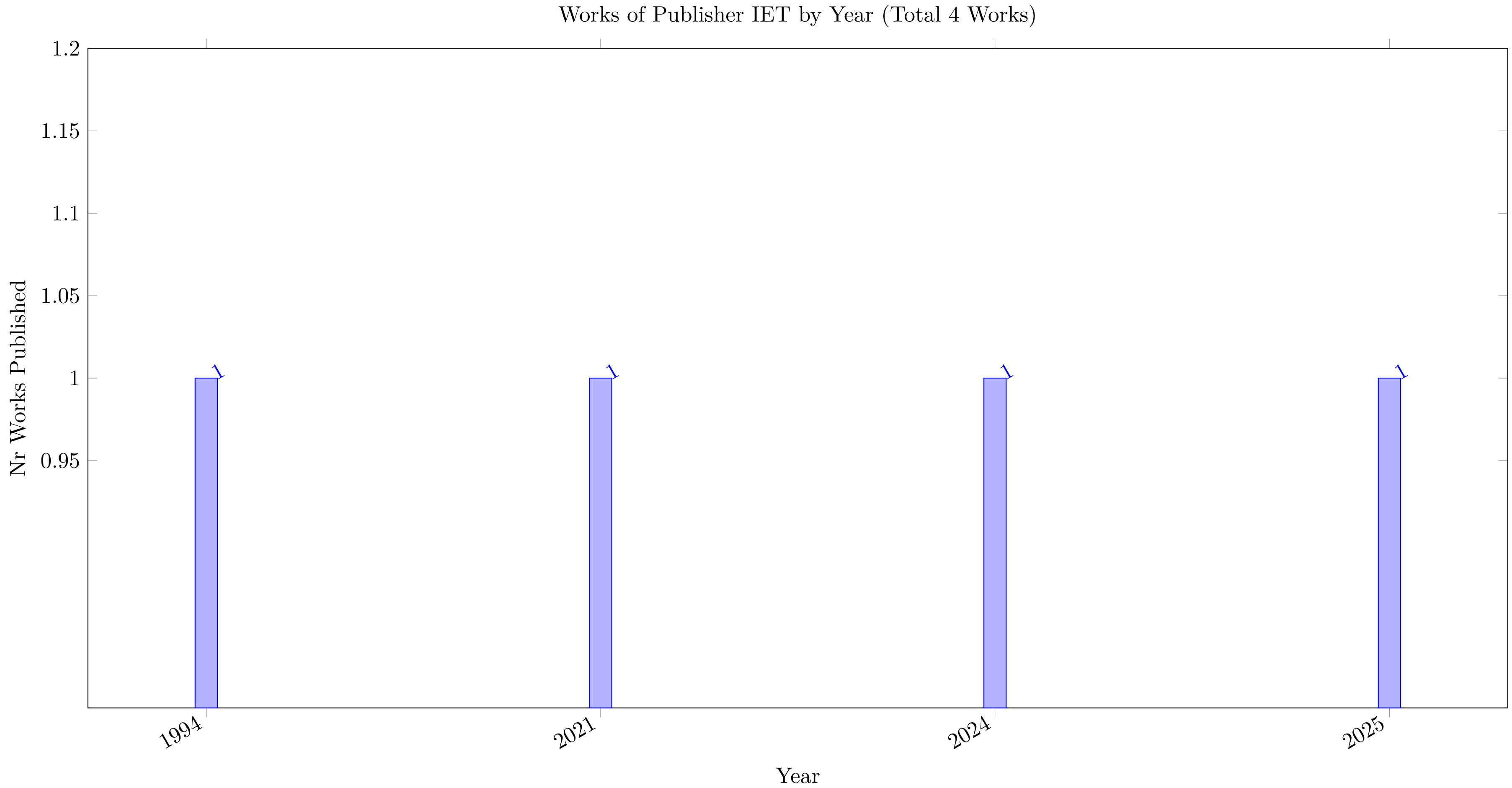
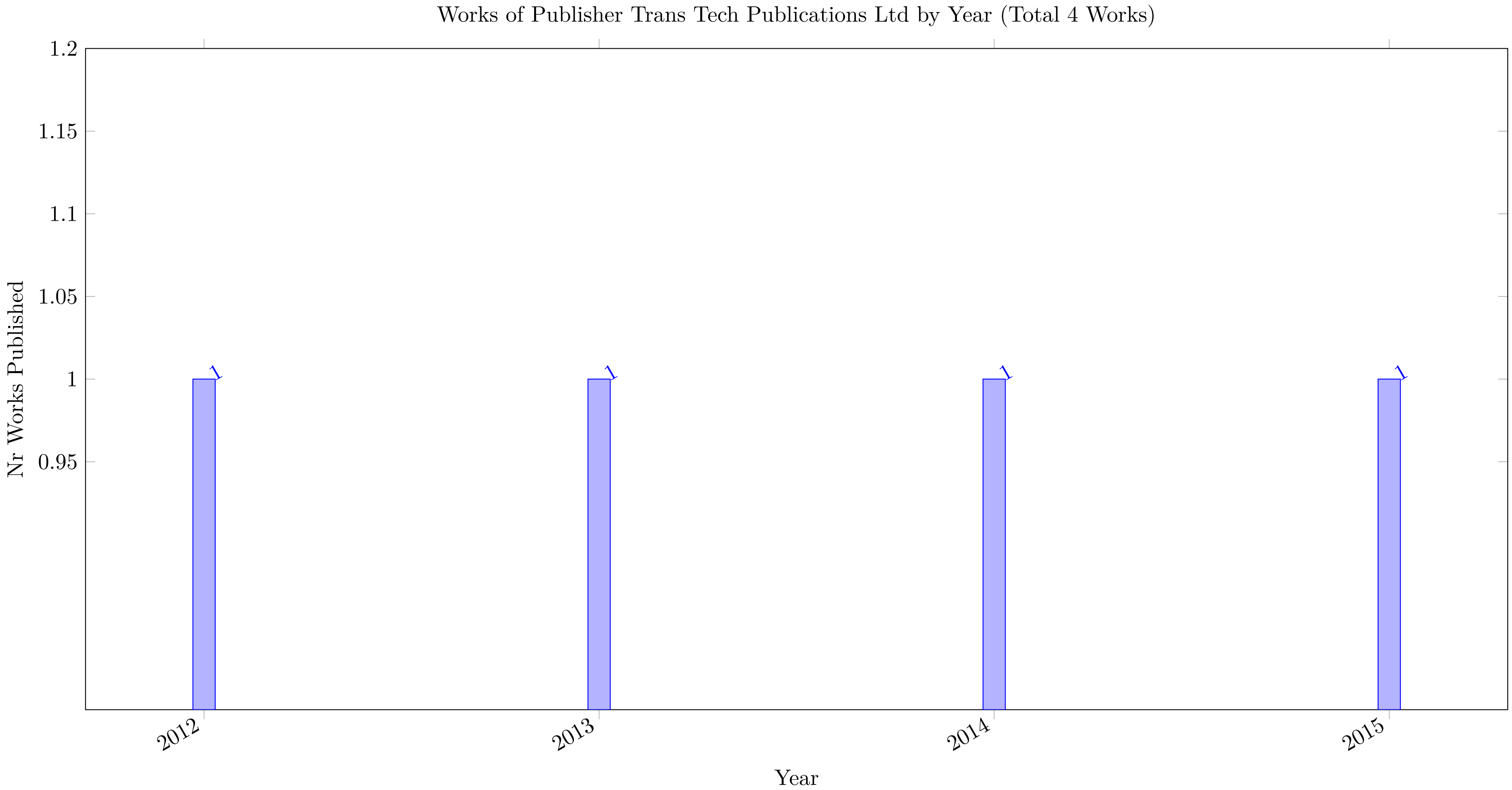
1.15
1.1
1.05
1
0.95
1
1
1
1
2011
2012
2013
2015
Year


Works of Publisher Professional Book Center by Year (Total 4 Works)
Nr Works Published
4.8
4.6
4.4
4.2
4
3.8
4
2005
Year


Works of Publisher IET by Year (Total 4 Works)
Nr Works Published
1.2
1.15
1.1
1.05
1
0.95
1
1
1
1
1994
2021
2024
2025
Year


Works of Publisher Trans Tech Publications Ltd by Year (Total 4 Works)
Nr Works Published
Year
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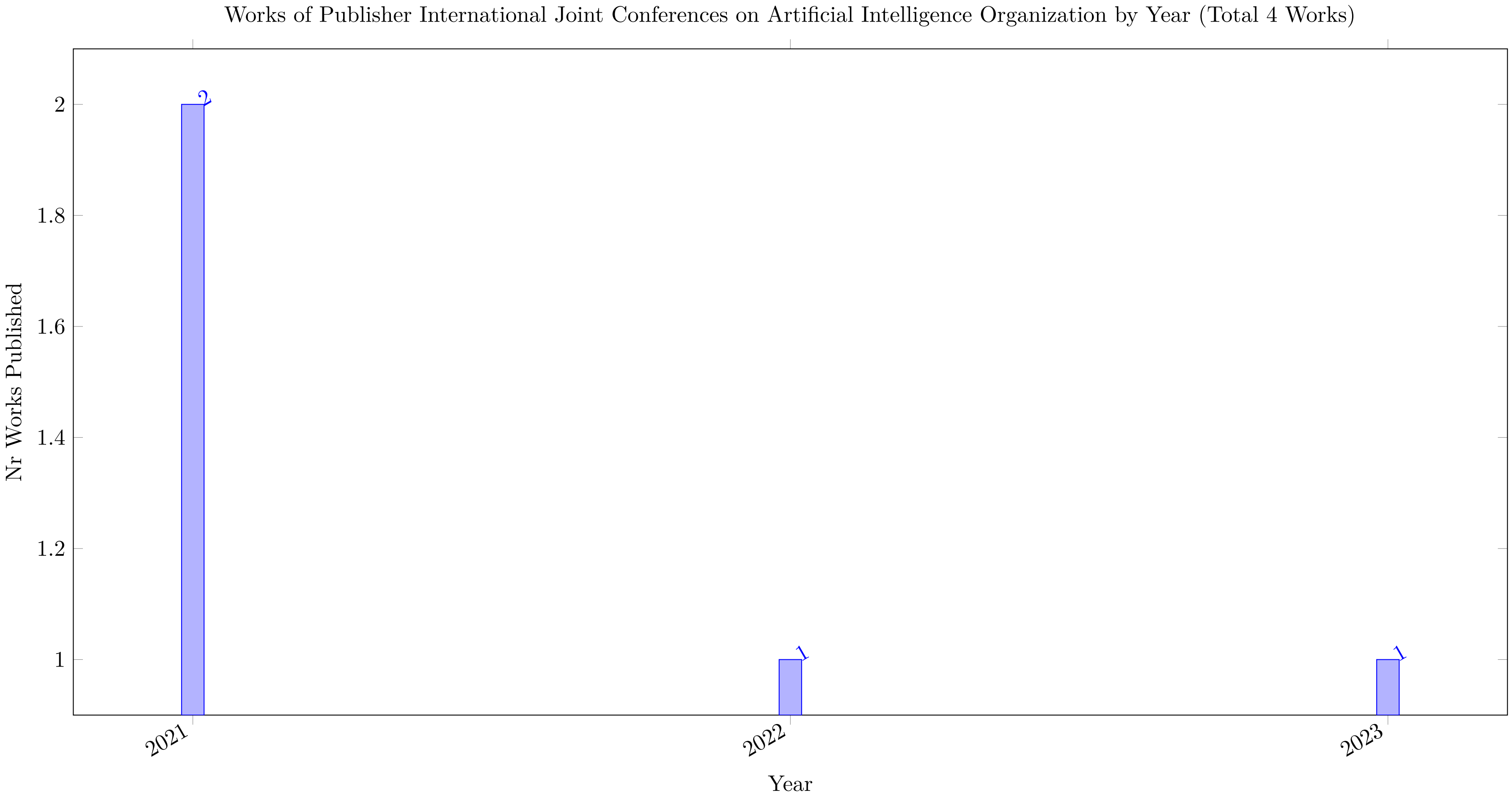
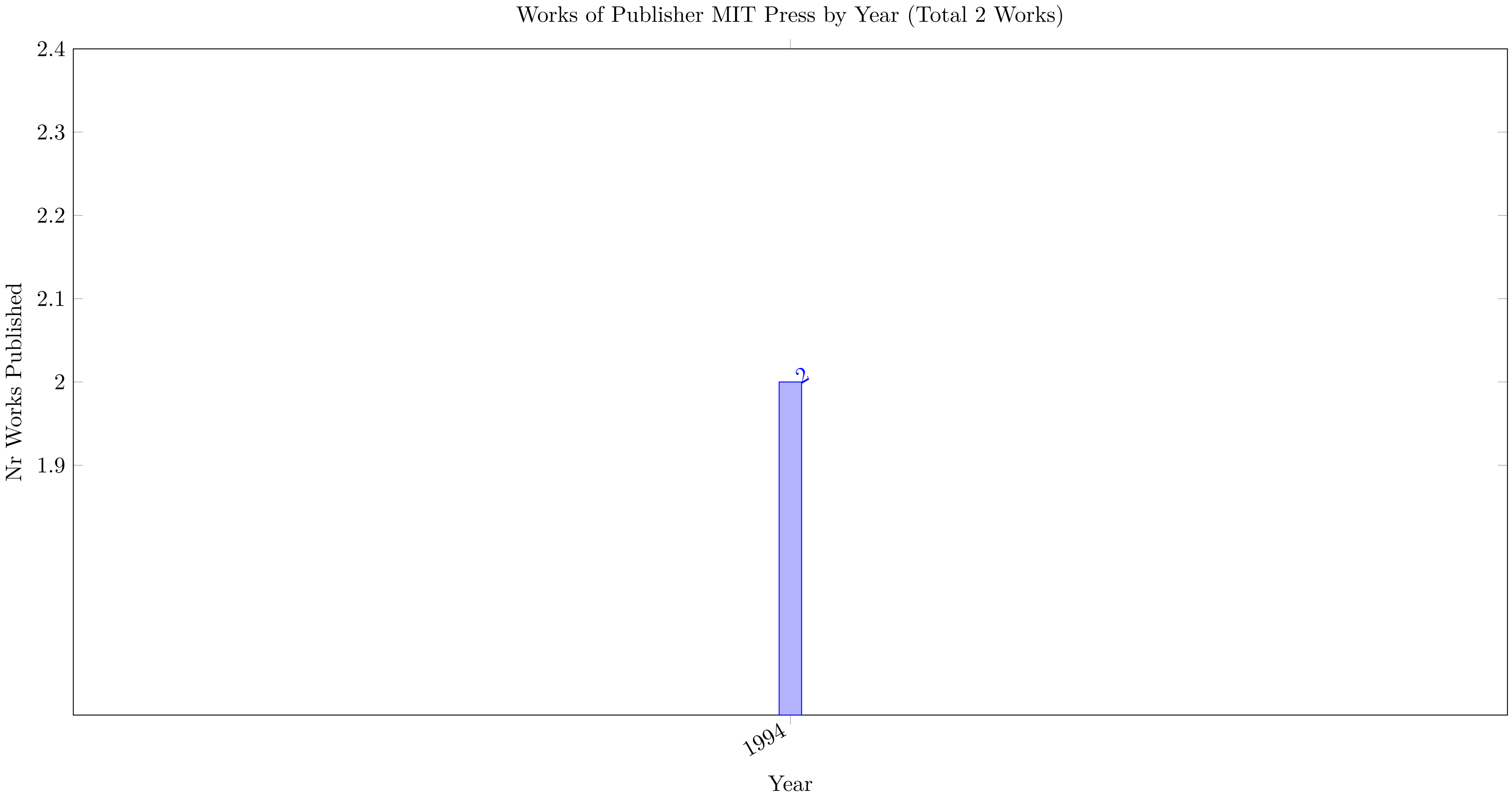
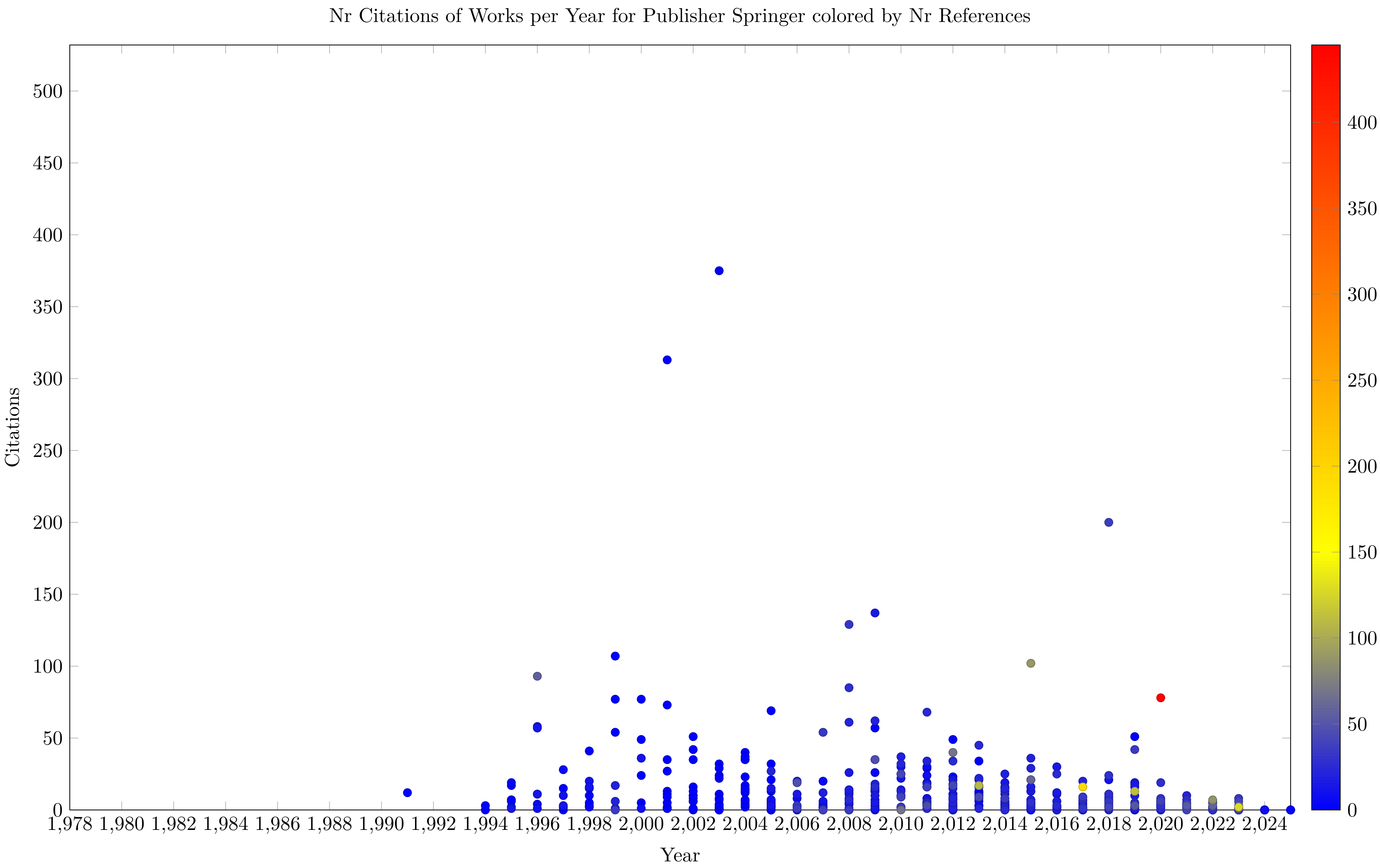
1.15
1.1
1.05
1
0.95
2012
2013
2014
2015
1
1
1
1


Works of Publisher International Joint Conferences on Artificial Intelligence Organization by Year (Total 4 Works)
Nr Works Published
Year
2
1.8
1.6
1.4
1.2
1
2021
2022
2023
2
1
1


Works of Publisher MIT Press by Year (Total 2 Works)
2.4
2.3
2.2
2.1
2
1.9
Nr Works Published
2
1994
Year


Nr Citations of Works per Year for Publisher Springer colored by Nr References
Citations
Year
0
50
100
150
200
250
300
350
400
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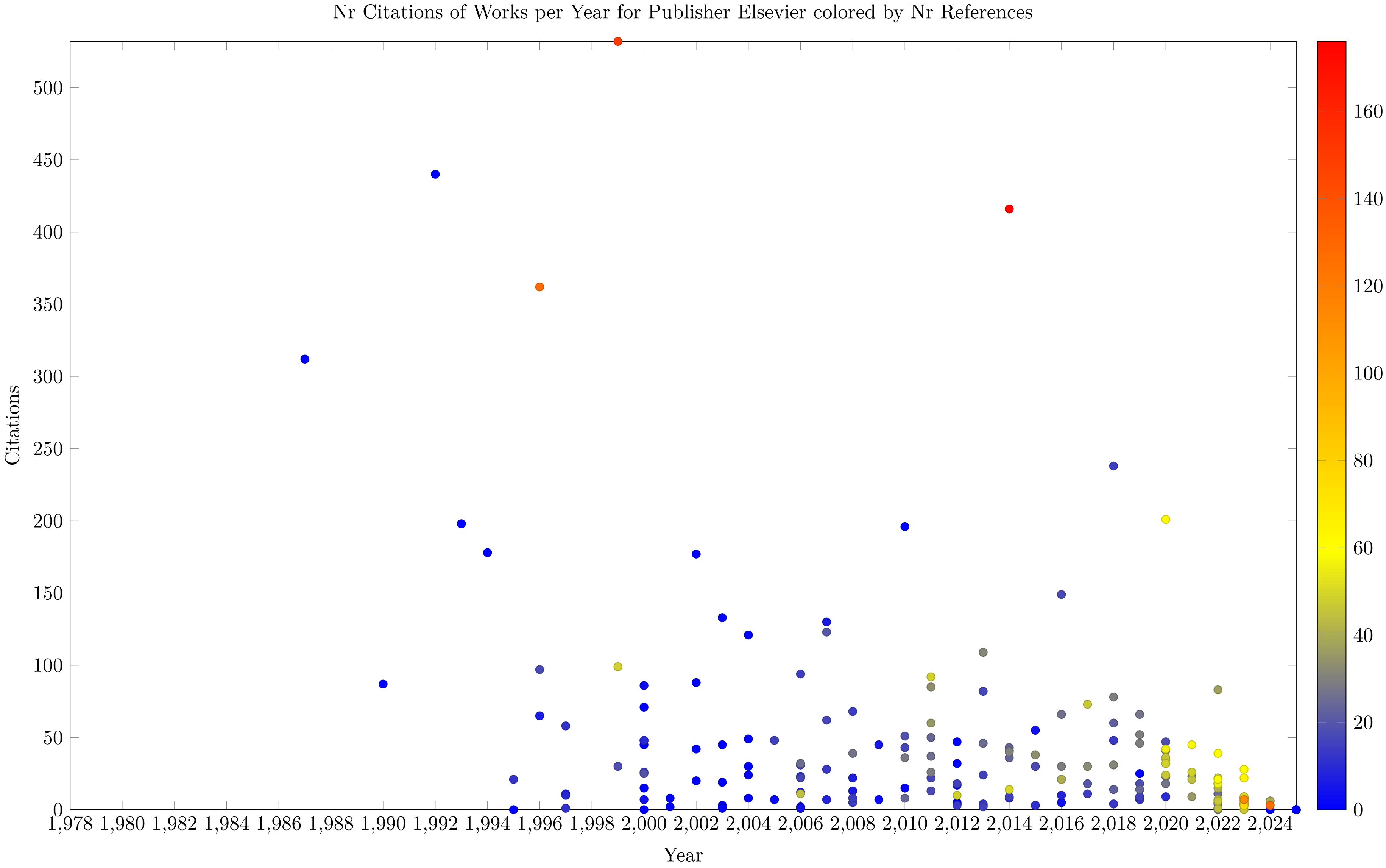
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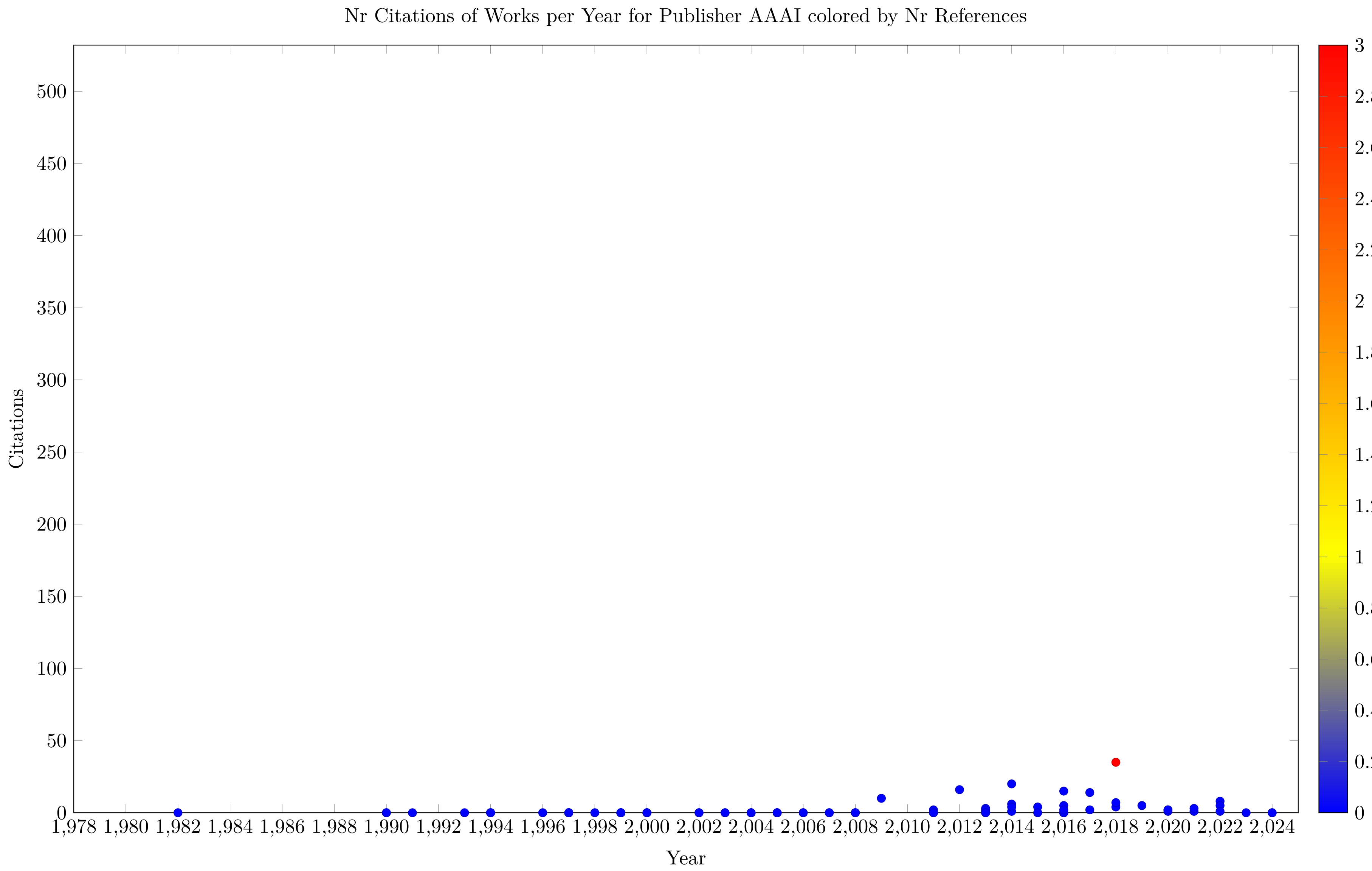
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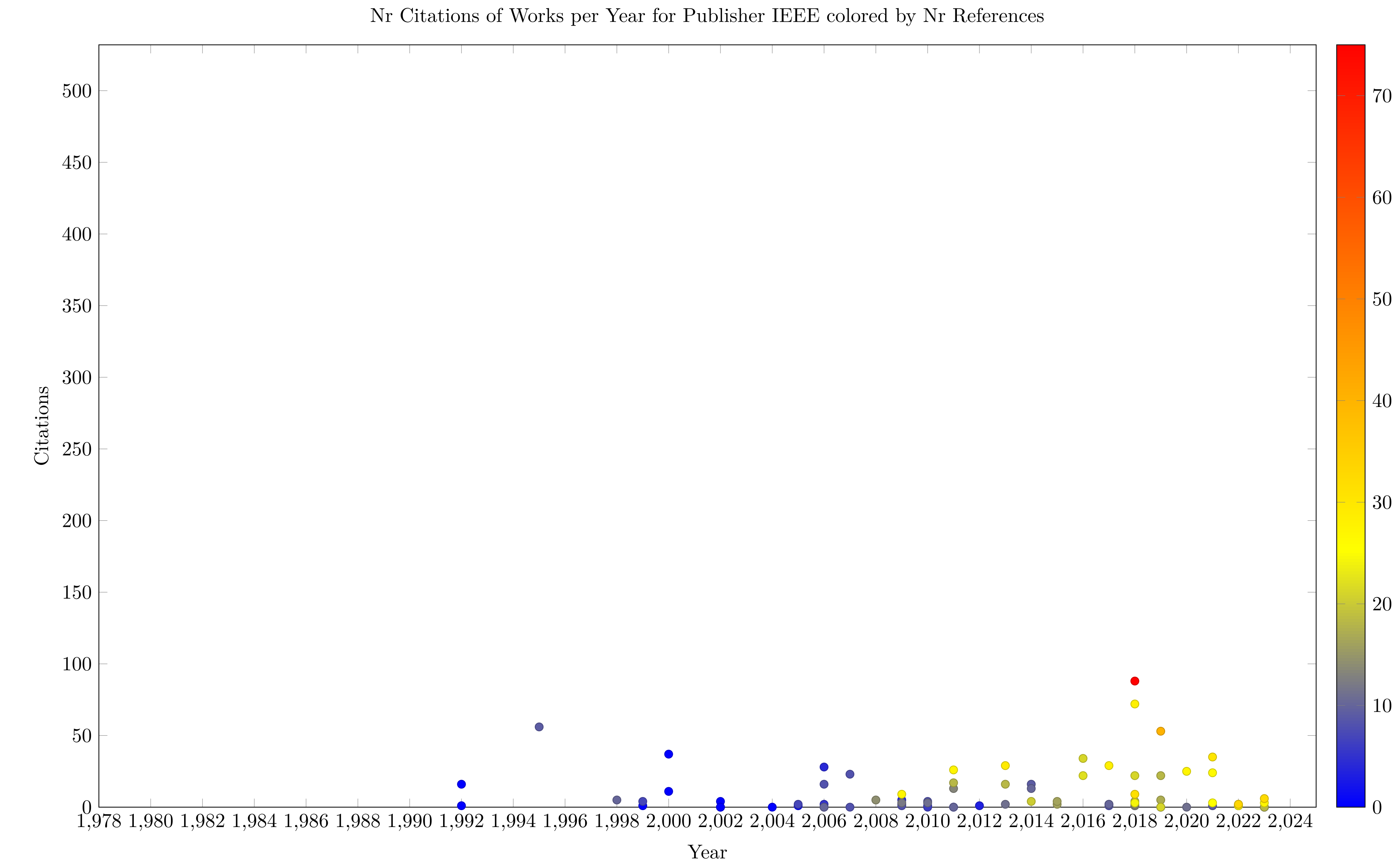
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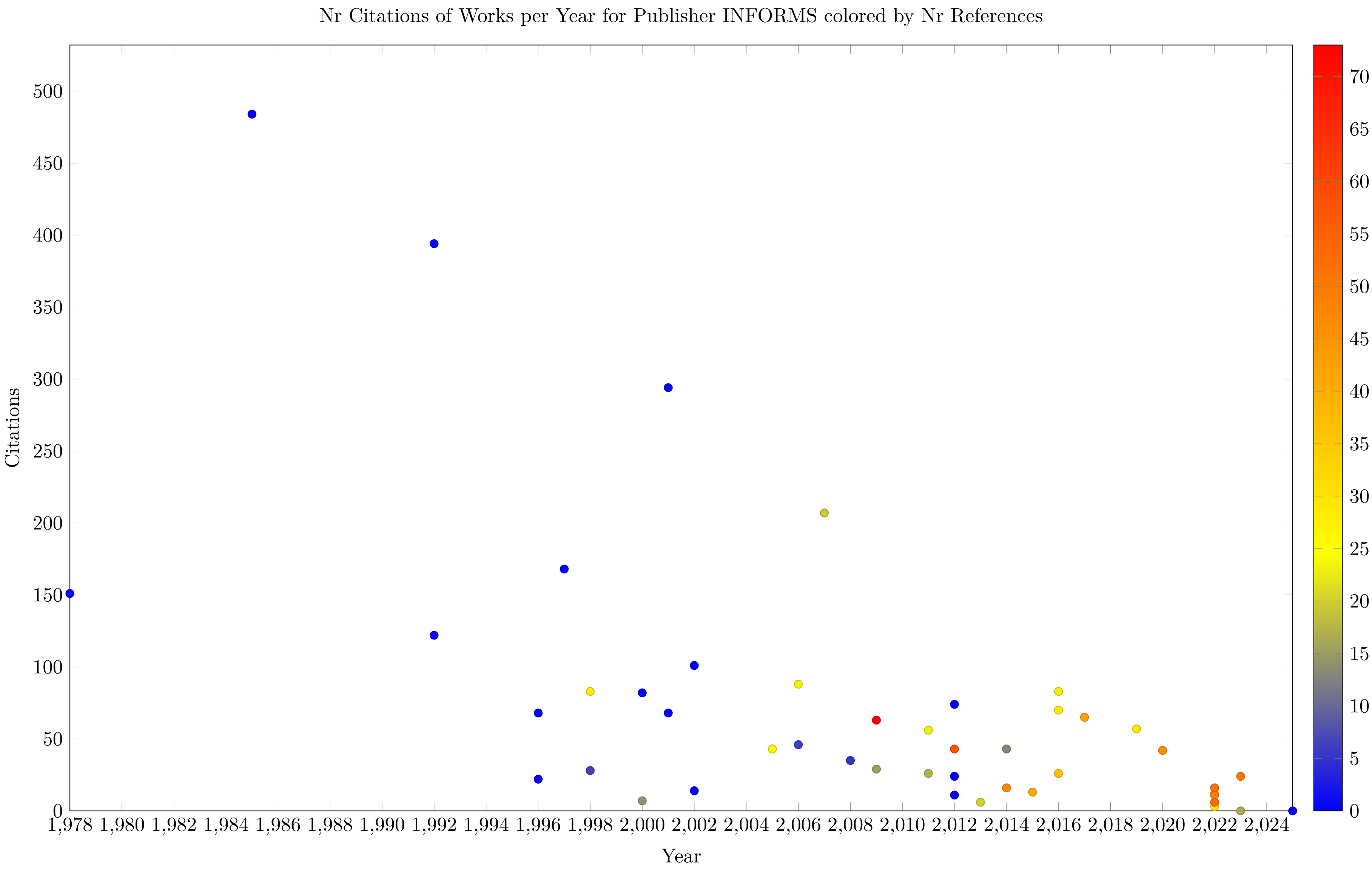
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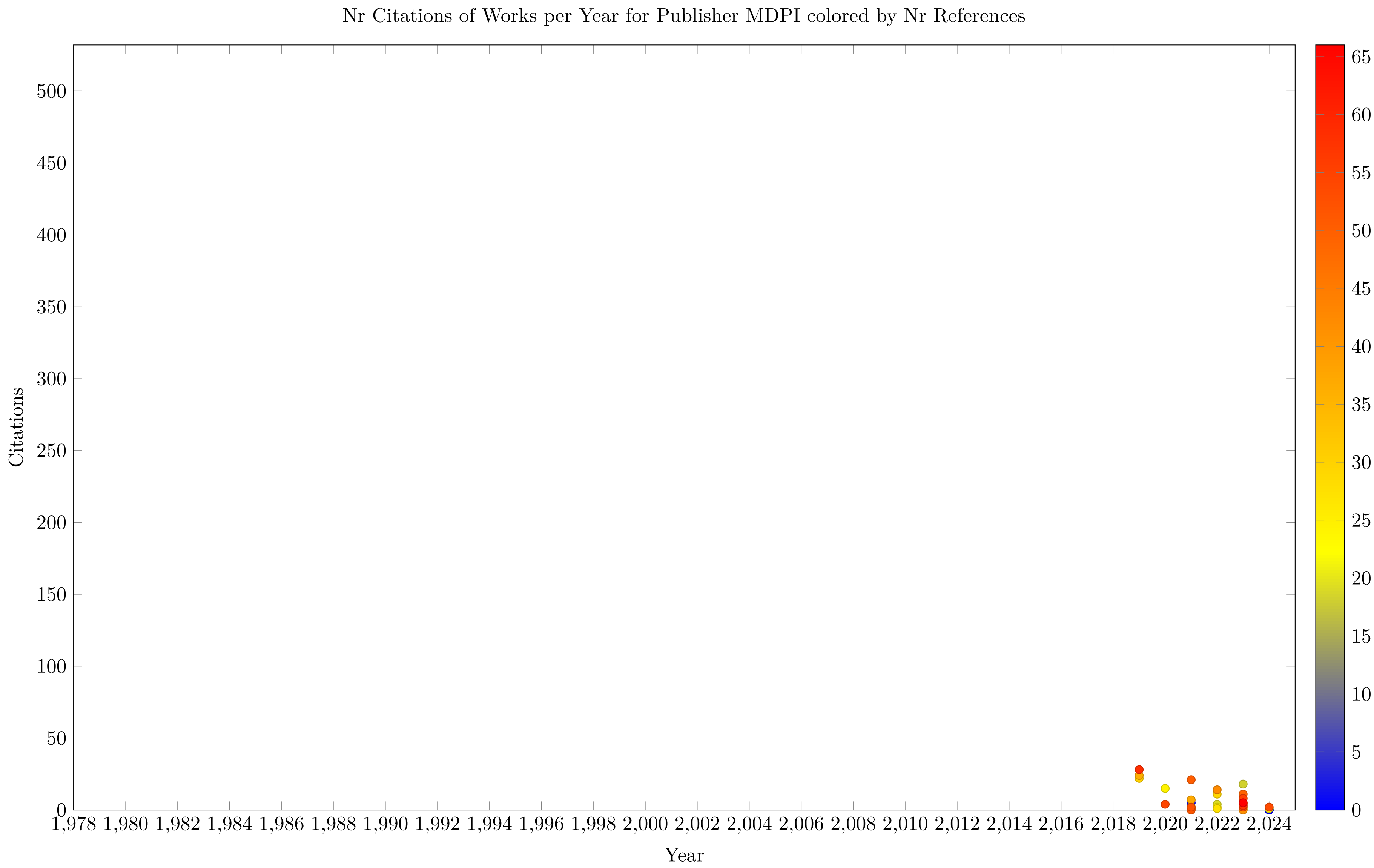
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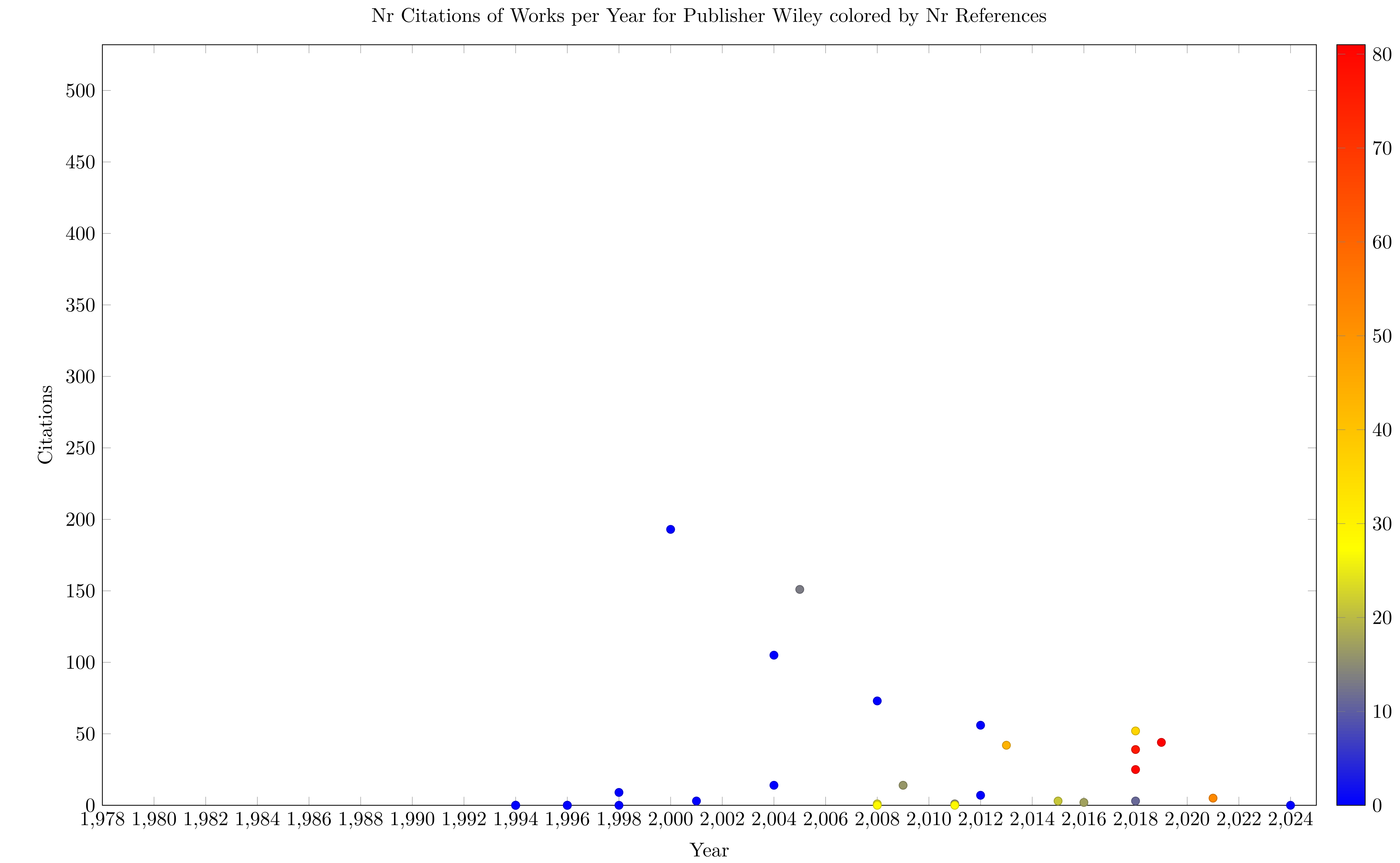
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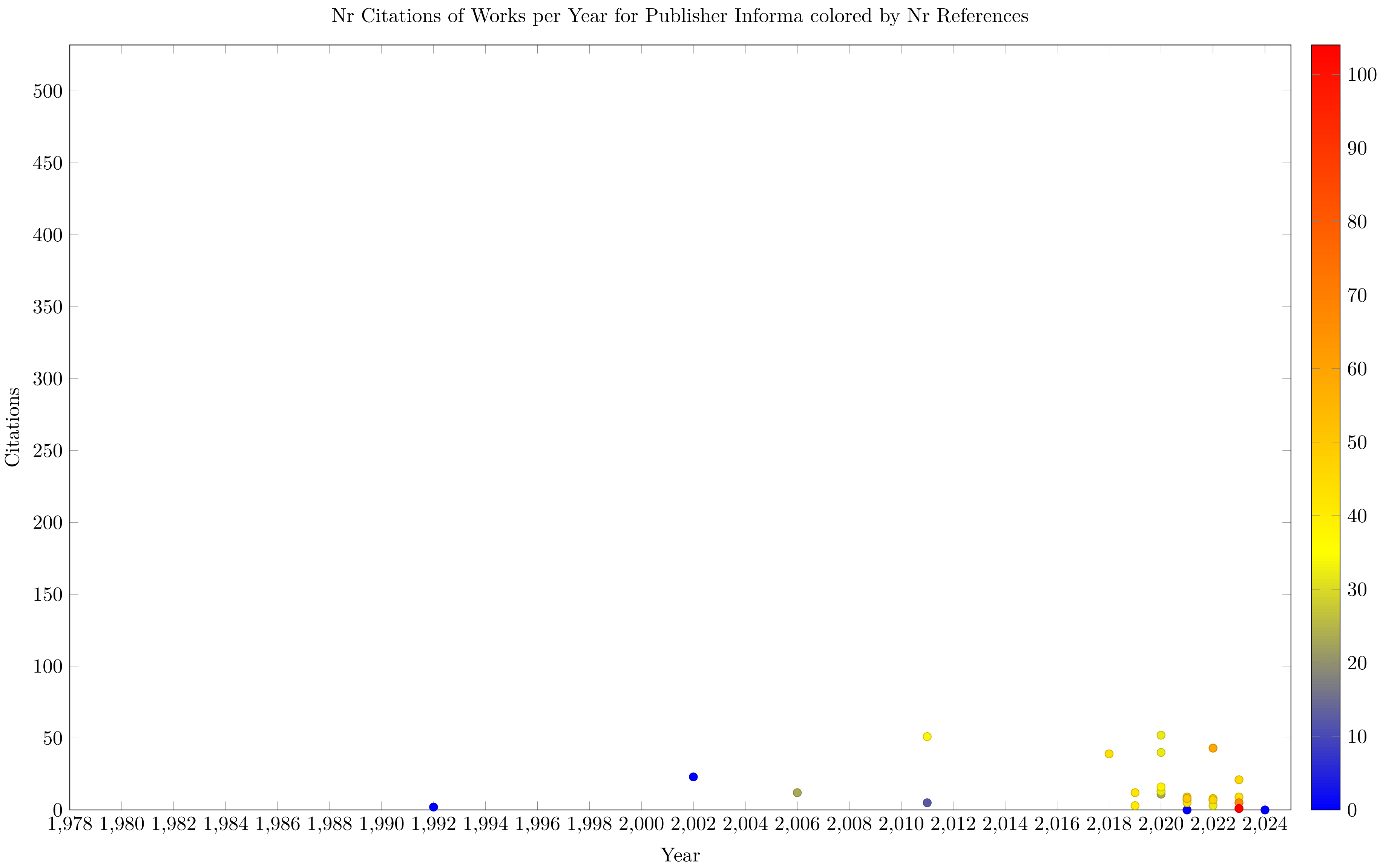
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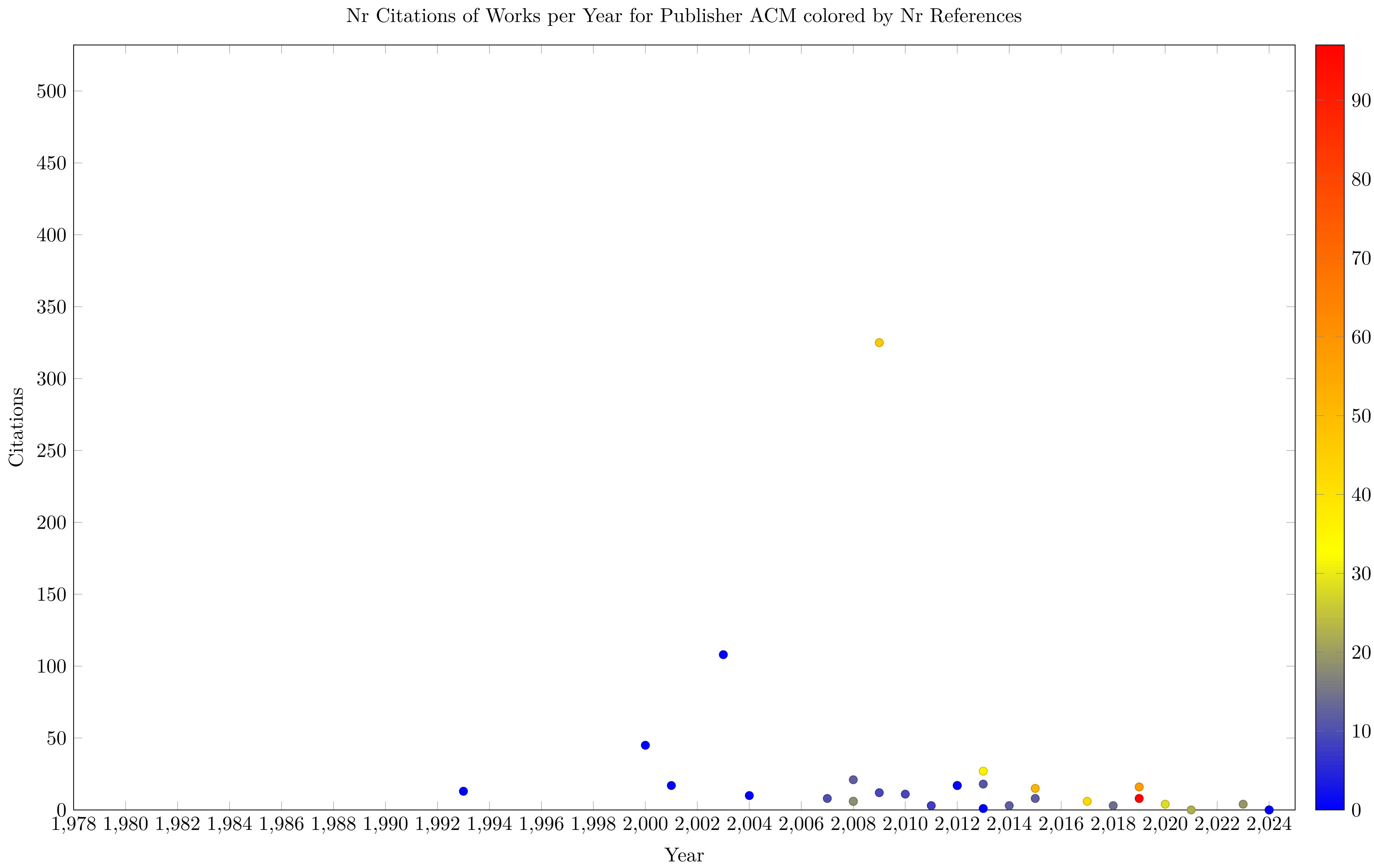

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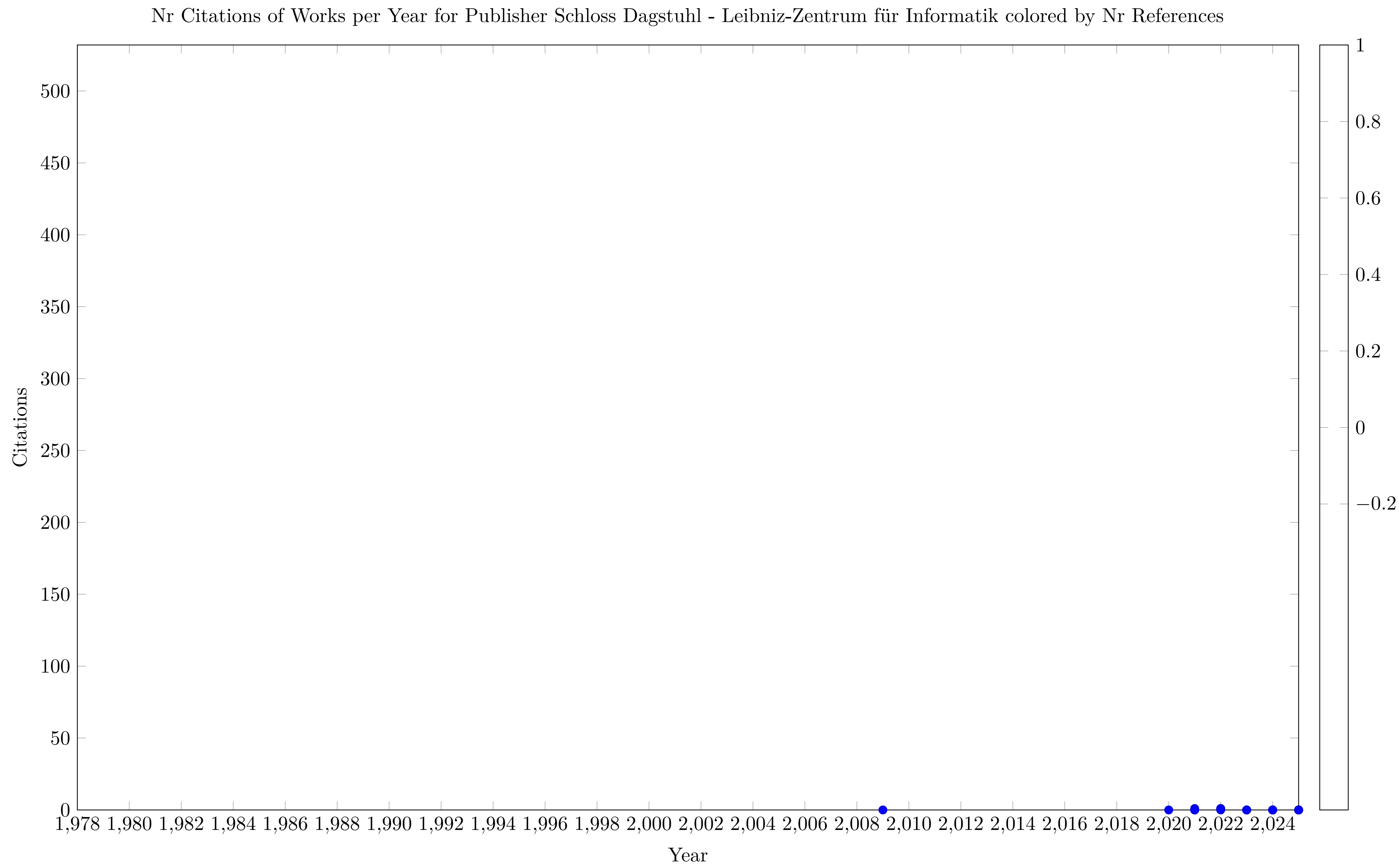
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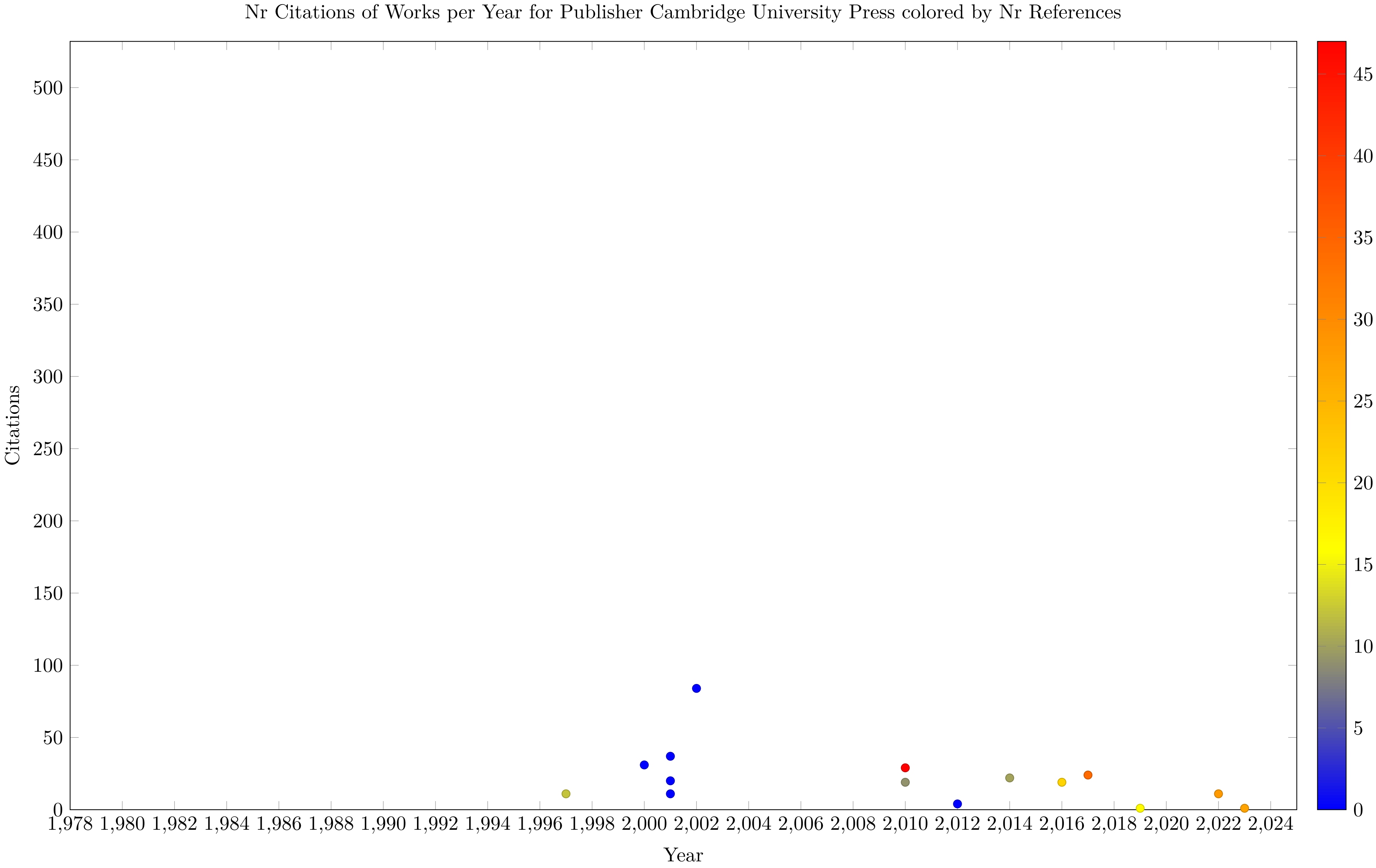
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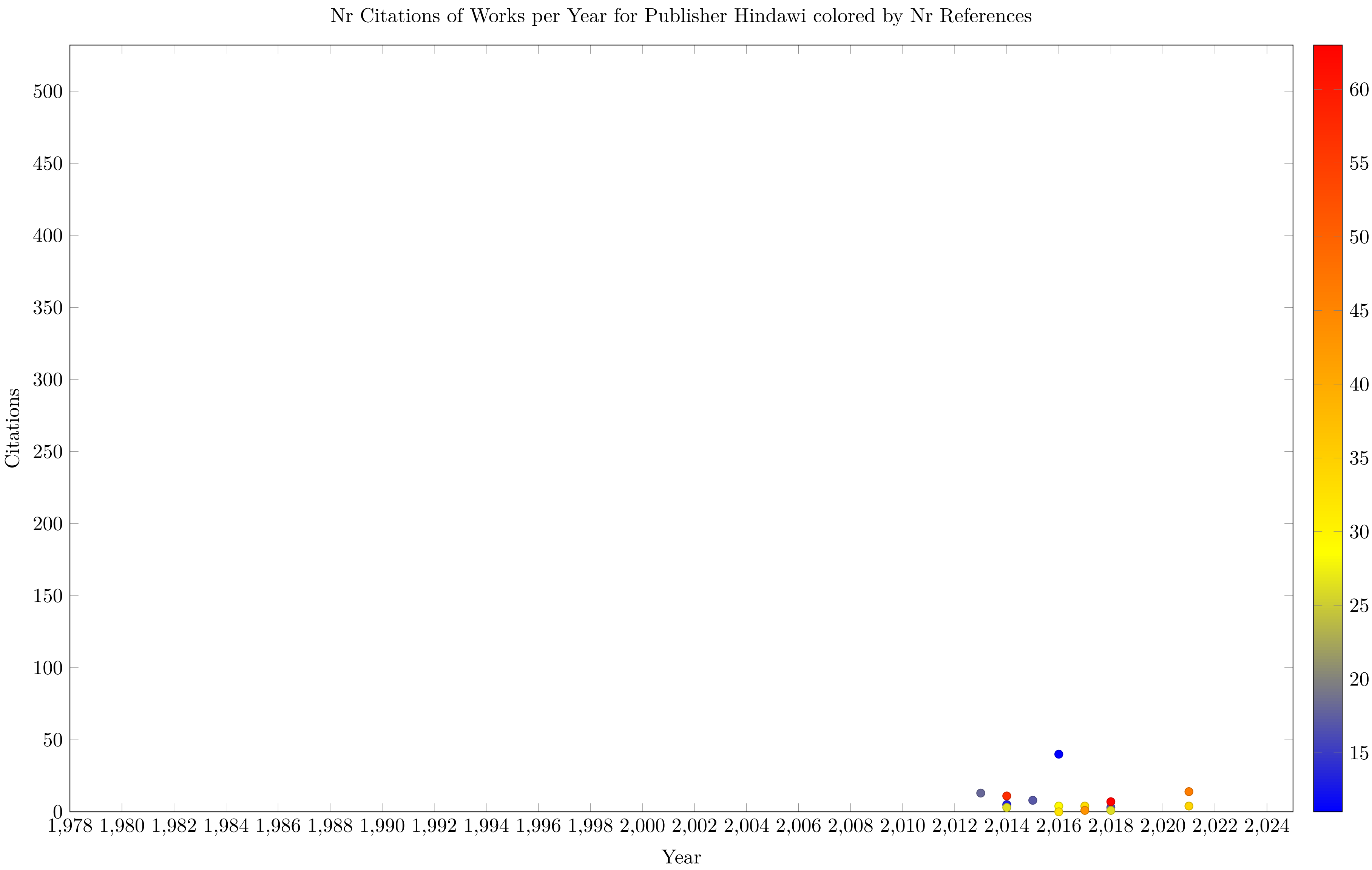
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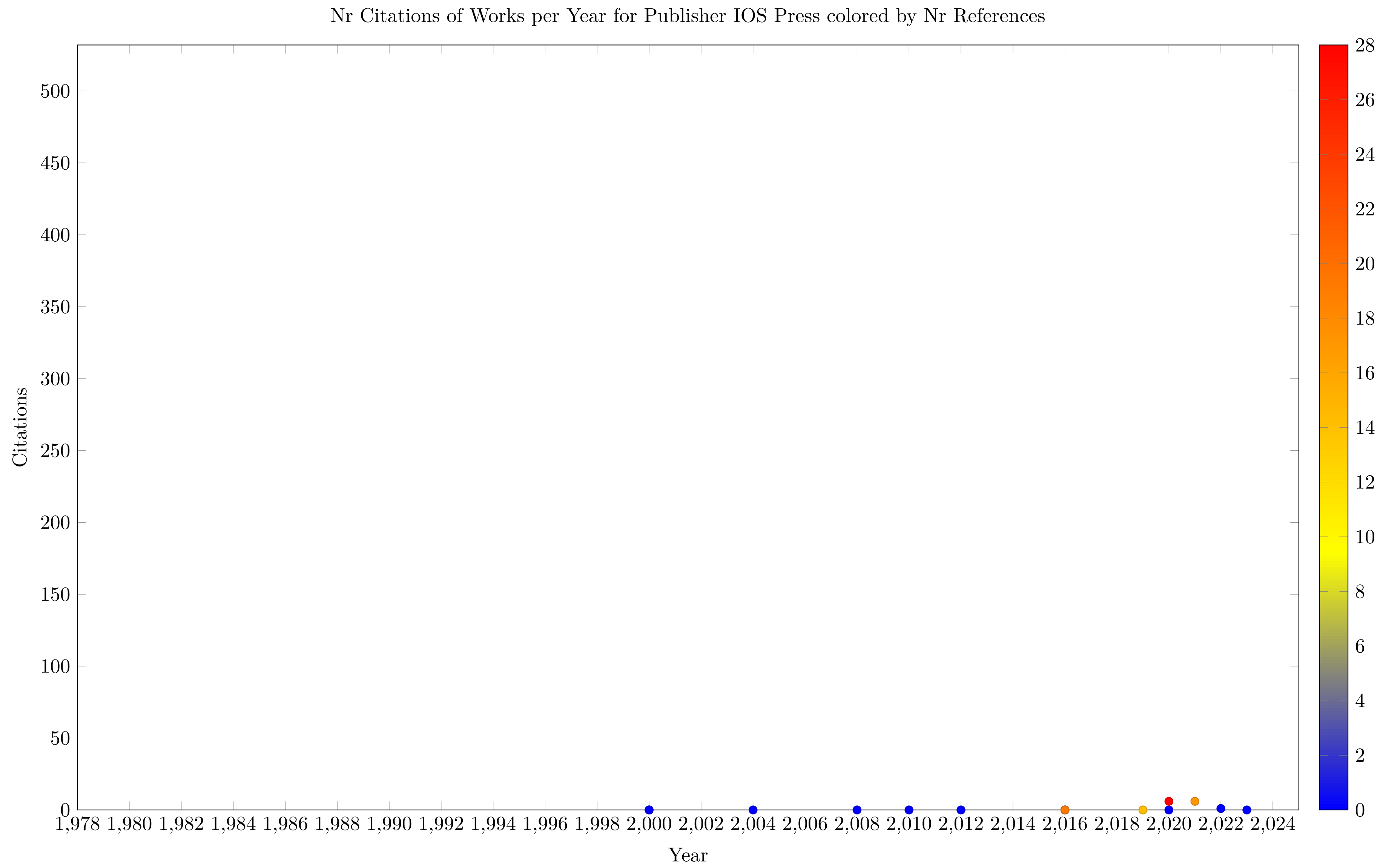
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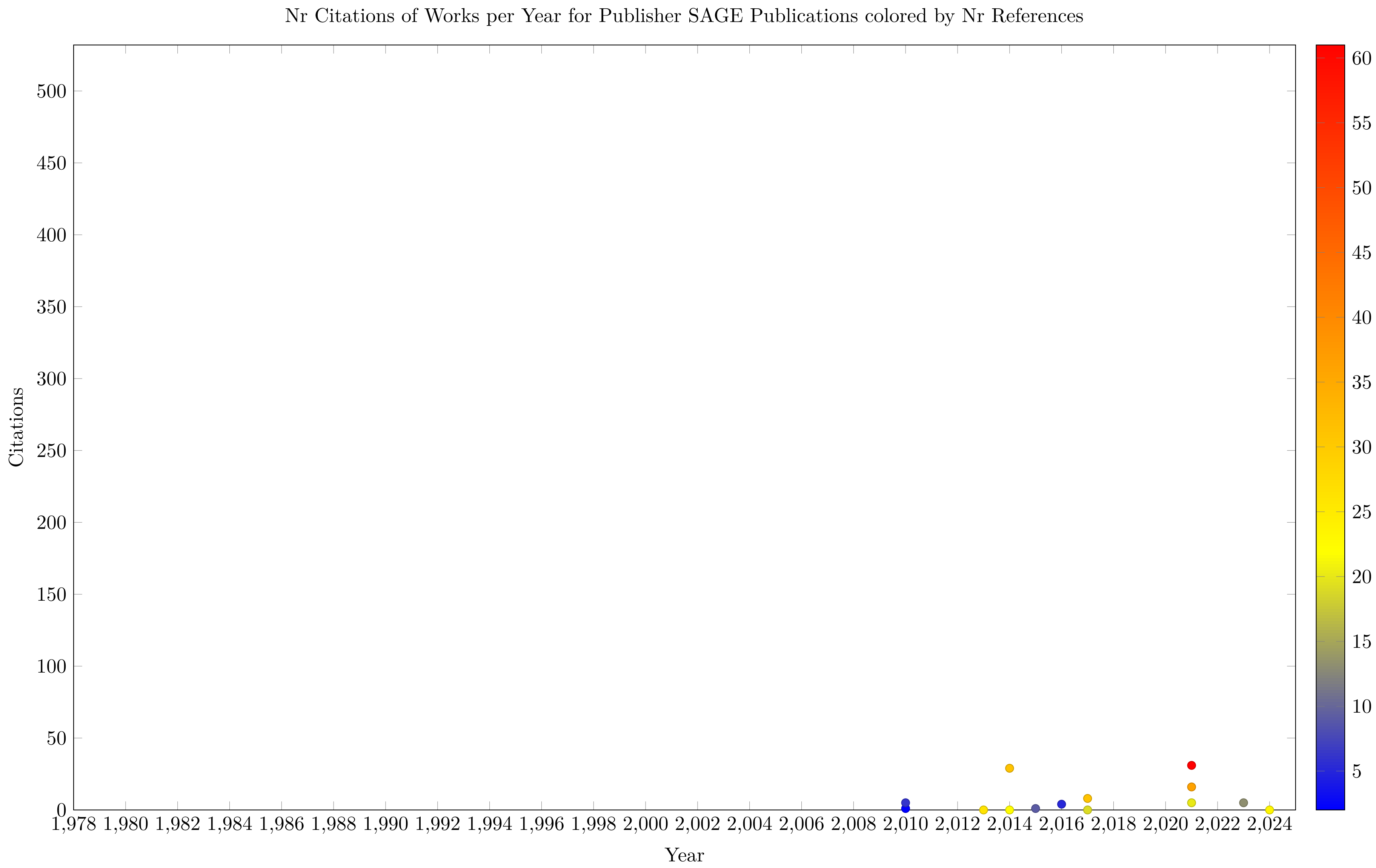
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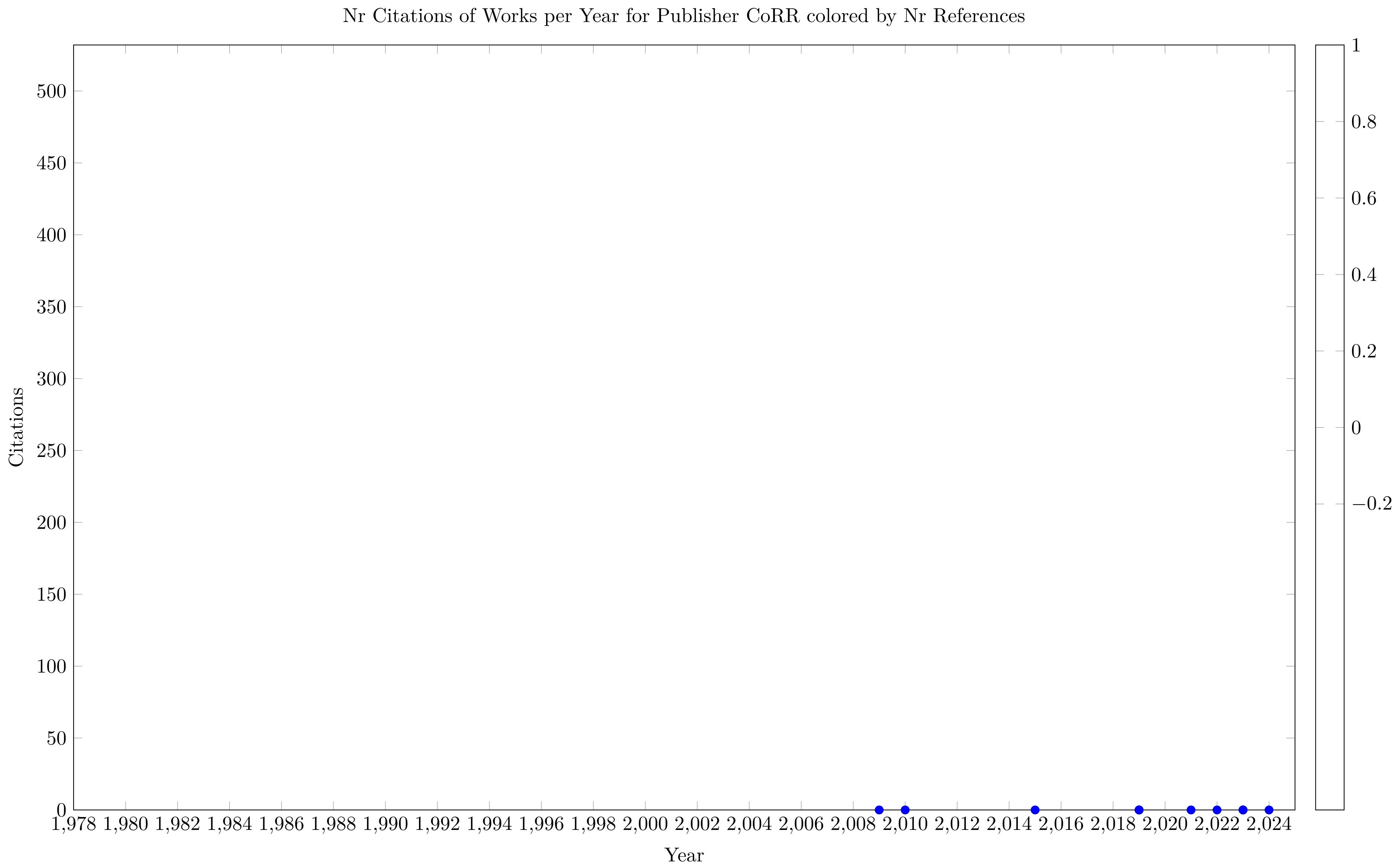
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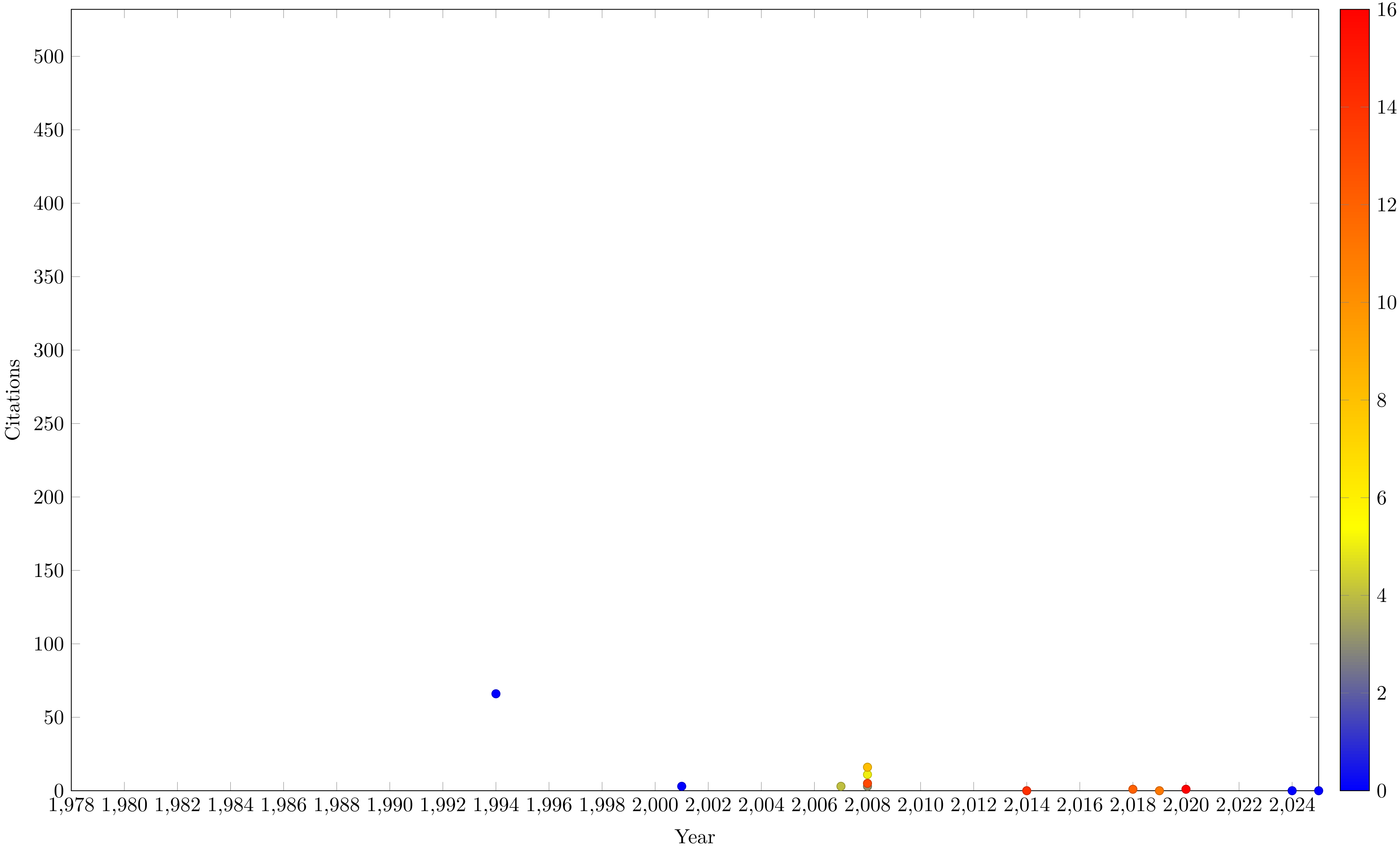

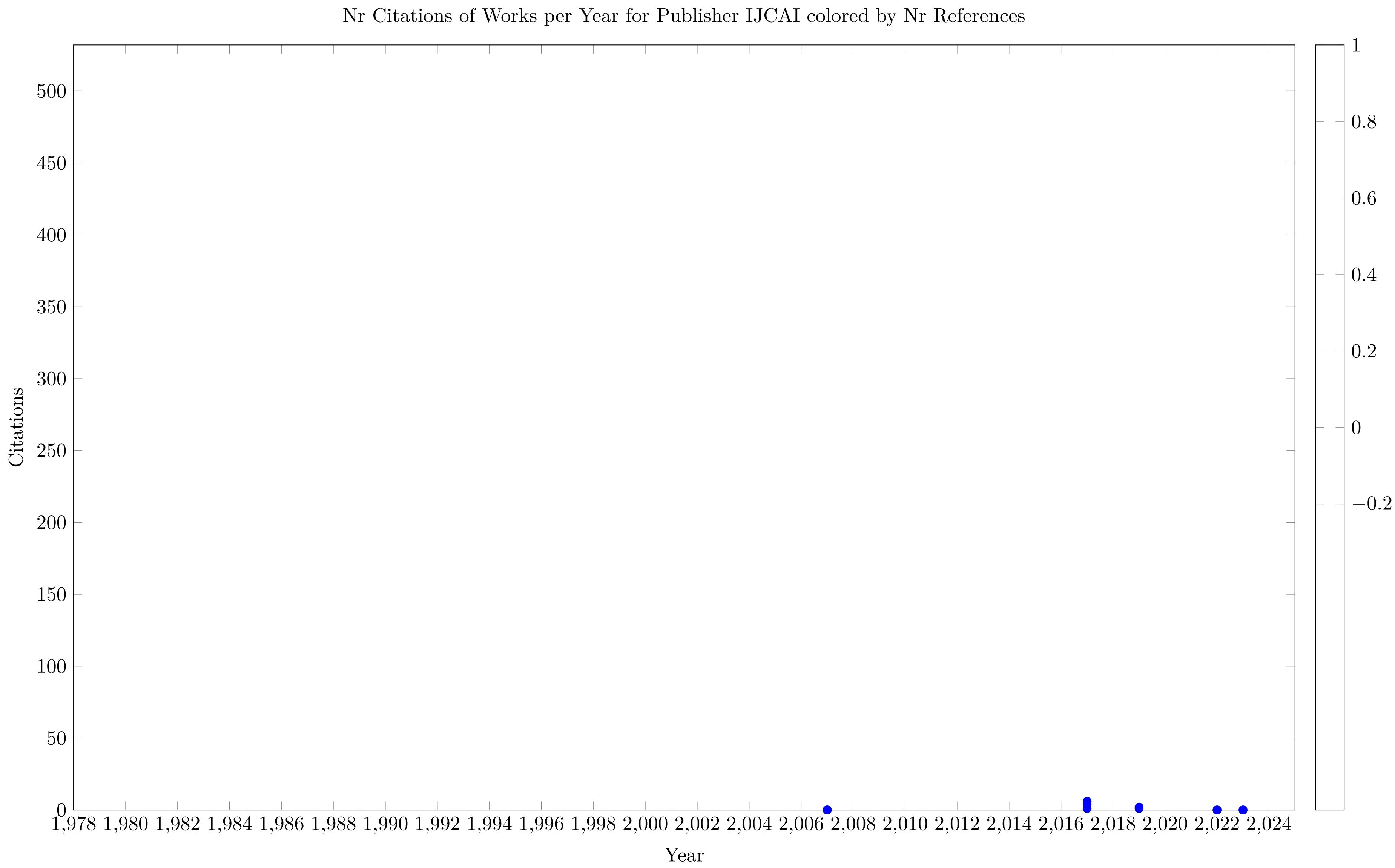
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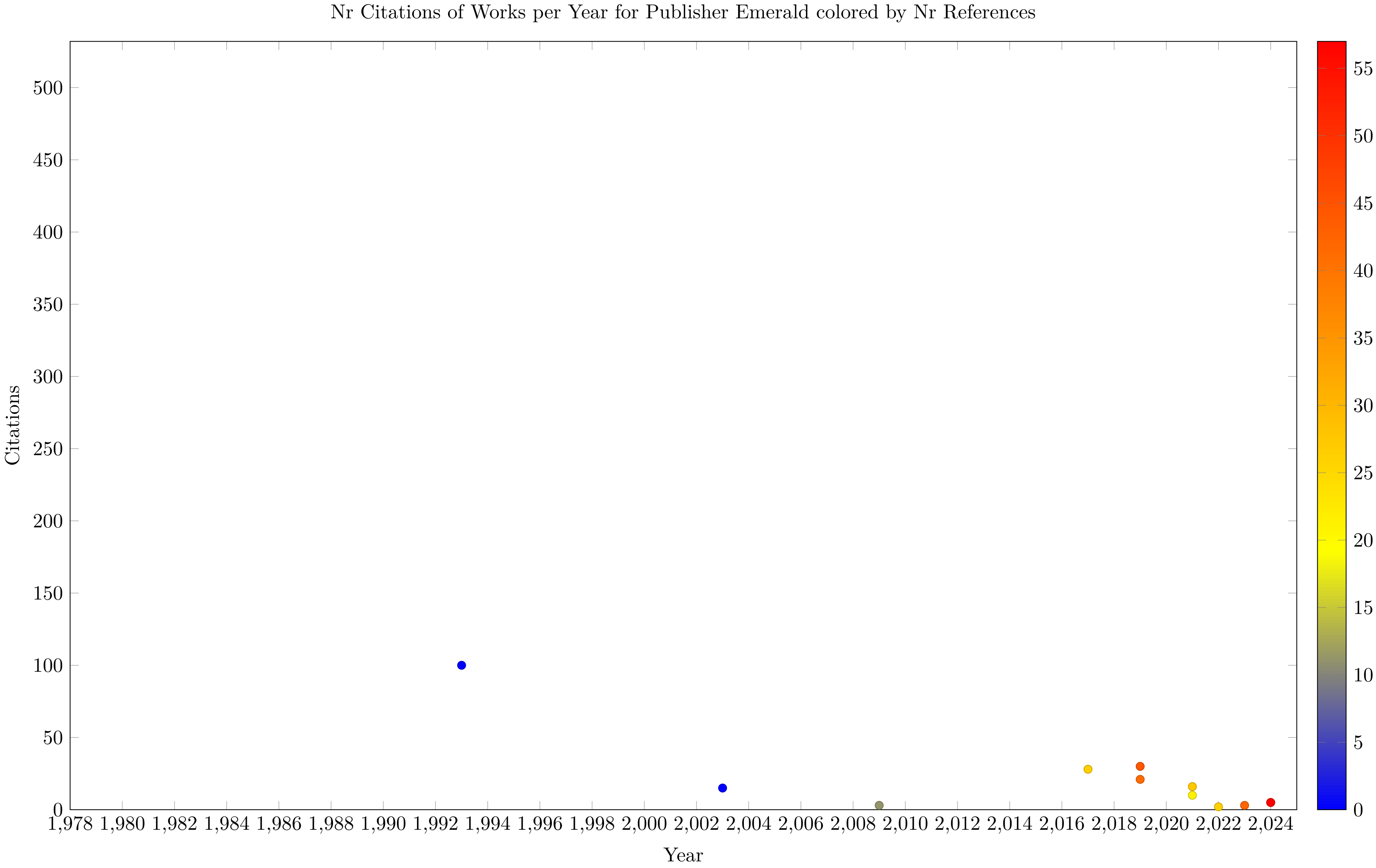
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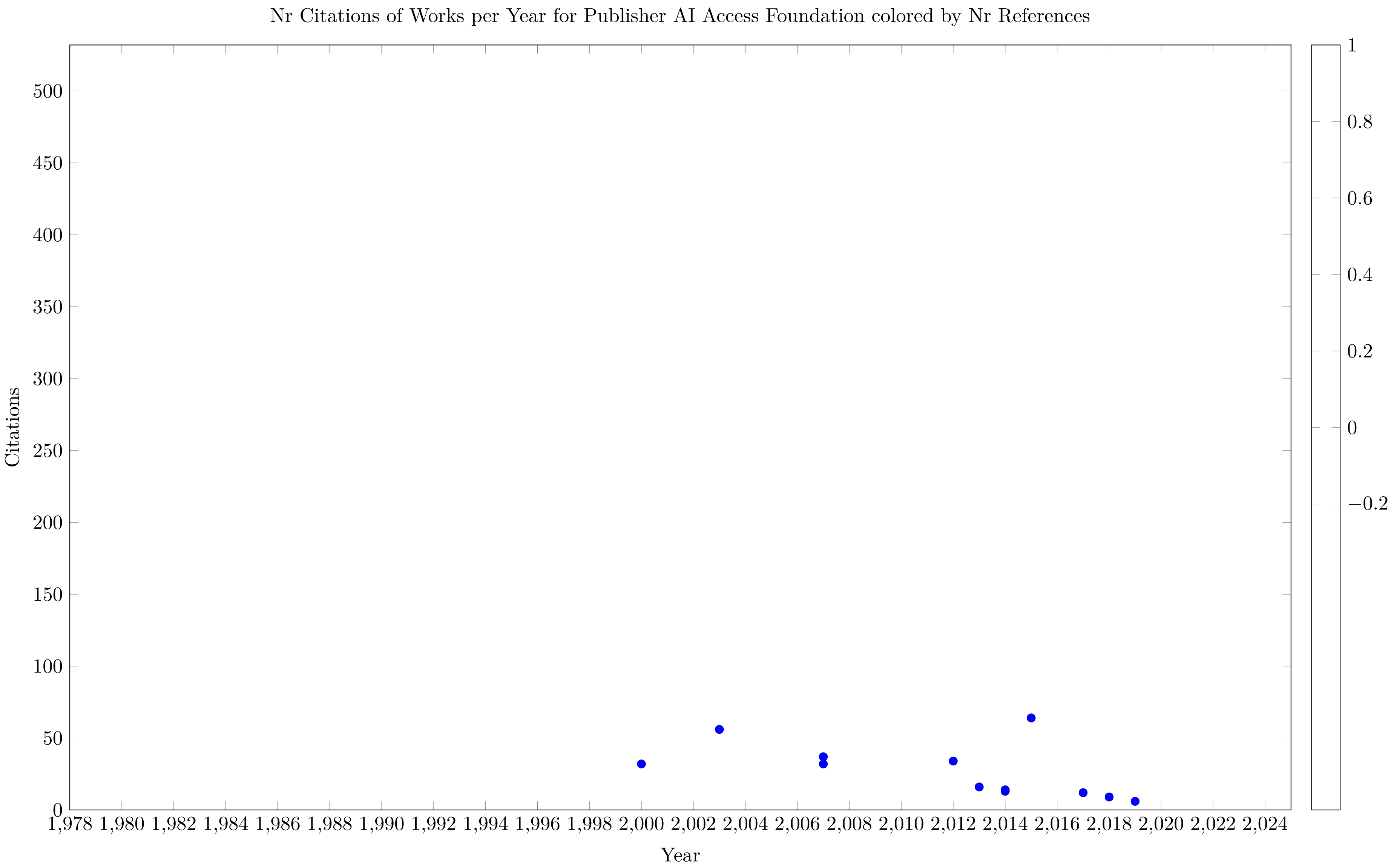
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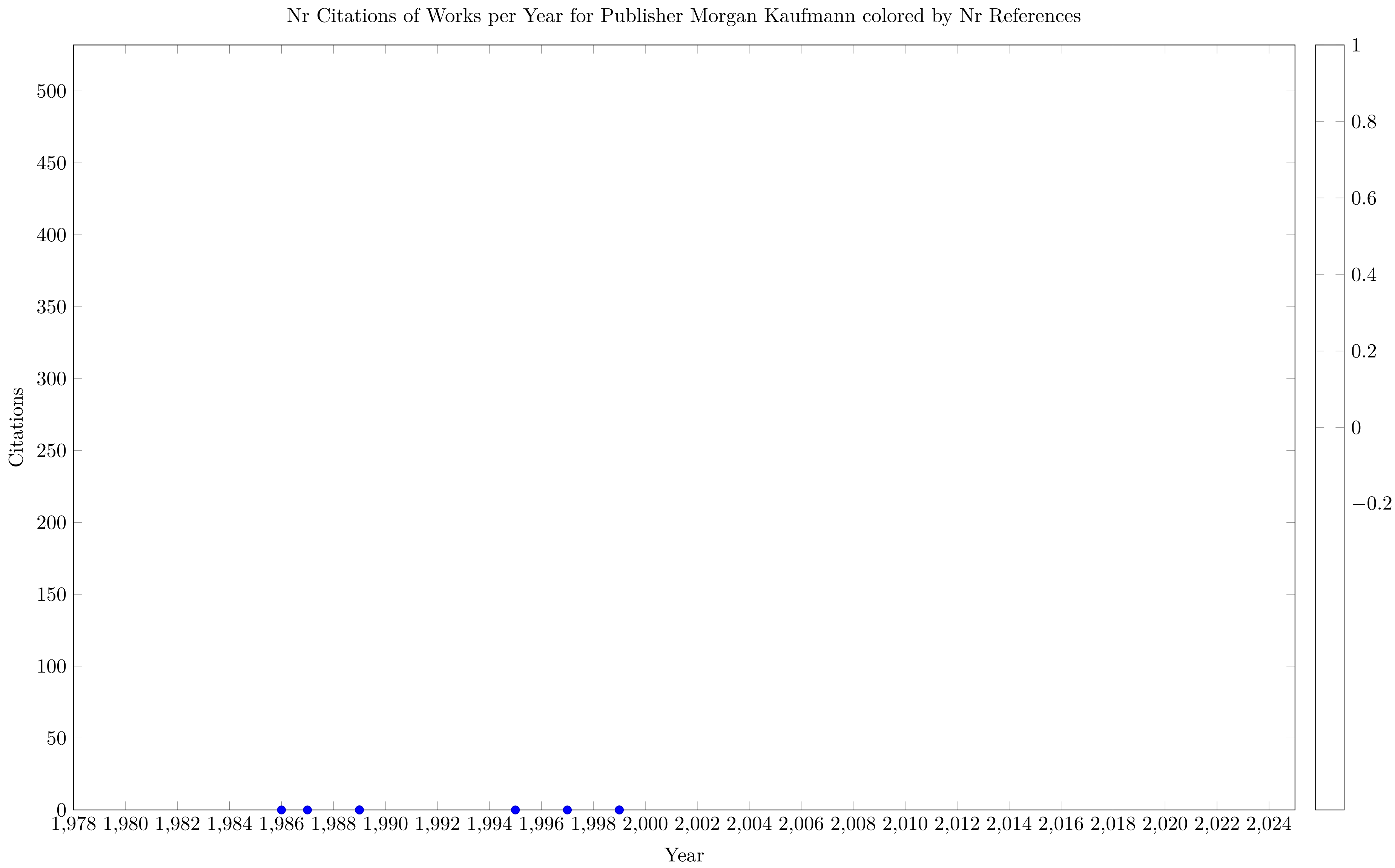
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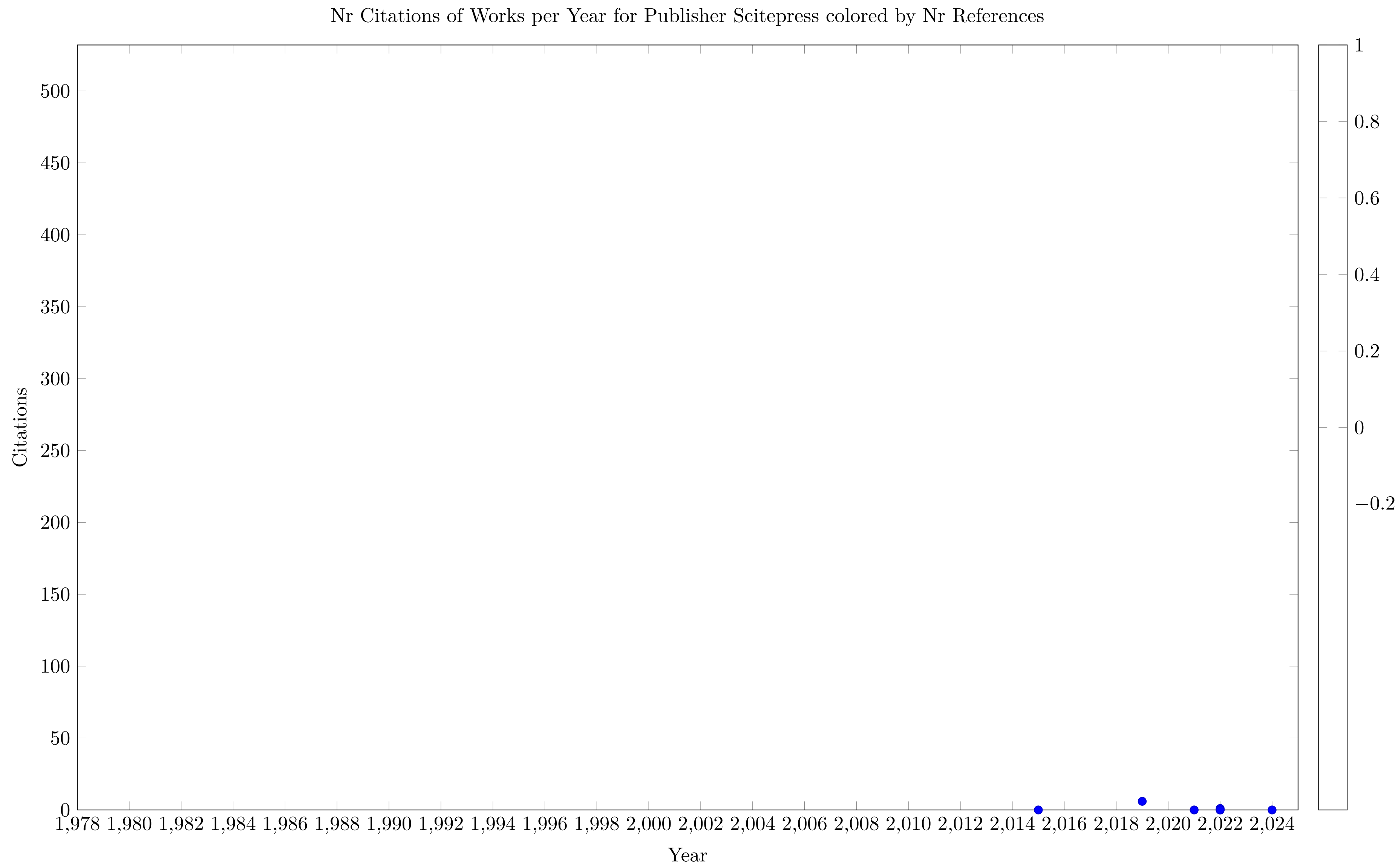
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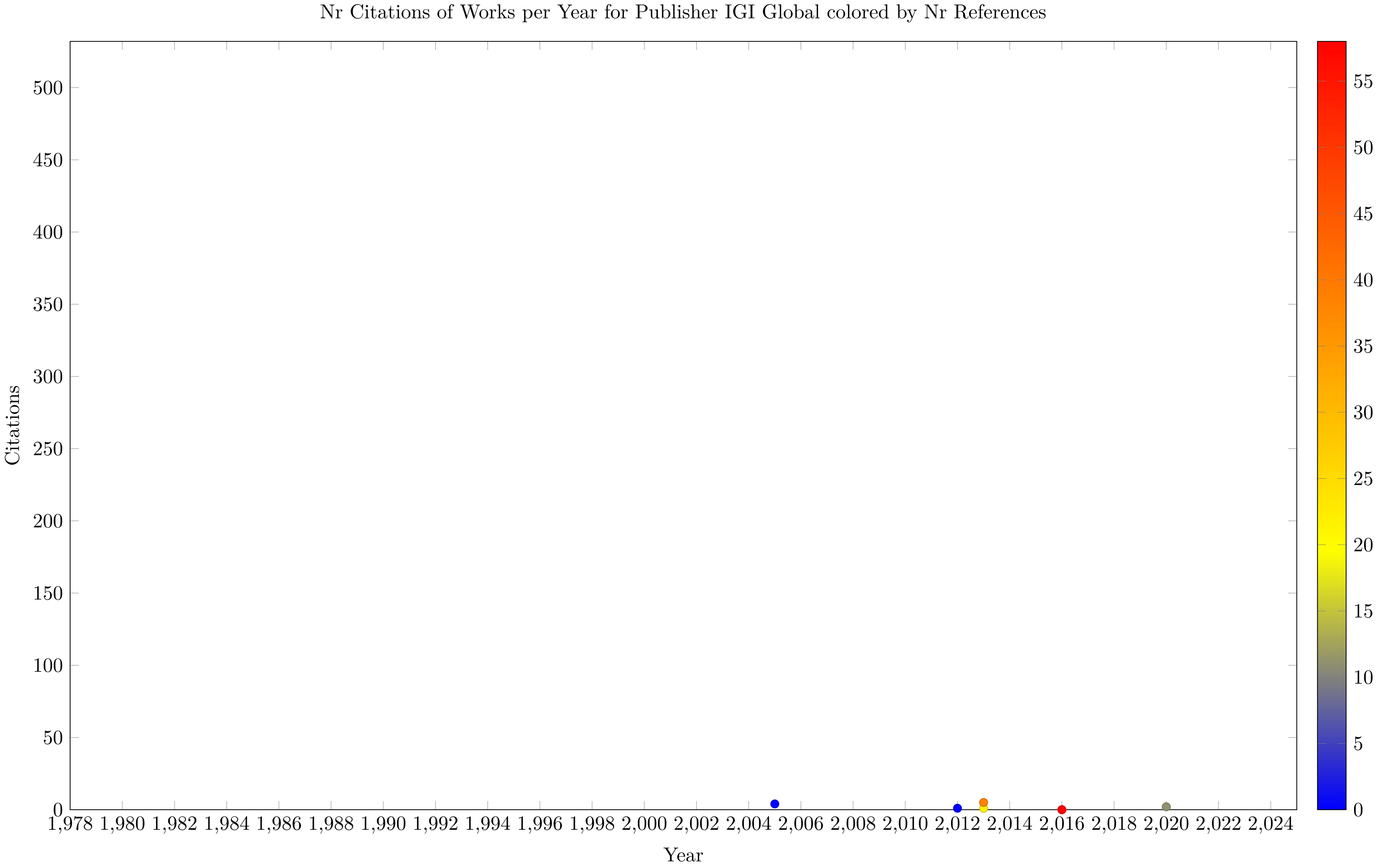
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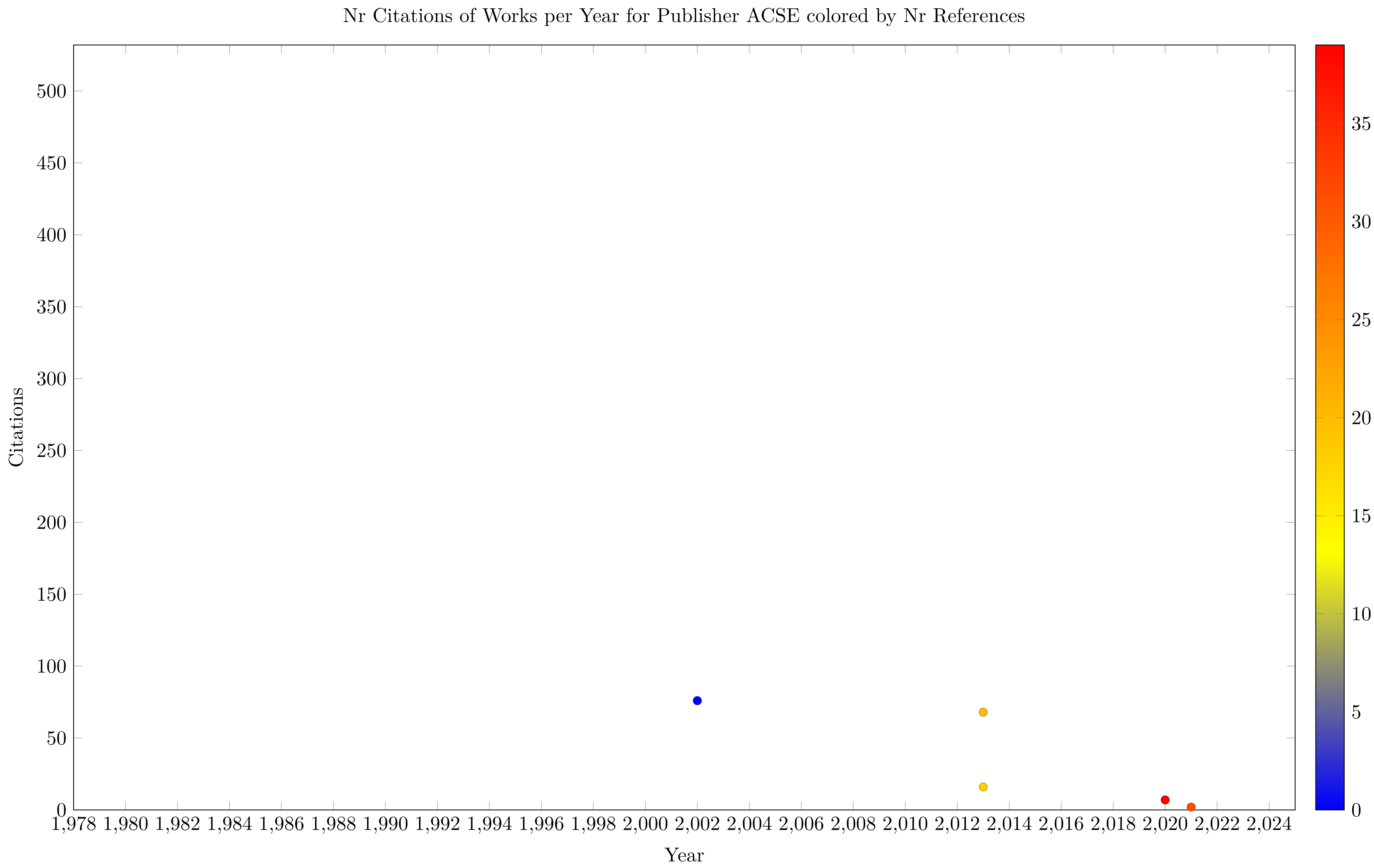
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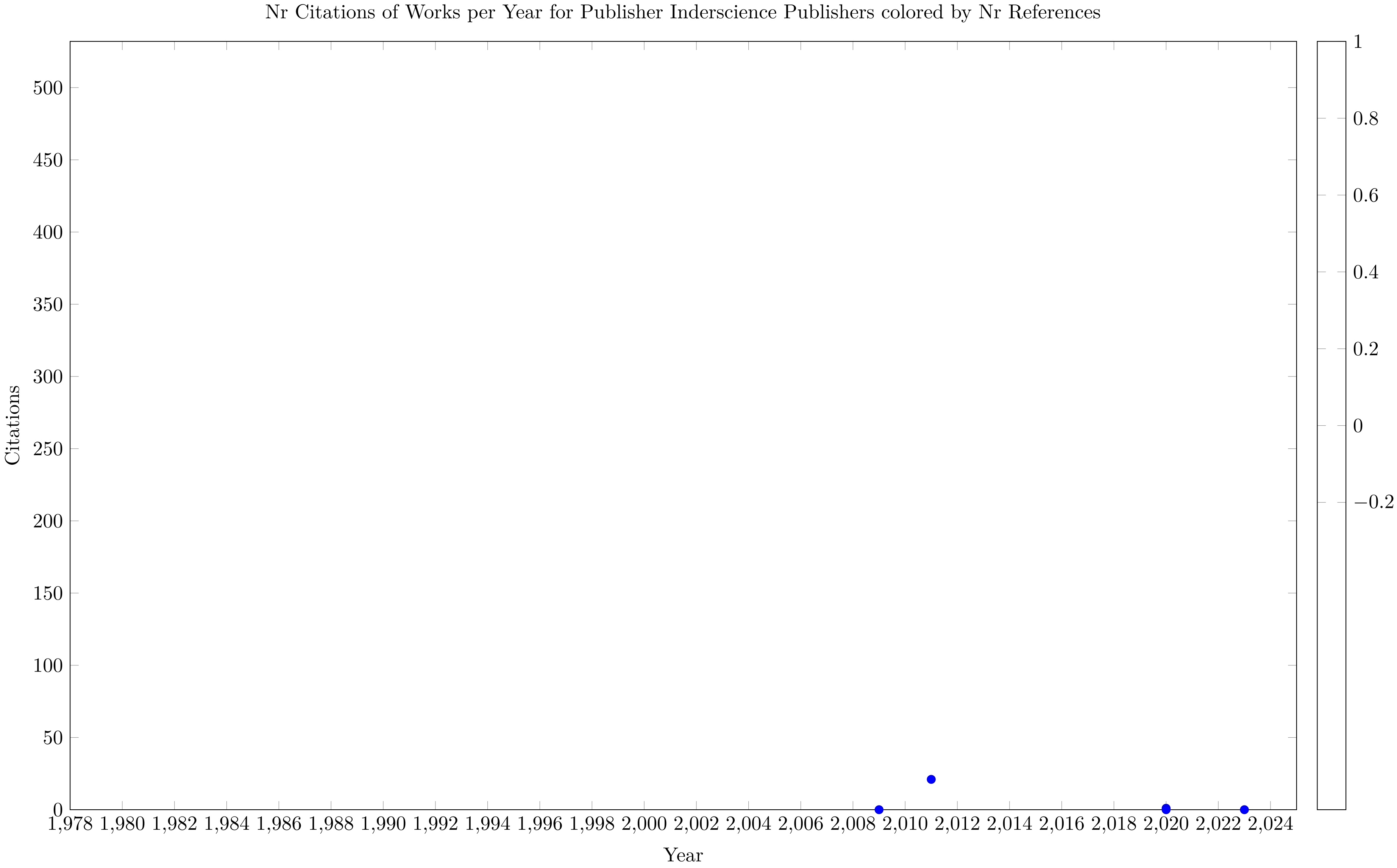
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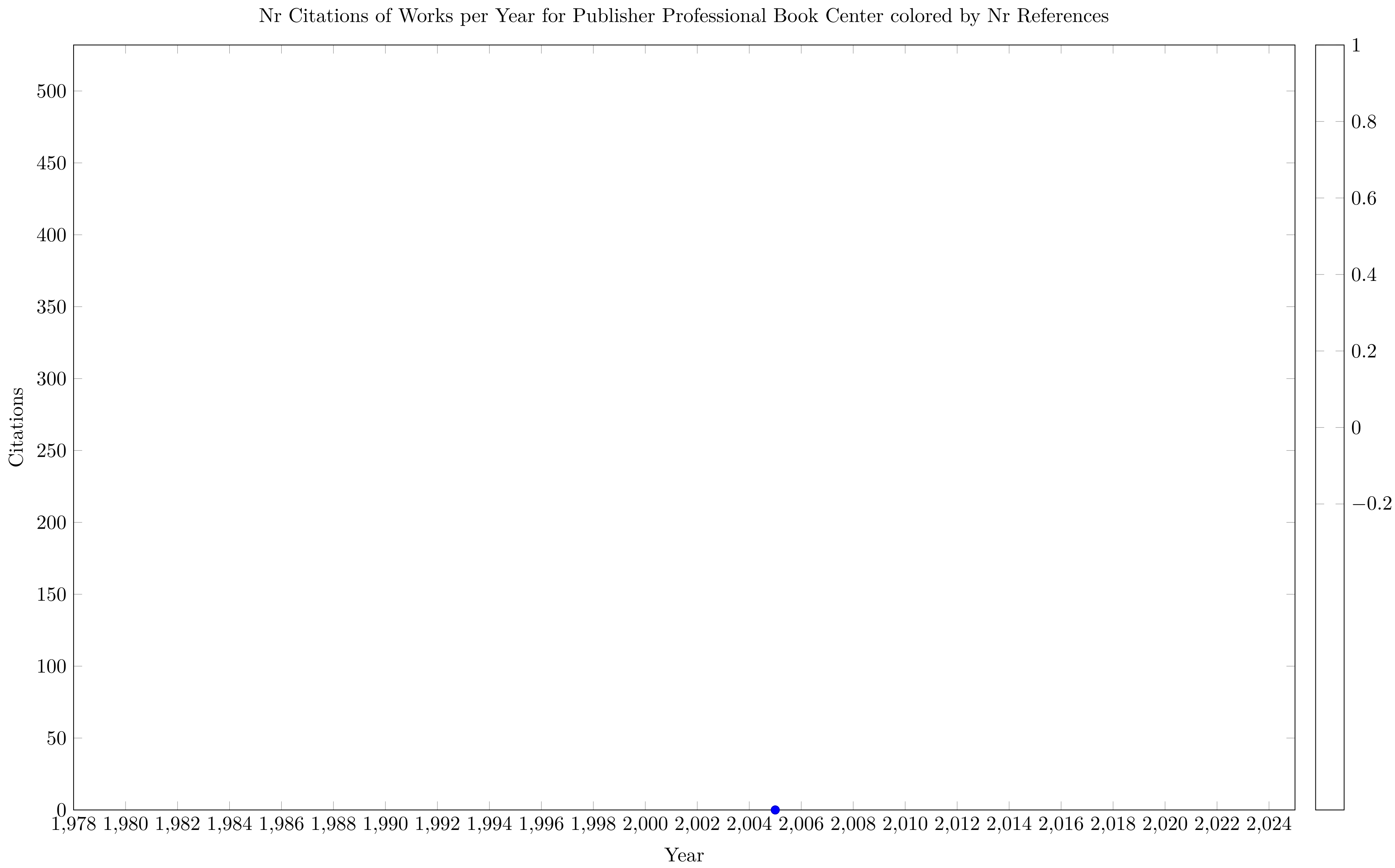
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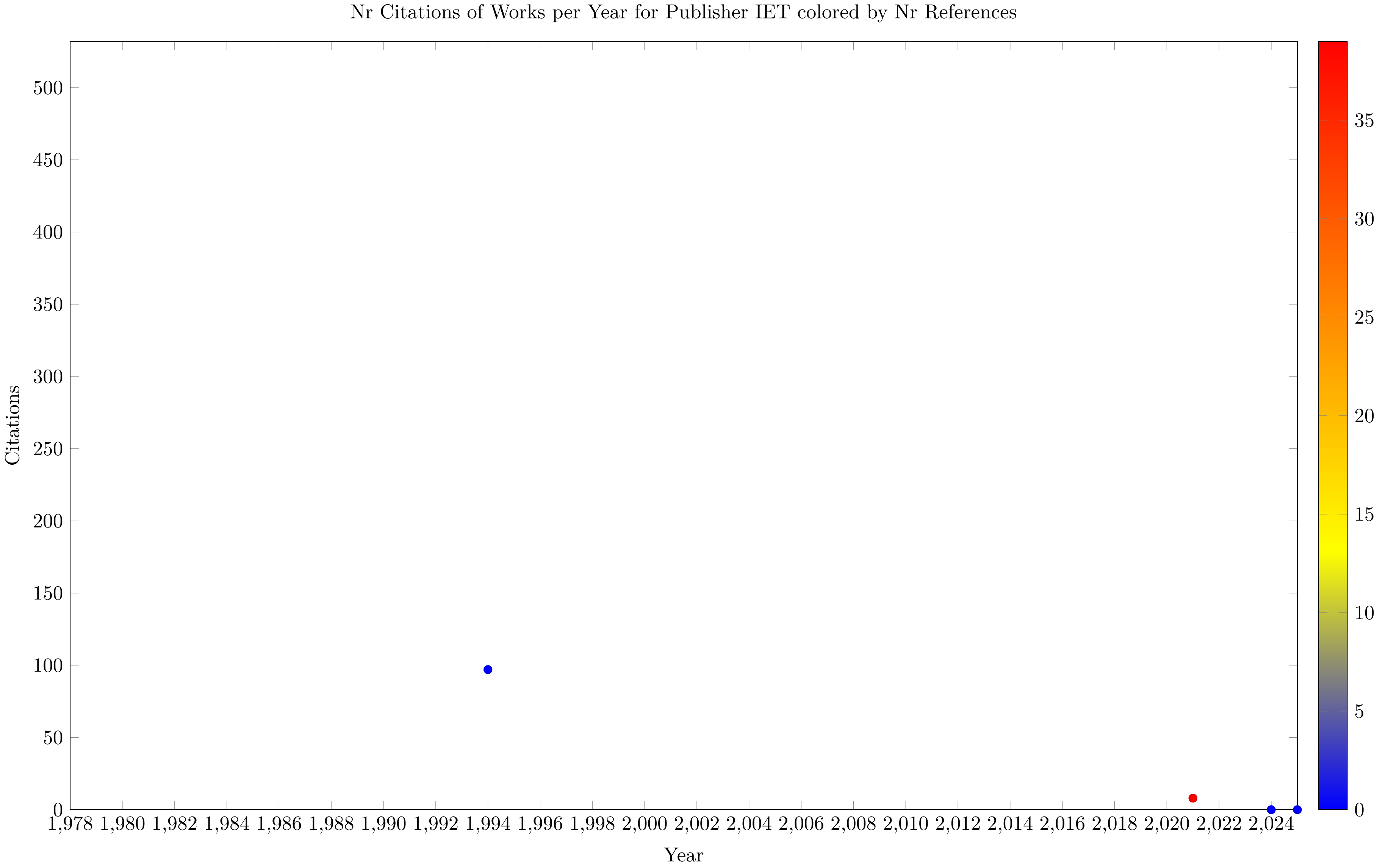
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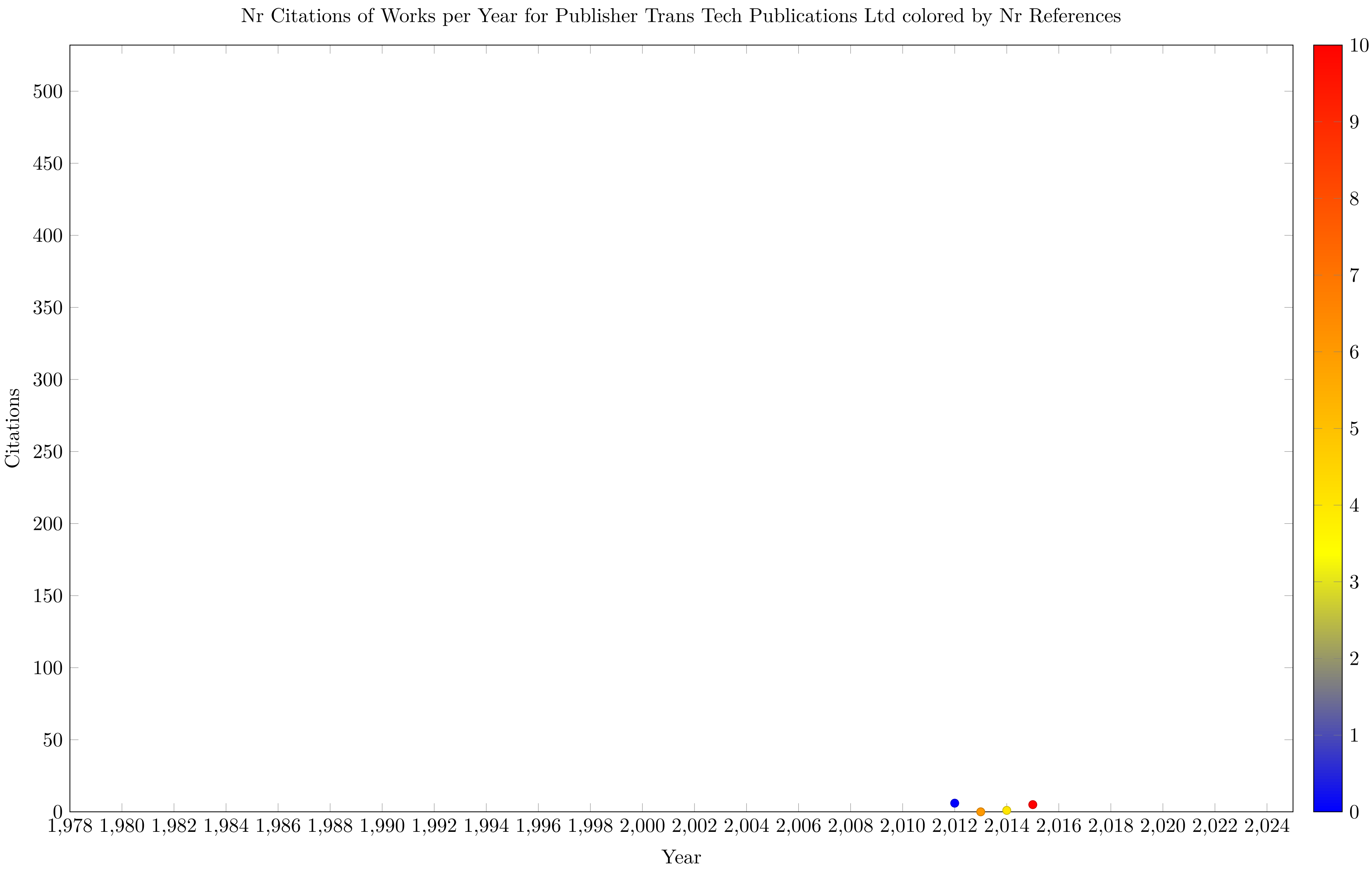
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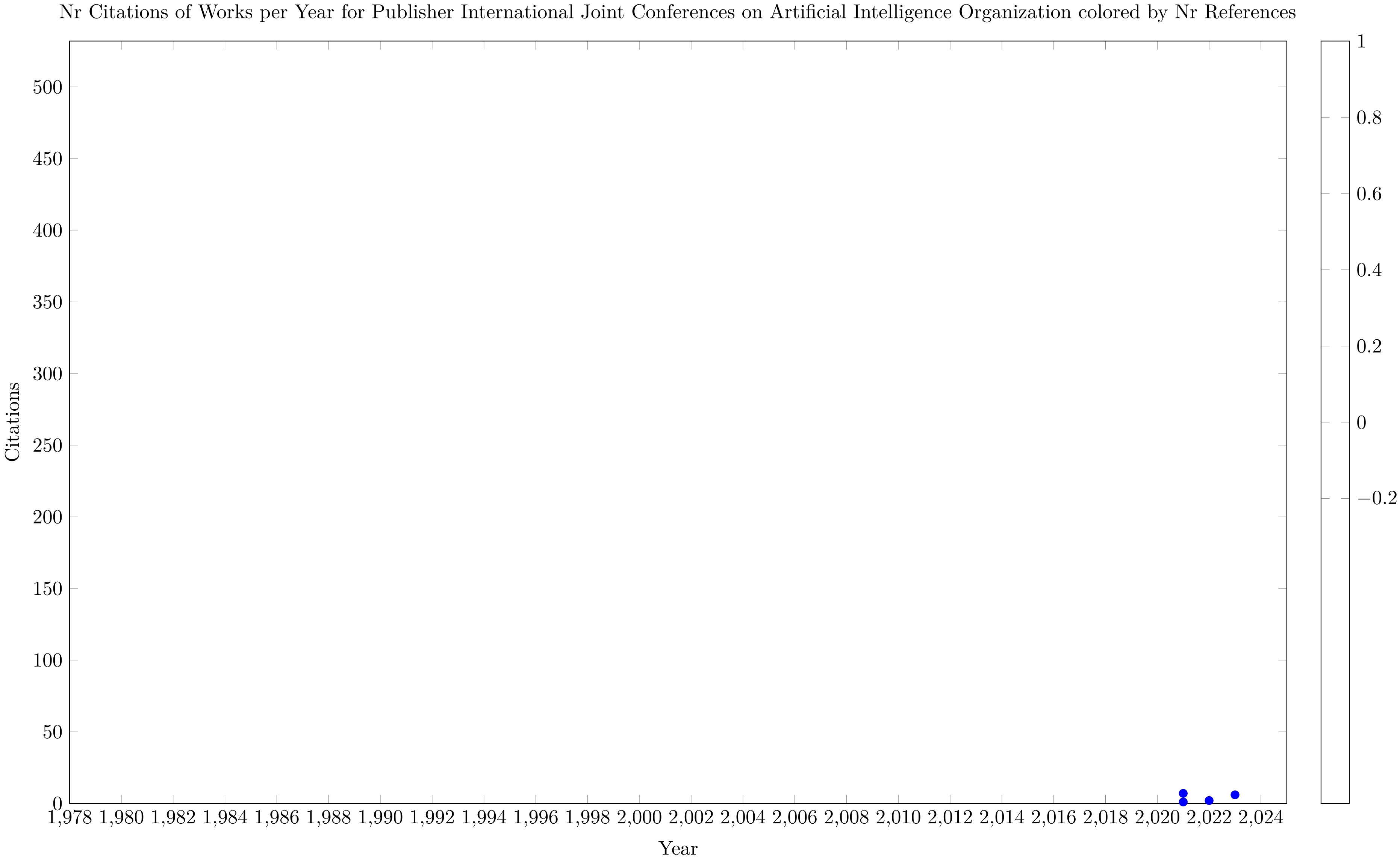
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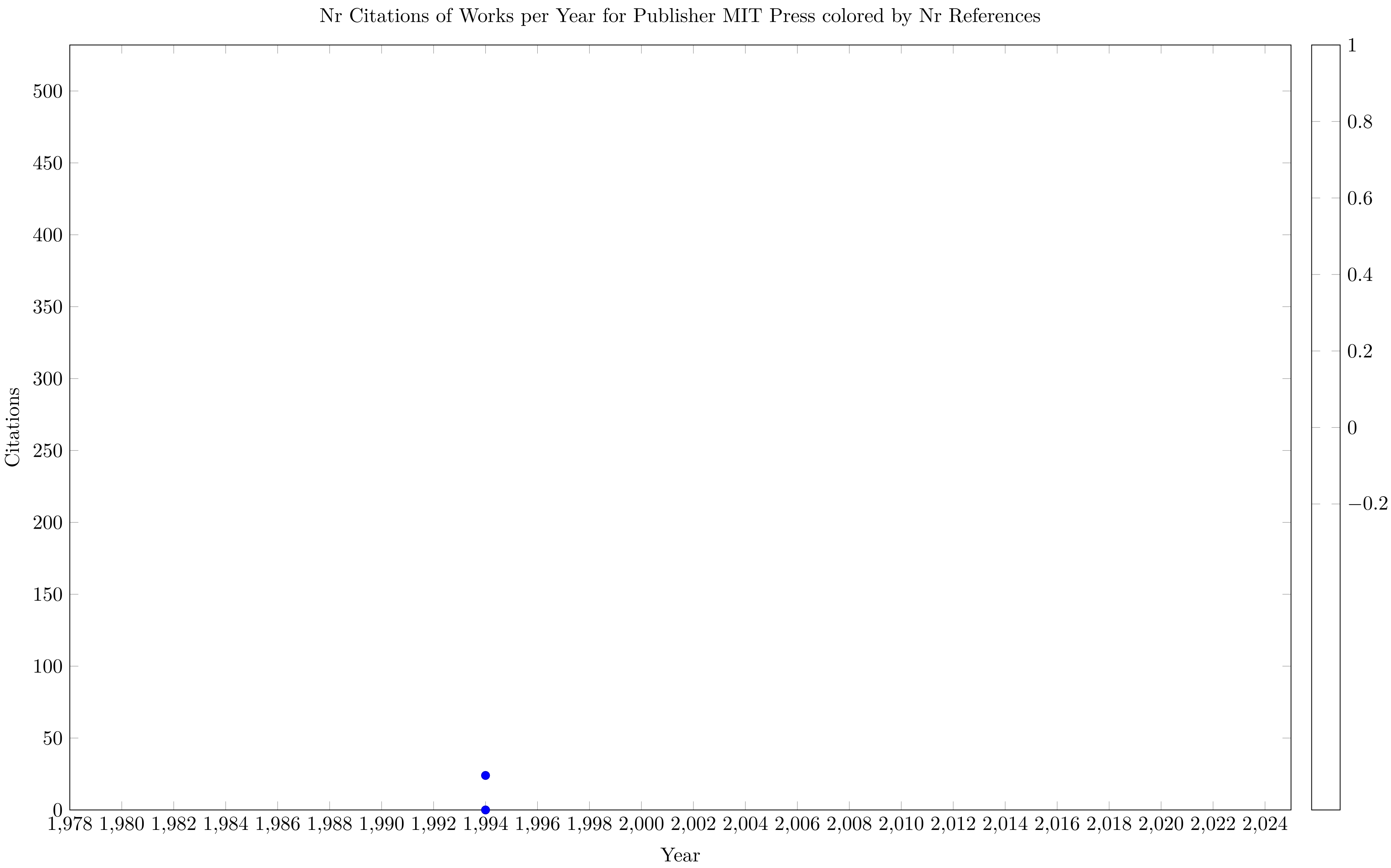
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